NOT MEASUREMENT SENSITIVE

MIL-HDBK-850
21 January 1994
SUPERSEDING
GLOSXMCGTERMS
FOURTH EDITION 1981

MILITARY HANDBOOK

Glossary of Mapping, Charting, and Geodetic Terms

AMSC N/A AREA MCGT

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This military handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DMA(TIJ), Stop A-10, 8613 Lee Highway, Fairfax, Virginia, 22031-2137, by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. The purpose of this glossary is to provide a comprehensive and authoritative source of current usage of mapping, charting, and geodetic terms for all levels of users to help them communicate effectively. This edition of the glossary in addition to the main glossary of terms, includes a list of MC&G related abbreviations, acronyms, and initials commonly used within DoD. It also includes a section on Mapping, Charting & Geodesy organizations.

4. The terms and definitions in this publication were selected from authoritative glossaries and dictionaries, and from technical publications and papers concerned with the many disciplines associated with mapping, charting, and geodesy. Numerous changes, additions, and deletions were made after a thorough review by Defense Mapping Agency components and by the Departments of the Army, Navy, and Air Force.

5. **This publication is not a substitute** for the Department of Defense Dictionary of Military and Associated Terms (JOINT PUB 1-02), formerly JCS PUB 1, which the Secretary of Defense has directed to be used throughout the Department of Defense. Terms included herein which are designated "(JCS)" were extracted from and defined as stated in JOINT PUB 1-02. In some instances, the JCS definition has been expanded to include more detailed or supplementary information. This additional matter is set off by brackets and is not to be construed as changing or conflicting with the JCS definition. JCS terms which have been accepted by NATO and by the Inter-American Defense Board are so designated in JOINT PUB 1-02. Only those NATO terms which do not appear in JOINT PUB 1-02 are so indicated in this glossary.

6. The designation "(USPLS)" indicates U.S. Public Land Survey terms similarly defined by the Bureau of Land Management, U.S. Department of Interior.

7. Alphabetization of terms in this glossary follows the standard A through Z order, except that multiword terms are alphabetized according to the initial word.

8. Multiple definitions for a single term are numbered and, wherever applicable, are identified with the appropriate science, discipline, or function in parentheses. At the end of some definitions the user's attention is directed to related terms by the expression "See also."

9. When two or more terms have identical meaning, the definition has been applied only to the preferred term, followed by the expression "Also called" and a list of the synonyms. The synonyms are shown in alphabetical order in the glossary, and are referenced to the preferred term. Antonyms are listed after the expression "Opposite of."

Any mention herein of a commercial product does not constitute endorsement by the United States Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
1.	SCOPE	1
1.1	Scope	1
1.2	Purpose	1
1.3	Applicability	1
1.4	Classification	1
2.	APPLICABLE DOCUMENTS	2
2.1	Government Documents	2
2.1.1	Specifications, standards , and handbooks	2
2.1.2	Other government documents, drawings, and publications	2
2.2	Non-Government publications	2
2.3	Order of precedence	2
3.	TERMS AND DEFINITIONS	3
4.	LIST OF ABBREVIATIONS, ACRONYMS, AND INITIALS	285
5.	MC&G ORGANIZATIONS	292
	INDEX	294
	CONCLUDING MATERIAL	295

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This handbook provides a comprehensive glossary of Mapping, Charting, and Geodesy (MC&G) terms commonly used within the Department of Defense.

1.2 Purpose. This Glossary applies uniformly to designers, producers, and users of MC&G products.

1.3 Applicability. These terms apply to both internal and contractual development efforts by the Military Departments, Office of the Secretary of Defense, Organization of the joint Chiefs of staff, and the Defense Agencies of the Department of Defense (DoD), collectively known as DoD Components, and to all levels involved in the preparation and maintenance of MC&G products.

1.4 Classification. This handbook is UNCLASSIFIED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. This section is not applicable to this handbook.

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

JOINT PUB 1-02 Department of Defense Dictionary of Military and Associated Terms.

(Copies of this publication are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2 Non-government publications. This section is not applicable to this handbook.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

3. TERMS AND DEFINITIONS

A

A1 time—A particular atomic time scale, established by the U.S. Naval Observatory, with the origin on 1 January 1958, at zero hours UT2 and with the unit (second) equal to 9,192,631,770 cycles of cesium at zero field. See also UT2 time.

ABAC—A nomogram for obtaining the conversion angle to apply when plotting great-circle bearings on a Mercator projection.

abnormal magnetic variation—Any anomalous deflection, whose cause is unknown, of the compass needle from the magnetic meridian.

abscissa—The horizontal coordinate of a set of rectangular coordinates. Also used in a similar sense in connection with oblique coordinates. Also called **total departures; *x*-coordinate.**

absolute accuracy—The evaluation of all errors in determining a position with respect to an absolute reference frame, such as the DoD World Geodetic System.

absolute error—Absolute deviation, the value taken without regard to sign, from the corresponding true value.

absolute gravity station—A marked point, usually in a laboratory, where the value of absolute gravity has been determined. See also **absolute gravity.**

absolute gravity—The acceleration of gravity directly determined by a device that measures time and length. See also **gravity.**

absolute orientation—The scaling and leveling to ground control (in a photogrammetric instrument) of a relatively oriented stereoscopic model or group of models. See also **relative orientation.**

absolute parallax—See **absolute stereoscopic parallax.**

absolute positioning—Determination of the position of a point with respect to the center of mass of the Earth as defined in the DoD World Geodetic System.

absolute stereoscopic parallax—Considering a pair of aerial photographs of equal principal distance, the absolute stereoscopic parallax of a point is the algebraic difference of the distances of the two images from their respective photograph nadirs, measured in a horizontal plane and parallel to the air base. Also called **absolute parallax; horizontal parallax; linear parallax; parallax; stereoscopic parallax; *x*-parallax.**

absolute term—A term (usually only one) in an equation, which represents a known numerical value and does not contain any unknown or variable elements.

absolute unit—Any unit in a system that is based directly upon associated fundamental units of length, mass, and time. See also **dynamic number.**

absolute value—A mathematical quantity taken without regard to its associated plus or minus sign. Used often with residuals.

absolute vector—A directed line segment whose end points are measured in absolute units from a point designated as the origin.

absorption—Conversion of radiant energy into other forms by passage through or reflection from matter.

acceleration of gravity—The acceleration of a freely falling body, approximately $9.8 m/sec^2$.

acceleration—1. The rate of change of velocity. 2. The act or process of accelerating, or the state of being accelerated.

accelerometer—1. A device that measures the rate of change of speed of an object. 2. An instrument, specially designed for carrying in aircraft or missiles, which measures the rate of change in velocity, direction, and/or altitude.

accidental error—See **random error.**

accommodation—1. The ability of the human eye to adjust itself to give sharp images of objects of different distances. In stereoscopy, the ability of the human eyes to bring two images into superimposition for stereoscopic viewing. 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

The limits or range within which a stereoplotting instrument is capable of operating. For example, the multiplex can adjust (or accommodate) for small tilts in the projectors ranging from approximately 10° about the *x*-axis to 20° about the *y*-axis.

accumulated discrepancy—The algebraic sum of the separate discrepancies which occur in the various steps of making a survey or of the computation of a survey.

accumulated divergence—(leveling) The algebraic sum of the divergences for the sections of a line of levels, from the beginning of the line to any section end at which it is desired to compute the total divergence.

accumulative error—See **systematic error.**

accuracy checking—The procurement of presumptive evidence of a map's compliance with specified accuracy standards. Accuracy checking generally indicates the relative (rather than the absolute) accuracy of map features.

accuracy evaluation—The comparison of the quality of an MC&G product with maintenance criteria to ascertain its adequacy with respect to its intended use.

accuracy method—The method used in determining the stated accuracy of a product. The method can range from a system employing highly sophisticated techniques to a highly subjective judgment, and is essential for proper use of the stated accuracy.

accuracy review—The comparison of an existing MC&G product against source material or data more accurate than that from which it was produced, for the purpose of determining the accuracy of its horizontal and vertical values.

accuracy testing—The procurement of confirmed evidence, on a sampling basis, of a map's compliance with specified accuracy standards. Accuracy testing is designed to indicate both the relative and absolute accuracy of map features.

accuracy—**1.** The degree of conformity with a standard, or the degree of perfection attained in a measurement. Accuracy relates to the quality of a result, and is distinguished from precision, which relates to the quality of the operation by which the result is obtained and can be repeated. **2.** The closeness of the best estimated value obtained by the measurements to the "true" value of the quantity measured.

accurate contour—A contour line, the accuracy of which lies within one-half of the basic vertical interval. Also called **normal contour.**

acetate—A nonflammable plastic sheeting used as a base for photographic films or as a drafting base for overlays where critical registration is not required.

achromatic color—Color that does not elicit hue.

achromatic lens—A lens that has been partly corrected for chromatic aberration. Such a lens is usually of a multi-element design to bring green and red light rays to approximately the same point of focus.

acoustic navigation—Navigation by means of sound waves whether or not they are within the audible range. Also called **sonic navigation.** See also **Doppler sonar navigation.**

actinic light—Light which is capable of causing photochemical changes in a sensitized emulsion.

active satellite—A satellite which transmits an electromagnetic signal. A satellite with the capability to transmit, repeat, or retransmit electromagnetic information. See also **passive satellite.**

active tracking system—A satellite tracking system which operates by transmission of signals to and receipt of responses from the satellite.

actual error—The difference between the true value and the measured value of a physical quantity.

acutance—An objective measure of the ability of a photographic system to show a sharp edge between contiguous areas of low and high illuminance.

adaptation—The faculty of the human eye to adjust its sensitivity to varying intensities of illumination.

additive color mixture— Super- imposition or other nondestructive combination of light of different chromaticities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

additive color viewer—Projector for positive transparencies obtained through multiband photography. Each image is superimposed by use of a different colored light.

additivity of luminance—Luminance produced with a mixture of light from several sources is the sum of the luminances produced by the light from lack of the sources acting separately.

adequate—A term used to describe a product which meets all of the accuracy and currency standards established by its most stringent use, and thus, is suitable for all its intended uses.

adjacency—A topological property which provides relationships between objects which abut or are located in close proximity. Also known as contiguity.

adjoining sheets—Adjacent maps to one or all sides and corners of a particular map sheet.

adjusted mapping support data (AMSD)—The post-mission refined version of those parameters used to position collector platforms, sensors, and sensor pointing angles.

adjusted value—A value of a quantity derived from observed data by some orderly process which eliminates discrepancies arising from errors in those data.

adjustment of observations—The determination and application of corrections corresponding to errors affecting the observations, making the observations consistent among themselves, and coordinating and correlating the derived data.

adjustment—1. (general) The determination and application of corrections to observations, for the purpose of reducing errors or removing internal inconsistencies in derived results. The term may refer either to mathematical procedures or to corrections applied to instruments used in making observations. 2. (leveling) The determination and application of corrections to orthometric differences of elevation or to orthometric elevations, to make the elevation of all bench marks consistent and independent of the circuit closures. 3. (cartography) Placing detail or control stations in their positions relative to other detail or control stations. See also **adjustment of observations; angle method of adjustment; balancing a survey; direction method of adjustment; figure adjustment; instrument adjustment; land-line adjustment; least squares; map adjustment.**

administrative map—(JCS) A map on which is graphically recorded information pertaining to administrative matters, such as supply and evacuation installations, personnel installations, medical facilities, collecting points for stragglers and prisoners of war, train bivouacs, service and maintenance areas, main supply roads, traffic circulation, boundaries, and other details necessary to show the administrative situation. See also **map.**

Advanced Very High Resolution Radiometer (AVHRR)—A five to six channel sensor with a resolution of 11 km. Applications include cloud temperature, sea surface temperature, land temperature, and vegetation indices.

Advanced Weapons and Systems Data Base (AWSDB)—A DMA relational data base used to assess and consolidate the MC&G product requirements for fielded, new, and emerging DoD weapons and systems.

aerial camera—A camera specifically designed for use from an airborne station.

aerial cartographic photography—See **mapping photography.**

aerial film speed (AFS)—A measure of speed for aerial film which replaces the formerly used aerial exposure index. It is defined as 3/2E, where E is the exposure in meter-candle-seconds at the point on the characteristic curve where the density is 0.3 above base plus fog density on black-and-white film.

aerial film—Specially designed roll film supplied in many lengths and widths, with various emulsion types for use in aerial cameras.

aerial imagery—Any remotely-sensed nonphotographic image taken from the air (as opposed to space).

aerial mosaic—See **mosaic,** *definition* 1.

aerial photogrammetry—The use of aerial photographs in the science of photogrammetry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

aerial photographic reconnaissance—The obtaining of information by aerial photography—divided into three types: (1) strategic photographic reconnaissance; (2) tactical photographic reconnaissance; and (3) survey/cartographic photography, which is aerial photography taken for survey/cartographic purposes and to survey/cartographic standards of accuracy.

aerial photography—The art, science, or process of taking aerial photographs. See also **mapping photography; reconnaissance photography.**

aerial photograph—Any photograph taken from the air. Also called **air photograph.**

aerial platform—A term referring to the support of an aerial camera at the air station.

aerial reconnaissance—The collection of information by visual, electronic, photographic, or other means from the air.

aerial survey—A survey utilizing photographic, electronic, or other data obtained from an airborne station.

aerial triangulation—See **phototriangulation.**

aerodetic—(JCS) Of or pertaining to, or determined by aerodesy.

aerolevelling—As applied to model orientation during phototriangulation, barometric height measurements of the camera air stations which have been recorded during the photographic mission are used to present the *Bz* values during the orientation of the successive models on the stereoplotting instrument. Only differences in altitude are required and these are provided by the statoscope. See also **orientation,** definition 7.

aerometeorograph—An instrument that records the pressure and temperature of the air, the amount of moisture in the air, and the rate of motion of the wind.

Aeronautical Data Maintenance (ADM)—A production element of DMA's DPS which involves the processes of collecting, compiling, and updating. See also **Digital Production System.**

aeronautical chart—(JCS) A specialized representation of mapped features of the Earth, or some part of it, produced to show selected terrain, cultural, and hydrographic features, and supplemental information required for air navigation, pilotage, or for planning air operations. Also called **navigation chart.**

aeronautical information overprint— (JCS) Additional information which is printed or stamped on a map or chart for the specific purpose of air navigation.

aeronautical pilotage chart—An aeronautical chart designed primarily for air navigation.

aeronautical planning chart—An aeronautical chart of small scale designed to satisfy long-range air navigation and mission planning requirements.

aeropause—(JCS) Region in which functional effects of the atmosphere on man and aircraft cease to exist.

aerospace—(JCS) Of, or pertaining to, the Earth's envelope of atmosphere and the space above it; two separate entities considered as a single realm for activity in launching, guidance, and control of vehicles which will travel in both entities.

aerotriangulation — See **phototriangulation.**

affine deformation—One in which the scale along one axis or reference plane is different from the scale along the other axis or plane.

affine transformation—A transformation in which straight lines remain straight and parallel lines parallel. Angles may undergo changes and differential scale changes may be introduced.

age of diurnal inequality—The time interval between the maximum semimonthly north or south declination of the Moon and the time that the maximum effect of the declination upon the range of tide or speed of the tidal current occurs. Also called **age of diurnal tide; diurnal age.**

age of diurnal tide—See **age of diurnal inequality.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

age of parallax inequality—The time interval between the perigee of the Moon and the maximum effect of the parallax (distance of the Moon) upon the range of tide or speed of tidal current. Also called **parallax age.**

age of phase inequality—The time interval between the new or full Moon and the maximum effect of these phases upon the range of tide or speed of tidal current. Also called **age of tide; phase age.**

age of the Moon—The elapsed time, usually expressed in days, since the last new Moon.

age of tide—See **age of phase inequality.**

agglomeration—A generalization process that groups two or more proximate features to form a single feature.

aggregation operations—The process of bringing together many distinct parts or categories of data into one grouping, usually as a composite display.

aggregation—The combining of data collected at point locations or smaller statistical units into larger units.

agonic line—(JCS) A line drawn on a map or chart joining points of zero magnetic declination for a specified year date. In nautical and aeronautical navigation, the term **magnetic variation** is used instead of **magnetic declination.**

aiming line—See **line of sight**, definition 2; **line of collimation.**

Air Almanac—A joint publication of the United States Naval Obeervatory and Her Majesty's Nautical Almanac Office. It covers a 6-month period. It contains tabulated values of the Greenwich hour angle and declination of selected celestial bodies, plus additional celestial data used in navigation.

Air Target Chart—(JCS) A display of pertinent air target intelligence on a specialized graphic base. It is designed primarily to support operations against designated air targets by various weapon systems. [The charts provide graphic overprint and textual data relative to radar return information and installations within the area. Air Target Charts are prepared at various scales and are produced under the Air Target Materials Program as a series of geographically integrated charts.]

Air Target Materials Program (ATMP)—(JCS) A DoD program under the management control of the Defense Mapping Agency established for and limited to the production of medium- and large-scale map, chart, and geodetic products which support worldwide targeting requirements of the unified and specified commands, the military departments, and allied participants. It encompasses the determination of production and coverage requirements, standardization of products, establishments of production priorities and schedules, and the production, distribution, storage, and release/exchange of products included under it.

air base—**1.** (photogrammetry) The line joining two air stations, or the length of that line. **2.** The distance, at the scale of the stereoscopic model, between adjacent perspective centers as reconstructed in the plotting instrument. See also **air station.**

air coordinates—See **rectangular space coordinates.**

air photograph—See **aerial photograph.**

air plot—(JCS) **1.** A continuous plot used in air navigation of a graphic representation of true headings steered and air distances flown. **2.** A continuous plot of the position of an airborne object represented graphically to show true headings steered and air distances flown. **3.** Within ships, a display which shows the positions and movements of an airborne object relative to the plotting ship.

air station—(JCS) In photogrammetry, the point in space occupied by the camera lens at the moment of exposure. Also called **camera station.** See also ***air base.***

air surveillance plotting board—(JCS) A gridded, small scale, air defense map of an appropriate area. It is maintained at the air control center. On it are posted current locations, number, and altitudes of all friendly or enemy aircraft within range of radar or ground observer facilities.

air target materials—See **target materials.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Airborne Control (ABC) system—A survey system for horizontal and vertical control surveys involving electromagnetic distance measurements and horizontal and vertical angle measurements from two or more known positions to a helicopter hovering over the unknown position. The elevation of the unknown position is determined by the use of a special plumb line cable.

Airborne Profile Recorder (APR)—See **Terrain Profile Recorder.**

airborne electronic survey control—Control surveys accomplished by electronic means from an airborne vehicle or platform, such as hiran and shoran.

airborne landing model—A specially designed assault model for use in briefing airborne troops and support personnel. These models emphasize the aspects of objects as seen from the air rather than from the ground.

airway—(JCS) A control area or portion thereof established in the form of a corridor marked with radio navigational aids.

Airy spheroid (ellipsoid)—A reference ellipsoid used in Great Britain and having the following dimensions: semimajor axis—6,377,563.396 meters; semiminor axis—6,356,256.910 meters; and the flattening or ellipticity—1/299.3249646.

Airy theory of isostasy—The theory that the continents and islands are resting hydrostatically on highly plastic or liquid material, with roots or projections penetrating the inner material of the Earth, just as icebergs extend downward into the water. The greater the elevation, the deeper the penetration. It has been called the **roots of mountain theory**, and has the support of some geologists. See also **Pratt-Hayford theory of isostasy.**

Aitoff equal-area map projection—A Lambert equal-area azimuthal projection of a hemisphere converted into a map projection of the entire sphere by a manipulation suggested by Aitoff. It is a projection bounded by an ellipse in which the line representing the Equator (major axis) is double the length of the line representing the central meridian (minor axis).

albedo—The ratio of radiant energy reflected to that received by a surface, usually expressed as a percentage; reflectivity. The term generally refers to energy within a specific frequency range, as the visible spectrum. Its most frequent application is to the light reflected by a celestial body.

Albers conical equal-area map projection—An equal-area projection of the conical type, on which the meridians are straight lines that meet in a common point beyond the limits of the map, and the parallels are concentric circles whose center is at the point of intersection of the meridians. Meridians and parallels intersect at right angles and the arcs of longitude along any given parallel are of equal length. The parallels are spaced to retain the condition of equal area. On two selected parallels, the arcs of longitude are represented in their true length. Between the selected parallels the scale along the meridians will be a trifle too large, and beyond them, too small.

albumin (albumen) process—A process of making photolithographic press plates utilizing bichromated albumin as the photosensitive coating, and requiring a true negative to make the printing plate. See also **plate.**

Aldis signaling lamp—A signaling lamp used in some cases for night observations of distant stations in triangulation.

alerts—An ephemeris prepared for one or more satellites, predicting rise and set times referred to universal time coordinated, maximum angle of elevation above the observer's horizon, and azimuth from the observer. Used to identify specific satellite passes. See also **look angles.**

algorithm—A statement of the steps to be followed in the solution of a problem.

alidade—*The part of a surveying instrument* which consists of a sighting device, with index, and reading or recording accessories. See also **peepsight alidade; pendulum alidade; photoalidade; telescopic alidade.**

alignment (alinement)—**1.** (cartography) Representation of a road, railway, etc., on a map or chart in relation to surrounding topographic detail. **2.** (general surveying) The placement or location of points along a straight line. **3.** (highway and route surveying) The ground plan showing the direction (center line) of the route to be followed, as distinguished from **profile,** which shows the vertical element.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

alignment correction—(taping) A correction applied to the measured length of a line to allow for the tape not being held exactly in a vertical plane containing the line.

almanac—A periodical publication of astronomic coordinates useful to a navigator. It contains less information than an ephemeris and values are generally given to less precision. See also **ephemeris.**

almucantar—See **parallel of altitude.**

along-track (A/T)—The direction of the tangent to a locus of points on the earth, a constant distance from the satellite ground path. Positive in the direction of satellite motion.

alphanumeric grid—See **atlas grid.**

alphanumeric string—A string of information consisting of both letters and numbers, and possible including other symbols such as punctuation marks and mathematical symbols. Frequently includes other select characters such as punctuation marks and mathematical symbols.

altazimuth instrument—An instrument equipped with both horizontal and vertical graduated circles, for the simultaneous observation of horizontal and vertical directions or angles. Also called **astronomic theodolite; universal instrument.**

alternate depository—A file of originals, duplicate copies, computer tapes, reproduced materials, etc., of selective, current, and evaluated MC&G data indexed and stored at an appropriately secure location, physically separated from basic libraries, but available for use in the event of destruction of the primary DoD library file.

altimeter—An instrument that indicates the height above a reference surface. See also **barometric altimeter; precision altimeter; radar altimeter; surveying altimeter.**

altimetry—The art and science of measuring altitudes by barometric means and interpreting the results.

altitude circle—See **parallel of altitude.**

altitude contour ratio—See **C-factor.**

altitude datum—(JCS) The arbitrary level from which vertical displacement is measured. The datum for height measurement is the terrain directly below the aircraft or some specified datum; for pressure altitude, the level at which the atmospheric pressure is 29.92 inches of mercury (1013.2 mbs); and for true altitude, mean sea level.

altitude difference—The difference between computed and observed altitudes, or between precomputed and sextant altitudes. Also called **altitude intercept.**

altitude hole—(JCS) The blank area at the origin of a radial display, on a radar tube presentation, the center of the periphery of which represents the point on the ground immediately below the aircraft. In side-looking airborne radar, this is known as the **altitude slot.**

altitude intercept—See **altitude difference.**

altitude slot—See **altitude hole.**

altitude tints—See **hypsometric tinting.**

altitude—1. The vertical distance of a point, or an object considered as a point, measured from a reference surface, as mean sea level (the geoid), ellipsoid, mean terrain. 2. Angular distance above the horizon; the arc of a vertical circle between the horizon and a point on the celestial sphere, measured upward from the horizon. See also **absolute altitude; angular altitude; apparent altitude; circummeridian altitudes; computed altitude; density altitude; elevation; ellipsoidal height; exmeridian altitude; flight altitude; geoidal height; high altitude; meridian altitude; negative altitude; observed altitude; orbital altitude; parallel of altitude; photo altitude; positive altitude; pressure altitude; radar altitude; sextant altitude; simultaneous altitudes; solar altitude; true altitude.**

American Standard Code for Information Interchange (ASCII)—A widely used ANSI standard code which uses seven bits to represent numbers, letters, and control characters. Since eight-bit codes are more common on computers than seven-bit codes, ASCII is commonly embedded in an eight-bit code. It is mainly used to transmit data between

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

digital devices.

amphibious assault landing model— See **assault landing model.**

amphidromic point—A no-tide or nodal point on a chart of cotidal lines from which the cotidal lines radiate.

amphidromic region—An area surrounding an amphidromic point in which the cotidal lines radiate from the no-tide point and progress through all hours of the tidal cycle.

amplitude of vibration—(pendulum) The length of the arc passed over by a pendulum in moving from its mean position to the position of maximum displacement.

amplitude—**1.** The maximum value of the displacement of a wave or other periodic phenomenon from a reference position. **2.** Angular distance north or south of the prime vertical; the arc of the horizon or the angle at the zenith between the prime vertical and a vertical circle, measured north or south from the prime vertical to the vertical circle. The term is customarily used only with reference to bodies whose centers are on the celestial horizon, and is prefixed "E" or "W," as the body is rising or setting, respectively, and suffixed "N" or "S" to agree with the declination. See also **compass amplitude; grid amplitude; magnetic amplitude; true amplitude.**

amplitude of vibration—**(pendulum)** The length of the arc passed over by a pendulum in moving from its mean position to the position of maximum displacement.

anaglyph—*A stereogram in which the two views* are printed or projected superimposed in complementary colors, usually red and blue. By viewing through filter spectacles of corresponding complementary colors, a stereoscopic image is formed.

analemma—**1.** A figure eight shaped diagram drawn across the Torrid Zone on a terrestrial globe to show the declination of the Sun throughout the year and also the equation of time. **2.** A sundial.

analog instruments—Devices that represent numerical quantities by means of physical variables, giving all values within a particular range, for example, by translation; by rotation, as in a mechanical gear system; and by voltage or current as in analog networks that use resistance to represent mechanical losses, capacitors and inductors to store energy and simulate the action of springs, etc. Stereoscopic plotters are examples of photogrammetric analog instruments.

analog—A form of data recording that works on the principle of continuous measurement, rather than discrete counting. For example a paper map with continuous lines is analog, a computer displayed map being made of bits (pixels) is digital.

analysis—A methodological investigation of a process by a consistent procedure, and its separation into related units for further detailed study.

Analytical Photogrammetric Positioning System (APPS)—A stereo photogrammetric work station used in conjunction with a Point Positioning Data Base (PPDB) for precise point positioning. See also **Point Positioning Data Base (PPDB).**

analytical aerotriangulation—A photo-triangulation procedure (using aerial photography) in which the spatial solution is obtained by computational routines.

analytical nadir-point triangulation—Radial triangulation performed by computational routines in which nadir points are utilized as radial centers.

analytical orientation—Those computational steps required to determine tilt, direction of principal line, flight height, preparation of control templets at rectification scale, angular elements, and linear elements in preparing aerial photographs for rectification. Developed data are converted to values to be set on circles and scales of rectifier or transforming printer.

analytical photogrammetry— Photo-grammetry in which solutions are obtained by mathematical methods.

analytical photography—Photography, either motion picture or still, accomplished to determine (by qualitative, quantitative, or any other means) whether a particular phenomenon does or does not occur.

analytical products—Products made with the use of analytic techniques.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

analytical radar prediction—Prediction based on proven formulas, power tables, graphs, and/or other scientific principles. An analytical prediction considers surface height, structural and/or terrain information and criteria for radar reflectivity together with the aspect angle and range to the target.

analytical radial triangulation—Radial triangulation performed by computational routines. See also **graphical radial triangulation.**

analytical three-point resection radial triangulation—A method of computing the coordinates of the principal points of overlapping aerial photographs by resecting on three horizontal control points appearing in the overlap area.

anastigmatic lens—A lens which has been corrected for astigmatism and, therefore, focuses vertical and horizontal lines with equal brightness and definition. Anastigmatic lenses are also free of most common aberrations.

anchorage chart—A nautical chart showing prescribed or recommended anchorages.

ancillary data—Auxiliary or supplementary data.

aneroid altimeter—See **barometric altimeter.**

aneroid barometer—A barometer which balances the atmospheric pressure against a mechanically elastic device. The usual form of an aneroid barometer consists of a thin box of corrugated metal, almost exhausted of air. When the atmospheric pressure increases, the box contracts; when the pressure lessens, the box expands. By mechanical means these movements are amplified and communicated to an index hand which registers the changes on a graduated dial.

angle equation—A condition equation which expresses the relationship between the sum of the measured angles of a closed figure and the theoretical value of that sum, the unknowns being the corrections to the observed directions or angles, depending on which are used in the adjustment.

angle method of adjustment—(triangulation and traverse) A method of adjustment of observations which determines correction to observed angles. The angle method of adjustment may be used where a chain of single triangles is to be adjusted.

angle of convergence—(JCS) The angle subtended by the eyebase of an observer at the point of focus. Also called **angular parallax; parallactic angle.**

angle of coverage—See **angle of field.**

angle of current—(hydrography) In stream gagging, the angle of current is the angular difference between 90° and the angle made by the current with a measuring section.

angle of depression—(JCS) 1. The angle in a vertical plane between the horizontal and a descending line [as from an observer to an object]. 2. In air photography, the angle between the optical axis of an obliquely mounted air camera and the horizontal. Also called **depression angle; descending vertical angle; minus angle.** See also **angle of elevation; tilt angle; true depression angle.**

angle of deviation—(optics) The angle through which a ray is bent by refraction.

angle of elevation—The angle in a vertical plane between the horizontal and an ascending line, as from an observer to an object. Also called **ascending vertical angle; plus angle.** See also **angle of depression.**

angle of field—A property of a lens. The angle subtended by lines that pass through the center of the lens and locate the diameter of the maximum image area within the specified definition of the lens. Lenses are generally classified according to their angles of coverage, as follows: narrow-angle; wide-angle; normal-angle; and superwide-angle or ultrawide-angle. Also called **angle of coverage; angular field.**

angle of incidence—(optics) As measured from the normal, the angle at which a ray of light strikes a surface.

angle of inclination—An angle of elevation or angle of depression.

angle of reflection—(optics) As measured from the normal, the angle at which a reflected ray of light leaves a surface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

angle of refraction—The angle which the refracted ray makes with the normal to the surface separating two transparent media.

angle of tilt—See **tilt.**

angle of view—(JCS) 1. The angle between two rays passing through the perspective center (rear nodal point) of a camera lens to two opposite corners of the format. 2. In photogrammetry, twice the angle whose tangent is one-half the length of the diagonal of the format divided by the calibrated focal length. Also called **covering power; field of view.**

angle of yaw—The angle between a line in the direction of flight and a plane through the longitudinal and vertical axes of an aircraft. It is considered positive if the nose is displaced to the right. Also called **yaw angle.**

angle point—A term applied to a marker at each point to indicate a change in the direction of a survey line.

angle to right—The horizontal angle measured clockwise from the preceding line to the following one. Also called **clockwise angle.**

angle-to-right traverse—In surveying, a technique applicable to either open or closed traverses, wherein all angles are measured in a clockwise direction after the transit has been oriented by a backsight to the preceding station.

angle—The inclination to each other of two intersecting lines, measured by the arc of a circle intercepted between the two lines forming the angle, the center of the circle being the point of intersection. See also **adjusted angle; altitude; azimuth; azimuth angle; break angle; conversion angle; counterclockwise angle; concluded angle; crab angle; critical angle; crossing angle; deflection angle; dihedral angle; dip angle; direct angle; direction angle; distance angle; double zenith distance; drift angle; Eulerian angles; Greenwich hour angle; grid magnetic angle; horizontal angle; hour angle; interlocking angle; local hour angle; locking angle; look angles; measured angle; meridian angle; oblique ascension; observed angle; parallactic angle; phase angle; reciprocal vertical angle; refraction angle; repetition of angles; right ascension; screen angle; sidereal hour angle; slope angle; solid angle; spherical angle; spheroidal angle; traverse angle; vectorial angle; vertical angle; zenith distance.**

angular altitude—A measure in degrees of a given object above the horizon, taken from a given or assumed point of observation, and expressed by the angle between the horizontal and the observer's line of sight.

angular calibration constants—In a multiple-lens camera, or multiple-camera assembly, the values of angular orientation of the lens axes of the several lens-camera units to a common reference line. For example, in a trimetrogon camera, the angular relationships of the wing camera axis with respect to the axis of the central (vertical) camera.

angular distance—1. The angular difference between two directions, numerically equal to the angle between two lines extending in the given directions. 2. The arc of the great circle joining two points, expressed in angular units. 3. Distance between two points, expressed in angular units of a specified frequency. It is equal to the number of waves between the points multiplied by 2π if expressed in radians, or multiplied by 360° if expressed in degrees.

angular distortion—1. (cartography) Distortion in a map projection because of nonconformality. 2. (optics) The failure of a lens to reproduce accurately in the image space the angle subtended by two points in the object space.

angular error of closure—See **error of closure,** definition 2.

angular field—See **angle of field.**

angular magnification—The ratio of the angle subtended at the eye by the image formed by an optical device, to the angle subtended at the eye by the object itself without the optical device. This is convenient where a distance in the object cannot be measured for expressing a linear magnification, as in using a telescope.

angular momentum—The quantity obtained by multiplying the moment of inertia of a body by its angular speed.

angular parallax—See **angle of convergence.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

angular rate—See **angular speed.**

angular speed—Change of direction per unit time. Also called **angular rate.**

angular velocity—A representation of the rate of rotation of a particle about the axis of rotation, with magnitude equal to the time rate of angular displacement of any point of the body.

angulator—An instrument for converting angles measured on an oblique plane to their corresponding projections on a horizontal plane. A rectoblique plotter and photoangulator are types of angulators. See also **equiangulator; topoangulator.**

annex point—A point used to assist in the relative orientation of vertical and oblique photographs, selected in the overlap area between the vertical and its corresponding oblique about midway between the pass points. Alternate sets of photographs only will contain annex points. See also **pass point.**

annotated photograph—A photograph on which hypsographic, geologic, cultural, hydrographic, vegetation, or place name information has been added to identify, classify, outline, clarify, or describe features that would not otherwise be apparent in examination of an unmarked photograph. Generally, the term does not apply to photographs marked only with geodetic control or pass points.

annotation overprint—The outline delimiting a target or installation, or a symbol which locates its position together with an identifying reference number as depicted on a target graphic.

annotation text—A descriptive text containing the identification, function, location, physical characteristics, and other information concerning a target or installation. Descriptive texts are also prepared for special areas, radar significant power lines, and precise radar significant location points.

annotation—(JCS) A marking placed on imagery or drawings for explanatory purposes or to indicate items or areas of special importance.

annual aberration—Aberration caused by the velocity of the Earth's revolution about the Sun.

annual change—See **magnetic annual change.**

annual inequality—Seasonal variation in water level or tidal current speed, more or less periodic, due chiefly to meteorological causes.

annual magnetic change—See **magnetic annual change.**

annual magnetic variation—See **magnetic annual variation.**

annual parallax—The angle subtended at a celestial body by the radius of the Earth's orbit. Also called **heliocentric parallax; stellar parallax.**

annual rate of change—See **magnetic annual change.**

annual rate—See **magnetic annual change.**

annular eclipse—An eclipse in which a thin ring of the source of light appears around the obscuring body. Annular solar eclipses occur, but never annular lunar eclipses.

anomalistic drift—The variation or drift of a frequency source. For example, the frequency changes of a crystal oscillator due to a variety of causes, such as temperature variation and component aging, none of which can be predicted in advance or completely controlled.

anomalistic month—The interval of time between two successive passages of the Moon in her orbit through perigee. The length of the anomalistic month is 27.55455 mean solar days.

anomalistic period—The interval between two successive perigee passes of a satellite in orbit about its primary. Also called **perigee-to-perigee period.**

anomalistic tide cycle—The average period of about 27 1/2 days, measured from perigee to perigee, during which the Moon completes one revolution around the Earth.

anomalistic year—The period of one revolution of the Earth around the Sun, from perihelion to perihelion, 365 days, 6 hours, 13 minutes, 53.16 seconds in 1955, and increasing at the rate of 0.002627 second annually.

anomalous magnetic variation—See **local**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

magnetic anomaly.

anomaly—1. (general) A deviation from the norm. 2. (geodesy) A deviation of an observed value from a theoretical value due to a corresponding irregularity in the Earth's structure at the area of observation. 3. (astronomy) The angle between the radius vector to an orbiting body from its primary and the line of apsides of the orbit, measured in the direction of travel, from the point of closest approach to the primary. This term is also called the **true anomaly** when it is desired to distinguish it from the **eccentric anomaly**, which is the corresponding angle at the center of the orbit; or from the **mean anomaly**, which is what the true anomaly would become if the planet had a uniformly angular motion. See also **Bouguer anomaly; deflection anomaly; free-air anomaly; gravity anomaly; Hayford gravity anomalies; height anomaly; isostatic anomaly; local magnetic anomaly; magnetic disturbance; mean free-air anomaly; point anomaly; surface anomalies.**

Antarctic Circle—The geographic parallel having a south latitude equal to the complement of the declination of the winter solstice. The obliquity of the ecliptic is steadily changing so that the winter solstice is not a point of fixed declination, and the Antarctic Circle, as defined, is not a line of fixed position. When the Antarctic Circle is to be shown on a map, however, it is desirable that it be treated as a line of fixed position, and that a conventional value be adopted for its latitude. For this purpose, the value 66°33' south latitude is used. Also called **south polar circle.**

antihalation coating—(photography) A light-absorbing coating applied to the backside of the support of a film or plate (or between the emulsion and the support) to suppress halation.

antipode—Anything exactly opposite to something else. Particularly, that point on the Earth 180° from a given place.

antisolar point—That point on the celestial sphere 180° from the Sun.

antivignetting filter—(JCS) A filter bearing a deposit which is graduated in density to correct for the uneven illumination given by certain lenses, particularly wide-angle types.

apareon—The point on a Mars-centered orbit where a satellite is at its greatest distance from Mars.

apastron—That point of the orbit of one member of a double star system at which the stars are farthest apart. Opposite of **periastron.**

aperiodic compass—Literally "a compass without a period," or a compass that, after being deflected, returns by one direct movement to its proper reading, without oscillation. Also called **deadbeat compass.**

aperture ratio—See **relative aperture.**

aperture stop—(optics) The physical element (such as a stop, diaphragm, or lens periphery) of an optical system which limits the size of the pencil of rays traversing the system. The adjustment of the size of the aperture stop of a given system regulates the brightness of the image without necessarily affecting the size of the area covered. Also called **stop.**

aperture—1. The opening in a lens diaphragm through which light passes. 2. The diameter of the objective lens of a telescope or other optical instrument, usually expressed in inches, but sometimes as the angle between lines from the principal focus to opposite ends of a diameter of the objective lens. See also **relative aperture.**

apex—See **vertex.**

aphelion—The point in the elliptical orbit of a planet which is the farthest from the Sun, when the Sun is the center of attraction. Opposite of **perihelion.**

aphylactic map projection—A map projection which does not possess any of the three special properties of equivalence, conformality, or equidistance. Also called **arbitrary projection.**

aplanatic lens—A lens which transmits light without spherical aberration.

apiune (apolune)—The point on the elliptical orbit of a satellite of the Moon which is farthest from the Moon. Also called **apocynthion.** Opposite of **periune; pericynthion.**

apoapsis—See **apocenter.**

apocenter—In an elliptical orbit, the point in the orbit which is the greatest distance from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

focus where the attracting mass is located. Also called **apoapsis; apofocus.** Opposite of **periapsis; pericenter; perifocus.**

apochromatic lens—A lens that has been corrected for chromatic aberration for three colors.

apocynthion—See **apiune (apolune).**

apofocus—See **apocenter.**

apogean tides—Tides of decreased range occurring when the Moon is near apogee.

apogee—(JCS) The point at which a missile trajectory or a satellite orbit is farthest from the center of the gravitational field of the controlling body or bodies.

aposphere—A mathematical surface of constant curvature applicable to the earth spheroid over a certain limited area. It may be either oblate or prolate in shape.

apparent altitude—The observed vertical angle of a celestial object corrected for instrumental errors, personal errors, and inaccuracies in the reference level (principally dip), but not for refraction, parallax, or semidiameter. Also called **rectified altitude.**

apparent horizon—(JCS) The visible line of demarcation between land/sea and sky. Also called **local horizon; topocentric horizon; visible horizon.**

apparent motion—Motion relative to a specified or implied reference point which may itself be in motion. The expression usually refers to movement of celestial bodies as observed from the Earth. Also called **relative motion.**

apparent noon—Twelve o'clock apparent time, or the instant the apparent sun is over the upper branch of the meridian.

***apparent* place**—(astronomy) See ***apparent* position.**

apparent position—An astronomic term applied to the observable position of a star, planet, or the Sun. The position on the celestial sphere at which a heavenly body (or a space vehicle) would be seen from the center of the Earth at a particular time. Also called **apparent place.** See also **astrometric position.**

apparent precession—(JCS) The apparent deflection of the gyro axis, relative to the Earth, due to the rotating effect of the Earth and not due to any applied force. Also called **apparent wander; wander.**

***apparent* sidereal time**—The local hour angle of the true vernal equinox. Also called **true sidereal time.**

apparent solar day—The interval of time from a transit of the apparent sun across a given meridian to its next successive transit across the same meridian.

apparent solar time—Time measured by the apparent diurnal motion of the true sun. Also called **apparent time; true solar time.**

apparent sun—The actual Sun as it appears in the sky. Also called **true sun.**

apparent time—See **apparent solar time.**

apparent wander—See **apparent precession.**

apparent—A term used to designate certain measured or measurable astronomic quantities to refer them to the observed position of celestial bodies.

appearance ratio—See **hyperstereoscopy.**

approach chart—An aeronautical chart providing essential information for making an approach to an airfield under either visual or instrument flight conditions.

approximate contour—A contour substituted for a normal contour whenever there is a question as to its reliability (reliability is defined as being accurate within one-half the contour interval).

appulse—The near approach of one celestial body to another on the celestial sphere, as in occultation or conjunction.

apse line—See **line of apsides.**

apsis—Either of the two orbital points nearest or farthest from the center of attraction, the perihelion and aphelion in the case of an orbit about the Sun, and the perigee and apogee in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the case of an orbit about the Earth.

arbitrary grid—Any reference system developed for use where no grid is available or practical, or where military security for the reference is desired.

arbitrary projection—See **aphylactic map projection.**

ARC Digitized Raster Graphics (ADRG)—Digital raster representations of paper graphic products. Maps/charts are converted into digital data by raster scanning and transforming the map image into the Equal Arc Second Raster Chart/Map (ARC) System frame of reference. Used for electronic map displays. See also **ARC Projection System; Equal Arc Second Raster Chart/Map.**

ARC Projection System—A coordinate system that divides the world into 18 latitudinal zones. See also **Equal Arc Second Raster Chart/Map; ARC Digitized Raster Graphics.**

arc correction—(pendulum) The quantity which is applied to the period of vibration of a pendulum to allow for the pendulum's departure from simple harmonic motion.

arc measurement—A survey method used to determine the size of the Earth. A long arc is measured on the Earth's surface and the angle which subtends this measured arc is determined. By assumptions and mathematical formula the size and shape of the Earth can then be determined.

arc navigation—A navigation system in which the position of an airplane or ship is maintained along an arc measured from a control station by means of electronic distance measuring equipment, such as **shoran.** See also **loran.**

arc of parallel—A part of an astronomic or geodetic parallel of latitude.

arc of visibility—The horizontal angular range through which a navigation light is visible form seaward. The angular range is defined by limiting bearings of direction.

arc triangulation—A system of triangulation of limited width designed to progress in a single general direction. Arc triangulation is executed for the purpose of connecting independent and widely separated surveys, coordinating, and correlating local surveys along the arc, furnishing data for the determination of a geodetic datum, providing a network of control points for a countrywide survey, etc.

Arctic Circle—The geographical parallel having a north latitude equal to the complement of the declination of the summer solstice. The obliquity of the ecliptic is steadily changing so that the summer solstice is not a point of fixed declination, and the Arctic Circle, as defined, is not a line of fixed position. When the Arctic Circle is to be shown on a map, however, it is desirable that it be treated as a line of fixed position, and that a conventional value be adopted for its latitude. For this value 66° 33' north latitude is used. Also called **north polar circle.**

arc—**1.** (topology) An individual line segment defined by a series of X, Y coordinate pairs. Nodes are at the ends of arcs and form the points of intersection between arcs. **2.** (geodesy) A portion of a geodetic triangulation network, between two high-order stations. **3.** Acronym for Equal Arc Second Raster Chart/Map.

Area Requirements and Product Status (ARAPS)—Data base which describes DMA's customer area requirements and forecasts for various MC&G products. ARAPS is used by DMA to plan production schedules.

area analysis intelligence—Intelligence data relative to a specific geographic area.

area coverage—**1.** Complete coverage of an area by aerial photography having parallel overlapping flight lines and stereoscopic overlap between exposures in the line of flight. **2.** When applied to shoran, the term implies that recorded shoran distances are available for each exposure. **3.** Complete coverage of a geographical area by maps or other graphic material.

area pattern screen—A photographic negative or positive containing repetitively arranged small feature symbols which have been designed to present a visual portrayal of a map or chart areal feature; i.e., swamp, orchard, sand, etc. See also **contact screen; line pattern.**

area survey—**1.** A survey of areas large enough to require a network of control stations. **2.** An extension and densification of survey control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

area target—(JCS) A target consisting of an area rather than a single point. See also **pinpoint target.**

area triangulation—A system of triangulation designed to progress in every direction. Area triangulation is executed to provide survey control points over an area, as of the city or county; or for filling in the areas between arcs of triangulation which form a network extending over a county or state. See also **survey net; triangulation net.**

area weighted average resolution (AWAR)—A single average value for the resolution over the picture format for any given focal plane.

areal feature—A topographic feature, such as sand, swamp, vegetation, etc., which extends over an area. It is represented on the published map or chart by a solid or screened color, by a prepared pattern of symbols, or by a delimiting line. 2. (digital mapping) Any area enclosed by a delimiting line that has any unique characteristic, e.g., forest, residential, etc. 3. (raster) A block of grid cells which represent a homogeneous portion of the earth.

area—A level of spatial measurement referring to a two-dimensional defined space. A polygon on the earth as projected onto a horizontal plane is an example of an area.

areodesy—(JCS) That branch of mathematics which determines by observations and measurements, the exact positions of points and the figures and areas of large portions of the surface of the planet Mars, or the shape and size of the planet Mars.

argument of latitude—In celestial mechanics, the angular distance measured in the orbit plane from the ascending node to the orbiting object; the sum of tha argument of perigee and the true anomaly.

argument of perigee—An orbital element defined as the angle at the center of attraction from the ascending node to the perigee point measured in the direction of motion of the orbiting body.

argument—In astronomy, an angle or arc, as in argument of perigee.

array—A systematic arrangement of elements in one or more dimensions.

artificial asteroid—A man-made object placed in orbit about the Sun.

artificial Earth satellite—A man-made Earth satellite, as distinguished from the Moon.

artificial horizon—(JCS) See **attitude indicator.**

artificial monument—A relatively permanent object used to identify the location of a survey station or corner. Objects include man-made structures such as abutments, stone markers, concrete markers, and railroad rails.

artwork prediction—See **experience radar prediction.**

Arundel method—A combination of graphical and analytical methods, based on radial triangulation, for point-by-point topographic mapping from aerial photographs.

ascending node—That point at which a planet, planetoid, or comet crosses the ecliptic from south to north, or a satellite crosses the equator of its primary from south to north. Opposite of **descending node.** Also called **northbound node.**

ascending vertical angle—See **angle of elevation.**

ascensional difference—The difference between right ascension and oblique ascension.

aspect categories—Classification of an aspect into a grouping based on cardinal compass direction or a range of degrees.

aspect change—(JCS) The different appearance of a reflecting object viewed by radar from varying directions. It is caused by the change in the effective reflecting area of the target.

aspect ratio—The numerical ratio of picture width to height.

aspects—The apparent positions of celestial bodies relative to one another; particularly the apparent positions of the Moon or a planet relative to the Sun.

aspect—Horizontal direction in which a slope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

faces, commonly expressed as the direction clockwise from north.

aspherical lens—A lens in which one or more surfaces depart from a true spherical shape.

assault landing model—A special form of assault model designed specifically for planning amphibious landings. Also called **amphibious assault landing model.** See also **assault models.**

assault models—Large-scale models giving a particular representation of vegetation, lesser landforms, prominent man-made features, and a detailed representation of specific or sensitive objectives such as airfields, radar installations, and the like. These models emphasize the aspects of objects as seen from surface approach .

associated Legendre function—A solution of the Legendre equation, which is a special case of the Laplace equation, in the form of a power series of a special kind; used in the spherical harmonic expansion of the gravitational potential.

assumed ground elevation—The elevation assumed to prevail in the local area covered by a particular photograph or group of photographs. Used especially to denote the elevation assumed to prevail in the vicinity of a critical point, such as a peak or other feature having abrupt local relief.

assumed latitude—The latitude at which an observer is assumed to be located for an observation or computation, as the latitude of an assumed position or the latitude used for determining the longitude by time sight.

assumed longitude—The longitude at which an observer is assumed to be located for an observation or computation, as the longitude of an assumed position or the longitude used for determining the latitude by meridian altitude.

assumed plane coordinates—A local plane coordinate system set up at the convenience of the surveyor. The reference axes are usually assumed so that all coordinates are in the first quadrant. The *y*-axis may be in the direction of astronomic north, geodetic north, magnetic north, or an assumed north.

assurance level—See **confidence interval.**

astatized gravimeter—A gravimeter, sometimes referred to as unstable, where the force of gravity is maintained in an unstable equilibrium with the restoring force. The instability is provided by the introduction of a third force which intensifies the effect of any change in gravity from the value in equilibrium.

asteroid—A minor planet; one of the many small celestial bodies revolving around the Sun, most of the orbits being between those of Mars and Jupiter. Also called **minor planet; planetoid.** See also **artificial asteroid.**

astigmatism—An aberration affecting the sharpness of images for objects off the axis in which the rays passing through different meridians of the lens come to a focus in different planes. Thus, an extra-axial point object is imaged as two mutually perpendicular short lines located at different distances from the lens.

astigmatizer—A lens which introduces astigmatism into an optical system. Such a lens is so arranged that it can be placed in or removed from the optical path at will. In a sextant, an astigmatizer may be used to elongate the image of a celestial body into a horizontal line.

astre fictif—Any of several fictitious stars assumed to move along the celestial equator at uniform rates corresponding to the speeds of the several harmonic constituents of the tide-producing force. Each astre fictif crosses the meridian at the instant the constituent it represents is at a maximum.

astro compass—(JCS) An instrument used primarily to obtain true heading or true bearing by reference to celestial bodies.

astrodynamics—The practical application of celestial mechanics, astroballistics, propulsion theory, and allied fields to the problem of planning and directing the trajectories of space vehicles.

astrogeodetic datum orientation—The position of a reference ellipsoid in relation to the geoid in a specified area of a geodetic network. It may be expressed by the astrogeodetic deflection and geoidal height at the datum point or by an astrogeodetic geoid chart of the area.

astrogeodetic deflection—The angle at a point between the normal to the geoid and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

normal to the ellipsoid of an astrogeodetically oriented datum. Also called **relative deflection.**

astrogeodetic leveling—A method to determine variations in the separation of the geoid and the ellipsoid using astrogeodetic deflections. Also called **astronomic leveling; geoidal height profile.**

astrogeodetic undulations—The separation between an astrogeodetic geoid, defined for a particular datum, and a specified ellipsoid surface. See also **geoidal height.**

astrograph mean time—A form of mean time, used in setting an astrograph. Astrograph mean time 1200 occurs when the local hour angle of Aries is 0°.

astrographic position—See **astrometric position.**

astrograph—**1.** A device for projecting a set of precomputed altitude curves onto a chart or plotting sheet, the curves moving with time such that if they are properly adjusted, they will remain in the correct position on the chart or plotting sheet. **2.** A telescope, usually of moderate focal length, which is designed specifically for the purpose of accurately recording the positions of celestial objects by photographic means.

astrogravimetric leveling—A concept whereby a gravimetric map is used for the interpolation of the astrogeodetic deflections of the vertical to determine the separation of the ellipsoid and the geoid in studying the figure of the Earth.

astrogravimetric points—Astronomic positions corrected for the deflection of the vertical by gravimetric methods.

astrolabe—**1.** (general) Any instrument designed to measure the altitudes of celestial bodies. **2.** (surveying) An instrument designed for very accurate celestial altitude measurements. See also **equilangulator; pendulum astrolabe; planispheric astrolabe; prismatic astrolabe.**

astrometric position—The position of a heavenly body (or space vehicle) on the celestial sphere corrected for aberration but not for planetary aberration. Astrometric positions are used in photographic observation where the position of the observed body can be measured in reference to the positions of comparison stars in the field of the photograph. Also called **astrographic position.** See also **apparent position.**

astrometry—The branch of astronomy dealing with the geometric relations of the celestial bodies and their real and apparent motions. The techniques of astrometry, especially the determination of accurate position by photographic means, are used in tracking satellites and space probes.

astronomic arc—The apparent arc described above (diurnal arc) or below (nocturnal arc) the horizon by the Sun or another celestial body.

astronomic azimuth mark—A marked point whose astronomic azimuth from a survey station is determined from direct observations on a celestial body. The mark may be a lamp or illuminated target placed especially for the purpose; it may be a well defined illuminated point on a permanent structural point.

astronomic azimuth—The angle between the astronomic meridian plane of the observer and the plane containing the observed point and the true normal (vertical) of the observer, measured in the plane of the horizon, preferably clockwise from north.

astronomic bearing—See **true bearing.**

astronomic constants—The elements of the orbits of the bodies of the solar system, their masses relative to the Sun, their size, shape, orientation, rotation, and inner constitution, and the velocity of light. See also **system of astronomic constants.**

astronomic control—A network of control stations the positions of which have been determined by astronomic observation. Latitudes and longitudes thus determined will normally differ from the geodetic latitudes and longitudes of the same stations by amounts corresponding to components of the deflection of the vertical.

astronomic coordinates—**1.** Quantities defining a point on the surface of the Earth, or of the geoid, in which the local direction of gravity is used as a reference. Also called **geographic coordinates; gravimetric coordinates; terrestrial coordinates. 2.** The coordinates of an astronomic body referred to a given equinox.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

astronomic date—Designation of epoch by year, month, day, and decimal fraction. For example, the astronomic date of December 21, 1978, 18^h UTC (universal time coordinated) is 1978 December 21.75 UTC. The astronomic date is also used in connection with the other time systems. The system commences every calendar year at 0^h on December 31 of the previous year. This epoch is denoted by January 0.0.

astronomic day—A mean solar day beginning at mean noon, 12 hours later than the beginning of the civil day of the same date. Astronomers now generally use the civil day.

astronomic equator—The line on the surface of the Earth whose astronomic latitude at every point is 0°. Due to the deflection of the plumb line, the astronomic equator is not a plane curve. However, the verticals at all points on it are parallel to one and the same plane, the plane of the celestial equator; that is, the zenith at every point on the astronomic equator lies in the celestial equator. When the astronomic equator is corrected for station error, it becomes the geodetic equator. Also called **terrestrial equator**. See also **geodetic equator**.

astronomic latitude—The angle between the plumb line and the plane of celestial equator. Also defined as the angle between the plane of the horizon and the axis of rotation of the Earth. Astronomic latitude applies only to positions on the Earth and is reckoned from the astronomic equator (0°) north and south through 90°. Astronomic latitude is the latitude which results directly from observations of celestial bodies, uncorrected for deflection of the vertical.

astronomic leveling—See **astrogeodetic leveling**.

astronomic longitude—The angle between the plane of the celestial meridian and the plane of an initial meridian, arbitrarily chosen. Astronomic longitude is the longitude which results directly from observations on celestial bodies, uncorrected for deflection of the vertical.

astronomic meridian plane—A plane that contains the vertical of the observer and is parallel to the instantaneous rotation axis of the Earth.

astronomic meridian—A great circle of the celestial sphere intersecting the north and south celestial poles. The local astronomic meridian is that meridian which intersects the zenith of the point.

astronomic parallel—A line on the surface of the Earth which has the same astronomic latitude at every point. Because the deflection of the vertical is not the same at all points on the Earth, an astronomic parallel is an irregular line, not lying in a single plane. See also **astronomic equator**.

astronomic position—1. A point on the Earth whose coordinates have been determined as a result of observations of celestial bodies. The expression is usually used in connection with positions on land determined with great accuracy for survey purposes. 2. A point on the Earth, defined in terms of astronomic latitude and longitude.

astronomic refraction error—See **astronomic refraction**.

astronomic refraction—The apparent displacement of an object that results from light rays from a source outside the atmosphere being bent in passing through the atmosphere. This results in all objects appearing to be higher above the horizon than they actually are. The magnitude of this displacement is greater when the object is near the horizon and decreases to a minimum assumed to be zero when the object is at the zenith. Also called **astronomic refraction error; celestial refraction**. See also **atmospheric refraction; refraction**.

astronomic station—A point on the Earth whose position has been determined by observations on celestial bodies.

astronomic surveying—The celestial determination of latitude and longitude. Separations are calculated by computing distances corresponding to measured angular displacements along the reference spheroid.

astronomic theodolite—See **altazimuth instrument**.

astronomic tidal constituent—See **constituent**.

astronomic time—Solar time in a day (astronomic day) that begins at noon. Astronomic time may be either apparent solar time or mean solar time. Since 1925, civil time is generally

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

used instead of astronomic time.

astronomic transit—See **transit**, definition 4.

astronomic triangle—The navigational triangle, either terrestrial or celestial, used in the solution of celestial observations. Referring to the celestial sphere it is the triangle formed by arcs of great circles connecting the celestial pole, the zenith, and a celestial body. The angles of the astronomic triangles are: at the pole, the hour angle; at the celestial body, the parallactic angle; at the zenith, the azimuth angle. The sides are: pole to zenith, the colatitude; zenith to celestial body, the zenith distance; and celestial body to pole, the polar distance. Also called **PZS triangle.**

astronomic unit—A unit of length equal to 149,600,000 kilometers (adopted 1960) used for measuring distances within the solar system. This distance approximates the mean distance of the Earth from the Sun.

astronomic year—See **tropical year.**

astronomic—Of or pertaining to astronomy, the science which treats of heavenly bodies, and the arts based on that science.

ASW Prediction Area Chart (ASW)—A standard DMA hydrographic chart (Antisubmarine Warfare) overprinted with homogeneous acoustic provinces characterized by a single velocity profile, a bottom loss class, and a bathymetric relief.

asymmetry of object (target)—Lack of symmetry in the visible aspect of an object as seen from a particular point of observation. A square or rectangular pole may so face the observer that the line bisecting its tangents does not pass through its geometric center. With a square cupola or tower, the error resulting from observing tangents and taking a mean may be quite large. The error caused by asymmetry of an observed object is of the same character and requires the same treatment as the error resulting from observing an eccentric object. See also **phase.**

asymptote—A straight line or curve which some curves of infinite length approach but never reach.

atlas grid—A reference system that permits the designation of the location of a point or an area on a map, photo, or other graphic in terms of numbers and letters. Also called **alphanumeric grid.**

atmosphere—(JCS) The air surrounding the Earth. See also **ionosphere; stratosphere; tropopause; troposphere.**

atmospheric drag—A major perturbation of close artificial satellite orbits caused by the resistance of the atmosphere. The secular effects are decreasing eccentricity, major axis, and period. Also called **drag.**

atmospheric refraction—The refraction of light passing through the Earth's atmosphere. Atmospheric refraction includes both astronomic refraction and terrestrial refraction.

atomic time—Time interval based on the frequency of atomic oscillators.

atran—An acronym for "automatic terrain recognition and navigation," a navigation system which depends upon the correlation of terrain images appearing on a radar cathode-ray tube with previously prepared maps or simulated radar images of the terrain.

attenuation—(JCS) Decrease in intensity of a signal, beam, or wave as a result of absorption of energy and of scattering out of the path of a detector, but not including the reduction due to geometric spreading, i.e., the inverse square of distance effect.

attitude indicator—(JCS) An instrument which displays the attitude of the aircraft by reference to sources of information which may be contained within the instrument or be external to it. When the sources of information are self-contained. The instrument may be referred to as an **artificial horizon.**

attitude—1. (JCS) The position of a body as determined by the inclination of the axes to some frame of reference. If not otherwise specified, this frame of reference is fixed to the Earth. 2. Grid bearing relative to the long axis of the target 3. (photogrammetry) The angular orientation of a camera, or of the photograph taken with that camera, with respect to some external reference system. Usually expressed as tilt, swing, and azimuth; or roll, pitch, and yaw.

attribute identifier—A three-alphanumeric character designator of an attribute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

attribute tagging—The process of assigning an attribute to a particular feature.

attribute value—A specific quality of quantity assigned to an attribute.

attribute—A characteristic of a site or phenomenon. May be physical, social, economic or titular in nature. For example, road types and road names are road attributes.

augmentation correction—A correction due to augmentation, particularly that sextant altitude correction due to the apparent increase in the semidiameter of a celestial body as its altitude increases.

augmentation—*The apparent increase in the* semidiameter of an azimuth celestial body as its altitude increases, due to the reduced distance from the observer. The term is used principally in *reference to the Moon.*

augmenting factor—A factor used in connection with the harmonic analysis of tides or tidal currents to allow for the difference between the times of hourly tabulation and the corresponding constituent hours.

Australian National spheroid—A reference spheroid having the following dimensions: semimajor axis—6,378,160.0 meters; and a flattening or ellipticity of 1/298.25.

austral—Of or pertaining to south.

authalic (equal-area) latitude—A latitude based on a sphere having the same area as the spheroid, and such that areas between successive parallels of latitude are exactly equal to the corresponding areas on the spheroid. Authalic latitudes are used in the computation of *equal-area map projections.*

authalic map projection—An equal-area map projection.

auto reflection—The focusing of an autocollimating theodolite on the surface of the mirror or prism rather than focused at infinity. The image seen is on the face of the reflector and the gauss image will follow the motion of the reticule. Whereas, in autocollimation, the instrument is focused at infinity and the gauss image moves in opposite directions to the reticule.

autocollimation—(surveying) The procedure used to determine or transfer azimuth to an instrument or device. This procedure requires use of a specially adapted telescope, capable of bisecting the real image of its own reticule as *reflected from a mirror or Porro prism. When such* bisection is accomplished, the line of sight of the telescope is perpendicular to the face of the mirror or apex edge of the prism.

autocollimator—A collimator provided with a means of illuminating its cross hairs so that, when a reflecting plane is placed normal to the *emergent light beam, the reflected image of the* cross hairs appears to be coincident with the cross hairs themselves. This device is used in calibrating optical and mechanical instruments and transferring direction.

autofocus rectifier—A precise, vertical photoenlarger which permits the correction of distortion in an aerial negative caused by tilt. The instrument's operations are motor driven and are interconnected by mechanical linkages to insure automatically maintained sharp focus.

Automated Air Facilities Information File (AAFIF)—Computer file of structured information defining validated attributes relative to facilities for each airfield of concern to DoD *flight operations.*

Automated Mapping / Facilities Management (AM/FM)—A computer aided *mapping system* which stores annotation and *limited attribute data.* AM/FM systems are commonly used by the utility industries and municipalities.

Automated Tactical Target Graphic (ATTG)—A tactical target materials item which provides aerial photographic coverage of a target and a limited area surrounding it at a scale permitting optimum identification of target detail. The ATTG also includes textual intelligence on a sheet separate from the graphic portion. Each part can be revised independent of the other. ATTG's cover single targets and are produced in two forms: a lithographic sheet and a miniaturized version in an aperture card.

automated imposition machine—(miso, misomex) A step and repeat machine used to expose multiple images onto either a printing plate or sheet of film.

Automatic Digital Annotation System

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

(ADAS)—A system used to record camera position and other information on film at time of exposure.

automatic clipping/joining—The system capability for copying small portions of a data base for movement and placement elsewhere in the data base without operator intervention.

automatic gage—See **self-registering gage.**

automatic level—See **pendulum level.**

automatic polygon centroid calculation—The system capability for determining the center of a polygon area without operator intervention. Usually associated with automatic label placement.

automatic rectifier—Any rectifier which employs mechanisms to insure automatic fulfillment of the lens law and the Scheimpflug condition. These devices, called **inversors**, provide a mechanical solution for the linear and angular elements of rectification. Essentially, this class of rectifier is a tilt analyzer using inversors to solve for the optical geometric elements needed for sharp focus.

automatic rod—See **tape rod.**

automatic snapping—The system capability for completing a line segment whose end approaches a predefined threshold of closeness to an intersection or node, without any operator intervention.

automatic traverse computer—

autoradar plot—See **chart comparison unit.**

autoreducing tachymeter—A class of tachymeter by which horizontal and height distances are read simultaneously. Horizontal distance is the intercept multiplied by 100 and the vertical distance is the midwire (curve) multiplied by a factor which appears in the optics.

autoscreen film—A photographic film embodying a halftone screen which automatically produces a halftone negative from continuous tone copy.

autumnal equinox—That point of intersection of the ecliptic and the celestial equator occupied by the Sun as it changes from north to south declination, about 23 September. Also called **first point of Libra; September equinox.**

auxiliary contour—See **supplementary contour.**

auxiliary guide meridian— Where guide meridians have been placed at intervals exceeding the distance of 24 miles, and new governing lines are required, a new guide meridian is established, and a local name is assigned, such as "Twelfth Auxiliary Guide Meridian West," or "Grass Valley Guide Meridian." Auxiliary guide meridians are surveyed, in the same manner as guide meridians. See also **guide meridian; principal meridian.**

auxiliary meander corner—An auxiliary meander corner is established at a suitable point on the meander line of a lake lying entirely within a quarter section or on the meander line of an island falling entirely within a section and which is found to be too small to subdivide. A line is run connecting the monument to a regular corner on the section boundary.

auxiliary station—Any station connected to the main scheme net and dependent upon it for the accuracy of its position.

average deviation—In statistics, the average or arithmetic means of the deviations, taken without regard to sign, from some fixed value, usually the arithmetic mean of the data. Also called **mean deviation.**

average terrestrial pole—The average position of the instantaneous pole of rotation of the Earth, averaged over a specified time period. See also **conventional international origin.**

averaging device—A device for averaging a number of readings, as on a bubble sextant.

axis of homology—The intersection of the plane of the photograph with the horizontal plane of the map or the plane of reference of the ground. Corresponding lines in the photograph and map planes intersect on the axis of homology. Also called the **axis of perspective; map parallel; perspective axis.** See also **ground parallel.**

axis of lens—See **optical axis.**

axis of level—See **spirit level axis.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

axis of perspective—See **axis of homology.**

axis of the level bubble—See **spirit level axis.**

axis of tilt—A line through the perspective center perpendicular to the principal plane. The axis of tilt could be any of several lines in space (e.g., the isometric parallel or the ground line), but the present definition is the only one which permits the concept of tilting a photograph without upsetting the positional elements of exterior orientation.

axis—See **camera axis; collimation axis; coordinate axes; equatorial axis; fiducial axes; horizontal axis; major axis; minor axis; optical axis; polar axis; semimajor axis; semiminor axis; spirit level axis; topple axis; transverse axis; vertical axis; *x*-axis; *y*-axis; *z*-axis.**

azimuth angle—**1.** (JCS) An angle measured clockwise in the horizontal plane between a reference direction and any other line. **2.** (astronomy) The angle 180° or less between the plane of the celestial meridian and the vertical plane containing the observed object, reckoned from the direction of the elevated pole. In astronomic work, the azimuth angle is the spherical angle at the zenith in the astronomic triangle which is composed of the pole, the zenith, and the star. In geodetic work, it is the horizontal angle between the celestial pole and the observed terrestrial object. **3.** (surveying) An angle in triangulation or in a traverse through which the computation of azimuth is carried. In a simple traverse, every angle may be an azimuth angle. Sometimes, in a traverse, to avoid carrying azimuths over very short lines, supplementary observations are made over comparatively long lines, the angles between which form azimuth angles. In triangulation, certain angles, because of their size and position in the figure, are selected for use as azimuth angles, and enter into the formation of the azimuth condition equation (azimuth equation).

azimuth bar—See **azimuth instrument.**

azimuth by altitude—An azimuth determined by solution of the navigational triangle with altitude, declination, and latitude given.

azimuth circle—A ring designed to fit snugly over a compass or compass repeater, and provided with means for observing compass bearings and azimuths.

azimuth equation—A condition equation which expresses the relationship between the fixed azimuths of two lines which are connected by triangulation or traverse.

azimuth error of closure—See **error of closure,** definition 3.

azimuth instrument—(magnetic) An instrument for measuring azimuths, particularly a device which fits over a central pivot in the glass cover of a magnetic compass. Also called **azimuth bar; bearing bar.**

azimuth line—(photogrammetry) A radial line from the principal point, isocenter, or nadir point of a photograph, representing the direction to a similar point of an adjacent photograph in the same flight line; used extensively in radial triangulation.

azimuth mark—A mark set at a significant distance from a triangulation or traverse station to mark the end of a line for which the azimuth has been determined, and to serve as a starting or reference azimuth for later use. See also **astronomic azimuth mark; geodetic azimuth mark; Laplace azimuth mark.**

azimuth resolution—(JCS) The ability of radar equipment to separate two reflectors at similar ranges but different bearings from a reference point. Normally the minimum separation distance between the reflectors is quoted and expressed as the angle subtended by the reflectors at the reference point.

azimuth transfer—Connecting, with a straight line, the nadir points of two vertical photographs selected from overlapping flights.

azimuth traverse—A survey traverse in which the direction of the measured course is determined by azimuth and verified by back azimuth. To initiate this type of traverse it is necessary to have a reference meridian, either true, magnetic, or assumed.

azimuthal chart—A chart on an azimuthal projection. Also called **zenithal chart.**

azimuthal equidistant chart—A chart on the azimuthal equidistant map projection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

azimuthal equidistant map projection—An azimuthal map projection on which straight lines radiating from the center or pole of projection represent great circles in their true azimuths from that center, and lengths along those lines are of exact scale. This projection is neither equal-area nor conformal.

azimuthal map projection—A map projection on which the azimuths or directions of all lines radiating from a central point or pole are the same as the azimuths or directions of the corresponding lines on the sphere. Also called **zenithal map projection.**

azimuthal orthomorphic map projection—See **stereographic map projection.**

azimuth—1. (JCS) Quantities may be expressed in positive quantities increasing in a clockwise direction or in *x-/y*-coordinates where south and west are negative. They may be referenced to true north or magnetic north depending on the particular weapon system used. 2. (surveying) The horizontal direction of a line measured clockwise from a reference plane, usually the meridian. Also called **forward azimuth** to differentiate from **back azimuth.** 3. (photogrammetry) Azimuth of the principal plane. See also **astronomic azimuth; azimuth by altitude; back azimuth; computed azimuth angle; direction method of determining astronomic azimuth; geodetic azimuth; grid azimuth; inertial azimuth; Laplace azimuth; magnetic azimuth; method of repetitions (determination of astronomic azimuth); micrometer method (determination of astronomic azimuth); normal section azimuth; true azimuth.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

B

Bache-Wurdem base-line measuring apparatus—A compensating base line measuring apparatus having a measuring element composed of a bar of iron and a bar of brass, each a little less than 6 meters in length, held together firmly at one end, with the free ends so connected by a compensating lever as to form a compensating apparatus.

back azimuth—1. (geodetic surveying) If the azimuth of point B from point A is given on a reference sphere or ellipsoid, the back azimuth is the azimuth of point A from point B. Because of the convergence of the meridians, the forward and backward azimuths of a line do not differ by exactly 180°, except where A and B have the same geodetic longitude or where the geodetic latitudes of both points are 0°. 2. (plane surveying) When referred to a plane rectangular coordinate system, same as above except forward and backward azimuths differ by exactly 180°. See also **azimuth**, definition 1.

back bearing—1. A bearing differing by 180°, or measured in the opposite direction from a given bearing. Also called **reciprocal bearing**. 2. The bearing at the opposite end of a line from the observer as measured from the true meridian at the opposite end of the line. The back bearing on all lines (other than north-south lines) are different from the bearing at the observer's station. They differ by the amount of convergency of the meridians between the two points.

back focal distance—See **back focal length.**

back focal length—The distance measured along the lens axis from the rear vertex of the lens to the plane of best average definition. Also called **back focal distance; back focus.**

back focus—See **back focal length.**

backsight—1. A sight on a previously established survey point or line. 2. (traverse) A sight on a previously established survey point, which is not the closing sight of the traverse. 3. (leveling) A reading on a rod held on a point whose elevation has been previously determined and which is not the closing sight of a level circuit; any such rod reading used to determine height of instrument prior to making a foresight. Also called **plus sight.**

backstep—The method of determining the offsets for the bottom latitude of a projection by measuring the appropriate distances down from the top latitude of a chart.

backup—An image printed on the reverse side of a map sheet already printed on one side. Also the printing of such images.

balancing a survey—Distributing corrections through any traverse to eliminate the error of closure, and to obtain an adjusted position for each traverse station. Also called **traverse adjustment.** See also **compass rule; distance prorate rule; transit rule.**

Baldwin solar chart—A chart designed for orienting a planetable by means of the Sun's shadow.

ballistic camera—A precision terrestrial camera, usually employing glass plates, used at night to photograph such objects as rockets, missiles, or satellites against a star background. Also called **tracking camera.** See also **BC-4 camera.**

band interleaved by line—A specific implementation of a multivariate raster dataset. For each line in the raster, the values of each of the variables or bands are stored in sequence, before the set for the succeeding line.

band sequential—A specific implementation of a multivariate raster data set. The complete data array for each separate variable or band is stored independently of the other variables.

band—A channel. A range of wavelengths of electromagnetic radiation.

bar check—A method of field calibrating the sounding equipment used in hydrographic survey by suspending a bar or disc beneath the transducer at various depths.

bar scale—See **graphic scale.**

Barlow leveling rod—A speaking rod marked with triangles each 0.02 foot in height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

barometer—An instrument for measuring atmospheric pressure. See also **aneroid barometer; cistern barometer; mercury barometer; siphon barometer.**

barometric altimeter—An instrument that indicates elevation or height above sea level, or some other reference height, by measuring the weight of air above the instrument. Also called **pressure altimeter; sensitive altimeter.** See also **aneroid altimeter.**

barometric elevation—An elevation determined with a barometer or altimeter.

barometric hypsometry—The determination of elevations by means of either mercurial or aneroid barometers.

barometric leveling—A method of determining differences of elevation from differences of atmospheric pressure observed with a barometer or barometric altimeter. A type of indirect leveling.

barycenter—The center of mass of a system of masses; as the barycenter of the Earth-Moon system.

basal coplane—(photogrammetry) The condition of exposure of a pair of photographs in which the two photographs lie in a common plane parallel to the air base. If the air base is horizontal, the photographs are said to be exposed in horizontal coplane.

basal orientation—The establishment of the position of both ends of an air base with respect to a ground system of coordinates. In all, six elements are required. These are essentially the three-dimensional coordinates of each end of the base. In practice, however, it is also convenient to express these elements in one of two alternative ways: (1) The ground rectangular coordinates of one end of the base and the difference between these and the ground rectangular coordinates of the other end of the base. (2) The ground rectangular coordinates of one end of the base, the length of the base, and two elements of direction such as base direction and base tilt.

basal plane—See **epipolar plane.**

base apparatus—(surveying) Any apparatus designed for use in measuring with accuracy and precision the length of a base line in triangulation, or the length of a line in first- or second-order traverse. See also **Bache-Wurdeman base-line measuring apparatus; compensating base-line measuring apparatus; duplex base-line measuring apparatus; Hassler base-line measuring apparatus; iced-bar apparatus; Jaderin wires (base apparatus); optical base-line measuring apparatus; Repsold base-line measuring apparatus; Schott base-line measuring apparatus.**

base chart—See **base map.**

base color—The first color printed of a polychrome map to which succeeding colors are registered.

base construction line—The bottom line of a map projection, at right angles to the central meridian, along which other meridians are established.

base data—The basic level of map data on which other information is placed for purposes of comparison or geographical correlation.

base direction—The direction of the vertical plane containing the air base, which might be expressed as a bearing or an azimuth. See also **basal orientation.**

base line—1. (JCS) (surveying) A surveyed line established with more than usual care, to which surveys are referred for coordination and correlation. 2. (JCS) (photogrammetry) The line between the principal points of two consecutive vertical air photographs. It is usually measured on one photograph after the principal point of the other has been transferred. 3. (JCS) (triangulation) The side of one of a series of connected triangles, the length of which measured with prescribed accuracy and precision, and from which the lengths of the other triangle sides are obtained by computation. [Base lines in triangulation are classified according to the character of the work they are intended to control, and the instruments and methods used in their measurement are such that prescribed probable errors for each class are not exceeded. These probable errors, expressed in terms of the lengths, are as follows: first-order base line, 1/1,000,000; second-order base line, 1/500,000; third-order base line 1/250,000.] Also called **triangulation base line.** 4. (USPLS) A line which is extended east and west

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

on a parallel of latitude from an initial point, and from which are initiated other lines for the cadastral survey of the public lands within the area covered by the principal meridian that runs through the same initial point. 5. (navigation) The line between two radio transmitting stations operating in conjunction for the determination of a line of position, as the two stations of a loran system.

base manuscript—See **compilation manuscript.**

base map—(JCS) A map or chart showing certain fundamental information, used as a base upon which additional data of specialized nature are compiled or overprinted. Also, a map containing all the information from which maps showing specialized information can be prepared. See also **chart base; map.**

base net—A small net of triangles and quadrilaterals, starting from a measured base line, and connecting with a line of the main scheme of a triangulation net.

base sheet—A sheet of dimensionally stable material upon which the map projection and ground control are plotted, and upon which stereotriangulation or stereocompilation is performed.

base station—1. (surveying) The point from which a survey begins. 2. (gravity) A geographic position whose absolute gravity value is known. In exploration, a reference station where quantities under investigations have known values or may be under repeated or continuous measurement in order to establish additional stations in relation to it.

base tape—A tape or band of metal or alloy, so designed and graduated and of such excellent workmanship that it is suitable for measuring lengths of lines (base lines) for controlling triangulation, and for measuring the lengths of first- and second-order traverse lines.

base tilt—The inclination of the air base with respect to the horizontal. See also **basal orientation.**

base-altitude ratio—The ratio between the air base length and the flight altitude of a stereoscopic pair of photographs. This ratio is referred to as the K-factor. More commonly called **base-height ratio.** Also indicated functionally as **B/H.**

base-height ratio—See **base-altitude ratio.**

base-line extension—(navigation) The continuation of the base line in both directions beyond the transmitters of a pair of radio stations operating in conjunction for determination of a line of position.

base-line levels—A level line run along a base line to determine and establish the elevation of the base-line stations.

base-line terminal stations—The monumented stations marking the end points of a base line.

basement contours—Contours on the surface of the basement complex or basic metamorphic and volcanic rocks underlying an area.

basic control—Horizontal and vertical control of third- or higher-order accuracy, determined in the field and permanently marked or monumented, that is required to control further surveys.

basic cover—(JCS) Coverage of any installation or area of a permanent nature with which later coverage can be compared to discover any changes that have taken place. See also **comparative cover.**

bathygraphic—Descriptive of ocean depths.

Bathymetric Archive Data (BAD)—In the DMA Digital Production System, the Bathymetric Archive Data (BAD) layer includes all soundings from all the surveys taken. The sounding data in the BAD layer will eventually be extremely dense. Also included is accuracy and source data, such as the time the survey was taken, where it was taken, and how it was taken.

Bathymetric Model Data (BMD)—In the DMA Digital Production System, soundings from the Bathymetric Archive Data layer are selected to form the Bathymetric Model Data (BMD) layer. The soundings in the BMD layer are the most critical of all the soundings taken. These soundings form the model of the ocean floor for the compilation of Nautical Charts. See also **Bathymetric Archive Data.**

Bathymetric Navigation Charts (BNC)—

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Charts depicting underwater topography in the form of bathymetric curves. BNCs portray detailed multibeam/swath bathymetry. BNC sheets conform to the basic worldwide index of Bottom Contour (BC) Charts for scale, coverage, and number determination.

Bathymetric Navigation Planning Charts (BNPC)—Charts depicting underwater topography in the form of bathymetric curves. BNPCs portray detailed multibeam/swath bathymetry collected in the Oceanographic Survey Program and the morphological interpretation of interswath data. BNPC sheets conform to the basic worldwide index of Bottom Contour (BC) Charts for scale, coverage, and number determination. Produced at various scales.

Bathymetric Recovery Area Charts (BRACs)—See **Precise Bathymetric Navigation Zone Charts.**

bathymetric chart—A topographic map of the floor of the ocean.

bathymetric contour—See **depth contour.**

bathymetric—Relating to the measurement of ocean depths.

bathymetry—The science of determining and interpreting ocean depths and topography.

battle map—(JCS) A map showing ground features in sufficient detail for tactical use by all forces, usually at a scale of 1:25,000. See also **map.**

baud rate—A measure of the speed of data transmission between a computer and other devices [equivalent to the number of discrete conditions or signal events per second].

BC-4 camera—A trade name for the ballistic or geodetic stellar camera consisting of a Wild Astrotar or Wild Cosmotar lens cone mounted on the modified lower part of the Wild T-4 astronomic theodolite. Originally designed for the recording of the trajectory of a rocket but since adapted for the photographic tracking of artificial Earth satellites for geodetic purposes.

beacon tracking—The tracking of a moving object by means of signals emitted from a transmitter or transponder within or attached to the object.

beam of light—A group of pencils of light, as those originating at the many points of an illuminated surface. A beam of parallel light rays is a special case in which each pencil is of such small cross section that it may be regarded as a ray.

Beaman arc—A specially graduated arc fitted to the vertical circle of a transit or alidade for the easy reduction of stadia observations. Also called **stadia circle.**

bearing angle—See **bearing,** definition 1.

bearing bar—See **azimuth instrument.**

bearing circle—A ring designed to fit snugly over a compass or compass repeater, and provided with vanes for observing compass bearings.

bearing line—A line extending in the direction of a bearing.

bearing of line—(plane surveying) The horizontal angle which a line makes with the meridian of reference adjacent to the quadrant in which the line lies. A bearing is identified by naming the end of the meridian (north or south) from which it is reckoned and the direction (east or west) of that reckoning. Thus, a line in the northeast quadrant making an angle of 50° with the meridian will have a bearing of N 50° E. In most survey work, it is preferable to use azimuths rather than bearings.

bearing tree—A marked tree used as a corner accessory; its distance and direction from the corner being recorded. Bearing trees are identified by prescribed marks cut into their trunks; the species and sizes of the trees are also recorded.

bearing—1. (JCS) (general) The horizontal angle at a given point measured clockwise from a specific datum to a second point. Also called ***bearing angle.*** See also **grid bearing; relative bearing.** 2. (navigation) The horizontal direction of one terrestrial point from another, expressed as the angular distance from a reference direction. It is usually measured from 0° at the reference direction clockwise through 360°. The terms **bearing** and **azimuth** are sometimes used interchangeably, but in navigation the former customarily applies to terrestrial objects and the latter to the direction of a point on the celestial

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

sphere from a point on the Earth. 3. (surveying) See **bearing of line.** See also **astronomic bearing; back bearing; compass bearing; computed bearing; curve of equal bearing; electronic bearing; false bearing; great-circle bearing; grid bearing; Lambert bearing; magnetic bearing; polar bearing; rhumb bearing; true bearing.**

Bell gravity meter—A single-axis, pendulous force rebalance accelerometer mounted on a stabilized platform and interfaced to a dynamic digital filter for measuring gravity aboard a survey platform.

bench mark (BM)—A marked vertical control point which has been located on a relatively permanent material object, natural or artificial, and whose elevation above or below an adopted datum has been established. It is usually monumented to include bench mark name or number, frequently its elevation, and the name of the responsible agency. Since elevations are computed at a later time, they are seldom added to newer control bench marks. A BM (aside from a vertical angle bench mark) seldom has a surveyed latitude or longitude. See also **first-order bench mark; junction bench mark; permanent bench mark; primary bench mark; second-order bench mark; temporary bench mark; tidal bench mark; vertical-angle bench mark.**

Bessel spheroid (ellipsoid)—A reference ellipsoid having the following approximate dimensions: semimajor axis—6,377,397.2 meters; semiminor axis—6,356,078.9 meters; and the flattening or ellipticity—1/299.15.

Bessel's method—See **triangle-of-error method.**

Besselian star numbers—Constants used in the reduction of a mean position of a star to an apparent position (used to account for short-term variations in the precession, nutation, aberration, and parallax).

Besselian year—See **fictitious year.**

bianglo screen—A photographic negative containing a composite of two dot screens, with the screen angles oriented 30° apart. These screens are used to print tones of color for chart features with thin lines.

Bilby steel tower—A triangulation tower consisting of two steel tripods, one within the other. The inner tripod holds the instrument platform, and the outer tripod holds the observer's platform. The tower can be easily erected and as easily disassembled and moved to a new location. See also **survey tower.**

bimargin format—The format of a map or chart on which the cartographic detail is extended to two edges of the sheet, normally north and east, thus leaving only two margins. See also **bleed; bleeding edge.**

binary digit—Usually called a Bit, a binary digit is the smallest unit of information which can be stored in the computer. See also **bit.**

binocular vision—Simultaneous vision with both eyes.

binocular—An optical instrument for use with both eyes simultaneously.

bit map—A pattern of bits within a grid, stored in memory, and used to generate an image on a raster scan display.

bit plane—A gridded memory in a graphics device used for storing information for display. Typically one color is assigned to each bit plane for display.

bit—A binary digit.

bivariate normal distribution function—Mathematical function describing the behavior of two-dimensional random errors (e.g., latitude, longitude; *x*, *y*; easting, northing). Also called **circular normal distribution.**

blackbody—An ideal surface or body that completely absorbs all radiant energy falling upon it. Blackbodies are used as models in the design and calibration of remote sensing systems.

blaze—A mark made upon a tree trunk usually at about breast height. The bark and a small amount of the live wood are removed with an axe or other cutting tool, leaving a flat, smoothed surface which forever brands the tree. On rough-barked tree monuments or bearing trees the appropriate marks are scribed into a smooth, narrow, vertical blaze the lower end of which is about 6 inches above the root crown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

The blaze should be just long enough to allow the markings to be made.

bleeding edge—(JCS) That edge of a map or chart on which cartographic detail is extended beyond the neatline to the edge of the sheet. See also **bimargin format; bleed.**

bleed—**1.** (lithography) A condition wherein ink pigment is dissolved by press fountain solution causing a light film of ink (scum) on the plate and impression. **2.** (cartography) Cartographic detail extending to the edge of a map or chart sheet.

blind image—See **blue line.**

blip—(JCS) The display of a received pulse on a cathode-ray tube. Also called **echo.**

blister—See **border break.**

block adjustment—The adjustment of strip coordinates or photograph coordinates for two or more strips of photographs. See also **strip adjustment.**

block out—See **opaque**, definition 4.

block—**1.** (digital storage) A group of records or words treated as a logical unit of information. For example, a VAX magnetic disk block equals 512 bytes, and an unspecified magnetic tape block equals 8192 bytes. A fixed or variable number of records. **2.** Synonymous with physical records--a sequence of words or characters written contiguously by a computer on an external storage medium. Typically, one block is written each time a WRITE command is executed. **3.** A set of entities in a CAD system which can be treated as a single compound object. **4.** (aerial photography) Two or more strips of overlapping photography. See also **flight block.**

bloomed lens—See **coated lens.**

blooming—**1.** The term used to describe localized overexposure caused by incoming radiant energy levels which exceed film emulsion latitude thereby causing the image to lack definition. **2.** A process for increasing the light transmission of lenses.

blow up—A photographic enlargement. Also used as a verb.

blue line—A nonreproducible blue image or outline usually printed photographically on paper or plastic sheeting, and used as a guide for drafting, stripping, or layout. Also called **blind image.**

blue magnetism—The magnetism displayed by the south-seeking end of a freely suspended magnet. This is the magnetism of the Earth's north magnetic pole. See also **red magnetism.**

blunder—A mistake generally caused by carelessness. A blunder may be large and easily detectable, or smaller and more dangerous, or very small and indistinguishable from a random error. Blunders are detected by repetition and by external checks, such as closing a traverse or substituting the solution of an equation in the original. See also **random error; systematic error.**

boat sheet—The worksheet used in the field for plotting details of a hydrographic survey as it progresses. See also **field sheet.**

Bonne map projection—A modified equal-area map projection of the so-called conical type having lines representing a standard parallel and a central meridian intersecting near the center of the map. The line representing the central meridian (geographic) is straight and the scale along it is exact. All geographic parallels are represented by arcs of concentric circles at their true distances apart, divided to exact scale, and all meridians, except the central one, are curved lines connecting corresponding points on the parallels.

Boolean operation—Any operation in which each of the operands and the result take one of two values.

border break—(JCS) A cartographic technique used when it is required to extend a portion of the cartographic detail of a map or chart beyond the sheet lines into the margin. Also called **blister.**

border data—See **marginal data.**

border information—See **marginal data.**

border matching—The process by which individual digital data sets are joined together through merging/smoothing utilizing computer applications software.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Boston leveling rod—A two-piece rod with fixed target on one end. The target is adjusted in elevation by moving one part of the rod on the other. Read by vernier. For heights greater than 5 1/2 feet, the target end is up; for lesser heights, the target end is down.

Bottom Contour Charts (BC)—Charts depicting detailed underwater topography in the form of bathymetric curves. BCs provide a general picture of the seafloor and its features, portrayed from survey data and foreign published charts. Designed for the use of submarines or ships by the method of bottom contour matching.

Bouguer anomaly—A difference between an observed value of gravity and a theoretical value at the point of observation, which has been corrected for the effect of topography and elevation only, the topography being considered as a plate of indefinite extent.

Bouguer correction—A correction made in gravity work to take account of the altitude (elevation) of the station and the density of the mass between an infinite plane though the point of observation and the infinite plane of the reference elevation.

Bouguer plate—An imaginary layer of infinite length and of thickness equal to the height of the observation point above the reference surface (usually the geoid). In applying the Bouguer correction, the attracting layer lessens the free air effect.

Bouguer reduction—Geophysically, the Bouguer reduction removes all masses above the reference surface (usually the geoid) and then reduces the gravity from the terrain to the reference surface.

boundary (de facto)—An international or administrative boundary whose existence and legality are not recognized by all concerned, or are not defined by appropriate documents, but which is a practical division between separate national or provincial administering authorities.

boundary (de jure)—An international or administrative boundary whose existence and legality are recognized.

boundary discontinuities—Different elevation values for common DMA standard digital terrain elevation data (DTED) matrix points in adjacent data files.

boundary line—A line of demarcation between contiguous political or geographical entities. The word "boundary" is some times omitted, as in "state line"; some times the word "line" is omitted, as in "international boundary," "county boundary," etc. The term **boundary line** is usually applied to boundaries between political territories, and "state boundary line," between two states. A boundary line between privately owned parcels of land is termed a **property line** by preference, or if a line of the United States public land surveys, is given the particular designation of that survey system, as "section line," "township line," etc.

boundary map—A map prepared specifically for the purpose of delineating a boundary line and adjacent territory.

boundary monument—A material object placed on or near a boundary line to preserve and identify the location of the boundary line on the ground.

boundary point averaging—Averaging the elevations of common points from two different data sets and reassigning each the average value; or resolving a discrepancy between two elevations for a common point by assigning the average elevation value to that point.

boundary point weighted averaging—Resolving a discrepancy between a series of overlapping elevations from two data sets by biasing one data set utilizing either the accuracy of a data set or the relative distance to the edge of the data.

boundary survey—A survey made to establish or to reestablish a boundary line on the ground or to obtain data for constructing a map or plat showing a boundary line. The term **boundary survey** is usually restricted to surveys of boundary lines between political territories. For the survey of a boundary line between privately owned parcels of land, the term **land survey** is preferred; except in United States public land surveys the term **cadastral survey** is used.

boundary vista—A lane cleared along a boundary line passing through a wooded area.

Bowie effect—The indirect effect on gravity due to the warping of the geoid, or the elevation of the geoid with respect to the spheroid of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

reference.

Bowie method of adjustment—A method for the adjustment of large networks of triangulation.

box compass—See **declinatoire.**

break angle—The deflection angle between the two vertical phases passing through the common nadir point and the principal points of the left and right oblique photographs.

break tape—See **broken tape measurement.**

break-circuit chronometer—A chronometer equipped with a device which automatically breaks an electric circuit, the breaks being recorded on a chronograph.

break-up—(JCS) 1. In detection by radar, the separation of one solid return into a number of individual returns which correspond to the various objects or structure groupings. This separation is contingent upon a number of factors including range, beam width, gain setting, object size, and distance between objects. 2. In imagery interpretation, the result of magnification or enlargement which causes the imaged item to lose its identity and the resultant presentation to become a random series of tonal impressions.

breakeway method—See **breakaway strip method.**

breakaway strip method—A technique used in photomosaicking when two or more sheets are prepared. The process involves placing an extra wide strip of masking tape along the outside edge of the neatline of one sheet before mosaicking photos. The mosaicked overedge is then cut along the neatline and transferred to the adjoining sheet. Also called **breakaway method.**

breaking tape—See **broken tape measurement.**

bridging—A photogrammetric method of establishing and adjusting control between bands of existing ground control, both horizontally and vertically. The term is usually qualified as horizontal or vertical according to its primary purpose. Also called **horizontal bridging; horizontal/vertical bridging; vertical bridging.**

brightness scale—(photography) The ratio of the brightness of highlights to the deepest shadow in the actual terrain, as measured from the camera stations, for the field of view under consideration .

brightness value—The amount of reflected or emitted energy exiting from the earth's surface as recorded by a remote sensing system. These data are stored as digital values on computer-compatible tapes (CCT) for digital image processing purposes. The greater the brightness of the scene (or return from the scene), the higher the digital value. Hence the values stored on the CCT are often referred to as brightness values. It is not proper to call them reflectance values unless the brightness values have been scaled to truly represent reflectance values.

British grid reference system—A system of rectangular coordinates devised or adopted by the British for use on military maps. There is no related global plan for the many grids, belts, and zones which make up the British grid system. It is being replaced by the Universal Transverse Mercator (UTM) grid system.

broadcast ephemeris—A set of parameters broadcast by satellite from which Earth-fixed satellite positions can be computed. In particular, the parameters for the Navy Navigation Satellites (NNS) are computed for each NNS by fitting 36- to 48-hour orbital arcs to Doppler data from four tracking stations and extrapolating this arc 12 to 24 hours beyond the last data used. The length of the arc fit and the extrapolation period depend on the upper atmospheric air density. The computed parameters are injected into the satellite memory and are transmitted along with time on each even minute. See also **Navy Navigation Satellite System.**

broken base—A base line for triangulation consisting of two or more lines that form a continuous traverse and have approximately the same general direction.

broken grade—(tape) The change in grade when the middle point of a tape is not on grade with its ends. If the middle support for the tape is not on the same grade as the end supports, the fact is noted with a reference "broken grade at—," naming the particular tape length which contains the broken grade.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

broken tape measurement—(surveying) The short distances measured and accumulated to total a full tape length when a standard 100-foot tape cannot be held horizontally without plumbing from above shoulder level. Also called **break tape; breaking tape.**

broken telescope transit—A precise astronomic transit in which the light entering the objective lens is reflected at right angles by a prism placed within the telescope, the reflected light ray passing to the eyepiece, which is in the horizontal axis of the telescope.

Brown gravity apparatus—An apparatus for measuring the acceleration of gravity which utilizes the Mendenhall pendulum, but has a clamping device for holding the pendulum in the receiver when being transported from station to station, and which utilizes an electrical pickup and amplifying device for recording the oscillations (pendulum) on the chronograph sheet.

browsing—System capability to find an undefined feature or set of features in a spatial data base.

Brunton compass—An instrument combining the features of both the sighting compass and the clinometer that can be used in the hand or upon a Jacob's staff or light tripod for reading horizontal and vertical angles, for leveling, and for reading the magnetic bearing of a line. Also called **Brunton pocket transit.**

Brunton pocket transit—See **Brunton compass.**

bubble axis—See **spirit level axis.**

bubble level—See **spirit level.**

bubble sextant—A sextant in which the bubble of a spirit level serves as the horizon.

bucking in—The act of aligning a theodolite to a collimator or autocollimating theodolite serving as collimator to make their lines of sight parallel and on the same plane.

bug—An error in a computer program or in a piece of electronic equipment that causes it to function improperly.

bull's-eye level—See **circular level.**

Bullard method of isostatic reduction—See **Hayford-Bullard (or Bullard) method of isostatic reduction.**

burn—(lithography) The process of exposing a pressplate.

byte—1. A group of adjacent bits that are operated on as one unit. Eight bits equal one byte. 2. A storage unit equivalent to an ASCII character of information in a computer system.

***Bz* curve method**—A method utilizing characteristics of the *Bz* curve for finding the displacement of true photo plumb points from indicated projector plumb points in multiplex strip orientation. The method also provides a means of strip leveling using only the barometric altimeter readings of the aircraft flying height.

***Bz* curve**—(photogrammetry) A graphical representation of the vertical errors in a stereotriangulated strip. In a *Bz* curve, the *x*-coordinates of the vertical control points, referred to the initial nadir point as origin, are plotted as abscissas, and the differences between the known elevations of the control points and their elevations as read in the stereotriangulated strip are plotted as ordinates; a smooth curve drawn through the plotted points is the *Bz* curve. The elevation read on any pass point in the strip is adjusted by the amount of the ordinate of the *Bz* curve for an abscissa corresponding to the *x*-coordinate of the point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C

C-constant—See **level constant.**

C-factor—An empirical value which expresses the vertical measuring capability of a given stereoscopic system; generally defined as the ratio of the flight height to the smallest contour interval accurately plottable. The C-factor is not a fixed constant, but varies over a considerable range, according to the elements and conditions of the photogrammetric system. In planning for aerial photography, the C-factor is used to determine the flight height required for a specified contour interval, camera, and instrument system. Also called **altitude contour ratio.**

cadastral map—A map showing the boundaries of subdivisions of land, usually with the bearings and lengths thereof and the areas of individual tracts, for purposes of describing and recording ownership. Also called **property map.** See also **plat.**

cadastral survey—A survey relating to land boundaries and subdivisions, made to create units suitable for transfer or to define the limitations of title. The term **cadastral survey** is now used to designate the surveys of the public lands of the United States, including retracement surveys for the identification and resurveys for the restoration of property lines; the term can also be applied properly to corresponding surveys outside the public lands, although such surveys are usually termed **land surveys** through preference.

cairn—An artificial mound of rocks, stones, or masonry usually conical or pyramidal, whose purpose is to designate or to aid in identifying a point of surveying or of cadastral importance.

calculated altitude—See **computed altitude.**

calendar day—The period from midnight to midnight. The calendar day is 24 hours of mean solar time in length and coincides with the civil day unless a time change occurs during the day.

calendar month—A division of the year as determined by a calendar, approximately one-twelfth of a year in length. While arbitrary in character, the calendar month is based roughly on the synodical month. The calendar month ranges in length from 28 to 31 mean solar days.

calendar year—A conventional year based on the tropical year and adjusted by "leap years" to fit the nonintegral length of the tropical year.

calibrated focal length—**1.** (JCS) An adjusted value of the equivalent focal length, so computed as to equalize the positive and negative values of distortion over the entire field used in a camera. See also **focal length. 2.** The distance along the lens axis from the interior perspective center to the image plane.

calibration card—A card having a list of calibration corrections or calibrated values.

calibration constants—The results obtained by calibration, which give the calibrated focal length of the lens-camera unit and the relationship of the principal point to the fiducial marks of a camera and give significant calibration corrections for lens distortions.

calibration correction—The value to be added to or subtracted from the reading of an instrument to obtain the correct reading.

calibration course—See **field comparator.**

calibration error—See **instrument error.**

calibration plate—A glass negative exposed with its emulsion side corresponding to the position of the emulsion side of the film in the camera at the time of exposure. This plate provides a record of the distance between the fiducial marks of the camera. Also called **flash plate; master glass negative.**

calibration table—A list of calibration corrections or calibrated values.

calibration templet—(photogrammetry) A templete of glass, plastic, or metal made in accordance with the calibration constants to show the relationship of the principal point of a camera to the fiducial marks; used for the rapid and accurate marking of principal points on a series of photographs. Also, for a multiple-lens camera, a template prepared from the calibration data and used in assembling the individual photographs into one composite photograph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

calibration—The act or process of determining certain specific measurements in a camera or other instrument or device by comparison with a standard, for use in correcting or compensating for errors or for purposes of record. See also **camera calibration; field calibration; shop calibration.**

Callippic cycle—A period of four Metonic cycles equal to 76 Julian years, or 27,759 days.

call—(USPLS) A reference to, or statement of, an object course, distance, or other matter of description in a survey or grant, requiring or calling for a corresponding object, or other matter of description, on the land.

camera axis direction—(JCS) Direction on the horizontal plane of the optical axis of the camera at the time of exposure. This direction is defined by its azimuth expressed in degrees in relation to true/magnetic north.

camera axis—(JCS) An imaginary line through the optical center of the lens perpendicular to the negative photo plane.

camera calibration—(JCS) The determination of the calibrated focal length, the location of the principal point with respect to the fiducial marks, and the lens distortion effective in the focal plane of the camera and referred to the particular calibrated focal length. [In a multiple-lens camera, the calibration also includes the determination of the angles between the component perspective units. The setting of the fiducial marks and the positioning of the lens are ordinarily considered as adjustments, although they are sometimes performed during the calibration process. Unless a camera is specifically referred to, distortion and other optical characteristics of a lens are determined in a focal plane located at the equivalent focal length and the process is termed **lens calibration.**]

camera lucida—A monocular instrument using a half-silvered mirror, or the optical equivalent, to permit superimposition of a vertical image of an object upon a plane. Also called **camera obscura.** See also **sketchmaster.**

camera magazine—(JCS) A removable part of a camera in which the unexposed and exposed portions of film are contained.

camera obscura—See **camera lucida.**

camera station—See **air base; air station.**

camera transit—See **photo- theodolite.**

camera window—(JCS) A window in the camera compartment through which photographs are taken .

camera—A lightproof chamber or box in which the image of an exterior object is projected upon a sensitized plate or film through an opening usually equipped with a lens or lenses, shutter, and variable aperture. See also **aerial camera; BC-4 camera; ballistic camera; continuous strip camera; convergent camera; copy camera; direct scanning camera; fan cameras; frame camera; geodetic stellar camera; horizon camera; mapping camera; metric cemera; multiple-camera assembly; multiple-lens camera; PC-1000 camera; panoramic camera; photogrammetric camera; positioning camera; precision camera; rectifier; rotating prism camera; split cameras; stellar camera; stereometric camera; terrestrial camera; trimetrogon camera; variable perspective camera system; zenith camera.**

Canadian grid—See **perspective grid.**

candela—A unit of luminous intensity.

cantilever extension— Phototriangulation from a controlled area to an area of no control. Also, the connection by relative orientation and scaling of a series of photographs in a strip to obtain strip coordinates. Also called **extension.**

Cape Canaveral datum—This special datum is defined with its origin at station Central on the John F. Kennedy Space Center, Cape Canaveral, Florida, with azimuth to Central SE Base. The geodetic coordinates of these two stations were identical to those on North American datum of 1927. Datum differences for other points may be determined by subtracting North American datum of 1927 values from the Cape Canaveral datum values as established by the USC&GS transcontinental traverse of the United States. See also **North American datum of 1927.**

Source: https://assist.dla.mil - Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

cardan link—A universal joint. An optical cardan link is a device for universal scanning about a point.

cardinal point effect—(JCS) The increased intensity of a line or group of returns on the radarscope occurring when the radar beam is perpendicular to the rectangular surface of a line or group of similarly aligned features in the ground pattern.

cardinal points—**1.** The directions: north, south, east, west. **2.** (optics) Those points of a lens used as reference for determining object and image distances. They include principal planes and points, nodal points, and focal points.

Carpentier inversor—One of the inversors which corrects for the Scheimpflug condition in a rectifier if the negative, lens, or easel planes are tilted and not parallel.

carrying contour—A single contour line representing two or more contours, used to show vertical or near-vertical topographic features, such as steep slopes and cliffs.

Cartesian coordinates—A coordinate system in which locations of points in space are expressed by reference to three mutually perpendicular planes, called coordinate planes. The three planes intersect in three straight lines called **coordinate axes**. [Also the values representing the location of a point in a plane in relation to two perpendicular intersecting straight lines, called **axes**. The point is located by measuring its distance from each axis along a parallel to the other axis.]

Cartographic Automatic Mapping (CAM)—A mainframe computer mapping program available from the Central Intelligence Agency. It works with World Data Bank (WDB II)coordinate files to create map projections and generate plots.

cartographic annotation—The delineation of additional data, new features, or deletion of destroyed or dismantled features on a mosaic to portray current details. Cartographic annotations may include elevation values for airfields, cities, and large bodies of water; new construction and destroyed or dismantled roads, railroads, bridges, dams, target installations, and cultural features of landmark significance.

cartographic compilation—See **compilation**, definition 1.

cartographic data base (CDB)—**1.** A data base of map graphics captured from a map or used to produce a map. A CDB incorporates a hierarchy for feature displacement. **2.** An internal DMA function which consists of actual digital data products, an automated directory defining the data availability, data base management software, computer hardware, and CDB operations staff and management. The CDB is resident at DMAAC.

cartographic feature—The natural or cultural objects shown on a map or chart. See also **topography**, definition 1.

cartographic film—Film with a dimensionally stable base, used for map negatives and/or positives. Usually referred to by trade name.

cartographic license—The freedom to adjust, add, or omit map features within allowable limits to attain the best cartographic expression. License must not be construed as permitting the cartographer to deviate from specifications.

cartographic photography—See **mapping photography**.

cartographic primitive—A type of primitive that does not participate in topology. Text is the only cartographic primitive. See also **geometric primitive**.

cartographic scanner—A device for strip-by-strip scanning of two-dimensional copy and for digital registration of the light/dark (black/white) parts as rectangular coordinates.

cartography—The art and science of expressing graphically, by maps and charts, the known physical and political/administrative features of the Earth, or of another celestial body.

cartometric scaling—The accurate measurement of geographic or grid coordinates on a map or chart by means of a scale. This method may be used for plotting the positions of points, or determining the location of points.

carving—The development of the model surface by carving away the steps of the plaster

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

step cast in the production of relief models.

cassette—(JCS) In photography, a reloadable container for either unexposed or exposed sensitized materials which may be removed from the camera or darkroom equipment under lightened conditions.

Cassini map projection—A conventional map projection constructed by computing the lengths of arcs along a selected geographic meridian and along a great circle perpendicular to that meridian, and plotting these as rectangular coordinates on a plane.

Cassini-Soldner map projection— Similar to a polyconic map projection except that it uses but one central meridian for a whole series. Best adapted for north-south belts and large-scale maps of small areas.

casting—The process of reproducing relief models in plaster or epoxy from the terrain base of the model, or after the surface of the model has been developed. Models are first cast negative, from which any number of positive castings may be made.

casual error—See **random error.**

catadioptric system—(optics) An optical system containing both refractive and reflective elements.

catenary correction—(taping) See **sag correction.**

cathode ray—**1.** One of the high-speed electrons projected in a stream from the heated cathode of a vacuum tube under the propulsion of a strong electric field. **2.** A stream of cathode-ray electrons.

cathode-ray tube—A vacuum tube in which cathode rays, usually in the form of a slender beam, are projected upon a fluorescent screen that serves as an anticathode where the rays produce a luminous spot.

cathode—The electrode at which electrons enter a device from the external circuit.

catoptric system—(optics) An optical system in which all elements are reflective (mirrors).

cautionary note—Information calling special attention to some fact, usually a danger area, shown on a map or chart.

celestial coordinates—Any set of coordinates used to define a point on the celestial sphere.

celestial equator system of coordinates—A set of celestial coordinates based on the celestial equator as the primary great circle; usually declination and hour angle or sidereal hour angle. Also called **equator system; equatorial system; equinoctial system of coordinates.**

celestial equator—The great circle on the celestial sphere whose plane is perpendicular to the axis of rotation of the Earth. Also called **equinoctial.**

celestial fix—A position established by means of observation on one or more celestial bodies.

celestial geodesy—The branch of geodesy which utilizes observations of near celestial bodies, including Earth satellites, to determine the size and shape of the Earth.

celestial horizon—That circle of the celestial sphere formed by the intersection of the celestial sphere and a plane through the center of the Earth and perpendicular to the zenith nadir line. Also called **rational horizon.**

celestial latitude—Angular distance north or south of the ecliptic; the arc of a circle of latitude between the ecliptic and a point on the celestial sphere, measured northward or southward from the ecliptic through 90°, and labeled "N" or "S" to indicate the direction of measurement. Also called **ecliptic latitude.**

celestial line of position—A line of position determined by means of the observation of a celestial body.

celestial longitude—Angular distance east of the vernal equinox, along the ecliptic; the arc of the ecliptic or the angle at the ecliptic pole between the circle of latitude of the vernal equinox and the circle of latitude of a point on the celestial sphere, measured eastward from the circle of latitude of the vernal equinox, through 360°. Also called **ecliptic longitude.**

celestial mechanics—The study of the theory of the motions of celestial bodies under the influence of gravitational fields.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

celestial meridian—An hour circle of the celestial sphere, through the celestial poles and the zenith. The two intersections of the celestial meridian with the horizon are known as the north and south points.

celestial observation—1. Observation of celestial phenomena. 2. (navigation) The measurement of the altitude or the azimuth, or both, of a celestial body. Also the data obtained by such measurement.

celestial parallel—See **parallel of declination.**

celestial pole—Either of the two points of intersection of the celestial sphere and the extended axis of the Earth.

celestial refraction—See **astronomic refraction.**

celestial sphere—(JCS) An imaginary sphere of infinite radius concentric with the Earth, on which all celestial bodies except the Earth are imagined to be projected. [For observations on bodies within the limits of the solar system, the assumed center is the center of the Earth. For bodies where the parallax is negligible, the assumed center may be the point of observation.]

celestial triangle—A spherical triangle on the celestial sphere, especially the navigational triangle.

cell—Variable size rectangular geographic area, often designated by latitude/longitude boundaries. DMA produces standard elevation (DTED) and feature (DFAD) data in one degree by one degree cells.

center line data (CLD)—Data which describes points, lines and areas as one or several coordinate points selected at the center of the phenomena being described. Attributes used with centerline data describe the feature in real world terms and not the display of graphic output. Symbolized graphic data, on the other hand, contains attribution for the display of graphic output.

center of gravity—The point in any body at which the force of gravity may be considered to be concentrated.

center of instrument—The point on the vertical axis of rotation of an instrument at the same elevation as the axis of collimation when that axis is in a horizontal position. In a transit or theodolite, it is close to or at the intersection of the horizontal and vertical axes of the instrument.

center of mass—The point at which all the given mass of a body or bodies may be regarded as being concentrated as far as motion is concerned.

center of oscillation—(pendulum) The position in a compound pendulum of the particle which corresponds to the heavy particle of an equivalent simple pendulum. The centers of suspension and oscillation are interchangeable. If the center of oscillation is made the center of suspension, the former center of suspension becomes the new center of oscillation. This principle is the basis of design of compound reversible pendulums.

center of projection—See **perspective center.**

center of radiation—See **radial center.**

center of suspension—(pendulum) The fixed point about which a pendulum oscillates. See also **center of oscillation.**

center point—See **radial center.**

centerline—1. (USPLS) The line connecting opposite corresponding quarter corners or opposite subdivision-of-section corners or their theoretical positions. 2. A line extending from the true center point of overlapping aerial photos through each of the transposed center points. 3. (engineering survey) The continuous center of a highway or railroad, with stationing indicating starting point, culverts, points of curvature, etc.

centimeter-gram-second (c.g.s.) system—A system of units based on the centimeter as the unit of length, the gram as the unit of mass, and the mean solar second as the unit of time. A part of the metric system.

central force field—The spatial distribution of the influence of a central force.

central force orbit—The theoretical orbit

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

achieved by a particle of negligible mass moving in the vicinity of a point mass with no other forces acting; an unperturbed orbit.

central force—A force which for purposes of computation can be considered to be concentrated at one central point with its intensity at any other point being a function of the distance from the central point. Gravitation is considered as a central force in celestial mechanics.

central meridian—The longitudinal line of symmetry of a map projection, and generally a base line for referencing the projection to the associated grid coordinate system. As such, grid north and true north are coincident along the central meridian, but at any other point of the projection, grid north and true north diverge as a function of geodetic latitude and longitudinal distance from the central meridian due to convergence of the meridians. Most map projections portray meridian convergence, but one notable exception is the Mercator Projection.

central point figure—A triangulation figure consisting of a polygon with an interior station, formed by a series of adjoining triangles with a common vertex at the interior station.

centrifugal force—The force with which a body moving under constraint along a curved path reacts to the constraint, acting in a direction opposite from the center of curvature of the path. For a body constrained to Earth, centrifugal force acting on the body due to Earth's rotation is perpendicular to and away from Earth's rotational axis. The vector sum of this centrifugal force and the gravitational force due to the total mass of the body and Earth is defined as the gravity force acting on the body. See also **centripetal force.**

centripetal force—The force directed towards the center of curvature, which constrains a body to move in a curved path. See also **centrifugal force.**

centroid—(digital mapping) A point interior to a polygon whose coordinates are the average of the corresponding coordinates for all nodal points which define the polygon. This point may be outside or inside the polygon and can be some point other than the geographic center of the polygon.

chain gage—See **tape gage.**

chain node—An arc-node topological model that stores the entity descriptions and spatial extents in separate records, and links them with record pointers. The basic entity is a chain or arc, a series of coordinates that start and end at a node. A node is a beginning, an end, an intersection, or a point represented by a coordinate pair. See also **arc, edge, link and node.**

chain-node data structure—A vector data structure in which the entity description and the spatial extent of each feature in a vector data set are stored in separate records, linked by record pointers. In addition, all intersections between features, as well as point features, are captured and stored as nodes.

chaining pin—See **pin.**

chaining—See **taping.**

chain—A device used by surveyors for measuring distance, or the length of this device as a unit of distance. The usual chain is 66 feet long, and consists of 100 links, each 7.92 inches long. See also **engineer's chain; Gunter's chain.**

challenger—See **interrogator responsor.**

chambered spirit level—A level tube with a partition near one end which cuts off a small air reservoir so arranged that the length of the bubble can be regulated.

change detection—An image enhancement technique which compares two images of the same area from different time periods. Identical picture elements are eliminated, leaving signatures which have undergone change.

characteristic curve—(photography) A curve showing the relationship between exposure and resulting density in a photographic image, usually plotted as the density (D) against the logarithm of the exposure (log E) in candle-meter-seconds. Also called **D log E curve; density exposure curve; H and D curve; sensitometric curve; time gamma curve.** See also **contrast; density,** definition 1.

character—The distinctive trait, quality, property, or behavior of man-made or natural features as portrayed by a cartographer. The more character applied to detail, the more closely it will resemble these features as they

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

appear on the surface of the Earth. See also **generalization.**

Chart Update Manual (CHUM) — A DMA published manual which provides aeronautical chart users with current information on a monthly basis concerning charts and hazards to flight safety for those charts. Future CHUMs may be available electronically (ECHUM).

chart base—(JCS) A chart used as a primary source for compilation or as a framework on which new detail is printed. Also called **topographic base**. See also **base map.**

chart comparison unit—A device permitting simultaneous viewing of navigational instrument presentation, such as a radarscope and a navigational chart, so that one appears superimposed upon the other. Also called **autoradar plot** when used with radar.

chart datum—See **hydrographic datum.**

charted depth—The vertical distance from the tidal datum to the bottom surface.

charting photography—See **mapping photography.**

chartlet—A small chart, such as those annexed to Notices to Mariners.

chart—1. A special-purpose map, generally designed for navigation or other particular purposes, in which essential map information is combined with various other data critical to the intended use. 2. To prepare a chart, or engage in a charting operation. See also **aeronautical chart; aeronautical pilotage chart; aeronautical planning chart; Air Target Chart; anchorage chart; approach chart; azimuthal chart; azimuthal equidistant chart; Baldwin solar chart; bathymetric chart; chartlet; coastal chart; combat chart; conformal chart; conic chart; conic chart with two standard parallels; Consol chart; cotidal chart; current chart; Decca chart; enroute chart; equatorial chart; firing chart; general chart; Global Navigation Chart; gnomonic chart; great-circle chart; harbor chart; historical chart; hydrographic chart; hypsographic map (or chart); ice chart; index chart; instrument approach chart; isobaric chart; isoclinic chart; isogonic chart; isogriv chart; isomagnetic chart; isoporic chart; Jet Navigation Chart; Lambert conformal chart; local chart; long-range navigation chart; loran chart; lunar chart; lunar earthside chart; lunar farside chart; magnetic chart; Marsden chart; mean chart; Mercator chart; meteorological chart; mileage chart; miscellaneous chart; modified Lambert conformal chart; new chart; oblique chart; oblique Mercator chart; obsolete chart; Operational Navigation Chart; orthographic chart; orthomorphic chart; perspective chart; pilot chart; pilotage chart; planning chart; plotting chart; polar chart; polyconic chart; radar chart; rectangular chart; route chart; sailing chart; search-and-rescue chart; secant conic chart; sectional chart; sextant chart; simple conic chart; star chart; stereographic chart; Tactical Pilotage Chart; tidal current chart; time zone chart; track chart; transverse chart; transverse Mercator chart; virtual PPI reflectoscope chart; visibility chart; weather map; World Aeronautical Chart.**

check point—1. (JCS) A predetermined point on the Earth's surface used as a means of controlling movement, a registration target for fire adjustment, or a reference for location. 2. (JCS) Geographical location on land or water above which the position of an aircraft in flight may be determined by observation or by electronic means. 3. A point, selected on obliques only, in the vicinity of each tie point and distant point for the purpose of checking the identification of these points.

check profile—A profile plotted from a field survey and used to check a profile prepared from a topographic map. The comparison of the two profiles serves as a check on the accuracy of the contours on the topographic map.

checked spot elevation—An elevation established in the field by: closed spirit leveling, trigonometric leveling by a closed circuit of barometric leveling, or any other method such that proof of accuracy is obtained.

checking positive—A composite printing on glass of the contour and drainage drawings used on the shadow projector for checking the horizontal accuracy of landforms to be developed on relief models.

chopping—(star or satellite trails) Interrupting the photographic image of a star or satellite trail

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

by a shutter or other device to provide precise timing and orientation data for geodetic observations of aerospace vehicles against a stellar background.

chord—1. (route surveying) Chord used in highway and other surveys to indicate a straight line between two points on a curve, regardless of the distance between them. 2. In surveying and geometry, a straight line joining any two points on an arc, curve, circumference, or surface.

chorographic map—Any map representing large regions, countries, or continents on a small scale. Atlas and small-scale wall maps belong in this class.

chromatic aberration—See **lateral chromatic aberration; longitudinal chromatic aberration.**

chromatic colors—Colors eliciting hue.

chromaticity coordinates—The proportions of standard components required for color match, used as an ordinate and abscissa to represent color in a chromaticity diagram.

chromaticity diagram—A plane diagram formed by plotting one of the chromaticity coordinates against another.

chromaticity—A composite of dominant wavelength and purity.

chronograph—An instrument for producing a graphical record of time as shown by a clock or other device. In use, a chronograph produces a double record: the first is made by the associated clock and forms a continuous time scale with significant marks indicating periodic beats of the timekeeper; the second is made by some external agency, human or mechanical, and records the occurrence of an event or of a series of events.

chronometer correction—See **clock correction.**

chronometer error—The amount by which the chronometer differs from the correct time.

chronometer rate—See **clock rate.**

chronometer—A portable timekeeper with compensated balance, capable of showing time with extreme precision and accuracy. See also **break-circuit chronometer; hack chronometer.**

cine theodolite—A photographic tracking instrument which records on each film frame the target and the azimuth and elevation angles of the optical axis of the instrument.

circle of confusion—(optics) The circular image of a distant point object as formed in a focal plane by a lens. A distant point object (e.g., a star) is imaged in a focal plane of a lens as a circle of finite size, because of such conditions as: (1) the focal plane's not being placed at the point of sharpest focus; (2) the effect of certain aberrations; (3) diffraction at the lens; (4) grain in a photographic emulsion; and/or (5) poor workmanship in the manufacture of the lens.

circle of declination—See **hour circle.**

circle of equal altitude—See **parallel of altitude.**

circle of equal declination—See **parallel of declination.**

circle of latitude—1. A great circle of the celestial sphere through the ecliptic poles, and hence perpendicular to the plane of the ecliptic. 2. A meridian along which latitude is measured.

circle of longitude—1. A circle of the celestial sphere, parallel to the ecliptic. 2. A circle on the surface of the Earth, parallel to the plane of the Equator; a parallel, along which longitude is measured. Also called **parallel of latitude.**

circle of perpetual apparition—That circle of the celestial sphere, centered on the polar axis and having a polar distance from the elevated pole approximately equal to the latitude of the observer, within which celestial bodies do not set. See also **circle of perpetual occultation.**

circle of perpetual occultation—That circle of the celestial sphere, centered on the polar axis and having a polar distance from the depressed pole approximately equal to the latitude of the observer, within which celestial bodies do not rise. See also **circle of perpetual apparition.**

circle of position—A small circle on the globe

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

(Earth) at every point of which, at the instant of observation, the observed celestial body (sun, star, or planet) has the same altitude and, therefore, the same zenith distance.

circle of right ascension—See **hour circle.**

circle of the sphere—A circle upon the surface of the sphere, specifically of the Earth or of the heavens, called a **great circle** when its plane passes through the center of the sphere; in all other cases, a **small circle.**

circle position—See **position,** definition 4.

circuit closure—(leveling) The amount by which the algebraic sum of the measured differences of elevation around a circuit fails to equal the theoretical closure of zero. See also **error of closure,** definition 4.

circuit—(leveling or traverse) A continuous line of levels, a series of lines of levels, or a combination of lines or parts of lines of levels, such that a continuous series of measured differences of elevation extends around the circuit or loop and then back to the starting point. Also applied to a continuous line of transit traverse in a similar manner.

circular cylindrical coordinates—See **cylindrical coordinates.**

circular error (CE)—An accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (e.g., horizontal position) will be within the given figure. Commonly used are CEP (50 percent), CE 1σ (68.3 percent), and CE (90 percent).

circular error probable (CEP)—The 50 percent error interval based on the bivariate normal distribution function. Also called **circular probable error.**

circular level—A spirit level having the inside surface of its upper part ground to spherical shape, the outline of the bubble formed being circular, and the graduations being concentric circles. This form of spirit level is used where a high degree of precision is not required, as in plumbing a level rod or setting an instrument in approximate position. Also called **bull's-eye level; universal level.**

circular map accuracy standard (CMAS)—The U. S. National Map Accuracy Standards used for domestic mapping. CMAS are specified separately for horizontal and vertical mapped features. See also **horizontal accuracy; vertical accuracy.**

circular near-certainty error (3.5σ)—The 99.78 percent error interval based on the bivariate normal distribution function.

circular normal distribution—See **bivariate normal distribution function.**

circular probable error (CPE)—See **circular error probable.**

circular sigma—See **circular standard error.**

circular standard error (σ_c)—The 68.3 percent error interval based on the bivariate normal distribution function. Also called **circular sigma.**

circulation map—See **traffic circulation map.**

circumferentor—A type of surveyor's compass having slit sights on projecting arms.

circumlunar—Around the Moon, generally applied to trajectories.

circummeridian altitudes—Exmeridian altitudes observed for determination of latitude when a heavenly body is close to transit.

circumpolar—Revolving about the elevated pole without setting. A celestial body is circumpolar when its polar distance is approximately equal to or less than the latitude of the observer.

cislunar—**1.** This side of the Moon. **2.** Of or pertaining to phenomena, projects, or activity in the space between the Earth and the Moon, or between the Earth and the Moon's orbit.

cistern barometer—A mercury barometer in which a column of mercury is enclosed in a vertical glass tube, the upper end of which is sealed and exhausted of air, and the lower end placed in a cistern or reservoir of mercury which is exposed to atmospheric pressure. The atmospheric pressure on the free surface of the mercury in the cistern determines the height to which the mercury will rise in the vertical tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

This may be measured, and the pressure reported in terms of that height, as in inches of mercury.

city graphic (CG)—Large-scale lithographic map of populated places and environs portraying streets and through-route information. Features include important buildings, airfields, military installations, industrial complexes, embassies, etc. Produced at 1:25,000 and larger scale. See also **city products.**

city plan—See **city products.**

city products—Large-scale maps of populated places and environs, usually portraying street and through-route information, important buildings and other urban features, airfields, port facilities, and relief, drainage, and vegetation when important. Several different types of city products are produced by DMA, among which are city graphics, city plans, city route graphics, and military city maps. Specifications for these maps vary according to particular military requirements.

city route graphics (CRG)—See **city products.**

city survey—A specialized type of land survey restricted to work completed primarily within the limits of a city.

civil day—A solar day beginning at midnight. The civil day may be based on either apparent solar time or mean solar time. See also **astronomic day.**

civil time—Solar time in a day (civil day) that begins at midnight. Civil time may be either apparent solar time or mean solar time; it may be counted in two series of 12 hours each, beginning at midnight, marked "am" (ante meridian), and at noon, marked "pm" (post meridian), or in a single series of 24 hours beginning at midnight.

Clairaut's theorem—A theorem that, in its original form, relates the value of centrifugal force at the Equator to the value of gravity at the Equator. Importance to physical geodesy is that the flattening of the Earth can be obtained from gravity measurements.

clamping error—A systematic error in observations made with a repeating theodolite caused by strains set up by the clamping devices of the instrument.

Clarke spheroid (ellipsoid) of 1866— A reference ellipsoid having the following approximate dimensions: semimajor axis—6,378,206.4 meters and the flattening or ellipticity—1/294.9786982.

Clarke spheroid (ellipsoid) of 1880— A reference ellipsoid having the following approximate dimensions: semimajor axis—6,378,249.145 meters; and the flattening or ellipticity—1/293.465.

classification copy—A specialized item of source material used as a guide by the compiler and/or draftsman in preparing a map or chart. Usually consists of detailed information pertaining to roads, railroads, city data, and the like that has been developed from field surveys. Usually furnished in the form of overlays, annotated maps, drawings, photographs, or field sheets.

classification survey—See **field inspection.**

Classified Hydrographic Information Processing System (CHIPS)—The DMA data base which provides navigation update information for classified charts and publications.

Classified Notice to Mariners—DMA publication which provides corrective data applicable to classifies charts and publications. This information is also accessible through the Classified Hydrographic Information Processing System (CHIPS).

clearing *y*-parallax—See **relative orientation.**

clinometer—A simple instrument used for measuring the degree of slope in percentage or in angular measure.

clipping—**1.** (GIS) The process of subdividing overlapping polygons to produce all combinations of separate unique polygons. Also called complexing. **2.** (computer graphics) The severing of that portion of a line segment which falls outside a predefined frame.

clock correction—The quantity which is added, algebraically, to the time shown by a clock to obtain the time of a given meridian. If the clock is slow, the correction is positive; if fast, negative. When applied to a chronometer it is called **chronometer correction.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

clock rate—*The amount* gained or lost by a clock in a unit of time. When applied to a chronometer, it is called **chronometer rate.**

clockwise angle—*See* **angle to right.**

closed traverse—A survey traverse which starts and ends upon the same station, or upon stations whose relative positions have been determined by other surveys of equal or higher order of accuracy.

closest approach—**1.** The event that occurs when two planets or other celestial bodies are nearest to each other as they orbit about the Sun or other primary focii. **2.** The place or time of such an event. **3.** (satellite surveying) The time and location of the satellite when it is closest to the observer/receiver antenna.

closing corner—A corner established where a survey line intersects a previously fixed boundary at a point between corners. The closing corner is located by law at the actual point of intersection without regard to its monumented location.

closing error—See **error of closure,** definition 1.

closing the horizon—Measuring the last of a series of horizontal angles at a station, required to make the series complete around the horizon. At any station, the sum of the horizontal angles between adjacent lines should equal 360°.

closing township corner— **1.** (USPLS) The point of intersection of a guide meridian or a range line with a previously fixed standard parallel, or a base line. **2.** The point of intersection of any township or range line with a previously fixed boundary at a point between previously established corners. See also **township corner.**

closing—The act of finishing a survey process so that the accuracy may be checked.

closure of horizon—See **error of closure,** definition 6.

closure of traverse—See **error of closure,** definition 8.

closure of triangle—See **error of closure,** definition 7.

closure—See **error of closure,** definition 1.

clustering operations—Operations allowing one to agglomerate (cluster) individual items or features into groups.

coaltitude—The complement of altitude, or 90 minus the altitude. The term has significance only when used in connection with altitude measured from the celestial horizon, when it is synonymous with **zenith distance.**

coast pilot—See **sailing directions.**

Coast-Survey method—See **triangle-of-error method.**

coastal chart—A nautical chart intended for offshore navigation of vessels having a need for positions relative to the coast. Depicted information supports coastal navigation and military operations. Produced from 1:100,000 to 1:1,000,000 scale.

coastal refraction—(JCS) The change of the direction of travel of a radio ground wave as it passes from land to sea or from sea to land. Also called ***land effect;*** **shoreline effect.**

coastlining—The process of obtaining data from which the coastline can be drawn on a chart.

coated lens—A lens whose air glass surfaces have been coated with a thin transparent film of such index of refraction as to minimize the light loss by reflection. This reflection loss for uncoated lenses amounts to approximately 4 percent per air glass surface. Also called **bloomed lens.**

codeclination—The complement of the declination; it carries the same sign as the declination; equal to the declination of the celestial pole nearest a celestial position minus the declination of the celestial positional.

coefficient of refraction—The ratio of the refraction angle at the point of observation to the angle at the center of the Earth which is formed by the observer, the center of the, Earth, and point observed.

cogeoid—See **compensated geoid.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

coincidence method—1. (theodolite) The procedure by which the circles of the theodolite are read. See **coincidence**, definition 1. 2. (pendulum) The determination of the period of a free-swinging pendulum by observing the time interval between coincidences with a clock pendulum or chronometer beat.

coincidence—1. In the measurement of angles with theodolites, the instant at which two diametrically opposed index marks on the circle are in perfect optical alignment and appear to form a continuous line across the dividing line of the circle. 2. (surveying) A prismatic arrangement common to leveling instruments wherein one-half of opposite ends of the leveling bubble are brought into view in a single image. Coincidence is achieved when the two halves of the bubble ends match. 3. (pendulum) An exact agreement in occurrence of a prescribed phase of the beat of a free-swinging pendulum and a prescribed phase of the beat of a clock or chronometer.

colatitude—The complement of the latitude, or 90° minus the latitude. Colatitude forms one side, zenith to pole, of the astronomic triangle. It is the side opposite the celestial body.

collation—1. The verification of the order, number, and date of maps. 2. The assembling of pages of publications in sequence.

collection (acquisition)—1. (JCS) The obtaining of information in any manner, to include direct observation, liaison with official agencies, or solicitation from official, unofficial, or public sources. 2. The process of arranging for and obtaining existing data from one or more sources for a library file or a specific mapping, charting, and geodetic production program.

collection requirement—An identified gap in information or material holdings, including general requirement statements, intended for field collection action. Not intended to apply to requirements of data available from existing Department of Defense data libraries.

collimate—1. (physics and astronomy) To render parallel to a certain line or direction; to render parallel, as rays of light; to adjust the line of sight or lens axis of an optical instrument so that it is in its proper position relative to the other parts of the instrument. 2. (photogrammetry) To adjust the fiducial marks of a camera so that they define the principal point. Also called **adjustment for collimation; collimation adjustment.** See also **collimating marks.**

collimating eyepiece—A prismatic eyepiece used with a collimator.

collimating marks—Index marks, rigidly connected with the camera body, which form images on the negative. These images are used to determine the position of the optical center or principal point of the imagery. Also called **fiducial marks.**

collimation adjustment—See **collimate**, definition 2.

collimation axis—In an optical instrument, the line through the rear nodal point of the objective lens that is precisely parallel with the center line of the instrument.

collimation error—The angle by which the line of sight of an optical instrument differs from its collimation axis. Also called **error of collimation.**

collimation plane—The plane described by the collimation axis of a telescope of a transit when rotated around its horizontal axis.

collimator—An optical device for artificially creating a target at infinite distance, a beam of parallel rays of light; used in testing and adjusting certain optical instruments. It usually consists of a converging lens and a target, a system or arrangement of cross hairs, placed at the principal focus of the lens. See also **autocollimator; collimating eyepiece; vertical collimator.**

color composite—A composite in which the component images are shown in different colors. See also **composite.**

color gradients—See **hypsometric tinting.**

color mixture curve—A graph representing tristimulus value for unit flux of spectral energy, shown as a function of wavelength.

color mixture data—Amounts of components required in a three-color colorimeter to match various wavelengths.

color plate—A general term for the press plate from which any given color is printed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Normally, the term is modified to reflect a special color or type of plate, such as brown plate or contour plate. See also **process plates.**

color proof process—A photo- mechanical printing process which makes possible the combining of negative separations by successive exposures to produce a composite color proof on a vinyl plastic sheet. The method is usually referred to by the manufacturer's trade name of the materials used.

color proof—A single or composite copy of all colors of a polychrome (multicolor) printing which may be produced by any method.

color registration guide—A visual display on a litho copy of a chart which accurately reflects the amount and direction of misregistration between the graticule and certain significant overprint. See also **register marks.**

color separation drawing—One of a set of drawings which contains similar or related features, such as drainage or culture. There are as many drawings as there are colors to be shown on the lithographed copy.

color separation guide—See **guide.**

color separation—1. The process of preparing a separate drawing, engraving, or negative for each color required in the production of lithographed map or chart. 2. A photographic process or electronic scanning procedure using color filters to separate multicolored copy into separate images of each of the thres primary colors

colorimeter—An instrument designed for the direct measurement of color.

colures—The hour circles through the equinoxes and the solstices. See also **equinoctial colure; solstitial colure.**

coma—An aberration affecting the sharpness of images off the axis in which rays from a point object off the axis passing through a given circular zone of the lens come to a focus in a circle rather than a point, and the circles formed by rays through different zones are of different sizes and are located at different distances from the axis. Therefore, the image of a point object is comet-shaped.

combat chart—A special-purpose chart of a land-sea area using the characteristics of a map to represent the land area and the characteristics of a chart to represent the sea area, with such special characteristics as to make the chart most useful in military operations, particularly amphibious operations. Produced at 1:50,000 scale. Also called **map chart.**

combination plate—Halftone and line work on one plate. Also, two or more subjects combined on the same plate. See also **process plates.**

Command, Control, Communications and Intelligence (C^3I)—Communication and automated data processing (ADP) systems and their associated facilities, equipment, personnel, and procedures, that support the functions of command, control, communications, and intelligence.

commercial-off-the-shelf (COTS)—Production items that are available from commercial stock and need not be either newly purchased or immediately manufactured. Also known as Contractor Off-the-Shelf.

common control—(JCS) Horizontal and vertical map or chart location of points in the target area and position area, tied in with the horizontal and vertical control in use by two or more units. May be established by firing, survey, or combination of both, or by assumption. See also **control point; *field control; ground control.***

common establishment—See **establishment of the port.**

communications satellite—(JCS) An orbiting vehicle which relays signals between communications stations. They are of two types: (1) **active communications satellite**—a satellite which receives, regenerates, and retransmits signals between stations; (2) **passive communications satellite**—a satellite which reflects communications signals between stations.

compact disc-interactive (CD-I)—A compact disc format containing prerecorded digital video, audio, and optical text data. Data cannot be erased or altered. CD-I complies with the Green Book Standard.

compact disc-read-only memory (CD-ROM)—A standard 12 cm plastic optical disc created by Philips and Sony to store large

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

amounts (about 600 megabytes) of digital information in microscopic pits which can be read by a laser beam. Recorded data cannot be erased or altered. CD-ROM complies with the Yellow Book Standard.

compacted data—Digital data that have been reduced in volume using a lossless technique. Lossless means no information is lost and all of the original data will be resident when unpacked. [These techniques are designed to remove redundancies and employ special techniques.]

comparative cover—(JCS) Coverage of the same area or object taken at different times to show any changes in detail. See also **basic cover; cover; coverage.**

comparator base—See **field comparator.**

comparator—**1.** An instrument or apparatus for measuring a dimension in terms of a standard. **2.** A precision optical instrument used to determine the rectangular coordinates of a point with respect to another point on any plane surface, such as a photographic plate. 3. (surveying) An instrument for comparing standards of length; for subdividing such standards; or for determining a standard length of measuring devices (bar, tape, etc.). See also **field comparator; monocomparator; stereocomparator; Vaisala comparator; vertical comparator.**

compass amplitude—Amplitude relative to compass east or west. See also **amplitude.**

compass bearing—**1.** (navigation) Bearing relative to compass north. See also **magnetic bearing,** definition 1. **2.** (surveying) See **magnetic bearing,** definition 2.

compass index error—The instrument error in the magnetic bearing given by readings of the needle.

compass north—(JCS) The uncorrected direction indicated by the north-seeking end of a compass needle. See also **magnetic north. [Compass north and magnetic north** differ in that the former may be determined by other influences than the Earth's magnetic field.]

compass rose—(JCS) A graduated circle, usually marked in degrees indicating directions and printed or inscribed on an appropriate medium.

compass rule—A method of balancing a traverse survey. Corrections corresponding to the closing errors in latitude and departure are distributed according to the proportion (length of line to total length of traverse). The compass rule is used when it is assumed that the closing errors are as much due to errors in observed angles as to errors in measured distances.

compass survey—A traverse survey which relies on the magnetic needle for orienting the sequence as a whole or for determining the bearings of the lines individually.

compass—An instrument for indicating a horizontal reference direction relative to the Earth. See also **aperiodic compass; astro compass; Brunton compass; circumferentor; declinatoire; declinometer; Earth inductor compass; gyro compass; gyromagnetic compass; lensatic compass; liquid hand compass; magnetic compass; peep sight compass; prismatic compass.**

compensated geoid—A surface derived from the geoid by application of computed values of the deflection of the vertical which depend upon the topographic and isostatic compensation. Also called **cogeoid.**

compensating backsights and foresights—When backsight and foresight distances are equal at a given position of a level instrument, the effects of curvature, refraction, and lack of adjustment of line of sight (if bubble is leveled when taking a rod reading) are compensated for. Backsight and foresight distances are commonly controlled by the use of pacing or stadia.

compensating base-line measuring apparatus—A base apparatus having a length element composed of two metals having different coefficients of thermal expansion, so arranged and connected that the differential expansion of its components will maintain a constant length of the element under all temperature conditions of use.

compensating error—An error that tends to offset a companion error and thus obscure or reduce the effect of each.

compensating lens—(photogrammetry) A lens introduced into an optical system to correct for radial distortion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

compensation plate—(photogrammetry) A glass plate having a surface ground to a predetermined shape, for insertion in the optical system of a diapositive printer or plotting instrument, to compensate for radial distortion introduced by the camera lens.

compilation history—Complete information regarding the development of a map or chart. It explains problems encountered and their solution, and aids in simplifying the research and analysis of source materials considered for compilation or revision of other maps or charts. The compilation history contains information on the planning factors, source materials utilized, control, compilation methods, drafting, reproduction, and edit procedures.

compilation instructions—Written directions describing cartographic sources and their use in determining information to be compiled. Compilation instructions are not to be confused with specifications.

compilation manuscript—The original drawing, or group of drawings, of a map or chart as compiled or constructed from various data on which cartographic and related detail is delineated in colors on a stable-base medium. A compilation manuscript may consist of a single drawing called a **base manuscript**, or because of congestion, several overlays may be prepared showing vegetation, relief, names, and other information. Since the latter is usually the case, the base and its appropriate overlays are collectively termed the **compilation manuscript.**

compilation scale—The scale at which a map or chart is delineated on the original manuscript. This scale may vary from that of the reproduction scale.

compilation—**1.** (JCS) Selection, assembly, and graphic presentation of all relevant information required for the preparation of a map or chart. Such information may be derived from other maps or charts or from other sources. **2.** (photogrammetry) The production of a new or recompiled map, chart, or related product from aerial photographs and geodetic control data by use of photogrammetric instruments. Also called **photogrammetric compilation; stereocompilation.** See also **recompilation.**

compiled map—A map incorporating information collected from various sources; not compiled from survey data made for the map in question.

complex feature—See **feature.**

component—**1.** One of the parts into which a vector quantity can be divided. For example, the Earth's magnetic force at any point can be divided into horizontal and vertical components. **2.** One of the parts of a complete system. See also **constituent.**

composite air photography—(JCS) Air photographs made with a camera having one principal lens and two or more surrounding and oblique lenses. The several resulting photographs are corrected or transformed in printing to permit assembly as verticals with the same scale.

composite feature—A feature composed of several related features. See also **superfeature.**

composite print—See **composite.**

composite—Reproduction from a successive series of images. A proof made by exposing color separation negatives one after the other on a single sheet of paper. Used in checking and editing. Also called **composite print.** See also **color composite; color proof; double burn.**

compound harmonic motion—The projection of two or more uniform circular motions on a diameter of the circle of such motion.

compound pendulum—Any actual pendulum. A compound pendulum may be considered as composed of an indefinitely large number of material particles, at different distances from the center of suspension, each constituting a simple pendulum. The period of vibration (oscillation) of the compound pendulum may be taken as a resultant of the periods of the simple pendulums of which it is composed.

Compressed Aeronautical Chart (CAC)—A data base of processed Arc-Second Raster Chart (ARC) Digitized Raster Graphics (ADRG) data produced by the U.S. Navy, and distributed by DMA. ADRG digital map images are converted from the Equal ARC system frame of reference into the Tesselated Spheroid Model-IV (TS) projection using a sline fit. The TS data are then compressed by a factor of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

approximately 48:1.

Compressed Raster Graphics--A data base of processed Equal Arc-Second Raster Chart/Map (ARC) Digitized Raster graphics (ADRG) data developed for the Army, and produced and distributed by DMA. CRG is intended to be the Army's map background standard, targeted at battlefield weapon and command and control systems where typically data storage and display device capabilities are limited. The CRG production technique compresses full resolution 24 bit, 254 dpi ADRG on CD-ROM to 4 bit, 127 dpi for a compression factor of 24:1. CRG data on CD-ROM will contain approximately 100 TLMs, 6 JOGs, and 1 ONC over the same geographic area.

compression ratio—A ratio of digital storage savings available by compressing data. **compression**—1. A method of reducing the volume of digital data such that the data can be restored (decompressed) to its original form. Lossless compression (e.g., run length encoding) retains all the information contained in the data prior to compression. Lossy compression (e.g., vector quantization) causes some reduction in the information content of the data. 2. See **flattening (of the Earth)**.

computed altitude—Altitude determined by computation, table, mechanical computer, or graphics.

computed azimuth angle—Azimuth angle determined by computation, table, mechanical device, or graphics for a given place and time.

computed bearing—Bearing angles determined by computation from known bearings.

computed data method—A method of rectification with an autofocus rectifier whereby tilt existing in an aerial photograph is computed and, from these computations, the instrument settings are established mathematically. Rectification is then accomplished without further comparison to template or other guide base.

Computer Aided Design/Drafting/ Computer Aided Manufacturing (CAD/CAM)—The use of computers to design and produce a product.

computer aided design (CAD)—Software with the capability of assisting the operator to perform standard engineering and architecture design functions. CAD generally involves layers of data consisting of graphic primitives and blocks of data.

computer aided mapping (CAM)—Software with the capability of assisting the operator to perform standard mapping functions. Also known as Computer Assisted Mapping.

computer assisted cartography (CAC)—The use of automated techniques in map/chart production.

computer assisted drafting (CAD)—Software which provides capabilities for drawing graphic primitives.

computer assisted drafting/design (CADD)—See **Computer Aided Design; Computer Assisted Drafting.**

computer assisted manufacturing (CAM)—The use of computers in manufacturing.

computer code—1. Source code, the text or instructions of a computer program (usually in a high level language). 2. Object module, a compiled version of the source code, it is machine readable, i.e., in binary.

computer compatible tape (CCT)—Magnetic tape containing data in computer readable digital format. Generally data is recorded as 6250 or 1600 bits per inch (bpi).

concave lens—See **negative lens.**

concept of operations (CONOPS)—(JCS) A verbal or graphic statement, in broad outline, of a developer's or commander's assumptions or intent in regard to an operation or series of operations. The concept of operations is frequently embodied in campaign plans and operation plans; in the latter case, particularly when the plans cover a series of connected operations to be carried out simultaneously or in succession. The concept is designed to give an overall picture of the operation. It is included primarily for additional clarity of purpose. Frequently, it is referred to as commander's concept or CONOPS.

concluded angle—(triangulation) The third angle of a triangle, not measured, but

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

computed from the two other angles.

condition equation—A set of adjustment equations wherein all variables representing parameters have been eliminated from the equations, leaving only variables representing the adjustments to the observed quantities in the equations. A least squares adjustment employing this type of equations is said to be by the condition method, as opposed to the observation method or variation of parameters method. Note that if it is desired to allow a previously established parameter to adjust, rather than hold it rigidly fixed, it should be considered as an observation with an appropriate weight. See also **angle equation; azimuth equation; correlate equation; latitude equation; length equation; longitude equation; normal equation; observation equation; perpendicular equation; side equation; side equation tests.**

conditions—A term used in adjustment computations to define parametric requirements of adjusted redundant observations.

cone angle banding—Technique used in analytical photogrammetry for reducing mensuration requirements on a photographic plate by segregating images into annular zones defined by specific bands subtended, usually by 5° of arc. Areas read are then only in certain outer bands depending on the calibration of the lens cone.

***confidence* interval**—A statement of accuracy based on a statistic whose distribution function is known; e.g., the normal distribution function or bivariate normal distribution function. Errors are stated as some percentage of the total probability of 100 percent; e.g., a 90 percent assurance level. Also called **assurance level; *error* interval; *probability* interval.**

configuration management (CM)—A procedure for applying technical and administrative direction and surveillance to (a) identify and document the functional and physical characteristics of an item or system, (b) control any changes to such characteristics and (c) record and report the change, process, and implementation status. The CM process must be carefully tailored to the capacity, size scope, and phase of the life cycle, nature and complexity of the system involved.

configuration of terrain—See **topographic expression.**

***conformal* chart**—A chart on a conformal projection.

conformal map projection—A map projection on which the shape of any small area of the surface mapped is preserved unchanged, and all angles around any point are correctly represented. Also called **orthomorphic *map* projection.**

conic chart with two standard parallels—A chart on the conic projection with two standard parallels. Also called **secant conic chart.**
conic chart—A chart on a conic projection.

conic map projection with two standard parallels—A conic map projection in which the surface of a sphere or spheroid, such as the Earth, is conceived as developed on a cone which intersects the sphere or spheroid along two standard parallels. The Lambert conformal projection is an example. Also called **secant conic map projections.**

conic map projection—A map projection produced by projecting the geographic meridians and parallels onto a cone which is tangent to (or intersects) the surface of a sphere, and then developing the cone into a plane. Conic map projections may be considered as including cylindrical map projection when the apex of the cone is at an infinite distance from the sphere, and projections on a tangent plane when that distance is zero. Conic map projections may be illustrated with a single cone which is tangent to the sphere or which cuts the sphere along two parallels; or they may be a series of tangent cones, all with apexes on an extension of the axis of the sphere, at constantly increasing (or decreasing) distances from the sphere. It is best used to show areas of large longitudinal rather than latitudinal distances. Also called **tangent conical map projection.**

conjugate distances—The corresponding distances of object and image from the nodal points of the lens.

conjugate image points—See **corresponding *image* points.**

conjugate image rays—See **corres-**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

ponding image rays.

conjugate points—The object and image points in an optical system. They are physically related according to the definition for **conjugate distances.**

conjunction—The situation of two celestial bodies having the same celestial longitude or the same sidereal hour angle. See also **inferior conjunction; opposition; superior conjunction.**

connecting traverse—A traverse which starts and ends at separate points whose relative positions have been determined by a survey of an equal- or higher-order of accuracy. Considered less subject to undetected error than a loop traverse.

connection—(geodesy) The systematic elimination of discrepancies between adjoining or overlapping triangulation networks for the purpose of establishing a common framework from which long-range measurements can betaken.

connectivity analysis—1. Analytical technique to determine whether a set of points (nodes) or lines are connected to each other. 2. (GIS) Function used to predict or track routings through a network.

consecutive mean—A smoothed representation of a time series derived by replacing each observed value with a mean value computed over a selected interval. Consecutive means are used in smoothing to eliminate unwanted periodicities or minimize irregular variations. Also called **moving average; overlapping mean; running mean.**

Consol chart—A chart showing Consol lines of position.

Consol—(JCS) A long-range radio aid to navigation, the emissions of which, by means of their radio frequency modulation characteristics, enable bearings to be determined.

constant error—A systematic error which is the same in both magnitude and sign throughout a given series of observations, such as an index error of an instrument. See also **accumulative error.**

constant of aberration—The maximum aberration of a star observed from the surface of the Earth, 20.496 seconds of arc.

constant of gravitation—The proportionality factor (equal to 6.67×10^{-5} m^3 kg^1 sec^{-2}) in the universal law of gravitation; i.e., every particle of matter attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of their distance apart. Also called **gravitational constant; law of universal gravitation.**

constant pressure chart—See **isobaric chart.**

constellation—An orbital group of satellites, e.g., the Global Positioning System(GPS).

constituent day—The duration of the Earth's daily rotation relative to a fictitious star which represents one of the periodic tide-producing forces; it approximates the length of the lunar or solar day and corresponds to the period of a diurnal constituent of twice the period of a semidiurnal constituent. The term is not applicable to the long-period constituents.

constituent—One of the harmonic elements in a mathematical expression for the tide-producing force and in the corresponding formulas for the tide or tidal current. Each constituent represents a periodic change or variation in the relative positions of the Earth, Moon, and Sun. Also called **astronomic tidal constituent; harmonic constituent; partial tide; tidal constituent.** See also **component; diurnal constituent; semidiurnal constituent.**

contact base-line measuring apparatus—A base apparatus composed of bars whose lengths are defined by the distance between their end faces or points. In use, the bars are laid end to end, one bar being kept in position while another bar is being moved ahead.

contact glass—See **focal plane plate.**

contact plate—See **focal plane plate.**

contact printer—A device which provides a light source and a means for holding the negative and the sensitive material in contact during exposure. Also, a specialized device for exposing diapositive plates at the same scale as that of the negative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

contact printing frame—In photography and platemaking, a device for holding the negative and the sensitive material in contact during exposure. The light source may or may not be a separate element. If the frame contains a vacuum pump to exhaust all air within the frame to insure perfect contact between the negative and the sensitive material, it is known as a **contact vacuum printing frame.**

contact print—(JCS) A print made from a negative or a diapositive in direct contact with sensitized material.

contact screen—**1.** A halftone screen made on a film base and used in direct contact with the film to obtain a halftone image from a continuous tone original. **2.** A pattern image on a film base contact with an open window negative to obtain a pattern image on film or plate. See also **area pattern screen; magenta contact screen.**

contact size—In reproduction, printing to the same size as the original. Also called **one-to-one (1:1) copy**. See also **scale of reproduction.**

contact slide base-line measuring apparatus—A modified contact base-line measuring apparatus consisting of two steel measuring bars (rods), each 4 meters in length, so mounted that contact is effected by coincidence of lines on a rod and a contact slide. Each rod forms a metallic thermometer with two zinc tubes, one on each side of the bar; opposite ends of the bar are fastened to the ends of the tubes, the other ends of which are free to move with changes of temperature.

contact vacuum printing frame—See **contact printing frame.**

contact vernier—The usual type of vernier, having the vernier scale and the graduated circle in physical contact.

contiguity analysis—An analytical technique to determine whether a set of areas (polygons) are situated next to each other. Sometimes referred to as adjacency analysis.

continental control network (CCN)—An accurate network of control points established over a large geographic area.

continuous processor—(JCS) Equipment which processes film or paper in continuous strips.

continuous strip camera—(JCS) A camera in which the film moves continuously past a slit in the focal plane, producing a photograph in one unbroken length by virtue of the continuous forward motion of the aircraft.

continuous strip photography—(JCS) Photography of a strip of terrain in which the image remains unbroken throughout its length along the line of flight.

continuous tone gray scale—A scale of tones from white to black or from transparent to opaque, each tone of which blends imperceptibly into the next without visible texture or dot formation. Also called **continuous wedge.** See also **step wedge.**

continuous tone—An image which has not been screened and contains unbroken, gradient tones from black to white, and may be either in negative or positive form. Aerial photographs are examples of continuous tone prints. See also **halftone; line copy.**

continuous wedge—See **continuous tone gray scale.**

contour finder—A stereomapping instrument of simple design for use with photographic prints. This instrument does not provide a method of compensating for scale changes in different parts of the model resulting from differences in relief.

contour interval—(JCS) Difference in elevation between two adjacent contour lines. [Occasionally, the interval may vary within an individual sheet.] See also **variable contour interval.**

contour line—(JCS) A line on a map or chart connecting points of equal elevation. See also **accurate contour; approximate contour; carrying contour; depression contour; depth contour; form lines; geoidal contour; index contour line; intermediate contour line; sea level contour; supplementary contour.**

contour map—A topographic map which portrays relief by the use of contour lines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

contour sketching—Freehand delineation of the surface relief on a map as seen in perspective view, but controlled by locations on the map corresponding to salient points on the ground.
contour value—A numerical value placed upon a contour line to denote its elevation relative to a given datum, usually mean sea level.

contour—An imaginary line on the ground, all points of which are at the same elevation above or below a specified datum surface, usually mean sea level.

contrast—(photography) The actual difference in density between the highlights and the shadows on a negative or positive. Contrast is not concerned with the magnitude of density, but only with the difference in densities. Also, the rating of a photographic material corresponding to the relative density difference which it exhibits. See also **characteristic curve; density**, definition 1.

control base—A surface upon which the map projection and ground control are plotted and upon which templates have been assembled or aerotriangulation has been accomplished and the control points thus determined have been marked.

control data card—A card containing positional data and descriptions of individual horizontal and/or vertical control points. Also called **geodetic data sheet**. See also **trig list**.

control flight—See **control strip**.

control marking—A note or other form of caveat shown on an MC&G product indicating a need for special handling and for controlled dissemination.

control net—See **survey net**.

control point photography— Electronically controlled aerial photography consisting of four flight lines flown in a cloverleaf pattern from the four cardinal directions and with the flights intersecting over a target or secondary control point.

control point—1. Any station of known coordinates in a horizontal or vertical control network that is identified in a data set or photograph and used for correlating the other station data The term is usually modified to reflect the type or purpose, such as **ground control point, horizontal control point, photocontrol point, picture control point, and vertical control point**. See also **control station; secondary control point; supplemental control point**. 2. (JCS) A point located by ground survey with which a corresponding point on a photograph is matched, as a check, in making mosaics.

control station—An object or mark on the ground of known position or elevation, or both, in a network of ground control. Control stations constitute the framework by which map details are fixed in their correct position, azimuth, elevation, and scale with respect to the Earth's surface. Also called **ground control point**. See also **control point**.

control strip—(aerial photography) A strip of aerial photographs taken to aid in planning and accomplishing later aerial photography, or to serve as control in assembling other strips. Also called **control flight; tie flight; tie strip**. See also **cross-flight photography**.

control survey classification—A series of designations to classify control surveys according to their field survey methods and accuracy. The highest prescribed order of control surveys is designated first order; the next lower prescribed classification, second order; etc.

control survey—A survey which provides positions (horizontal and vertical) of points to which supplementary surveys are adjusted. The fundamental control survey of the United States provides the geographic positions and plane coordinates of triangulation and traverse stations and the elevations of bench marks which are used as the bases for hydrographic surveys of the coastal waters, for the control of the topographic survey of the United States, and for the control of many state, city, and private surveys.

control-station identification—See **photoidentification**.

controlled map—(JCS) A map with precise horizontal and vertical ground control as a basis. Scale, azimuth, and elevations are accurate. See also **map**.

controlled mosaic—(JCS) A mosaic

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

corrected for scale, rectified, and laid to ground control to provide an accurate representation of distances and direction. See also **mosaic; semicontrolled mosaic; uncontrolled mosaic.**

controlling depth—The least depth in the approach or channel to an area, such as a port or anchorage, governing the maximum draft of craft that can enter.

control—1. Previously established positions and/or orientation data used for determining additional position data used for discrete topographic points, or for controlling cartographic representations to the specified datum. 2. A collective term for a system of marks or objects on the Earth or on a map or a photograph, whose positions or elevation, or both, have been or will be determined. See also **astronomic control; basic control; common control; electronic control; geodetic control; ground control; horizontal control; La place control; level control; photogrammetric control; recover; starting control; supplemental control; vertical control.**

conventional international origin (CIO)—The average terrestrial pole of the period 1900 to 1905. Often used as the origin to which the coordinates of the instantaneous pole of rotation of the Earth are referred. In 1967, the IUGG recommended that the CIO be used to define the direction of the geodetic north pole. Abbreviated to OIC in French publications. See also **average terrestrial pole.**

convergence constant—The angle at a given latitude between meridians 1° apart. Sometimes loosely called **convergency**, a term which more properly is the equivalent of convergence.

convergence of meridians—The angular drawing together of the geographic meridians in passing from the Equator to the poles. At the Equator, all meridians are mutually parallel; passing from the Equator, they converge until they meet at the poles, intersecting at angles that are equal to their differences of longitude. The term **convergence of meridians** is used to designate also the relative difference of direction of meridians at specific points on the meridians. Thus, for a geodetic line, the azimuth at one end differs from the azimuth at the other end by 180° plus or minus the amount of the convergence of the meridians at the end points.

convergent camera—An assembly of two aerial cameras which take simultaneous photographs and maintain a fixed angle between their optical axes. The effect is to increase the angular coverage in one direction, along the longitudinal axis of the aircraft.

convergent model datum—See **model datum.**

convergent photography— Photography taken with a convergent camera. In photogrammetry, the angle of convergence of the two lenses axes is usually 40° maintaining a 1:1 base-height ratio.

convergent position—A split camera installation so positioned that the plane containing the camera axis is parallel to the line of flight.

converging lens—See **positive lens.**

conversion angle—(JCS) The angle between a great-circle (orthodromic) bearing and a rhumb line (loxodromic) bearing of a point, measured at a common origin.

conversion factor—A quantity by which the numerical value in one system of units must be multiplied to arrive at the numerical value in another system of units.

conversion scale—(JCS) A scale indicating the relationship between two different units of measurement (e.g., meters to feet). See also **scale.**

conversion—The changing of one system of measurement to another; e.g., converting meters to feet. Conversion is usually accomplished by the use of conversion factors, scales, and tables.

convertible lens—A lens containing two or more elements which can be used individually or in combination.

convex lens—See **positive lens.**

cooperative mapping agreement— A formal agreement between national governments specifying responsibilities for MC&G activities such as procurement of aerial photography, execution of geodetic control surveys, and production of maps, charts, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

related products. See also **map exchange agreement.**

coordinate axes—In a rectangular coordinate system, the axes of reference which intersect at right angles at the point of origin.

coordinate conversion—Changing the coordinate values from one type to those of another with the coordinate reference systems or datums not changing; e.g., geographic coordinates to Transverse Mercator grid coordinates.

coordinate pair—A set of cartesian coordinates describing the two-dimensional location of a point, line, or areas (polygon) feature in relation to the common coordinate systems of the data base.

coordinate protractor—A square-shaped protractor having graduations on two adjacent edges with the center at one corner. It is equipped with a movable arm turning about the center, and graduated to show linear quantities on a given scale. The protractor is covered with a grid of the same scale and units as the arm.

coordinate reference notation—Grid coordinates are given in terms of linear measurement in meters. Geographic coordinates are given in terms of angular measurement, usually in degrees, minutes, and seconds but occasionally in grads.

coordinate reference notation—Grid coordinates are given in terms of linear measurement, usually meters but occasionally in yards, feet, or other units. Geographic coordinates are given in terms of angular measurement, usually in degrees, minutes, and seconds, but occasionally in grads.

coordinate transformation—1. A mathematical or graphic process of obtaining a modified set of coordinates by some combination of rotation of coordinate axes at their point of origin, translocation of the point of origin, modification of scale along coordinate axes, or change of the size or geometry of the reference space. 2. The set of parameters used to accomplish this process. See also **affine transformation; Cartesian coordinates; datum transformation; rectification; Universal Polar Stereographic grid; Universal Transverse Mercator grid.**

Coordinated Universal Time (CUT)—Also called **universal time coordinated (UTC).** See **Universal Time.**

coordinated series—A series of geographically integrated target charts and other graphics of a uniform scale and format developed to provide continuous and complete coverage of a large area. Also called **series.**

coordinates—(JCS) Linear or angular quantities which designate the position that a point occupies in a given reference frame or system. Also used as a general term to designate the particular kind of reference frame or system, such as plane rectangular coordinates or spherical coordinates. See also **assumed plane coordinates; astronomic coordinates; Cartesian coordinates; celestial equator system of coordinates; chromaticity coordinates; curvilinear coordinates; cylindrical coordinates; Earth-fixed coordinate system; ecliptic system of coordinates; galactic system of coordinates; geocentric coordinates; geocentric geodetic coordinates; geodetic coordinates; geographic coordinates; geomagnetic coordinates; grid coordinates; grid coordinate system; ground-space coordinate system; horizon system of coordinates; hour angle system (of coordinates); inertial coordinate system; local coordinate system; model coordinates; oblique coordinates; origin of coordinates; photograph coordinates; plane polar coordinates; plane rectangular coordinates; plate coordinates; polar coordinates; rectangular coordinates; rectangular space coordinates; relative coordinate system; right ascension system; selenocentric coordinates; space coordinates; space polar coordinates; spherical coordinates; state coordinate systems; strip coordinates; topocentric coordinates; topocentric equatorial coordinates; universal space rectangular coordinate system; Universal Transverse Mercator coordinates; vertical coordinates.**

coordination—(surveying) The placing of all survey data on the same coordinate system or datum. Coordination does not imply the adjustment of observations to remove discrepancies. Two field surveys over the same area may be coordinated by computation on the same datum, but there may remain between

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

them discrepancies that can be removed only by correlation.

coordinatograph—An instrument used to plot in terms of plane coordinates. It may be an integral part of a stereoscopic plotting instrument whereby the planimetric motions (*x* and *y*) of the floating mark are plotted directly. Also called **rectangular coordinate plotter.**

coplanar—Lying in the same plane.

copy (copying) camera—A precision camera used in the laboratory for copying purposes. Also called **process camera.**

copy—The manuscript or text furnished for reproduction. See also **continuous tone; line copy; tone copy.**

corange line—A line through points of equal tidal range.

Coriolis correction—A correction applied to an assumed position, celestial line of position, celestial fix, or to a computed or observed attitude to allow for apparent acceleration due to Coriolis force.

Coriolis force—See **Coriolis.**

Coriolis—A fictitious force used to explain the horizontal departure from a straight line of a moving object on or near the Earth's surface caused by viewing the trajectory of the moving object while the observer is stationary with respect to the rotating Earth. This "force" causes deflections to the right in the Northern Hemisphere and to the left in the Southern Hemisphere Coriolis deflects objects to the west if they are moving toward the Equator and to the east if they are moving away from the Equator. It affects air (wind) and water (current), and introduces an error in the bubble sextant observations made from a moving craft; the effect increasing with higher latitude and greater speed of the object. Also called **Coriolis Force; deflecting force.**

corner accessories—Nearby physical objects to which corners are referenced for their future identification or restoration. Accessories include bearing trees, mounds, pits, ledges, rocks, and other natural features, to which distances or directions (or both) from the corner or monument are known. Such accessories are actually a part of the monumentation. See also **bearing tree.**

corner joins—The location where three or more contiguous map sheets come together.

corner marks—See **register marks.**

corner ticks—See **register marks.**

corner—1. A point on a land boundary, at which two or more boundary lines meet. 2. A point on the surface of the Earth, determined by the surveying process, which defines an extremity on a boundary of the public lands. See also **auxiliary meander corner; closing corner; closing township corner; double corners; existent corner; found corner; indicated corner; lost corner; meander corner; obliterated corner; quarter section corner; quarter-quarter section corner; section corner; sixteenth section corner; special meander corner; standard corner; theoretical corner; township corner; witness corner.**

corrected establishment—The mean high water interval for all stages of the tide.

correction code—A code consisting of letters, numbers, and symbols which are used to indicate edit corrections on maps or on overlays attached thereto.

correction for datum—A conversion factor used in the prediction of tides to resolve the difference between chart datum of the reference and a secondary station.

correction for inclination of tape—See **grade correction.**

correction for inclination of the horizontal axis—A correction applied to an observed horizontal direction to eliminate any error that may have been caused by the horizontal axis of the instrument not being exactly horizontal.

correction for run of micrometer—A correction applied to an observed reading of a graduated circle made with a micrometer microscope to compensate for run of micrometer.

correction line—See **standard parallel,** definition 1.

correction notices—A variety of notices (e.g., Notice to Mariners, Notice to Airmen, errata

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

notices, chart update manuals, target material bulletins, etc.) utilized to transmit correction data which the user applies to an existing MC&G product.

correction overlay—A transparent material on which edit corrections are noted. The method permits an immediate location of features to be revised without the necessity of marking the drawing or map.

correction—A quantity, equal in absolute magnitude but opposite in sign to the error, added to a calculated or observed value to obtain the true or adjusted value. See also **arc correction; augmentation correction; Bouguer correction; clock correction; Coriolis correction; curvature correction; dynamic correction; dynamic temperature correction; eccentric reduction; Eötvös correction; field correction; free-air correction; grade correction; height-of-eye correction; index correction; ionospheric correction; latitude correction; length correction; level correction; orthometric correction; Polaris correction; rod correction; sag correction; semidiameter correction; slope correction; surface corrections; tape corrections; temperature correction; tension correction; terrain correction; tidal correction; timing correction; transit micrometer contact correction; velocity correction.**

correlate equation—An equation derived from an observation or condition equation, using undetermined multipliers, and expressing the condition that the sum of the squares of the residuals (or corrections) resulting from the application of these multipliers to the observation or condition equations shall be a minimum See also **condition equation; normal equation.**

Correlation Tracking and Triangulation (COTAT)—A trajectory measuring system composed of several antenna base lines, each separated by large distances, used to measure direction cosines to an object. From these measurements its space position is computed by triangulation.

correlation—**1.** (general) The statistical interdependence between two quantities (e.g., in geodesy, gravity anomalies are correlated with other gravity anomalies, with elevation, with elevation differences, and with geology, etc.). **2.** (surveying) The removal of discrepancies that exist among survey data so that all parts are interrelated without apparent error. The terms **coordination** and **correlation** are usually applied to the harmonizing of surveys of adjacent areas or of different surveys over the same area. Two or more such surveys are coordinated when they are computed on the same datum; they are correlated when they are adjusted together.

correspondence—(stereoscopy) The condition that exists when corresponding images on a pair of photographs lie in the same epipolar plane; the absence of *y*-parallax.

corresponding image points—The images on two or more overlapping photographs of a single object point. Sometimes incorrectly called **conjugate image points.**

corresponding image rays—Rays connecting each of a set of corresponding image points with its particular perspective center.

corresponding images—A point or line in one system of points or lines homologous to a point or line in another system. Sometimes incorrectly called **conjugate points.**

cotidal chart—A chart of cotidal lines that show approximate locations of high water at hourly intervals measured from a reference meridian, usually Greenwich.

cotidal hour—The average interval expressed in solar or lunar hours between the Moon's passage over the meridian of Greenwich and the following high water at a specified place.

cotidal line—A line on a chart passing through all points where high water occurs at the same time. The lines show the lapse of time, usually in lunar hour intervals, between the Moon's transit over a reference meridian (usually Greenwich) and the occurrence of high water for any point lying along the line.

counter-etch—To remove, with certain diluted acids, impurities from a lithographic plate, making it receptive to an image.

counterclockwise angle—A horizontal angle measured in a counterclockwise direction; used primarily for the measurement of deflection angles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

county map—A map of the area of a county as a unit.

course—**1.** (land surveying) The bearing of a line; also the bearing and length of a line. **2.** (traverse) The azimuth and length of a line, considered together. **3.** (navigation) The azimuth or bearing of a line along which a ship or aircraft is to travel or does travel, without change of direction; the line drawn on a chart or map as the intended track. The direction of a course is always measured in degrees from the true meridian, and the true course is always meant unless it is otherwise qualified; e.g., as a magnetic or compass course. See also **track.** **4.** (geography) A route on the Earth along which a river flows; the river itself.

covariance—A mathematical quantity σxy related to the coefficient of correlation pxy between two variables $\sigma xy = pxy\ \sigma x \sigma y$, where $\sigma^2 x$ and $\sigma^2 y$ are the variances of x and y, respectively. Used in the variance-covariance matrix of a least squares solution.

cover search—(JCS) In air photographic reconnaissance, the process of selection of the most suitable existing cover for a specific requirement.

cover trace—(JCS) (reconnaissance) One of a series of overlays showing all air reconnaissance sorties covering the map sheet to which the overlays refer.

coverage—(JCS) The ground area represented on imagery, photomaps, mosaics, maps, and other geographical presentation systems.

covering power—See **angle of view.**

cover—(JCS) Photographs or other recorded images which show a particular area of ground. See also **basic cover; comparative cover.**

crab angle—(JCS) The angle between the aircraft track or flight line and the fore and aft axes of a vertical camera, which are in line with the longitudinal axis of the aircraft.

crab—**1.** (aerial photography) The condition caused by failure to orient a camera with respect to the track of the aircraft. In vertical photography, crab is indicated by the edges of the photographs not being parallel to the air base lines. **2.** (air navigation) See **yaw,** definition 1.

Critical Design Review (CDR)—A review in the development/acquisition process to assure that the recommended design of hardware and software is as close to the production configuration as possible and ready for software coding and hardware procurement.

critical angle—The minimum angle of incidence at which a ray of radiant energy impinging on the surface of a transparent medium is completely reflected, no part of it entering the medium.

critical deficiency—An item or the condition of information or data which, because of omission, misidentification, mislocation, or other such significant error, could cause serious adverse impact on navigation safety or operational mission accomplishment.

critical elevation—The highest elevation in any group of related and more-or-less contiguous relief formations on a map or chart. See also **highest elevation.**

critical range—The spread of ranges in which there is an element of uncertainty of interpretation of values.

crop—To trim or cut off parts of a photograph in order to eliminate superfluous portions and thus improve balance or composition. Usually accomplished by masking the image area during printing.

cross hairs—A set of wires or etched lines placed on reticle held in the focal plane of a telescope. They are used as index marks for pointings of the telescope such as in a transit or level when pointings and readings must be made on a rod.

cross tilt—An error introduced into stereotriangulation due to the inability to recover the exact camera stations for successive pairs. This condition is generally due to variations in equipment, materials, or to imperfect relative orientation.

cross-check lines—A series of data lines which cross the principal lines of development, preferably at right angles, which provides verification of, or reveals discrepancies in, the principal lines of the survey development.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Cross-Country Movement Map (CCM)—A DMA map depicting cross-country mobility estimates for the M-1 tank expressed in terms of GO, RESTRICTED, SLOW, VERY SLOW, and NO GO with corresponding average speed range predictions. Open water and urban areas are also portrayed, as are ground movement obstacles and hydrologic hindrances. Produced at 1:50,000 and 1:250,000 scale.

cross-country movement study—A graphic or series of graphics and supporting text or tables portraying off-road movement conditions for specific vehicles or a group of vehicles. It is usually overprinted on a medium- or large-scale topographic map base.

cross-flight photography—Single photographic strips having stereoscopic overlap between exposures and having a flight direction at right angles to that of coexistent area coverage photography. When applied to shoran, the term implies that each of the cross-flight exposures is accompanied by recorded shoran distances. See also **control strip.**

cross-section—A horizontal grid system laid out on the ground for determining contours, quantities of earthwork, etc., by means of elevations of the grid points.

crossing angle—The angle at which two lines of position, course lines, etc., intersect.

crossline glass screen—See **halftone screen.**

crystal clock—A device for keeping accurate time. It consists essentially of a generator of constant frequency controlled by a resonator made of quartz crystal, with suitable methods for producing continuous rotation to operate time indicating and related mechanisms.

culmination—The position of a heavenly body when at highest apparent altitude (zenith). Known as **upper culmination;** also, for a heavenly body which is continually above the horizon, the position of lowest apparent altitude, known as **lower culmination.** Culmination occurs when the body transits the local meridian. See also **lower transit; transit; upper transit.**

cultural details—See **culture.**

cultural features—See **culture.**

culture—(JCS) Features of the terrain that have been constructed by man. Included are such items as roads, buildings, and canals; boundary lines, and, in a broad sense, all names and legends on a map. Also called **cultural details; cultural features; man-made features.**

currency review—The comparison of an existing MC&G product against source material of later date than that from which the product was produced, for the purpose of determining the products currency.

current chart—A map of a water area depicting current speeds and directions by current roses, vectors, or other means.

current cycle—A complete set of tidal current conditions, as those occurring during a tidal day, lunar month, or Metonic cycle.

current diagram—A graphic presentation showing the speed of the flood and ebb currents and the times of slack and strength over a considerable stretch of the channel of a tidal waterway. The times being referred to as tide or current phases at some reference stations.

current meter—A device for determining the velocity of flowing water by ascertaining the speed at which a stream of water rotates a vane or wheel.

current rose—A graphic presentation of currents for specified areas, utilizing arrows at the cardinal and intercardinal compass points to show the direction toward which the prevailing current flows and the percent frequency of set for a given period of time The arrows on some presentations may be further subdivided (by thickness or pattern) to designate categories of current speeds.

curvature correction—**1.** (astronomy) A correction applied to the mean of a series of observations on a star or planet to take account of the divergence to the apparent path of the star or planet from a straight line. **2.** (geodesy) The correction applied in some geodetic work to take account of the divergence of the surface of the Earth (spheroid) from a plane. In geodetic spirit leveling, the effects of curvature and of atmospheric refraction are considered together, and tables have been prepared from which combined corrections can be taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

curvature of Earth—1. (obstruction to line of sight) The offset from the tangent to the curve, as a result of the curvature of the Earth and refraction combined. 2. The divergence of the Earth's surface from a plane.

curvature of field—An aberration affecting the longitudinal position of images off the axis in such a manner that objects in a plane perpendicular to the axis are imaged in a curved or dish-shaped surface.

curve of alignment—A line connecting two points on the surface of the spheroid and defined by the condition that at every point the azimuths of the two end points of the line differ by exactly 180°. A curve of alignment is a line of double curvature slightly less in length than the normal section lines connecting its two end points.

curve of equal bearing—A curve connecting all points at which the great-circle bearing of a given point is the same.

curve to spiral—(JCS) The common point between the circular arc and the tangent spiral.

curved path error—The difference between the length of a ray refracted by the atmosphere and the straight-line distance between the ends of the ray.

curve—See **color mixture curve; degree of curve; distortion curve; exterior to a curve; interior to a curve; isoperimetric curve; latitudinal curve; middle point; point of compound curvature; point of curvature; point of cusp; point of inflection; point of intersection; point of reverse curvature; point of tangency; point of vertical curve; point of vertical tangent; spiral curve; transition curve; vertical curve.**

curvilinear coordinates—Any linear coordinates which are not Cartesian coordinates. Frequently used curvilinear coordinates are **polar coordinates** and **cylindrical coordinates.**

cut line—The guide line sketched on a photograph to indicate where it should be cut or torn in order to form the best possible match of detail with the photographs immediately adjacent to it when laying a mosaic.

cut tape—See **subtracting tape.**

cutoff cylinder—An accessory apparatus, used in standardization operations to refer the end of a base tape or bar standard to a ground mark.

cutoff line—A survey line run between two or more stations on a linear traverse for the purpose of producing a closed traverse of that part of the survey.

cutting positive—A printing on glass of the contour drawing used to make the etched zinc plate. A preliminary step in relief model production.

cut—1. An observation between two points, one of which is known. Also a graphic representation of such an observation. See also **intersection; resection.** 2. A printed sheet of specific symbols used in cartography, such as swamp, sand, route markers, etc. 3. A section of the right-of-way of a line of communication, such as a road or railroad, that has been excavated in order to reduce the grade (vertical inclination) or to allow sufficient lateral clearance for the roadbed, as on the side of a hill. 4. A notch, passage, or channel worn by natural action, as of water.

cylindrical coordinates—A system of curvilinear coordinates in which the position of a point in space is determined by (1) its perpendicular distance from a given line, (2) its distance from a selected reference plane perpendicular to this line, and (3) its angular distance from a selected reference line when projected onto this plane. Also called **circular cylindrical coordinates; cylindrical polar coordinates.**

cylindrical equal-area map projection—A cylindrical map projection upon a cylinder tangent to a sphere, showing the geographic meridians as a family of equal-spaced parallel straight lines perpendicular to a second family of parallel straight lines which represent the geographic parallels, and which are so spaced as to produce an equal-area map projection. The equal-area condition preserves a constant ratio between corresponding ground and map areas. This projection must not be confused with the Mercator projection to which it bears some general resemblance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

cylindrical equal-spaced map projection—A cylindrical map projection upon a cylinder tangent to a sphere, showing the geographic meridians as a family of equal-spaced parallel straight lines perpendicular to a second family of equal-spaced parallel straight lines which represent the geographic parallels. The spacing of the parallels need not be the same as that of the meridians.

cylindrical lens—A lens in which the surfaces are segments of cylinders.

cylindrical map projection—A map projection produced by projecting the geographic meridians and parallels onto a cylinder which is tangent to (or intersects) the surface of a sphere, and then developing the cylinder into a plane.

cylindrical polar coordinates—See **cylindrical coordinates.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

D

D log E curve—See **characteristic curve.**

daily aberration—See ***diurnal aberration.***

daily inequality—See **diurnal inequality.**

danger line—1. A line drawn on a chart to indicate the limits of safe navigation for a vessel of specific draft. 2. A line of small dots used to draw the navigator's attention to a danger which would not stand out clearly enough if it were represented on the chart solely by specific symbols.

Data Extraction Segment (DE/S)—Segment of DMA's Digital Production System which will provide the capability to extract terrain elevation and feature data from stereo soft copy (digital) imagery. Terrain data will be extracted automatically using digital correlation techniques as well as interactively by an operator. Feature data will be extracted interactively utilizing automatic feature recognition techniques. Digital image processing techniques will aid the analyst in the interpretation and extraction of features. See also **Digital Production System; MARK 85; MARK 90.**

Data Integration Segment (DI/S)—Segment of DMA's DPS which will provide support for DMA planning and production management activities as well as an automated management system for source library holdings and other data bases. Assignment generation, scheduling tools, and production reporting capabilities will be available to all levels of DMA management. See also **Digital Production System; MARK 90.**

Data Services Segment (DS/S)—Segment of DMA's Digital Production System which will provide data management and communication services for the MARK 90 phase. It will also provide for data base management and archiving of digital data as well as much of the management data required for operation of the DPS. This segment will have the capability to store one million gigabits of data. It is one of the largest data bases in existence. In addition to its other functions, Data Services will provide an electronic communication network for the DPS. See also **Digital Production System; MARK 90.**

data acquisition station—A ground station used for performing the various functions necessary to control satellite operations and to obtain data from the satellite.

data bank/hierarchy—A formalized structure which provides for the organization of data at various quantitative levels; e.g., data bank, data base, file, record, field (element); character, bit.

data base system—A collection of people, machines, and methods organized to accomplish all of the functions of a data base.

data base—1. A set of data, part or the whole of another set of data, and consisting of at least one file that is sufficient for a given purpose or for a given data processing system. 2. A collection of data fundamental to a system. 3. (digital mapping) An organized set of evaluated MC&G data stored in either graphic, textual, or digital form. A data base may contain one file of data, e.g. terrain elevation data, or several data files, e.g., cartographic, geodetic. There are three kinds of data bases: 1) those that contain information about information; 2) those that contain information about data; and 3) those that contain the actual data.

data category (layer)—Refers to data having similar characteristics being contained in the same data set, eg., roads, rivers. Usually information contained in one data category is related and is designed to be used with other categories.

data chamber—The portion of a cartographic aerial camera where ancillary data is recorded along the film margins. Data usually includes time, altitude, frame number, and other information required for identification and correlation purposes. It may include a character data generator block and extensive binary coded information to include aircraft position and camera orientation data.

data dictionary—Repository of information about the definition, structure, and usage of data. It does not contain the actual data.

data element—(JCS) A basic unit of information having a unique meaning and subcategories (data items) of distinct units of values.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

data encoding—To apply a code, frequently one consisting of binary numbers, to represent individual or groups of data. Sometimes used as data capture.

data entry—The process of loading data into a computer-compatible format directly into a data base.

data fusion—(digital mapping) The cartographic registration, or fit, of two or more MC&G data sets. Fusion may occur between different MC&G data forms such as raster and vector, and between data sets of the same data form. The degree of fit required is related to intended use and data set resolution.

data link—(JCS) The means of connecting one location to another for the purpose of transmitting and receiving data.

data manipulation—The performance of those data processing chores common to most users, such as sorting, input/output operations, and report generations.

data quality—Refers to the degree of excellence exhibited by the data in relation to its actual portrayal of the actual phenomena.

data reduction—Process of transforming masses of raw data into useful ordered, or simplified, basic information..

data set—A collection of similar and related data records that are recorded for use by a computer.

data smoothing—A process which reduces the frequency content of digital data. The information eliminated may be real or artificially introduced from collection processes. The intent is to eliminate and replace inconsistent digital data to provide a more understandable portrayal of the earth's surface.

data spacing—The distance between centers of digital data elements in any digital data matrix or evenly spaced array along a given direction. Since the spacing between elements in one direction can be different from that between elements in the other, a complete matrix definition usually requires an expression of both data spacings.

data structure—A syntactic structure of symbolic expressions and their storage allocation characteristics. See also **chain node; link and node; spaghetti vector; topological entities.**

data topology—Refers to the order or relationship of specific items of data or other items of data.

data transfer modules (DTM)—Storage devices, commonly magnetic tape cartridges, used for data exchange or transfer between planning system computers and weapon system computers.

date line—See **International date line.**

datum centered ellipsoid—The ellipsoid that gives the best fit to the astrogeodetic network of a particular datum, and hence does not necessarily have its center at the center of the Earth.

datum level—(JCS) A surface to which elevations, heights, or depths on a map or chart are related. Also called **datum plane; reference level; reference plane.** See also **altitude.**

datum line—See **reference line.**

datum plane—See **datum level.**

datum point—(JCS) Any reference point of known or assumed coordinates from which calculations or measurements may be taken. Also called **reference point.**

datum shifts—Three constants which correlate the relationship between the origins of two coordinate reference systems.

datum transformation—**1.** The systematic determination of differences in origin, orientation and scale between adjoining or overlapping datums or triangulation networks. **2.** The act of transforming the coordinates of one datum to another.

datum—**1.** (JCS) Any numerical or geometrical quantity or set of such quantities which may serve as a reference or base for other quantities. Where the concept is geometric, the plural form is "datums" in contrast to the normal plural "data." **2.** (geodesy) In geodesy, datum refers to the geodetic or horizontal datum. The classical datum is defined by five elements

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

giving the position of the origin (two elements), the orientation of the network (one element), and the parameters of a reference ellipsoid (two elements). More recent definitions express the position and orientation as functions of the deviations in the meridian and in the prime vertical, the geoid-ellipsoid separation, and the parameters of a reference ellipsoid. The World Geodetic System (WGS) is a geocentric system that provides a basic reference frame and geometric figure for the earth, models the earth gravimetrically, and provides the means for relating positions on various datums to an earth-centered, earth fixed coordinate system. Also called **horizontal datum; horizontal geodetic datum.** See also **horizontal control datum. 3.** (leveling) A level surface to which elevations are referred, usually, mean sea level but may also include mean low water, mean lower low water, or an arbitrary starting elevation(s). Also called **vertical datum.** See also **altitude datum; Cape Canaveral datum; Department of Defense World Geodetic System; European datum; hydrographic datum; Indian datum; international low water; local datum; low water datum; low water springs datum; lower low water datum; Mercury datum; model datum; National Geodetic Vertical datum of 1929; North American datum of 1927; North American datum of 1983; photographic datum; preferred datum; Pulkovo 1932 datum; reference datum; sounding datum; tidal datum; Tokyo datum; vertical control datum; World Geodetic System.**

day—The duration of one rotation of the Earth, or occasionally another celestial body, on its axis. It is measured by successive transits of a reference point on the celestial sphere over the meridian, and each type takes its name from the reference used. See also **apparent solar day; astronomic day; calendar day; civil day; constituent day; Julian day; lunar day; mean solar day; modified Julian day; sidereal day; solar day.**

de-integration—Definition of feature or thematic singles or sets from an integrated data base.

deadbeat compass—See **aperiodic compass.**

Decca chart—A chart showing Decca lines of position.

Decca—A trade name for a radio phase comparison system which uses master and slave stations to establish a hyperbolic lattice and provide accurate position fixing facilities.

December solstice—See **winter solstice.**

declination difference—The difference between two declinations, particularly between the declination of a celestial body and the value used as an argument for entering a table.

declination of grid north—See **grid declination.**

declination of the Moon—See **lunar declination.**

declination of the Sun—See **solar declination.**

declination—1. In a system of polar or spherical coordinates, the angle at the origin between a line to a point and the equatorial plane, measured in a plane perpendicular to the equatorial plane. 2. The arc between the Equator and the point measured on a great circle perpendicular to the Equator. 3. (JCS) (astronomy) The angular distance to a body on the celestial sphere measured north or south through 90° from the celestial equator along the hour circle of the body. Comparable to latitude on the terrestrial sphere. 4. Often used as a shortened term for **magnetic declination** although this use is not preferred. See also **grid declination; grid magnetic angle; lunar declination; parallel of declination; solar declination.**

declinatoire—A combined magnetic compass and straight-edge, suitable for use on a planetable to mark the magnetic meridian. Also called **box compass; trough compass.**

declinometer—A magnetic instrument similar to a surveyor's compass, but arranged so that the line of sight can be rotated to conform with the needle or to any desired setting on the horizontal circle. Used in determining the magnetic declination.

decompress—The process by which compressed data is expanded to its former file size.

definition—In imagery interpretation, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

degree of clarity and sharpness of an image.

deflecting force—See **Coriolis force.**

deflection angle traverse—A survey, usually an open traverse, in which the measurement is made on the deflection angle of each course or leg from the direction of the preceding leg. See also **deflection angle,** definition 1.

deflection angle—1. (surveying) A horizontal angle measured from the prolongation of the preceding line to the following line. Deflection angles to the right are positive; those to the left are negative. 2. (photogrammetry) A vertical angle, measured in the vertical plane containing the flight line by which the datum of any model in a stereotriangulated strip departs from the datum of the preceding model.

deflection anomaly—The difference between an uncorrected value of the deflection of the vertical as determined by observation and the value after being corrected in accordance with certain assumptions made with reference to the physical condition of the geoid.

deflection of the plumb line—Deflection of the plumb line has the same value as the deflection of the vertical except the sign of the value is reversed. See also **deflection of the vertical.**

deflection of the vertical—The angular difference, at any place, between the upward direction of a plumb line (the vertical) and the perpendicular (the normal) to the reference spheroid. This difference seldom exceeds 30 seconds except in mountainous terrain or great depths of the sea. Often expressed in two components, meridian and prime vertical. Also called **deflection of the plumb line; station error.** See also **astrogeodetic deflection; gravimetric deflection; topographic deflection.**

degaussing range—A station for determining magnetic signatures of ships and other marine craft. Such signatures are used to determine required degaussing coil current settings and other required corrective action.

degenerate amphidromic system—A system of cotidal lines whose center or nodal (no-tide) point appears to be located on land rather than in the open ocean.

degree of curve—The number of degrees of angular measure at the center of a circle subtended by a chord 100 feet in length. In highway surveying, a 100-foot arc is sometimes used instead of a 100-foot chord in defining degree of curve.

degrees of freedom—The number of independent equations that may be written from a redundant set of observations in excess of the number of equations required for a unique solution of the unknown parameter. That is, if from a set of redundant observations, n observation equations can be written to associate m number of unknown parameters ($n \geq m$), then the independent set of equations ($n-m$) is known as degrees of freedom in any least squares adjustment of these redundant observations The special, but very common, case of establishing one parameter as the mean of observations establishes $u=1$ and $m-n$, i.e., there is only one parameter and it may be uniquely determined by any one of the observations. (In this case, each observation is an equation, e.g., $x_1=10.2$). Therefore, in this special case the degrees of freedom equal ($n-1$).

delay—(JCS) 1. (radar) The ground distance from a point directly beneath the aircraft to the beginning of the area of radar scan. 2. The electronic delay of the start of the time base used to select a particular segment of the total.

densification network—Triangulation stations based on a long-line triangulation scheme but with shorter station-to-station distances established to provide more easily available control of higher accuracy to local users.

densitometer—An instrument which is used for measuring light (either transmitted or reflected) in terms of density tolerances. Transmission densitometers measure the full density range of negatives and the reflection densitometer measures the reflection range (density) of opaque copy. A numbered scale or digital display allows accurate readings of specific areas for comparisons or control purposes.

density altitude—(JCS) An atmospheric density expressed in terms of the altitude which corresponds with that density in the Standard Atmosphere.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

density exposure curve—See **characteristic curve.**

density—1. (photography) A measure of the degree of blackening of an exposed film, plate, or paper after development, or of the direct image (in the case of a printout material). It is defined strictly as the logarithm of the optical opacity, where the opacity is the ratio of the incident to the transmitted (or reflected) light. It varies with the use of scattered or specular light. See also **characteristic curve; contrast.** 2. (cartography) The amount of detail shown on a map or chart. Density varies with scale and the nature of the area being compiled. 3. (surveying) The number of control points in a given survey or area.

Department of Defense libraries—Those Department of Defense libraries designated as responsible for maintaining files and providing library services related to specific MC&G data of common DoD interest, and designated as the primary library of that data with the responsibility for providing service for all authorized agencies.

Department of Defense World Geodetic System (DoD WGS)—A unified world datum based on a combination of all available astrogeodetic, gravimetric, and satellite tracking observations. Previous World Geodetic Systems were WGS 59, WGS 60, WGS 66, and WGS 72. The current system is WGS 84. The system is revised as new geodetic, gravimetric, and satellite data materials change the currently accepted values.

departure—1. (plane surveying) The orthographic projection of a line on an east-west axis of reference. The departure of a line is the difference of the meridian distances or longitudes of the ends of the line. It is east or positive, and sometimes termed the **easting**, for a line whose azimuth or bearing is in the northeast or southeast quadrant; it is west or negative, and sometimes termed the **westing**, for a line whose azimuth or bearing is in the northwest or southwest quadrant. Also called **longitude difference.** 2. (navigation) The distance between two meridians at any given parallel of latitude, expressed in linear units, usually nautical miles; the distance to the east or west made good by a craft in proceeding from one point to another.

dependent resurvey—A resurvey for accomplishing a restoration based on the original conditions according to the records. The dependent resurvey is made, first, by identifying existing corners and other recognized and acceptable points of control of the original survey and, second, by restoring the missing corners by proportionate measurements in harmony with the original survey. This type of resurvey is used where there is a fair agreement between the conditions on the ground and the records of the original survey. Titles, areas, and descriptions should remain unchanged. See also **independent resurvey.**

depressed pole—The celestial pole below the horizon. Opposite of **elevated pole.**

depression angle—See **angle of depression.**

depression contour—A closed contour delimiting an area of lower elevation than the surrounding terrain. Directional ticks extend from the contour in a downhill direction.

depth contour—(JCS) A line connecting points of equal depth below the hydrographic datum. Also called **bathymetric contour; depth curve; isobath.**

depth curve—See **depth contour.**

depth number—A numerical value placed upon a depth contour to denote its depth relative to a given datum. Also a depth sounding as it appears on a chart.

depth of field—The distance between the points nearest and farthest from the camera which are imaged with acceptable sharpness.

depth of focus—The distance that the focal plane can be moved forward or backward from the point of exact focus, and still give an image of acceptable sharpness. Also called **focal range.**

depth of isostatic compensation—The depth below sea level at which the condition of equilibrium known as **isostasy** is complete.

depth—(JCS) The vertical distance from the plane of the hydrographic datum to the bed of the sea, lake, or river.

descending node—The point at which a planet, planetoid, or comet crosses the ecliptic from north to south, or a satellite crosses the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

equator of its primary from north to south. Opposite of **ascending node**. Also called **southbound node.**

descending vertical angle—See **angle of depression.**

description—1. A term for the formal published data of each triangulation station, bench mark, etc. The data include information of the location and type of mark and enable anyone to go to the immediate locality and identify the mark with certainty. 2. (cadastral surveying) A document listing the metes and bounds of a property.

descriptive name—(JCS) Written indication on maps or charts used to specify the nature of a feature (natural or artificial) shown by a general symbol. See also **geographic name, place name, and toponym.**

Design Concept Review (DCR)—A milestone in the development of a system where the review team determines if the requirements imposed are fully understood by the developer and are completely accounted for and determines if the System Design Architecture is reasonable and feasible. The requirements must be allocated to hardware, software, and operations.

detail points—Selected identified points, especially on oblique photographs, used to assist in correctly positioning features displaced as a result of elevation.

detailing—(surveying) The process of tying topographic details to the control net. Objects to be located in a survey may range from single points to meandering streams and complex geological formations.

detection—In imagery interpretation, the discovering of the existence of an object but without recognition of the object.

develop (development)—In photography, to subject to the action of chemical agents for the purpose of bringing to view the invisible or latent image produced by the action of light on a sensitized surface; also, to produce or render visible in this way.

developable—A surface that can be flattened to form a plane without compressing or stretching any part of it, such as a cone or cylinder.

deviation—(JCS) The angular difference between magnetic and compass headings. Also called **magnetic deviation.** See also **error; residual deviation.**

diagnostic point—A point within a data set with accurate and known geographic location that is used to verify the adequacy or inadequacy of transformation result to the data set. (The diagnostic point is not part of the transformation solution.)

diagonal check—Measurements made across the opposite corners of the basic frame of a map projection to insure the accuracy of its construction, or to establish and/or check the scale of reproduction.

diagram on the plane of the celestial equator—See **time diagram.**

diagram on the plane of the celestial meridian—A diagram in which the local celestial meridian appears as a circle with the zenith at the top, and the horizon as a horizontal diameter. See also **time diagram.**

diagram on the plane of the equinoctial—See **time diagram.**

diameter enlargement—A term used to indicate the degree of enlargement of original copy. A one-diameter enlargement of a 4- by 5-inch original would be 8 by 10 inches. See also **times (X) enlargement.**

diameter—(magnification) See **magnifying power.**

diaphragm stop—See **relative aperature.**

diapositive printer—A photographic device for producing diapositives from aerial negatives. Also called **reduction printer.** See also **fixed-ratio projection printer.**

diapositive—(JCS) (photogrammetry) A positive photograph on a transparent medium. The term is generally used to refer to a transparent positive on a glass plate used in a plotting instrument, a projector, or a comparator.

difference of elevation—The vertical distance between two points, or the vertical distance between the level surfaces that pass through the two points.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

difference of latitude—1. The shorter arc of any meridian between the parallels of two places, expressed in angular measure. 2. (plane surveying) The difference of latitude of the two ends of a line is frequently called **latitude of the line**, and defined as the orthographic projection of the line on a reference meridian. The latitude (as above defined) of the middle of a line is also referred to as **latitude of the line**.

difference of longitude—The smaller angle at the pole or the shorter arc of a parallel between the meridians of two places, expressed in angular measure.

differential aberration—The difference between the aberration of stars and that of a moving object (e.g., a satellite). Also called **parallactic aberration**.

differential distortion—The resultant dimensional changes in length and width in any medium. See also **differential shrinkage**.

differential leveling—The process of measuring the difference of elevation between any two points by spirit leveling. See also **direct leveling**.

differential shrinkage—The difference in unit contraction along the grain structure of the material as compared to the unit contraction across the grain structure; frequently refers to photographic film and papers and to map materials in general.

differential temperature—A natural error in surveying whereby temperature variations in the instrument cause reading errors that cannot be detected. It is usually caused by direct sunlight on the instrument which can be minimized by shading the instrument during survey operations.

diffraction—(optics) The bending of light rays around the edges of opaque objects. Due to diffraction, a point of light seen or projected through a circular aperture will always be imaged as a bright center surrounded by light rings of gradually diminishing intensity. Such a pattern is called a **diffraction disk, airy disk**, or **centric**.

diffuse reflection—Any reflection process in which the reflected radiation is sent out in many directions usually bearing no simple relationship to the angle of incidence. See also **diffusion; specular reflection**.

diffusion—The scattering of light rays either when reflected from a rough surface or during the transmission of light through a translucent medium.

Digital Aeronautical Flight Information File (DAFIF)—A DMA flight information data base containing airport, runway, navigational aid, and enroute data. Both the high altitude (18,000 feet and above) and low altitude (below 18,000 feet) enroute structures are included.

Digital Bathymetric Data Base (DBDB)—A gridded bathymetric data base portraying depths in uncorrected meters for each 5 minutes of latitude and longitude. Available on magnetic tape.

Digital Chart of the World (DCW) Project—A DMA R&D project designed to develop, refine, and establish a suite of standards to support future Defense Mapping Agency digital data products and enhance the utility of digital information in vector format. See also **Vector Product Format**.

Digital Chart of the World (DCW)—DCW is the first DMA product employing the Vector Product Standard (VPS). It provides global coverage of the earth's land surface at a 1:1,000,000 scale information content which includes all the information found on the base map for the Operational Navigational Chart Series. Application software is provided to allow data base queries by feature/attribute or spatial selection. Data output is to hard disk, text file, or plot file. See also **Vector Product Format**.

Digital Cities Data Base (DCDB)—A DMA standard product consisting of the outlines of cities digitized from Operational Navigation Charts (ONC).

Digital Comparator Segment (DC/S)—A segment of DMA's MARK 85 modernization process. The Digital Comparator is a digital photogrammetric system used to measure point locations on digital imagery. Points may be measured interactively by the operator or automatically using digital correlation techniques. This segment also has the capability to extract elevation data in soft copy. See also **MARK 85**.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Digital Elevation Model (DEM)—A numerical model of the elevations of points on the earth's surface. Digital records of terrain elevations for ground positions at regularly spaced horizontal intervals. Data are available for some USGS 7.5 minute topographic quadrangles and 1:250,000 scale maps. See also **Digital Terrain Elevation Data; Digital Terrain Model.**

Digital Feature Analysis Data (DFAD)—A data base consisting of selected natural and cultural planimetric features, type classified as point, line, or area features as a function of their composition and size. Each feature is assigned a code and further described with limited attribution. The data are stored in polygon format and segregated into 1 degree tiles. Primary applications are radar return, simulation, navigation, targeting and terrain obstruction studies. When combined with DTED, an off-line data base is created for use by simulators needing line-of-sight, obstruction, and perspective views. DFAD may also be used for sensor correlation, target recognition, lines of communication analysis, or for map production activities.

Digital Geographic Information (DGI)—The digital representation of elevation and sounding information, geographic feature geometry and feature attribute information, information concerning the appearance and status of the earth's surface and its features in the electromagnetic spectrum (e.g., radar, infrared, etc.), Military Geographic Information together with other ancillary information, and other digital information.

Digital Geographic Information Exchange Standard (DIGEST)—A set of standards developed by NATO's Digital Geographic Information Working Group designed to support the exchange of digital geographic data between co-producers. These standards apply to geodetic, geographic, geological, and geophysical data of military interest, as well as positioning, navigation, simulation, target recognition, map and chart automated production, and data display and manipulation applications.

Digital Landmass Blanking System (DLMB)—A land/water matrix of elevation values, where a value of 200 meters is arbitrarily assigned to land and a value of 0 meters is assigned over water. These matrix data are used by the E-3A Airborne Warning and Control Systems (AWACS), operating in a maritime/coastal environment to filter out unwanted radar returns from land and allow for better detection of ship and aircraft movement near the shoreline.

Digital Landmass System (DLMS)—An off-line cartographic data base consisting of two basic types of digital data files, one containing Terrain Elevation Data and the other containing cultural and natural feature data, maintained by DMA to support various user systems e.g., simulators and weapon systems. Formerly known as Digital Radar Landmass Simulator (DRLMS). See also **DFAD; DTED.**

Digital Line Graph (DLG)—Line map information produced by the U.S. Geological Survey in digital form. Data are available at several scales, for several themes, and in several formats.

Digital Line Graph-Enhanced (DLG-E)—An extension of the DLG topological model which builds a cartographic feature layer upon the topology. Nonlocational characteristics of feature objects are linked with the spatial or locational objects. See also **Digital Line Graph.**

Digital Production System (DPS)—DMA's production system for accurate and timely digital or softcopy products and services. DPS includes hardware and software developments as well as an MC&G digital data base which supports the generation of multiple products over any geographic area. The first phase, MARK 85, was to achieve maximum effective enhancement of existing DMA production capabilities. The MARK 90 phase integrates portions of MARK 85 with additional hardware and software for digital or soft copy operations. See also **Data Integration Segment; Source Acquisition Segment; Hardcopy Exploitation Segment; Universal Rectifier Segment; Production Management Segment; Data Services Segment; Source Preparation Segment; Data Extraction Segment; Product Generation Segment; MARK 85; MARK 90.**

Digital Terrain Analysis Data (DTAD)—Data in discrete digital (numerical) form representing natural and manmade features. Feature data are attributed and typically grouped in files for transportation, vegetation,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

surface configuration, surface materials, surface drainage, and abstacle analysis. The various data base products comprised of DTAD are used in terrain analysis, targeting, battlefield modeling and simulation.

Digital Terrain Elevation Data (DTED)—A uniform matrix of terrain elevation values produced by DMA. DTED provides basic quantitative data for all military training, planning, and operating systems requiring terrain elevation, slope, and/or surface roughness information. Level 2 post spacing is 1 arc second latitudinally. Level 1 post spacing is 3 arc second latitudinally; for both, longitudinal spacing varies with latitude.

Digital Terrain Matrix (DTM)—A regular grid of geographic or model coordinates that is the primary output of the collection process. The data contained therein consists of elevation data referred to a grid (row and column occurrences) system. Grid spacing may vary according to surface model accuracy requirements.

digital data base (off-line)—A digital data base maintained in a common format that supports different user systems, e.g., simulator or weapon systems. Normally the data must be transferred before it can be used by a specific user system. This term is commonly expressed as the off-line data base.

digital data base (on-line)—A digital data base in the format needed by a user system, e.g., simulator or weapon system, and which can be loaded into the user system. This term is commonly referred to as the on-line data base.

digital data—These are data represented in alphanumeric format readable by a computer.

digital feature data—1. (cultural data) Man-made, natural and landscape features in digital form, data in discrete digital (numerical) form, including all man-made features on the Earth's surface, e.g., lines of communication, built-up areas (cities), transmission lines and landmark structures. 2. (hydrographic data) Data derived from the measurement and description of physical features of the oceans, lakes, rivers and other waters, and their adjoining coastal areas (with particular reference to navigational usage) in digital form. 3. (landscape feature data) Data of all natural features and man's alteration to those features, e.g., lake with dam, agricultural features, etc., in digital form. 4. (natural feature data) Natural features on the Earth's surface which are not man-made, e.g. vegetation, water bodies, desert, etc. in digital form.

digital image—An image digital format obtained by partitioning the area of the image into a finite two-dimensional array of uniformly shaped, mutually exclusive regions called **pixels** (picture elements), and assigning a descriptive tone or shade to each such spatial region.

digital map—1. A map expressed and stored in digital form. 2. A representation in digital form, discrete points on the Earth's surface. Also called **numerical map.**

digital synthesized data—Data derived from the analysis of two or more different types of data such as Digital Terrain Elevation Data, Digital Natural Feature Data, and Digital Cultural Feature Data. An example is the derivation of data which can assess the effects of soil types, vegetation slope, drainage and microrelief on cross-country movement of vehicles.

digital synthetic data—Data in numerical format that describes either totally or partially fictitious information generated from lower resolution/frequency input.

digital terrain model (DTM)—A statistical representation of the continuous surface of the ground by a large number of selected points with known rectangular coordinates in an arbitrary coordinate field. See also **Digital Elevation Model.**

Digitized/Digital Point Positioning Data Base (DPPDB)—The digitized PPDB is a raster-scanned digital rendition of the current analog PPDB product. DMA is investigating the potential for a completely digital PPDB product.

dihedral angle—The angle between two intersecting planes.

dimensional stability—1. Ability to maintain size. 2. Resistance to dimensional changes caused by changes in moisture content and temperature.

DIN—The German Industrial Standard (Deutsche Industrie-Norm), a European system of standardization for mechanical, engineering, and scientific manufactured products. In MC&G, the most notable application is the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

measurement of film speed.

diopter—A unit of measurement of the power of a lens, especially a spectacle type lens. The power in diopters equals the reciprocal of the focal length in meters; thus, a lens whose focal length is 20 cm has a power of 5 diopters.

dioptric system—(optics) An optical system containing only refractive elements (lenses).

dip angle—1. (surveying) The vertical angle of the observation point between the plane of the true horizon and a sight line to the apparent horizon. 2. (photogrammetry) The vertical angle, at the air station, between the true and the apparent horizon, which is due to flight height, Earth curvature, and refraction.

dip circle—An instrument for measuring magnetic dip. It consists essentially of a dip needle, or magnetic needle, suspended in such a manner as to be free to rotate about a horizontal axis.

dip correction—See **height-of-eye correction.**

dip equator—See **magnetic equator.**

dip needle—See **magnetic dip needle,** definition 1.

dip of the horizon—See **dip,** definition 1.

dip pole—See **magnetic pole.**

dip—1. The vertical angle, at the eye of an observer, between the horizontal and the line of sight to the visible horizon. Also called **dip of the horizon.** 2. The angle between the horizontal and the lines of force of the Earth's magnetic field at any point. Also called **inclination; magnetic dip; magnetic inclination; magnetic latitude.** 3. The first detectable decrease in the altitude of a celestial body after reaching its maximum altitude on or near meridian transit.

direct angle—An angle measured directly between two lines, as distinguished in transit traverse from a deflection angle.

direct leveling—The determination of differences of elevation by means of a continuous series of short horizontal lines. Vertical distances from these lines to adjacent ground marks are determined by direct observations on graduated rods with a leveling instrument equipped with a spirit level. See also **differential leveling.**

direct measurement—Any measurement obtained by applying a tape to a line or a protractor to an angle, or by turning an angle with a transit; especially applicable to surveying. See also **indirect measurement.**

direct motion—The apparent motion of a planet or other object eastward among the stars.

direct observation—A measure of the quantity whose value is desired, such as a single measure of a horizontal angle.

direct photography—Photography in which the image of a subject is recorded directly by the camera in the conventional manner.

direct positive—A positive image obtained directly without the use of a negative.

direct problem—The determination of the geodetic position of the end point and the back azimuth at position two when the given information is the latitude and longitude of position one, along with the forward azimuth and the distance between the two points.

direct radial plot—See **direct radial triangulation.**

direct radial triangulation—A graphic radial triangulation made by tracing the directions from successive radial centers directly onto a transparent plotting sheet rather than laying the triangulation by the templet method. Also called **direct radial plot.**

direct scanning camera—A type of panoramic camera wherein the lenses swing or rotate about the rear nodal point at a given rate.

direct telescope—A telescope is said to be direct when it is in its normal position.

direct vernier—A vernier scale which has spaces or divisions slightly shorter than those of the primary scale. The numbers on the vernier scale coincide with the numbers on the primary scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

direction angle—In tracking, the angle between the antenna base line and an imaginary line connecting the center of the base line with the target .

direction instrument theodolite—A theodolite in which the graduated horizontal circle remains fixed during a series of observations, the telescope being pointed on a number of signals or objects in succession, and the direction of each read on the circle, usually by means of micrometer microscopes. Direction instrument theodolites are used almost exclusively in first- and second-order triangulation. Also called **direction theodolite; triangulation theodolite.**

direction method of adjustment—(triangulation and traverse) A method of adjustment of observations which determines corrections to observed directions. The direction method is used in the adjustment of first- and second-order survey observations.

direction method of determining astronomic azimuth—The determination of the astronomic azimuth of a line by measuring, with a direction theodolite, the horizontal angle between a selected star and a suitable mark, and applying that angle to the azimuth of the star computed for the epoch of the observation.

direction method of measuring horizontal angles—See **direction method of observation.**

direction method of observation—A method of observing angular relationships wherein the graduated circle is held in a fixed position, and the directions of the various signals are observed around the horizon. Thus, directions are pointings whereby angles are found by the differences in directions. Also called **direction method of measuring horizontal angles.**

direction of gravity—See **direction of the force of gravity.**

direction of relative movement—The direction of motion relative to a reference point, itself usually in motion.

direction of the force of gravity—The direction indicated by a plumb line. It is perpendicular to the surface of the geoid. Also called **direction of gravity.**

direction of tilt—The direction (azimuth) of the principal plane of a photograph. Also, the direction of the principal line on a photograph.

direction theodolite—See **direction instrument theodolite.**

directional radar prediction—(JCS) A prediction made for a particular heading.

directional reflectance—Reflectance measured for a specific mode of irradiation and collection.

direction—1. (JCS) A term used by a spotter or observer in a call for fire to indicate the bearing of the spotting line. 2. The position of one point relative to another without reference to the distance between them. Direction may be either two-dimensional or three-dimensional, the horizontal being the usual plane of the latter. Direction is usually indicated in terms of its angular distance from a reference direction. See also **great-circle direction; grid direction; horizontal direction; magnetic direction; Mercator direction; reference direction; relative direction; true direction; x-direction.**

discrepancy—A difference between results of duplicate or comparable measures of a quantity. The difference in computed values of a quantity obtained by different processes using data from the same survey.

disk operating system (DOS)—One of many operating systems for computers which act as an interface between a user and the computer hardware. DOS was specifically designed for personal computers (PCs).

dispersion—(optics) The separation of light into its component colors by its passage through a diffraction grating or by refraction such as that provided by a prism.

displacement—1. (cartography) The horizontal shift of the plotted position of a topographic feature from its true position, caused by required adherence to prescribed line weights and symbol sizes. 2. Any shift in the position of an image on a photograph which does not alter the perspective characteristics of the photograph (i.e., shift due to tilt of the photograph, scale change in the photograph, and relief of the objects photographed). See also **refraction displacement; relief**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

displacement; tilt displacement; *x*-displacement; *y*-displacement.

display—The graphic presentation of the output data of any device on system.

dissolve—(digital) The process of removing shared common attributes by eliminating the shared boundaries when merging two or more polygons.

distance angle—An angle in a triangle opposite a side used as a base in the solution of the triangle, or a side whose length is to be computed.

distance measuring equipment (DME)—See **electronic distance measuring equipment.**

distance prorate rule—A method of balancing a survey. A rule for holding angles to their recorded values and prorating the lengths of the lines in the traverse; operated by locating the bearings to a convenient meridian, preferably the closing line of the traverse, and prorating the dimensions through a trigonometric process.

distance—The spatial separation of two points, measured by the length of a line joining them. See also **angular distance; double meridian distance; double zenith distance; ecliptic polar distance; electrical distance; external distance; falling; front focal distance; great-circle distance; grid length; ground distance; hyperfocal distance; interocular distance; lunar distance; meridian distance; meridional distance; plus distance; polar distance; principal distance; projection distance; relative distance; rhumb line distance; slant range; Sun-zenith distance; tangent distance; time distance; zenith distance.**

distant points—Similar to tie points but which appear only on the obliques facing outward on the perimeter of a compilation. Distant points unite the sets of a strip into a flight unit; but, unlike tie points, do not serve to join several flight strips together.

distortion compensation—(photogrammetry) In a double projection direct viewing plotter system, that correction applied to offset the effect of radial distortion introduced in an original negative by the objective lens of an aerial camera.

distortion curve—A curve representing the linear distortion characteristics of a lens; it is plotted with image radial distance from the lens axis as abscissas and image radial displacements as ordinates.

distortion—Lens aberrations affecting the positions of images from their true relative positions. See also **aberration,** definition 2; **angular distortion,** definition 2; **differential distortion; film distortion; image distortion; image motion compensation distortion; lens distortion; linear distortion; panoramic distortion; radial distortion; scan positional distortion; tangential distortion; tipped panoramic distortion.**

distributed data base—A data base with unique components in geographically dispersed locations linked through a telecommunications network.

distribution map—A map which shows the geographic arrangement of a specific product, commodity, or formation.

disturbing function—See **disturbing potential,** definition 2.

disturbing potential—1. (geodesy) The difference between the gravity potential of the actual Earth and the potential function of the normal gravity. Also called **potential disturbance; potential of disturbing masses; potential of random masses.** 2. (astronomy) The difference between the total gravitational potential and the potential pertaining to a spherical mass distribution. Also called **disturbing function.**

diurnal aberration—Aberration resulting from the rotation of the Earth on its axis. The value of diurnal aberration varies with the latitude of the observer and ranges from zero at the poles to 0.32 second of arc at the Equator. A correction for diurnal aberration is applied to astronomic observations for longitude and azimuth. Also called **daily aberration.**

diurnal age—See **age of diurnal inequality.**

diurnal arc—See **astronomic arc.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

diurnal circle—The apparent daily path of a celestial body, approximating a parallel of declination.

diurnal constituent—Any tide constituent whose period approximates that of a lunar day (24.84 solar hours). See also **constituent.**

diurnal inequality—The difference in heights and durations of the two successive high waters or of the two successive low waters of each day; also, the difference in speed and direction of the two flood currents or the two ebb currents of each day. Also called **daily inequality; low water inequality.**

diurnal motion—The apparent daily motion of a celestial body.

diurnal parallax—See **geocentric parallax.**

diurnal variation—That component of a determinable magnitude which passes through a complete cycle in one day.

diurnal—Having a period of, occurring in, or related to a day.

divergence—(leveling) The difference between the numerical values of two runnings over the same section of a line of levels.

diverging lens—See **negative lens.**

DMA Automated Publication Production System (DAPPS)—An electronic imaging device used to create text and manipulate graphic images. Output is either to a laser printer or a phototypesetter.

dodging—(photography) The process of holding back light from certain areas of sensitized material to avoid overexposure of these areas.

domestic map—A map of an area within the limits of the United States.

dominant wavelength—The wavelength of a spectrally pure energy that if mixed with white light would match a color.

Doppler cone angle—The angle between the reference velocity vector and the reference range vector.

Doppler effect—(JCS) The phenomenon evidenced by the change in the observed frequency of a sound or radio wave caused by a time rate of change in the effective length of the path of travel between the source and the point of observation. Also called **Doppler shift.**

Doppler navigation—1. A system which measures ground speed and drift of an aircraft, based on the Doppler effect of electronically generated signals emitted from the craft and reflected from the terrain. 2. A system which determines positions based on the Doppler effect of satellite signals. See also **broadcast ephemeris; Doppler sonar navigation; Navy Navigation Satellite System; point positioning; precise ephemeris; translocation.**

Doppler satellite survey system—A receiver/antenna combination capable of receiving signals broadcast by satellites of the Navy Navigation Satellite System (NNSS). The position of the antenna is computed by point positioning, translocation, or short arc geodetic adjustment (SAGA), using the collected signals from the satellite passes.

Doppler shift—See **Doppler effect.**

Doppler sonar navigation—A system by which speed and drift of a marine craft are determined from the Doppler effect of sonar generated signals emitted from the craft and reflected from the ocean bottom or suspended particulate in the seawater. See also **Doppler navigation; sonar.**

dot screen—A photographic negative containing equal-sized dots which are equally spaced in parallel and perpendicular rows. The printed area of coverage determines the screen percent. The percentage of the printed area of coverage is measured by transmission density and/or by dot diameter. Dot screens are used to print tones of a color. Also called **flat tint screen.**

double optical projection stereoplotter—See **double-projection direct-viewing stereoplotter.**

double burn—The intentional exposure of two or more line and/or halftone negatives in succession and register on the same sensitized surface. Not to be confused with **double exposure**, which is usually unintentional. Also

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

called **double shooting**. See also **composite.**

double center theodolite—See **repeating theodolite.**

double centering—A method of prolonging a line from a fixed point whereby the backsight is taken with the telescope direct. The instrument is plunged, and the foresight is made with the telescope inverted. The point at which the vertical cross hair intersects the hub is then marked. The transit is then rotated to take a backsight on the fixed point with the telescope inverted, the foresight is made with the telescope direct, and a second projected point is marked on the hub. A point midway between the two marked points is the true point on the prolonged line. Also called **double reversing; double reversion; double sighting; reversing in azimuth and altitude; wiggling-in on a line; working-in on a line.**

double corners— Normally the two sets of corners along a standard parallel; the standard township, section, and quarter section corners placed at regular intervals of measurement; additionally, the closing corners established on the line at the points of intersection of the guide meridians, range, and section lines of the surveys brought in from the south. In other cases, not fully in conformity with the rectangular plan, two corners, each common to two townships only, instead of one corner of the four townships. Similarly, two corners, each common to two sections, and two quarter section corners, each referring to one section only. The term is sometimes used incorrectly to denote two lines established on the ground although the field note record indicates only one line, thus creating a hiatus or overlap.

double exposure—Two photographic exposures on the same sensitized surface.

double interpolation—Deriving a value from tabulated computations where two or three proportions may be required, as in an hour angle observation of Polaris for azimuth, where an interpolation is made suitable to the time factor, another to arrive at a result conforming with the latitude of the observing station.

double meridian distance—The algebraic sum of the perpendicular distances from the two ends of any line of a traverse to the initial, or reference, meridian.

double model stereotemplet—A templet representative of the horizontal plot of two or three adjacent stereoscopic models that have been adjusted to a common, though random, scale.

double proportionate measurement— A method for restoring a lost corner of four townships or a lost interior corner of four sections. It is based on the principle that monuments north and south should control the latitudinal position of a lost corner, and monuments east and west should control the longitudinal position. In this method the influence of one identified original corner is balanced by the control of a corresponding original corner upon the opposite side of a particular missing corner which is to be restored, each identified original corner being given a controlling weight inversely proportional to its distance from the lost corner.

double reversing—See **double centering.**

double reversion—See **double centering.**

double shooting—See **double burn.**

double sighting—See **double centering.**

double target leveling rod—Any target rod having graduations on two opposite faces.

double zenith distance—A value of twice the zenith distance of an object, obtained by observation and not by mathematical process.

double-projection direct-viewing stereoplotter—A class of stereo-plotters employing the principle of projecting the images of two correctly oriented overlapping aerial photographs onto a reference datum so the resultant images may be viewed directly without additional optical system support. Also called **double optical projection stereoplotter.**

double-rodded line—A line of differential levels wherein two sets of turning points, one high and one low, are used to give independent measures between bench marks.

doubly azimuthal map projection—An azimuthal map projection having two poles.

dove prism—A prism which reverts the image but does not deviate or displace the beam. A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

given angular rotation of the prism about its longitudinal axis causes the image to rotate through twice the angle. Also called **rotating prism.**

draconic month—See **nodical month.**

drafting guide—See **guide.**

drafting—The art of drawing from given specifications.

drag—(theodolite) A slight movement of the graduated circle of a theodolite produced by the rotation of the alidade. See also **atmospheric drag.**

drainage pattern—The pattern or overall appearance made by the network of drainage features on a map or chart.

drainage—In mapping, all features associated with water, such as shorelines, rivers, lakes, marshes, etc.

drift angle—(JCS) The angle measured in degrees between the heading of an aircraft or ship and the track made good. [Drift angle is designated right or left to indicate the direction of drift.]

drift station—A term sometimes used in shoran operations to designate the ground station about which the aircraft flies during arc navigation. The second ground station is then referred to as the **rate station.**

drift—**1.** (JCS) In ballistics, a shift in projectile direction due to gyroscopic action which results from gravitational and atmospherically induced torques on the spinning projectile. **2.** The lateral shift or displacement of a ship or aircraft from its course, due to the action of wind or other causes. **3.** Aerial photography obtained under this condition produces successive photographs whose edges are parallel but sidestepped. **4.** (precession) See **total drift.**

dummy pendulum—A pendulum of similar construction to the working pendulums except that it is equipped with a thermometer and is fastened rigidly in the receiver so that it cannot swing during observations. The dummy pendulum is subject to the same temperature conditions as the working pendulums, and is used in determining their temperature when in use.

dummy—**1.** A preliminary drawing or layout showing the position of illustrations and text as they will appear in the final reproduction. **2.** A set of blank pages made up to show the size, shape, and general style of a book, booklet, or pamphlet.

dumpy level—A leveling instrument which has its telescope permanently attached to the leveling base, either rigidly or by a hinge that can be manipulated by a micrometer screw.

duplex base-line measuring apparatus—A contact base-line measuring apparatus, composed of two disconnected bars, one of brass and the other of steel, each 5 meters in length and so arranged as to indicate the accumulated difference of length of the measures from the brass and steel components.

duplicate level line—A line of spirit leveling composed of two single lines run over the same route, but in opposite directions, and using different turning points.

duplicate negative—(JCS) A negative reproduced from an original negative or diapositive. [The duplicate negative may be a true reproduction of the original or a reproduction possessing greater or less contrast. With direct positive materials, chemical reversal process, and duplicating film it is not always necessary to make a positive to obtain a duplicate negative.]

DX-90—An IMO/IHO proposed standard for the exchange of digital data.

dylux—A light-sensitive paper or film used for proofing, usually has a blue positive reading image with a blue background.

dynamic correction—The quantity that must be added to the orthometric elevation of a point to obtain its dynamic number.

dynamic elevation—Elevation expressed in length units, but determined by dynamic number.

dynamic gravity meter—A type of gravity instrument in which the period of oscillation is a function of gravity and is the quantity directly observed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

dynamic height—A height derived by dividing the geopotential number by a constant, usually the value of normal gravity at 45° latitude.

dynamic number—The work required to raise a unit mass from sea level to a given point, expressed in absolute units.

dynamic temperature correction—(pendulum) The correction to the observed period of a pendulum for the rate of change of its temperature.

dynamical mean sun—A fictitious sun conceived to move eastward along the ecliptic at the average rate of the apparent sun.

dyne—The dyne is the unit of force of the c.g.s. systems of units. When acting on a mass of 1 gram, a dyne imparts to that mass an acceleration of 1 centimeter per second per second. Until about 1930, the dyne was used by the U.S. Coast and Geodetic Survey (now the National Geodetic Survey) in stating values of gravity. Since that time, gravity has been reported in terms of the gal, the c.g.s unit of acceleration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

E

11/10 peg adjustment—See **peg adjustment.**

Earth fixed coordinate system—Any coordinate system in which the axes are stationary with respect to the Earth.

Earth inductor compass—A compass depending for its indications upon the current generated in a coil revolving in the Earth's magnetic field.

Earth inductor—An instrument designed for use in magnetic surveys to determine magnetic dip. In principal, the instrument is a small dynamo by which the electrical flow can be introduced to the coil of the instrument. The presence or absence of current is indicated by a galvanometer which provides for direct reading of magnetic dip.

Earth Resources Observation System Data Center (EROS or *EDC*)—A user services and sales office established by the U.S. Department of Interior. EDC also processes LANDSAT data using the EROS Digital Imaging Processing System (EDIPS) and the EROS Digital Image Enhancement System (EDIES).

Earth satellite—A body that orbits about the Earth; specifically, an artificial satellite placed in orbit.

Earth tide—A periodic movement of the Earth's crust caused by tide-producing forces of the Moon and Sun.

Earth-centered ellipsoid—A reference ellipsoid whose geometric center coincides with the Earth's center of gravity and whose semiminor axis coincides with the Earth's rotational axis.

easement curve—See **spiral curve.**

east point—See **prime vertical plane.**

easting—**1.** (grid) Eastward (that is left to right) reading of grid values on a map. See also **false easting. 2.** (plane surveying) See **departure,** definition 1.

ebb tide—The portion of the tide cycle between high water and the following low water. Also called **falling tide.**

eccentric anomaly—See **anomaly,** definition 3.

eccentric error—*Centering error.*

eccentric reduction—(triangulation) The correction which must be applied to a direction observed by an instrument with either the instrument or signal (swing), or both, eccentric, to reduce the observed value to what it would have been if there had been no eccentricity. Also called **eccentricity correction.**

eccentric signal—A signal (target) which is not in the same vertical line with the station which it represents.

eccentric station—A survey point over which an instrument is centered and observations made, and which is not in the same vertical line with the station which it represents and to which the observations will be reduced before being combined with observations at other stations. In general, an eccentric station is established and occupied when it is impracticable to occupy the station center, or when it becomes necessary in order to see points which are not visible from the station center.

eccentricity correction—(triangulation) See **eccentric reduction.**

eccentricity of alidade—The distance between the center of figure of the index points on an alidade and the center of figure of the graduated circle. See also **eccentricity of instrument.**

eccentricity of circle—The distance between the center of figure of a graduated circle and its center of rotation. See also **eccentricity of instrument.**

eccentricity of ellipse—The ratio of the distance between the center and a focus of an ellipse to the length of its semimajor axis.

eccentricity of instrument—The combination of eccentricity of circle and eccentricity of alidade.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

eccentricity of spheroid (ellipsoid) of revolution—The eccentricity of an ellipse forming a meridian section of the spheroid.

eccentricity—1. Amount of deviation from a center. *See also* **eccentric station.** 2. (surveyor's compass) An effect caused by one or a combination of the following conditions: (1) a straight line through the ends of the magnetic needle fails to pass through the center of rotation of the needle; (2) the center of rotation of the needle is not coincident with the center of figure of the graduated circle; (3) the line of sight fails to pass through the vertical axis of the instrument. 3. The ratio of the distances from any point of a conic section to a focus and the corresponding directrix.

eccentric—Not having the same center.

echo sounder—An echo sounding instrument used for depth measurements in water.

echo sounding—A method for measuring depths by recording the time interval required for sound waves to go from a source of sound near the surface to the bottom and back again.

echo timing—The measurement of time required for a short train of energy waves to travel the round trip path from an originating station to a reflector or transponder.

echogram—A graphic record of depth measurements obtained by echo sounding equipment. Called **fathogram** when obtained from a Fathometer echo sounder.

echo—*See* **blip.**

eclipse year—The interval between two successive conjunctions of the Sun with the same node of the Moon's orbit.

eclipse—1. The reduction in visibility or disappearance of a nonluminous body by passing into the shadow cast by another nonluminous body. 2. The apparent cutting off, wholly or partially, of the light from a luminous body by a dark body coming between it and the observer. *See also* **annular eclipse; lunar eclipse; solar eclipse.**

ecliptic coordinate system—*See* **ecliptic system of coordinates.**

ecliptic latitude—*See* **celestial latitude.**

ecliptic longitude—*See* **celestial longitude.**

ecliptic meridian—A circle on the celestial sphere containing points of the same celestial longitude.

ecliptic node—*See* **node.**

ecliptic parallel—A circle on the celestial sphere containing points of the same celestial latitude.

ecliptic polar distance—The complement of the celestial latitude.

ecliptic pole—On the celestial sphere, either of the two points 90° from the ecliptic.

ecliptic system of coordinates—A system of curvilinear celestial coordinates which uses the ecliptic as the primary reference plane and the ecliptic meridian through the vernal equinox as the secondary plane. The points 90° from the ecliptic are the north and south ecliptic poles. Angular distance north or south of the ecliptic analogous to latitude, is celestial latitude. Celestial longitude is measured eastward along the ecliptic from the vernal equinox through 360°.

ecliptic—The great circle formed by the intersection of the plane of the Earth's orbit around the Sun (or apparent orbit of the Sun around the Earth) and the celestial sphere.

edge matching—The comparison and graphic adjustment of features to obtain agreement along the edges of adjoining map sheets.

edge mating—*See* **matching.**

edge—1. (imagery) The boundary of an object in a photograph or image, usually characterized by a drastic change in the brightness value from the immediate interior of the boundary to the immediate exterior of the boundary. 2. (topology) A line used to represent the location of a linear feature and the borders of faces. Edges may be topologically linked to nodes, edges, and faces.

editing—The process of checking a map or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

chart in its various stages of preparation to insure accuracy, completeness, and correct preparation from and interpretation of the sources used, and to assure legible and precise reproduction. Edits are usually referred to by a particular production phase, such as compilation edit, scribing edit, etc.

effective area—For any aerial photograph that is one of a series in a flight strip, that central part of the photograph delimited by the bisectors of overlaps with adjacent photographs. On a vertical photograph, all images within the effective area have less displacement than their conjugate images on adjacent photographs.

effective Earth radius—See **effective radius of the Earth.**

effective focal length (EFL)—See **principal distance**, definition 1.

effective radius of the Earth—A fictitious value for the radius of the Earth, used in place of the geometrical radius to correct for atmospheric refraction when the index of refraction in the atmosphere changes linearly with height.

Egault level—A French instrument having the spirit level attached to a level bar which also carries wyes in which the telescope rests.

electrical distance—Length measured in terms of the distance traveled by radio waves in unit time.

electrical survey net adjuster (ESNA)— A device used for obtaining least squares adjustments of level, traverse, and vertical angle nets through the medium of an analogous electrical net.

electromagnetic spectrum—The entire range of wavelengths or frequencies of electromagnetic radiation extending from gamma rays to the longest radio waves and including visible light. Most remote sensing systems are designed to operate within the electromagnetic spectrum.

Electronic Chart Display and Information System (ECDIS)—A system which in its most complete design integrates all essential aids to navigation into one system including chart information, positional information and ships' parameters such as course and speed. Echosounder, readings, radar, complete or selective and other inputs may also be included.

Electronic Chart Update Manual (ECHUM)—The electronic version of the Chart Update Manual (CHUM). See **Chart Update Manual.**

electronic bearing—A bearing obtained by means of electronic equipment.

electronic control—Control obtained by electronic devices.

electronic distance-measuring (EDM) equipment—Devices that measure the phase difference between transmitted and returned (i.e., reflected or retransmitted) electromagnetic waves, of known frequency and speed, or the round trip transit time of a pulsed signal, from which distance is computed. A wide range of such equipment is available for surveying and navigational use. Also called **distance measuring equipment (DME).** See also **electronic position indicator; Electrotape; Geodimeter; laser; sonar; Tellurometer.**

electronic line of position—A line of position established by means of electronic equipment.

electronic map data (EMD)—See **electronic map.**

electronic map—Consists of the information of a map or chart, that is captured, stored, and displayed by electronic means. See also **raster; raster data structure; Raster Product Standard(s) (RPS); vector; vector data; vector Product Format (VPF); Vector Product Standard (VPS).**

electronic position indicator (EPI)— A type of electronic distance measuring equipment used primarily in hydrographic surveying. The offshore range has been extended from the limits of shoran distances to more than 500 miles by use of EPI.

electronic refraction—The refraction due to the effects of the atmosphere and the ionosphere, which introduce appreciable changes in the quantities measured by means of electronic devices, such as in the phase differences measured with interferometers, in the rate of change of phase measured with the Doppler systems, and in the change in phase

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

between the times of transmitting and receiving a signal by the ranging instruments.

electronic sketchmaster—See **universal analog photographic rectification system.**

electronic survey—Any survey utilizing electronic equipment.

electronic telemeter—An electronic device that measures the phase difference or transit time between a transmitted electromagnetic impulse of known frequency and speed and its return.

Electrotape—A trade name for a precise electronic surveying device which transmits a radio frequency signal to a responder unit which in turn retransmits the signal back to the interrogator unit. The time lapse between original transmission and receipt of return signal is measured and displayed in a direct digital readout for eventual reduction into a precise linear distance. It operates on the same principal as the Tellurometer.

elements of a fix—The specific values of the coordinates used to define a position.

elevated pole—The celestial pole above the horizon. Opposite of **depressed pole.**

elevation angle—See **angle of elevation.**

elevation density category—A term used throughout sizing of digital elevation data transactions within the MARK 90 system. Elevation density category 1 represents elevation post spacing at 3 arc second intervals. A product having a scale smaller than 1:50,000 would typically be represented by this density. Elevation density category 2 represents elevation post spacing at 1 arc second intervals. A product having a scale of 1:50,000 or larger would typically be represented by this density.

elevation matrix—A series of height values for points spaced at regular intervals and usually expressed in the form of a two-dimensional array of numbers.

elevation meter—A mechanical or electromechanical device on wheels that measures slope and distance, and automatically and continuously integrates their product into difference of elevation.

elevation posts—A point with a known horizontal and vertical position with respect to some defined reference system. Post points are common elevation points between two different digital terrain files. Digital Terrain Elevation Data are elevation posts in digital form. A digital elevation matrix consists of elevation posts evenly distributed in a rectangular pattern.

elevation tints—See **hypsometric tinting.**

elevation—Vertical distance from a datum, usually mean sea level, to a point or object on the Earth's surface; not to be confused with **altitude** which refers to points or objects above the Earth's surface. In geodetic formulas, elevations are heights: h is height above ellipsoid, H is height above the geoid or local datum. Occasionally the h and H may be reversed. See also **adjusted elevation; assumed ground elevation; barometric elevation; checked spot elevation; critical elevation; difference of elevation; dynamic elevation; field elevation; fixed elevation; highest elevation; mean ground elevation; optimum ground elevation; orthometric elevation; preliminary elevation; spot elevation; standard elevation; supplemental elevation; unchecked spot elevation.**

ellipsoid of revolution—See **ellipsoid of rotation.**

ellipsoid of rotation—The surface generated by an ellipse rotating about one of its axes. Also called **ellipsoid of revolution.** See also **oblate spheroid; prolate spheroid.**

ellipsoidal height—The height above the reference ellipsoid, measured along the ellipsoidal outer normal through the point in question. Also called **geodetic height.**

ellipsoidal reflector—A mirror surface which conforms to a portion of an ellipsoid of revolution. Principally employed in several types of stereoplotter projectors such as the ER-55 .

ellipsoid—An ellipsoid is a mathematical figure generated by the revolution of an ellipse about one its axes. The ellipsoid that approximates the geoid is an ellipse rotated about its minor axis, or an oblate spheroid. See also **datum-centered ellipsoid; Earth-centered**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

ellipsoid; Fischer ellipsoid of 1960; triaxial ellipsoid.

ellipticity of an ellipse—The ratio between the difference in length of the semi-axes of an ellipse and its semimajor axis.

ellipticity of the spheroid—*See* **flattening (of the Earth).**

elongation—1. (surveying) The position of a celestial body relative to the observer's meridian, is such that the apparent azimuthal movement is at a minimum. 2. The angular distance of a body of the solar system from the Sun; the angle at the Earth between lines to the Sun and another celestial body of the solar system. *See* also **greatest elongation.**

emergency run—See **tide over run.**

emergent nodal point—See **nodal point,** definition 1.

emissivity—The amount of energy given off by an object relative to the amount given off by a "blackbody" at the same temperature. Normally expressed as a positive number between zero and one.

empirical orientation (rectification)— The composited rectified adjustments of magnification, swing, easel tilt, *y*-displacement, and *x*-displacement used to correctly recreate the exact conditions in the projected image that existed in the negative at the instant of exposure.

emulsion-to-base—A contact exposure in which the base of the copying film is in contact with the emulsion side of the sheet being copied. See also **emulsion-to-emulsion.**

emulsion-to-emulsion—A contact exposure in which the emulsion of the copying film is in contact with the emulsion of the sheet being copied. See also **emulsion-to-base.**

emulsion—A suspension of either light-sensitive silver salts, Diazos, or photopolymers, in a colloidal medium which is used for coating films, plates, and papers.

encode—To convert data by the use of a code or a coded set in such a manner that reconversion to the original form is possible.

end lap—See **overlap,** definition 1.

end node—1. The second node of an edge (an edge is traversed from the start node to the end node).2. The last node of a feature with several edges. 3. The last node of an areal feature which is coincident with the first node.

Engineer Topographic Laboratories (ETL)—See **Topographic Engineering Center (TEC).**

engineer's chain—Similar to a Gunter's chain except that it is 100 feet in length and contains 100 links, each 1 foot long.

engineer's level—Any of a group of precision leveling instruments for establishing a horizontal line of sight, used to determine differences of elevation.

engineering map—A map showing information that is essential for planning an engineering project or development and for estimating its cost.

engineering survey—A survey executed for the purpose of obtaining information that is essential for planning an engineering project or development and estimating its cost. The information obtained may, in part, be recorded in the form of an engineering map.

engraver subdivider—A specially designed scribing instrument which permits the selection of uniform tick spacing in subdividing or ticking map projections.

engraver—See **scriber.**

enlargement factor—See **scale of reproduction.**

enlargement/reduction diagram—Chart showing the necessary lens extension and copy board extension required for various enlargements and reductions.

enlargement—A negative, diapositive, or paper print made at a larger scale than the original. Also called **blow up.**

enroute chart—A chart of air routes in specific areas that shows the exact location of electronic aids to navigation, such as radio direction finder stations, radio and radar marker beacons, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

radio range stations. Also called **radio facility chart.**

entity node—A primitive used to represent features that are truly zero dimensional. Entity nodes are topologically linked to their containing face when faces are present.

entity point—Points used to represent features that are zero-dimensional (such as survey points). Entity points may be topologically linked to a containing face.

entity set—An attribute table or group of attributes.

entrance pupil—The image of the aperture stop formed by all the lens elements on the object side of the aperture stop.

entrance slit—Slit through which energy enters a spectroscopic instrument.

entrance window—The image of the field stop formed by all the lens elements on the object side of the field stop.

Eötvös correction—Component of uncorrected observed gravity obtained on a moving platform that is attributed to the east-west component of the velocity that either increases or decreases the effect of the centrifugal force caused by the Earth's rotation. The formula for the correction usually includes the term for the secondary velocity correction.

Eötvös effect—A vertical force experienced by a body moving in an east-west direction on the rotating Earth. In gravity measurements, a positive correction is applied if moving eastward, and a negative correction if moving westward.

Eötvös unit—The unit of gravitational gradient 10^{-9} meters per second per second per meter.

ephemeris time—The uniform measure of time defined by the laws of dynamics and determined in principal from the orbital motions of the planets, specifically in the orbital motion of the Earth as represented by Newcomb's "Tables of the Sun." Ephemeris time for close-Earth satellites, or more correctly the time associated with satellite ephemerides, is observation dependent. For example, if the observations were made in universal time (UT), the ephemeris time is UT; if the observations were made in international atomic time (IAT), the ephemeris time is in IAT.

ephemeris—(JCS) A publication giving the computed places of the celestial bodies for each day of the year, or for other regular intervals. [A publication giving similar information in a form suitable for use by a navigator is called an **almanac.** An ephemeris is also a statement, not necessarily in a publication, presenting a correlation of time and position of celestial bodies or man-made satellites. A short ephemeris of future location/time of satellite is known as a **set of alerts.**] See also **broadcast ephemeris; precise ephemeris.**

epipolar plane—(photogrammetry) Any plane which contains the epipoles, therefore, any plane containing the air base. Also called **basal plane.**

epipolar ray—The line on the plane of a photograph joining the epipole and the image of an object. Also expressed as the trace of an epipolar plane on a photograph.

epipoles—In the perspective setup of two photographs (two perspective projections), the points on the planes of the photographs where they are cut by the air base (extended line joining the two perspective centers). In the case of a pair of truly vertical photographs, the epipoles are infinitely distant from the principal points.

epoch—**1.** A particular instant for which certain data are given. **2.** A given period of time during which a series of related acts or events takes place. **3.** An arbitrary moment in time to which measurements of position for a body or orientation for an orbit are referred.

Equal Arc Second Raster Chart/Map (ARC)—A rectangular coordinate and projection system at any scale for the entire ellipsoid based on the World Geodetic System 1984 (WGS84). The ARC system divides the ellipsoid surface into 18 overlapping latitudinal band scaled zones.

equal energy—Spectral distribution characterized by equal flux per unit wavelength interval.

equal-altitude observations of the Sun—Azimuth observations that consist of measuring horizontal angles from a southerly reference point to the Sun's limbs at an identical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

vertical angle, if measured to the right limb in the a.m., then the angle should be measured to the left limb in the p.m. The same limb should be observed in vertical angle. The mean of the two horizontal angles, with small correction for the change in the Sun's declination during the interval from the a.m. to the p.m. readings, gives a resulting horizontal angle to the meridian.

equal-altitude observations—Observations of celestial objects at a fixed altitude (such as by an astrolabe) taken at more or less uniformly spaced azimuths around the horizon

equal-area map projection—A map projection having a constant area scale. Such a projection is not conformal and is not used for navigation. Also called **authalic map projection; equivalent map projection.**

equation of the equinox—The difference between the mean and true right ascensions of a body on the Equator, thus the difference between mean and apparent sidereal time. Also called **nutation in right ascension.**

equation of time—The algebraic difference in hour angle between apparent solar time and mean solar time, usually labeled "+" or "-" as it is to be applied to mean solar time to obtain apparent solar time.

equation—See **angle equation; azimuth equation; condition equation; correlate equation; error equation; Euler's equation; hydrostatic equation; Laplace equation; latitude equation; length equation; longitude equation; lunar equation; normal equation; observation equation; parametric equations; perpendicular equation; personal equation; side equation; side equation tests.**

equator system—See **celestial equator system of coordinates.**

equatorial axis—**1.** The diameter of the Earth described between two points on the Equator. **2.** (astronomy) A telescope mounting axis oriented parallel to the Earth's rotational axis.

equatorial bulge—The excess of the Earth's equatorial diameter over the polar diameter.

equatorial chart—**1.** A chart of equatorial areas. **2.** A chart on an equatorial projection.

equatorial cylindrical orthomorphic chart—See **Mercator chart.**

equatorial cylindrical orthomorphic map projection—See **Mercator map projection.**

equatorial diameter—The diameter of the Earth at the great circle comprising the terrestrial equator.

equatorial gravity value—The mean acceleration of gravity at the Equator, approximately equal to 9.780 3 m/sec^2.

equatorial horizontal parallax—The angle at a celestial object subtended by the equatorial semidiameter of the Earth used to indicate the distance of the object from the Earth.

equatorial intervals—The angles, expressed in units of time, between the various lines which compose the reticle of an astronomic transit and the mean position of those lines.

equatorial map projection—A map projection centered on the Equator.

equatorial node—Either of the two points where the orbit of the satellite intersects the equatorial plane of its primary.

equatorial radius—The radius assigned to the great circle comprising the terrestrial equator.

equatorial satellite—A satellite whose orbit plane coincides, or almost coincides, with the Earth's equatorial plane.

equatorial stars—Stars having declinations close to zero and whose diurnal path is a parallel of declination close to the Equator. Equatorial stars, because of their apparently greater speed of travel, are preferred for time and longitude determinations.

equatorial system—See **celestial equator system of coordinates.**

Equator—The great circle on the Earth midway between the poles and in a plane perpendicular to the Earth's axis of rotation. It is the line of 0°

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

latitude. See also **astronomic equator; celestial equator; fictitious equator; galactic equator; geodetic equator; geomagnetic equator; grid equator; lunar celestial equator; magnetic equator; oblique equator; transverse equator.**

equiangular spiral—See **rhumb line.**

equianguiator—An optical instrument, employing a 60° prism, used to determine astronomic latitude and longitude by equal altitudes of heavenly bodies when the time of the prime meridian (Greenwich) is known at the place of observation.

equigeopotential surface—See **equipotential surface.**

equilibrium spheroid—The shape that the Earth would attain if it were entirely covered by a tideless ocean of constant depth. See also **geoid.**

equilibrium theory—A hypothesis which assumes an ideal Earth which has no continental barriers and is uniformly covered with water of considerable depth. It also assumes that the water responds instantly to the tide-producing forces of the Moon and Sun to form a surface in equilibrium and moves around the Earth without viscosity or friction. See also **geoid.**

equilibrium—A state of balance between forces. A body is said to be in equilibrium when the vector sum of all forces acting upon it is zero.

equinoctial colure—The hour circle through the equinoxes.

equinoctial day—See **sidereal day.**

equinoctial point—See **equinox.**

equinoctial system of coordinates— See **celestial equator system of coordinates.**

equinoctial year—See **tropical year.**

equinoctial—See **celestial equator.**

equinox—One of the two points of intersection of the ecliptic and the celestial equator, occupied by the Sun when its declination is 0°. Also called **equinoctial point.** See also **autumnal equinox; mean equinox; vernal equinox.**

equipotential surface—A surface having the same potential of gravity at every point. Also called **level surface.** See also **geoid; geop; geopotential surface.**

equiscalar—A surface along which a scalar quantity has a constant value.

equivalent focal length—(JCS) The distance measured along the optical axis of the lens from the rear nodal point to the plane of best average definition over the entire field used in a camera. See also **focal length.**

equivalent map projection—See **equal-area map projection.**

equivalent scale—The relationship which a small distance on a graphic bears to the corresponding distance on the Earth, expressed as an equivalence, such as 1 inch (on the graphic) equals 1 mile (on the ground). Also called **verbal scale.**

equivalent vertical photograph—A theoretically, truly vertical photograph taken at the same camera station with a camera whose focal length is equal to that of a camera taking a corresponding tilted photograph.

erasable optical disc (EOD)—A class of compact discs which allow users to rerecord data onto the disk.

erect image—An image that appears upright, or in the same relative position as the object.

erecting telescope—An observer sees objects right side up when looking through an erecting telescope and upside down when looking through an inverting telescope. The eyepiece in the optical system of an erecting telescope usually has four lenses, and the eyepiece in the optical system of an inverting telescope has two lenses. See also **inverting telescope.**

erratic error—An error caused by an incomplete element in an instrument, such as backlash in a gear train. See also **instrument error.**

error budget—A correlated set of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

individual major error sources with the quantified error or uncertainty which each contributes to a total system accuracy or probable error.

error equation—The probability equation which expresses the laws of the occurrence of random errors. This equation expresses the relationship between observed values, plus first-order correction terms, and theoretical values. The error equation is the basis of the method of least squares, used in the adjustment of observations for determining the most probable value of a result from those observations.

error interval—See **confidence interval.**

error of closure—1. (general) The amount by which a quantity obtained by a series of related measurements differs from the true or fixed value of the same quantity. Also called **closing error; closure.** 2. (angles) The amount by which the actual sum of a series of angles fails to equal the theoretically exact value of that sum. Also called **angular error of closure.** 3. (azimuth) The amount by which two values of the azimuth of a line, derived by different surveys or along different routes, fail to be exactly equal to each other. Also called **azimuth error of closure.** 4. (leveling) The amount by which two values of the elevation of the same bench mark, derived by different surveys or through different survey routes or by independent observations, fail to be exactly equal to each other. Also called **leveling error of closure.** See also **circuit closure.** 5. (loop) The error in the closure of a survey on itself. Loops do not protect against systematic errors in distance measurement or blunders in starting position or azimuth. Also called **loop error of closure.** 6. (horizon) The amount by which the sum of a series of adjacent measured horizontal angles around a point fails to equal exactly 360°. Measurement of the last angle of the series is called **closing the horizon;** also called **closure of horizon; horizon closure.** 7. (triangle) The amount by which the sum of the three observed angles of a triangle fails to equal exactly 180° plus the spherical excess of the triangle. Also called **closure of triangle; triangle closure; triangle error of closure.** 8. (traverse) The amount by which a value of the position of a traverse station, as obtained by computation through a traverse, fails to agree with another value of the same station as determined by a different set of observations or routes of survey. Also called **closure of traverse; error of survey; horizontal closure error; traverse error of closure.** See also **linear error of closure; relative error of closure.**

error of collimation—See **collimation error.**

error of observation—The difference between an observed value of a quantity and a value adopted as representing the ideal or true value of that quantity.

error of run—(micrometer) The difference, in seconds of arc, between the intended value of one turn of the micrometer screw and its actual value as determined by measuring the space between two adjacent graduation marks of the circle with the micrometer. Also called **run; run of micrometer.**

error of survey—See **error of closure,** definition 8.

error of the mean—See **standard error of the mean.**

error—1. The difference between an observed or computed value of a quantity and the ideal or true value of that quantity. 2. An error is generally classified as one of three types: (1) a blunder (mistake) which can be identified and corrected; (2) a systematic error, either constant or variable, which must be compensated for; and (3) a random error, one of the class of small inaccuracies due to imperfections in equipment, surrounding conditions, or human limitations. See also **absolute error; actual error; accumulative error; blunder; chronometer error; circuit closure; circular error; circular error probable; circular near-certainty error; circular standard error; clamping error; collimation error; compass index error; compensating error; constant error; curved-path error; eccentric error; erratic error; external error; graduation error; gross error; index error; inherited error; instrument error; law of propagation of error; linear error; mean square error; modulation error; natural error; near-certainty error; orthometric error; parallactic error; periodic errors; personal error; pointing error; positional error; principal-distance error; principal-point error; prismatic error; probable error; random error; residual error; resultant error; scale error; shade error; standard error; systematic error; theoretical error.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

escape and evasion graphic—A map, chart or other graphic, usually produced on a lightweight durable material, specifically designed to guide personnel to safety from enemy held territory.

establishment of the port—The average interval between upper and lower lunar transit near time of new and full Moon and the next high water. Also called **common establishment; high water full and change: vulgar establishment.** See also **lunitidal interval.**

establishment—See **lunitidal interval.**

etch slip—A pencil-shaped abrasive used in removing unwanted marks on a metal pressplate. Also called **snakeslip.**

etched zinc plate—An etched copy of the contour drawing of the base map, used as the guide in cutting the stepped terrain base of a model for making relief models.

etch—1. To remove selected areas of the emulsion either chemically or manually. 2. Chemical treatment of a lithographic plate to make nonprinting areas grease-repellent and water-receptive or to produce the image on deep etch plates. 3. An acid solution mixed with the dampening fountain water on an offset press to help control ink on the pressplate.

Euler's equation—A relation in a parabolic orbit involving two radius vectors, their chord, and the time interval between them.

Euler's theorem—A mathematical expression to obtain the radius of curvature of a normal section in any azimuth on the reference ellipsoid. The azimuth angle, and the radius of curvature in the meridian and in the prime vertical must be known.

Eulerian angles—A system of three angles which uniquely defines with reference to one coordinate system (e.g. Earth axis) the orientation of a second coordinate system (e.g. the axis of an orbit).

European datum—The initial point of this system is located at Potsdam, Germany. Numerous national systems have been joined into a large datum based upon the international ellipsoid which was oriented by the astrogeodetic method. The European and African triangulation chains have been connected and the gap of the African arc measurement from Cairo to Cape Town has been filled. Thus, all of Europe, South Africa, and North Africa are molded into one system. Through common survey stations, it was also possible to convert data from the Russian Pulkovo 1932 system to the European datum and, as a result, the European datum includes triangulation as far east as the 84th meridian. Additional ties across the Middle East have permitted connection of the Indian and European datums. See also **preferred datum.**

evection—A perturbation of the Moon in its orbit due to the attraction of the Sun. This results in an increase in the eccentricity of the Moon's orbit when the Sun passes the Moon's line of apsides and a decrease when perpendicular to it. See also **lunar inequality,** definition 1.

Everest spheroid (ellipsoid)—A reference ellipsoid having the following approximate dimensions: semimajor axis—6,377,276.3 meters; flattening or ellipticity—1/300.80. Used in India, Burma, Pakistan, Laos, Cambodia, Thailand, and Vietnam.

exaggerated stereo—See **hyperstereoscopy.**

existent corner—A corner whose position can be identified by verifying the evidence of the monument or its accessories, by reference to the description that is contained in the field notes; or where the point can be located by an acceptable supplemental survey record, some physical evidence, or testimony.

existing data—Source material and/or information assumed or known to be in the possession of a given source and subject to "off-shelf" collection, as in contrast to data obtained by operational field surveys.

exit pupil—The image of the aperture stop formed by all the lens elements on the image side of the aperture stop.

exit window—The image of the field stop formed by all the lens elements on the image side of the field stop.

exmeridian altitude—An altitude of a celestial body near the celestial meridian of the observer to which a correction is to be applied to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

determine the meridian altitude.

exmeridian observation—Measurement of the altitude of a celestial body near the celestial meridian of the observer, for conversion to a meridian altitude; or the altitude so measured.

experience radar prediction—The determination of size, shape, and relative intensity of radar returns and a determination of radar shadow and no-return areas based primarily on the radar knowledge and experience of the individual making the prediction rather than on proven formulas, power tables, or graphs. Also called **artwork prediction.**

experimental map—A sample of a new map product prepared either to obtain user approval of the adequacy of content and symbolization or to disclose any problems which may occur in the various production stages. Also called **prototype.** See also **pilot sheet.**

explement—The difference between an angle and 360°.

exploitation—The process of obtaining usable data from imagery.

exploratory survey—A survey executed for the purpose of obtaining general information concerning areas about which such information is not a matter of record.

exposure interval—The time required between successive exposures of a series of photographs for the purpose of obtaining desired forward lap.

exposure station—See **air station.**

exposure time—The time during which a light-sensitive material is subjected to the action of light.

exposure—1. The total quantity of light received per unit area on a sensitized plate or film; may be expressed as the product of the light intensity and the exposure time. 2. The act of exposing a light-sensitive material to a light source. 3. One individual picture of a strip of photographs, usually called **frame.**

extended color—See **bleed.**

extension of control—Execution of additional control from existing control by any method.

extension—1. (surveying) See **prolongation.** 2. (photogrammetry) Extending existing control from a controlled area into an area without control. The term is usually further qualified as horizontal or vertical according to the primary purpose. Also called **horizontal extension; horizontal/vertical extension; vertical extension.** See also **cantilever extension.**

exterior orientation—The determining (analytically or in a photogrammetric instrument) of the position of the camera station and the attitude of the taking camera at the instant of exposure. In stereoscopic instrument practice, exterior orientation is divided into two parts, relative and absolute orientation. Also called **outer orientation.** See also **resection,** definition 3.

exterior perspective center—See **perspective center.**

exterior to a curve—Any area adjacent to a curve lying toward its convex side, the area not included within the circle of which the curve is part of the circumference.

external distance—The distance from the vertex of a circular curve to the middle point of the curve.

external error—The repeatability of a measurement with any condition extraneous to the measuring method itself changed; contrasted to internal error. See also **standard error.**

extra foresight—(leveling) The rod reading made at an instrument station in a line of levels and on a leveling rod standing on a bench mark or another point not in the continuous line of levels. In spirit leveling there may be one or more extra foresight from a single instrument station or setup, but there can be only one backsight and one foresight from any one instrument station.

extrapolation—The process of estimating the value of a quantity beyond the limits of known values by assuming that the rate or system of change between the last few known values continues.

eye base—See **interocular distance.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

eyepiece micrometer—See **ocular micrometer.**

eyepiece—In an optical device, the lens group which is nearest the eye and with which the image formed by the preceding elements is viewed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

F

f-number—See **relative aperture.**

face—1. (imagery) The emulsion side of a negative or layout plate, or the printing surface of a plate. 2. (topology) A two-dimensional topological entity defined as an undivided are surrounded by topological edges representing all or part of the extent of a feature or of an attributed area.

facsimile chart—See **modified facsimile chart.**

factored transparency—A system of radar simulation which utilizes a pair of photographic images on a glass plate or plates to store topographic and radar reflection data. The data are scanned by a flying-spot scanner cathode-ray tube, and the density of the images is read by two photomultiplier tubes. The two planar dimensions of the two images are the *x*- and *y*-dimensions of the topographic and reflectance data respectively. The densities of the reflectance images are used to store the intrinsic strength of radar target reflectance. The images are identical in their x- and y-values but separated in one dimension by the optical spacing of the dual readout system. Also called **land mass simulator plate.**

factory acceptance test (FAT)—A formal test of the system under procurement in the manufacturer's environment in order to verify successful production of hardware or software. Recipient personnel witness the test to verify requirements and successful product generation before the system is shipped to the recipient.

FAF block—A block of 1024 by 1024 pixels of image data. See also **fast access format.**

falling tide—See **ebb tide.**

falling—The distance by which a random line falls to the right or left of a corner on which the true line is too close. Usually the direction of falling is expressed as cardinal.

false bearing—The difference between the true bearing and the back bearing caused by the convergence of meridians.

false color—(photogrammetry) See **infrared.**

false easting—A value assigned to the origin of eastings, in a grid coordinate system, to avoid the inconvenience of using negative coordinates. See also **grid coordinates.**

false fix probability—A statistical value or ratio which reflects the likelihood of a false match occurring between prestored digital cartographic data and data or imagery acquired by electronic aerial sensor systems. See also **image correlation; terrain contour matching (TERCOM); terrain correlation.**

false horizon—A line resembling the visible horizon, but above or below it.

false northing—A value assigned to the origin of northings, in a grid coordinate system, to avoid the inconvenience of using negative coordinates. See also **grid coordinates.**

false origin—(JCS) A fixed point to the south and west of a grid zone from which grid distances are measured eastward and northward. See also **grid origin.**

false parallax—(JCS) The apparent vertical displacement of an object from its true position when viewed stereoscopically, due to movement of the object itself as well as to change in the point of observation.

false stereo—An imaginary impression of stereoscopic relief. See also **pseudoscopic stereo.**

fan camera photography—(JCS) Photography taken simultaneously by an assembly of three or more cameras, systematically installed at fixed angles relative to each other so as to provide wide lateral coverage with overlapping images. See also **tricamera photography.**

fan cameras—(JCS) An assembly of three or more cameras systematically disposed at fixed angles relative to each other so as to provide wide lateral coverage with overlapping images.

fast access format (FAF)—A filing scheme for breaking digital imagery into smaller chunks of data. See also **FAF block.**

fathogram—A graphic record of depth

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

measurements obtained by echo sounding equipment. Also called **echogram.**

fathom curve—See **depth contour.**

fathom line—See **depth contour.**

Fathometer—A trade name for an echo sounder.

Faye anomaly—See **free-air anomaly.**

Faye correction—See **free-air correction.**

featheredging—1. (cartography) The technique of progressively dropping contours, to avoid congestion on steep slopes, and tapering the line weight near the end of the contour to be dropped. Also called **feathering.** 2. (photomosaicking) The thinning of overlapping edges of photographs before assembling into a mosaic in order to make match lines less noticeable. When overlapping edges are feathered, shadows and sharp changes in contrast are reduced or eliminated. Also called **feathering.**

feathering—See **featheredging.**

Feature and Attribute Coding Catalog (FACC)—DMA Glossary which contains the Feature and Attribute Coding Standard (a modification of DMA's FACS). See **Feature/Attribute Coding Standard.**

Feature Extraction Segment (FE/S)—A segment of DMA's MARK 85 which implements a computer supported, analytical stereoplotter system capable of extracting feature data. This segment integrates, on a single system, all of the various steps involved in the mostly manual feature extraction process prior to DPS. See also **MARK 85; Digital Production System.**

feature analysis code (FAC) number—A unique number (usually sequential) assigned to each area or feature portrayed on the feature manuscript and used to relate feature analysis data table (FADT) information to the digital information which portrays the shape of the feature. See also **feature analysis data table.**

feature analysis data table (FADT)—A table containing the feature analysis code numbers and the numeric codes which represent the physical characteristics of features selected for portrayal. See also **feature analysis code (FAC) number.**

feature analysis—The process of locating, examining, and classifying the physical characteristics of cultural features on the earth's surface.

feature attribute—A property of a feature.

feature class—A set of features sharing a consistent set of attribute types. A feature class is implemented by using a set of tables that includes at least one primitive table and at least one attribute table. A feature class has the same columns (attribute types) of attribute information for each feature. Classes of features are created in order to allow the relational model to operate on them as sets. Every feature class has one and only one feature table.

feature code—A unique identifier assigned to a feature.

feature extraction—The art of extracting and classifying features contained in an image.

feature identification data (FID)—Information pertaining to the classification of a feature as to kind, function, and description, e.g., heavy fabrication industry with saw-tooth roof; truss bridge; powerline pylon, etc. Each classification has a unique feature identification code. See also **feature analysis data table, feature analysis code; feature analysis.**

feature table—A table made up of the primary-keyed rows of the features in a feature class. These rows collectively form the feature table for that feature class.

feature type—A classification of features into three categories: (1) point of feature—an object whose location can be described by a single set of coordinates; (2) linear (or lines) feature—portrayed by a line that does not represent an area; (3) areal feature—a topographic feature, such as sand, swamp, vegetation, etc., which extends over an area or in digital mapping any area enclosed by a delimiting line that has any unique characteristic, e.g., industrial area, forest, residential area, etc. Feature type is thus an attribute defining the topologic property of a feature (point, line, or area).

feature-oriented—A term used to describe

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the approach for encoding geographic entities in which a series of interrelated geographic objects are organized into a network-like structure of data and predefined relationships.

Feature/Attribute Coding Standard (FACS)—DMA standard for assigning features, attributes, values, and relationships to all digital entities under the Digital Production System. See also **Feature and Attribute Coding Catalog.**

feature—A set of phenomena with common attributes and relationships. The concept of feature encompasses both entity and object. See also **Primary row.**

Federal Information Processing Standards (FIPS)—Official source within the Federal Government for information processing standards. FIPS are developed by the Institute for Computer Sciences and Technology at the National Institute of Standards and Technology (NIST, formerly called the National Bureau of Standards).

felt side—The top or smooth side of paper that is contacted by the felt belt for extraction of moisture during manufacture. This is the correct side of the paper for printing.

fence—1. A line of readout or tracking stations for pickup of signals from an orbiting satellite. 2. A line or network of radar or radio stations for detection of a satellite in orbit.

fermenting dough theory—See **Pratt-Hayford theory of isostasy.**

ferrotype—To burnish photographic prints by squeegeeing wet upon a japanned sheet of iron or stainless plate and allowing to dry. This produces a harder, glossier surface on the photographic print.

fiber optics—A device for relaying an image by means of a large number of transparent fibers (filaments) by multiple total internal reflection. The fibers are most commonly glass and less often a highly transparent plastic. Each fiber carries only one element of the image, so that the image is a mosaic in which the cell size is the fiber cross section rather than a continuous picture.

fictitious equator—A reference line serving as the origin for measurement of fictitious latitude.

fictitious graticule—The network of lines representing fictitious parallels and fictitious meridians on a map or chart. See also **oblique graticule; transverse graticule.**

fictitious latitude—Angular distance from a fictitious equator. It may be called transverse, oblique, or grid latitude depending upon the type of fictitious equator.

fictitious longitude—The arc of fictitious equator between the prime fictitious meridian and any given fictitious meridian. It may be called **transverse, oblique,** or **grid longitude** depending upon the type of fictitious meridian.

fictitious loxodrome—See **fictitious rhumb line.**

fictitious loxodromic curve—See **fictitious rhumb line.**

fictitious meridian—One of a series of great circles or lines used in place of a meridian for certain purposes. It may be called **transverse, oblique,** or **grid meridian** depending upon the type of fictitious meridian. See also **prime fictitious meridian.**

fictitious parallel—A circle or line parallel to a fictitious equator, connecting all points of equal fictitious latitude. It may be called **transverse, oblique,** or **grid parallel** depending upon the type of fictitious equator.

fictitious pole—One of the two points 90° from a fictitious equator. It may be called **transverse** or **oblique pole** depending upon the type of fictitious equator.

fictitious rhumb line—A line making the same oblique angle with all fictitious meridians. It may be called **transverse, oblique,** or **grid rhumb line** depending upon the type of fictitious meridian. Also called **fictitious loxodrome; fictitious loxodromic curve.**

fictitious sun—A fictitious point termed the **mean sun,** which is imagined to move at a uniform rate along the Equator, its rate of motion being such that it makes one apparent revolution around the Earth in the same time as the actual Sun—that is, in 1 year.

fictitious year—The period between

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

successive returns of the Sun to a sidereal hour angle of 80° (about January 1). The length of the fictitious year is the same as that of the tropical year, since both are based upon the position of the Sun with respect to the vernal equinox. Also called **Besselian year.**

fictitious—In cartography, pertaining to or measured from an arbitrary reference line.

fidelity—The degree with which a system accurately reproduces the data input into it.

fiducial axes—The lines joining opposite fiducial marks on a photograph. The *x*-axis is generally considered to be the one nearly parallel with the line of flight.

fiducial mark(s)—**1.** (surveying) An index line or point. A line or point used as a basis of reference. **2.** (JCS) (photogrammetry) See **collimating marks. 3.** Also, markers in any instrument which define the axes whose intersection fixes the principal point of a photograph and fulfills the requirements of interior orientation.

field calibration—A term generally applied where only a combination of field and office computer techniques are available to check instrument accuracy. Adjustments, other than normal operator adjustments, cannot be made during field calibration.

field check—The operation of checking a map compilation manuscript on the ground. See also **field classification.**

field classification—Field inspection and identification of features which a map compiler is unable to delineate; identification and delineation of political boundary lines, place names, road classifications, buildings hidden by trees, and so forth. Field classification may be included as part of the control survey effort and normally is completed prior to the actual stereocompilation phase. See also **field inspection.**

field comparator—A short line whose length is measured with accuracy and precision, and is used to check the lengths of apparatus (tapes) used in the actual field operations. Also called **calibration course; comparator base.**

field completion—A combination of field inspections or surveys, either before or after compilation, to classify and complete the map content, correct erroneous data, and add information such as names, civil boundaries, and similar classification data. Its purpose is to fill in or confirm that portion of a map manuscript prepared by stereocompilation.

field contouring—Contouring a topographic map by field methods accomplished by planetable surveys on a prepared base or by stadia survey. Generally, this operation applies to terrain unsuitable for contouring by photogrammetric methods. Also used in limited areas when engineering design (drainage) requires 1-foot contours. See also **contour sketching.**

field control—(JCS) A series of points whose relative positions and elevations are known. These positions are used in basic data in mapping and charting. Normally, these positions are established by survey methods and are sometimes referred to as **trig control** or **trigonometrical net (work).** See also **common control (artillery); control point; ground control.**

field correction copy—A map or tracing prepared in the field, delineating corrections for subsequent reproduction of a map.

field correction—Adjustments made to field measurements, such as angles or distances, to correct for geometric or length discrepancies.

field elevation—An elevation taken from the field computation of a line of levels.

field inspection—The process of comparing aerial photographs with conditions as they exist on the ground, and of obtaining information to supplement or clarify that which is not readily discernible on the photographs themselves. Also called **classification survey.**

field intensity—See **field strength.**

field of view—(JCS) In photography, the angle between two rays passing through the perspective center (rear nodal point) of a camera lens to the two opposite sides of the format. Not to be confused with **angle of view.** See also **angle of view.**

field position—A position computed while field work is in progress to determine the acceptability of the observations or to provide a preliminary position for other purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

field sheet—The hydrographer's or topographer's work sheet; it presents a graphic display of all surface and subsurface features in the area being surveyed. See also **boat sheet.**

field standardization of tape—The comparison of the length of a tape to be used for survey measurements with the length of a standard tape, to determine the true length of the former.

field stop—The physical element (such as a stop, diaphragm, or lens periphery) of an optical system which limits the field of view covered by the system. See also **aperture stop.**

field strength—For any physical field, the flux density, intensity, or gradient of the field at the point in question. Also called **field intensity.**

field—An individual data element. In an attribute table, a field is a single attribute value of a single entity.

figure adjustment—(surveying) The adjustment of a single chain of triangles made to satisfy the requirement that the sum of the angles in each triangle equals 180°, and in the case of a quadrilateral that the sum of the angles equal 360°. An office computation.

figure of the Earth—See **geoid.**

filar micrometer—A device attached to a telescope or microscope, consisting of a wire thread (filament) connected with a screw in such manner that as the screw is turned, the wire moves through a continuous succession of parallel positions, all in the focal plane of the instrument .

file structuring—*The logical form of a file that results from applying a particular file organization and layout to a group of records.*

film base—A thin, flexible, transparent sheet of stable plastic material to which a light-sensitive emulsion may be applied.

film distortion—The dimensional changes which occur in photographic film with changes in humidity or temperature, or from aging, handling, or other causes.

film mosaic—See **panel base.**

film negative—*See* **negative,** definition 1.

film positive—See **positive,** definition 1.

film titling—See **titling.**

film—A film base which is coated with a light sensitive emulsion for use in a camera or printing frame. See also **aerial film; autoscreen film; cartographic film; infrared film; stable-base film; strip film; topographic base film.**

filtering—The removal of certain spectral or spatial frequencies to enhance features in the remaining image.

filter—Any transparent material which, by absorption, selectively modifies the light transmitted through an optical system.

final composite—A composite of the principal color separations made after all corrections have been completed.

FIREFINDER Operational Data Base (FODB)—Digitized triaxial coordinates collected on the preferred datum (usually WGS 84) at 125 meter post spacing within 100 by 125 km areas. Values are collected from DTED and transformed to UTM coordinates, with vertical values assigned within 256 elevation bands defined between minimum and maximum values occurring in the area. Source tapes are a special-purpose DMA product that require dubbing onto field cassettes prior to use by the FIREFINDER system. The FIREFINDER provides high speed computation of mortar and field artillery firing positions from radar intercepts of projectile trajectories.

firing chart—(JCS) Map, photomap, or grid sheet showing the relative horizontal and vertical positions of batteries, base points, base point lines, check points, targets, and other details needed in preparing firing data.

first approximation chart—*See* **historical chart.**

first of Aries—See **first point of Aries; vernal equinox.**

first point of Aries— Also called **first of Aries.** See **vernal equinox.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

first point of Cancer—See **summer solstice**, definition 1.

first point of Capricornus—See **winter solstice**, definition 1.

first point of Libra—See **autumnal equinox.**

first-order bench mark—A bench mark connected to the datum (usually mean sea level) by continuous first-order leveling.

first-order leveling—Spirit leveling conforming to the specifications of the current "Classification, Standards of Accuracy and General Specifications of Geodetic Control Surveys." Formerly known as **precise leveling** and **leveling of high precision.** Recommended for primary National Networks, as a basis for all subordinate elevation determinations, scientific studies such as crustal movement over large regions, extensive engineering projects such as hydroelectric dams. Such leveling generally includes the determination of geopotential values through simultaneous gravity measurements.

first-order level—A leveling instrument which meets the following criteria: (1) the sensitivity of the level bubble vial must be 10" of arc or less per division of 2-mm; (2) the instrument must be constructed of low expansion metal to minimize the effect of unequal heating; (3) the objective lens must have an effective opening of at least 40 mm and a magnification of 40X.

first-order traverse—A survey traverse which extends between adjusted positions of other first-order control surveys and conforms to the current specifications of first-order traverse, per "Classification, Standards of Accuracy and General Specifications of Geodetic Control Surveys."

first-order triangulation—First-order triangulation was at one time known as primary triangulation; changed in 1921 to precise triangulation; and in 1925 to first-order triangulation. These surveys conform to the current "Classification, Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for primary National Networks, as a basis for all subordinate surveys; metropolitan area surveys, where high value is attached to land and its line of communication frontage; and in scientific studies, such as crustal movement and space exploration.

first-order work—The designation given survey work of the highest prescribed order of precision and accuracy. Such surveys were formerly called **primary.**

Fischer ellipsoid of 1960—A reference ellipsoid with two primary uses. In the Mercury datum it has the approximate dimensions of semimajor axis—6,378,166.0 meters, and the flattening or ellipticity—1/298.3. In the South Asia datum the semimajor axis is 6,378,155.0 meters, and the flattening or ellipticity is 1/298.3.

Fischer level—A dumpy level capable of first-order leveling.

fixed elevation—An elevation which has been adopted, either as a result of tide observations or previous adjustment of spirit leveling, and which is held at its accepted value in any subsequent adjustment.

fixed position—See **adjusted position.**

fixed satellite—See **synchronous satellite.**

fixed-length records—A computer data element format in which all records have the same length. These records allow direct access without searching or indices. See also **variable-length records.**

fixed-ratio pantograph—See **pantograph.**

fixed-ratio projection printer—A diapositive printer having an optical system in which a lens is placed between the negative and the diapositive plate, the object and image distances being set at nominal values according to the laws of optics, but with freedom of adjustment within narrow limits, to produce diapositives whose scale is at a predetermined ratio to the negative scale.

fixer network—(JCS) A combination of radio or radar direction finding installations which, operating in conjunction, are capable of plotting the position relative to the ground of an aircraft in flight.

fixing—The process of rendering a developed photographic image permanent by removing the unaffected light-sensitive material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

fix—A relatively accurate position determined without reference to any former position, from terrestrial, electronic, or astronomic data Also, the point thus established.

flare triangulation—A method of triangulation in which simultaneous observations are made on parachute flares. This method is used for extending triangulation over lines too long to be observed by ordinary methods.

flash apparatus—An auxiliary apparatus used in timing a pendulum during observations for intensity of gravity.

flash plate—See **calibration plate.**

flat model—Any spatial model which is capable of being leveled. See also **warped model.**

flat stock—**1.** Charts or maps which are not folded and kept for filling official and sales orders. **2.** Flat sheets of map paper as opposed to roll paper.

flat tint screen—See **dot screen.**

flattening (of the Earth)—The ratio of the difference between the equatorial and polar radii of the Earth (semimajor and semiminor axes of the spheroid) to its equatorial radius (semimajor axis). Also called **compression; ellipticity of the spheroid.** See also **eccentricity of ellipse; eccentricity of spheroid of revolution; ellipticity of ellipse.**

flat—**1.** (lithography) An assembly of photographic negatives or positives on goldenrod paper or vinyl acetate for contact exposure with a sensitized metal press plate. May contain illustrations as well as text. See also **key flat; layout. 2.** (photography) Lacking in contrast. **3.** (optics) See **optical flat.**

flexure—**1.** (pendulum) The bending of a swinging pendulum, due to its lack of perfect rigidity. **2.** (pendulum support) The forced movement of a pendulum support caused by the motion of the swinging pendulum.

flicker method—**1.** The alternate projection of corresponding photographic images onto a tracing table platen or projection screen, or into the optical train of a photogrammetric instrument. **2.** (stereoscopy) The alternate blinking of the eyes and mentally comparing the appearance of images in a stereoscopic pair to determine differences between the two photos.

flight altitude—The vertical distance above a given datum, usually mean sea level, of an aircraft in flight.

flight block—An adjustable unit of photographic coverage consisting of overlapping strips of photography. A minimum size block consists of at least three overlapping flight strips.

flight chart—See **route chart,** definition 2.

flight information and air facilities data—Data concerning airfields and seaplane stations and related information required for the operation of aircraft.

flight line spacing—The distance between adjacent tracks in a series of parallel aerial photographic flight strips.

flight line—(JCS) In air photographic reconnaissance, the prescribed ground path over which an air vehicle moves during the execution of its photo mission.

flight map—A map on which are indicated the proposed lines of flight and/or positions of exposure stations. Flight data are plotted on the best available map of the area. Generally used for planning purposes.

flight strip—A succession of overlapping aerial photographs taken along a single course. Also called **strip.**

flipping—The act of superimposing and comparing identical areas of two overlapping vertical photographs as an aid in laying an uncontrolled mosaic.

float gage—Any of the tide or stream gages which permit direct reading of changes of water height by the action of a float, contained within a restricted pipe or channel, attached to a graduated tape or chain.

floating lines—(JCS) In photogrammetry, lines connecting the same two points of detail on each print of a stereo pair, used to determine whether or not the points are intervisible. The lines may be drawn directly onto the prints or superimposed by means of strips of transparent

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

material.

floating mark—(photogrammetry) A mark seen as occupying a position in the three-dimensional space formed by the stereoscopic fusion of a pair of photographs and used as a reference mark in examining or measuring the stereoscopic model.

floating—(cartography) The technique of making minor adjustments of detail in order to maintain their proper relative position.

flood control map—A special map, or set of maps, designed for study and planning the control of areas subject to inundation.

flood tide—The portion of the tide cycle between low water and the following high water. Also called **rising tide.**

flowline—The slope extending from the heights along the neatline to the model datum, at an angle no greater than 45°, to preclude forming the plastic sheet at a 90° angle at the neatline of a plastic relief map.

fluorescent map—A map reproduced with fluorescent ink or on fluorescent paper, which enables the user to read the map in darkness under ultraviolet light.

flux-gate magnetometer—An instrument designed to measure the Earth's magnetic field. Also called **saturable reactor.**

fluxmeter—An instrument for measuring the intensity of a magnetic field.

fly leveling—See **flying levels.**

fly-by method—(surveying) A technique of determining approximate elevations where extremely rugged terrain is encountered. The principle is identical to the two-base method except the roving barometers are air transported and read in the aircraft as it passes on a level with the topographic feature whose elevation is required.

flying levels—**1.** A level line run at the close of a working day to check the results of an extended line run in one direction only. Longer sights and fewer setups are used as the purpose is to detect large mistakes. Also called **fly leveling. 2.** Level lines run with the engineer's ordinary leveling equipment but with a distinctly low order of accuracy. Error of closure may be perhaps one foot (or more) times the square root of the distance in miles.

focal length—A general term for the distance between the center, vertex, or rear node of a lens (or the vertex of a mirror) and the point at which the image of an infinitely distant object comes into critical focus. The term must be preceded by an adjective such as "equivalent" or "calibrated" to have a precise meaning. See also **back focal length; calibrated focal length; effective focal length; equivalent focal length; nominal focal length.**

focal plane—(photography) The plane, perpendicular to the axis of the lens in which images of points in the object field of the lens are focused.

focal point—See **focus.**

focal range—See **depth of focus.**

focal-plane plate—A glass plate set in the camera so that the surface away from the lens coincides with the focal plane. Its purpose is to position the emulsion of the film in the focal plane when the film is physically pressed into contact with the glass plate. Also called **contact glass; contact plate.**

focus—The point toward which rays of light converge to form an image after passing through a lens. Also defined as the condition of sharpest imagery. Also called **focal point; principal focus.** See also **hyperfocal distance; sidereal focus.**

folded optics—(photogrammetry) Any optical or lens system containing reflecting components which reduces the physical length of a photographic or sensing system, or changes the path of an optical axis.

folding vernier—A single vernier so constructed and numbered that it may be read in either direction.

foot-meter rod—A stadia rod, marked in feet and tenths on one side, and meters and hundredths on the other side, used to determine distances and elevations in one unit of measurement and to check them by readings in a different unit.

force function—See **potential.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

foresight—1. An observation of the distance and direction to the next instrument station. 2. (*transit traverse*) A point set ahead to be used for reference when resetting the transit on line or when verifying the alignment. 3. (leveling) The reading on a rod that is held at a point whose elevation is to be determined. Also called **minus sight**. See also **backsight**.

forestry map—A map prepared principally to show the size, density, kind, and value of trees on a given area.

form lines—(JCS) Lines resembling contours, but representing no actual elevations, which have been sketched from visual observation or from inadequate or unreliable map sources, to show collectively the configuration of the terrain.

format—Predetermined arrangement of characters, fields, lines, punctuation, page number, etc.

forming machine—The equipment for forming, by heat and vacuum, preprinted plastic maps over a mold representing the terrain of the area.

formula for theoretical gravity—A formula expressing gravity on the spheroid of reference in terms of geographic position, it being assumed that the spheroid of reference is a level surface.

forward azimuth—*See* **azimuth**, *definition* **1**.

forward lap—See **overlap**, definition 1.

found corner—*A term adopted by the U.S.* Geological Survey to designate an existent corner of the public-land surveys which has been recovered by field investigation.

four-pole chain—See **Gunter's chain.**

***four-rod* chain**—See **Gunter's chain.**

fourth-order traverse—A survey traverse of an accuracy less than third-order traverse or which fails to meet third-order official standards. In fourth-order traverse, angles are observed with a transit or sextant or are determined graphically, and distances are measured with tape or stadia .

fractal—A family of mathematical functions which are sometimes used to describe natural phenomena and shapes such as coastlines, mountains, river patterns, etc.

fractional scale—*See* **representative fraction.**

fractional section— A section containing an area appreciably different from 640 acres, usually as a result of an invasion by a segregated body of water, or by other land which cannot properly be surveyed or disposed of as part of that section. See also **section.**

fractional township— A township containing less than 36 normal sections, usually because of invasion by a segregated body of water, or by other land which cannot properly be surveyed as part of that township, or by closing the public land surveys on State boundaries, or other limiting lines. **Half ranges** and **half townships** are fractional townships by definition. See also **township.**

frame camera—A camera in which an entire frame or format is exposed through a lens that is fixed relative to the focal plane. See also **panoramic camera.**

framework of control—See **survey net.**

frame—(JCS) In photography, any single exposure contained within a continuous sequence of photographs.

free-air anomaly—(JCS) The difference between observed gravity and theoretical gravity which has been computed for latitude and corrected for elevation of the station above or below the geoid, by application of the normal rate of change of gravity for change of elevation, as in free air. Also called **Faye anomaly.**

free-air correction—*Correction factor*, usually expressed as milligals per meter, which is applied to observed gravity to reduce the value to sea level. Also called **Faye correction.**

free-swinging pendulum—A pendulum moving wholly under the influence of gravity and an initial momentum imparted to it by mechanical or other means. In gravity work, the initial momentum may be imparted by drawing the pendulum slightly out of plumb and then releasing it.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Fresnel lens—A lens which consists of a thin stepped disc with each step having the curvature of a much thicker lens. A similar design would be the roof of a factory that looks like the teeth of a saw. This pattern, embossed in plastic, is used to distribute image (or light) brightness over a given area.

frilling—The separation, along the edges, of the photographic emulsion from its base.

front element—See **lens element.**

front focal distance—The distance measured from the vertex of the front surface of the lens to the front focal point.

front nodal point—See **nodal point,** definition 1.

front surface mirror—An optical mirror on which the reflecting surface is applied to the front surface of the mirror instead of to the back; i.e., to the first surface of incidence.

Fulcrum—A videodisc-based electronic map display system for the presentation and analysis of multisource data including maps, data bases, imagery, gazetteers, and graphic symbols. Developed by Interactive Television Company.

fundamental circle—See **primary great circle.**

fundamental star places—The apparent right ascensions and declinations of standard comparison stars obtained by leading observatories and published annually.

fundamental tables, deformation of the geoid and its effect on gravity—Tables giving the deformation of the geoid and its effect on gravity, computed for masses of unit density extending to various distances above and below the surface of the geoid. Fundamental tables serve as the basis for the preparation of special tables corresponding to particular assumptions respecting density, isostasy, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

G

galactic circle—See **galactic equator.**

galactic equator—A great circle of the celestial sphere, inclined 62° to the celestial equator and coinciding approximately with the center line of the Milky Way, constituting the primary great circle for the galactic system of coordinates. It is everywhere 90° from the galactic poles. Also called **galactic circle.**

galactic latitude—Angular distance north or south of the galactic equator; the arc of a great circle through the galactic poles, between the galactic equator and a point on the celestial sphere, measured northward or southward from the galactic equator through 90° and labeled *"N" or "S" to indicate the direction of* measurement.

galactic longitude—Angular distance east of sideraal hour angle (SHA) 94.4° along the galactic equator; the arc of the galactic equator or the angle at the galactic pole between the great circle through the intersection of the galactic equator and the celestial equator in Sagittarius (SHA 94.4°) and a great circle through the galactic poles measured eastward from the great circle through SHA 94.4° through 360°.

galactic pole—On the celestial sphere, either of the two points 90° from the galactic equator.

galactic system of coordinates—An astronomic coordinate system using latitude measured north and south from the galactic equator and longitude measured in the sense of increasing right ascension from 0° to 360°. The system was originally defined such that the pole was at RA = 12^h40^m, Dec = + 28°; however, in 1958 the International Astronomical Union (IAU) introduced the IAU galactic system which defined the pole at RA = 12^h49^m, Dec = 27°24'.

galley proof—A proof from type on a galley before it is made up in pages; also, such proofs.

gal—A unit of acceleration equal to 1 centimeter per second per second, or 1000 milligals, used in measuring the acceleration of gravity.

gamma—**1.** (photography) The tangent of the angle which the straight-line portion of the characteristic curve makes with the log-exposure axis. It indicates the slope of the straight-line portion of the curve and is a measure of the extent of development and the contrast of the photographic material. **2.** (geomagnetism) A small unit of magnetic field intensity sometimes used in describing the Earth's magnetic field. It is defined as being equal to 10^{-5} oersted. See also **nanotesla.**

gap—(JCS) (imagery) Any space where imagery fails to meet minimum coverage requirements. This might be a space not covered by imagery or a space where the minimum specified overlap was not obtained. See also **holiday.**

Gauss-Kruger grid—See **transverse Mercator grid.**

gauss—A centimeter-gram-second electromagnetic unit of magnetic induction equal to 104 tesla. See also **tesla.**

gazetteer—An alphabetical list of place names giving feature identification and geographic and/or grid coordinates.

General Navigation Charts—Small scale (1:1,000,000 or smaller) charts designed for open sea surface navigation. Depicted information supports voyage planning, ocean navigation, and military operations.

general chart—A nautical chart intended for offshore coastwise navigation. A general chart is of smaller scale than a coast chart, but of larger scale than a sailing chart.

general map—(JCS) A map of small scale used for general planning purposes. See also **map.**

general precession—Tha motion of the equinoxes westward along the ecliptic at the rate of about 50.13" per year. See also **lunisolar precession; planetary precession; precession in declination; precession in right ascension; precession of the equinoxes.**

general-purpose map—A map which provides a broad range of information and which

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

satisfies the needs of a broad range of users.

generalization—1. The processes of reducing raw data for transformation into a data base or map. Generalization involves aggregation, agglomeration (combination), selection, resolution feature elimination (omission), classification, smoothing, resampling, simplification, exaggeration, displacement, compacting variables, measurement level reduction, dimensionality change, and symbolization. 2. (line generalization) Commonly refers to coordinate thinning or line thinning and/or smoothing. See also **line thinning.**

generation—(photography) The preparation of successive positive/negative reproductions from an original negative (first generation). The first positive produced is a second-generation product; the negative made from this positive is a third-generation product, and the next positive or print from that negative is a fourth-generation product. With each successive generation, quality deteriorates.

generic term—That part of a name which describes the kind of feature to which the name is applied, and which has the same meaning in current local usage. For example, the generic term "wan" in "Tokyo-wan" means "bay."

geo-referenced data—Spatial data that pertain to a location on the earth's surface.

Geoceiver—Trade name for an antenna-receiver capable of receiving signals from the Navy navigation satellites, from which three-dimensional positions can be computed for the antenna location. See also **Doppler navigation,** definition 2.

geocentric coordinate system—See **geocentric coordinates.**

geocentric coordinates—(terrestrial) Coordinates that define the position of a point with respect to the center of the Earth. Geocentric coordinates can be either Cartesian (x,y,z) or spherical (geocentric latitude and longitude, and radial distance). Also called **geocentric coordinate system; geocentric position.**

geocentric diameter—The diameter of a celestial body measured in seconds of arc as viewed from the Earth's center.

geocentric geodetic coordinates—Geodetic coordinates referred to a geocentric reference ellipsoid.

geocentric gravitational constant— The product of the Earth's mass and the gravitational constant. This product is known to a far greater precision than either factor.

geocentric horizon—The plane through the center of the Earth, parallel to the topocentric horizon.

geocentric latitude—The angle at the center of the Earth between the plane of the celestial equator and a line to a point on the surface of the Earth. Geocentric latitude is used as an auxiliary latitude in some computations in astronomy, geodesy, and cartography, in which connection it is defined as the angle formed with the major axis of the ellipse (meridional section of the spheroid) by the radius vector from the center of the ellipse to the given point. In astronomic work, geocentric latitude is also called **reduced latitude,** a term that is sometimes applied to parametric latitude in geodesy and cartography. The geocentric and isometric latitudes are approximately equal.

geocentric longitude—See **geodetic longitude.**

geocentric parallax—The difference in the apparent direction or position of a celestial body as observed from the center of the Earth and a point on its surface. This varies with the body's altitude and distance from the Earth. Also called **diurnal parallax.**

geocentric position—See **geocentric coordinates.**

geocentric radius vector—The vector from the center of the Earth to the point in question. See also **geocentric coordinates.**

geocentric station position—The location of a station defined in terms of geocentric coordinates.

geocentric zenith—The point where a line from the center of the Earth through a point on its surface meets the celestial sphere.

geocentric—Relative to the Earth as a center; measured from the center of the Earth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

geodesic line—A line of shortest distance between any two points on any mathematically defined surface. A geodesic line for an ellipsoid, in general, is a line of double curvature, and usually lies between the two normal section lines which the two points determine. If the two terminal points are in nearly the same latitude, the geodesic line may cross one of the normal section lines. It should be noted that, except along the Equator and along the meridians, the geodesic line is not a plane curve and cannot be sighted over directly. However, for conventional triangulation the lengths and directions of geodesic lines differ inappreciably from corresponding pairs of normal section lines. Also called **geodesic; geodetic line.**

geodesic—See **geodesic line.**

geodesy—The science which deals with the determination of the size and figure of the Earth.

Geodetic Reference System 1980 (GRS 80)—This geodetic reference system was adopted by the XVII General Assembly of the International Union of Geodesy and Geophysics at Canberra, Australia in 1979. The GRS 80 is defined by the four fundamental constants: semi-major axis 6,378,137m, gravitational constant of the Earth (including the atmosphere) 3,986,005 x 10^8 m^3 s^{-2}, second degree zonal coefficient 108,263 x 10^{-8}, and angular rotational velocity 7,292,115 x 10^{-11} radians.

geodetic and geophysical data (G&G)—Information or earth data, pertaining to the sciences of geodesy and geophysics. Typically includes gravity information, geodetic point positioning data, datum definition, etc.

geodetic and geophysical data reduction—The process of enhancing the value of geodetic and geophysical data by analysis, evaluation, computation, and adjustment. The process includes (1) transforming unadjusted survey data and observations into an adjusted form with reliability statements; (2) establishing basic frameworks of horizontal and vertical control in advance of map, chart, and target materials production, and publishing of trig lists; and (3) analysis, evaluation, and computation of geodetic and geophysical data obtained by surface, airborne, or satellite techniques to establish, extend, connect, and transform datums and to relate datums to the Department of Defense World Geodetic System.

geodetic anomaly—See **anomaly,** definition 2.

geodetic astronomy—The branch of geodesy which utilizes astronomic observations to extract geodetic information.

geodetic azimuth mark—A marked point established in connection with a triangulation (or traverse) station to provide a starting azimuth for dependent surveys.

geodetic azimuth—The angle between the geodetic meridian and the tangent to the geodesic line at the observer, measured in the plane perpendicular to the ellipsoid normal of the observer; preferably clockwise from north. Although older surveys, particularly by the Coast and Geodetic Survey (now National Geodetic Survey), used south, the Department of Defense now uses north.

geodetic control data—Information concerning the precise horizontal and vertical geodetic location of points on the surface of the Earth and celestial bodies, including points obtained by photogrammetric techniques.

geodetic control—A system of horizontal and/or vertical control stations that have been established and adjusted by geodetic methods and in which the shape and size of the Earth (geoid) have been considered in position computations.

geodetic coordinates—The quantities of latitude, longitude, and height (ellipsoid), which define the position of a point on the surface of the Earth with respect to the reference spheroid. Also imprecisely called **geographic coordinates.**

geodetic data sheet—See **control data card.**

geodetic datum—See **datum,** definition 2.

geodetic equator—The line of zero geodetic latitude; the great circle described by the semimajor axis of the reference ellipsoid as it is rotated about the minor axis. See also **astronomic equator.**

geodetic height—See **ellipsoidal height.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

geodetic latitude—The angle which the normal at a point on the reference spheroid makes with the plane of the geodetic equator. Geodetic latitudes are reckoned from the Equator, but in the horizontal control survey of the United States they are computed from the latitude of station Meades Ranch as prescribed in the North American datum of 1927. The new North American datum of 1983 will be Earth-mass centered. A geodetic latitude differs from the corresponding astronomic latitude by the amount of the meridian component of the deflection of the vertical. Also called **topographical latitude.**

geodetic leveling—Spirit leveling of a high order of accuracy, usually extended over large areas, to furnish accurate vertical control as a basis for the control in the vertical dimension for all surveying and mapping operations. Spirit leveling follows the geoid and its associated level surfaces which are irregular, rather than any mathematically determined spheroid or ellipsoid and associated regular level surfaces.

geodetic line—See **geodesic line.**

geodetic longitude—The angle between the plane of the geodetic meridian and the plane of an initial meridian, arbitrarily chosen. A geodetic longitude can be measured by the angle at the pole of rotation of the reference spheroid between the local and initial meridians, or by the arc of the geodetic equator intercepted by those meridians. In the United States, geodetic longitudes are numbered from the meridian of Greenwich, but are computed from the meridian of station Meades Ranch as prescribed in the North American datum of 1927. The new North American datum of 1983 will be Earth-mass centered. A geodetic longitude differs from the corresponding astronomic longitude by the amount of the prime vertical component of the local deflection of the vertical divided by the cosine of the latitude. Also called **geocentric longitude.**

geodetic meridian plane—A plane that contains the normal to the reference ellipsoid at a given point and the rotation axis of the reference ellipsoid.

geodetic meridian—A line on a reference ellipsoid which has the same geodetic longitude at every point. Also called **geographic meridian.**

geodetic parallel—A line on the reference spheroid which has the same geodetic latitude at every point. A geodetic parallel, other than the Equator, is not a geodesic (geodetic) line. In form, it is a small circle whose plane is parallel with the plane of the geodetic equator.

geodetic position—A position of a point on the surface of the Earth expressed in terms of geodetic latitude, geodetic longitude, and geodetic height A geodetic position implies an adopted geodetic datum.

geodetic satellite—Any satellite whose orbit and payload render it useful for geodetic purposes.

geodetic stellar camera—A precision terrestrial camera, usually employing glass plates, used to photograph elevated illuminated objects against a star background.

geodetic survey—A survey in which the figure and size of the Earth is considered. It is applicable for large areas and long lines and is used for the precise location of basic points suitable for controlling other surveys.

geodetic zenith—The point where the normal (to the reference spheroid) extended upward, meets the celestial sphere.

Geodimeter—A trade name for an instrument that measures distance by precise electronic phase comparison of modulated light waves which travel to a reflector and return.

geoelectric survey—A survey to determine the electricity or resistivity potential of the Earth rocks.

Geographic Base File/Dual Independent-Map-Encoding (GBF/DIME)—Topologically structured files developed by the Bureau of the Census which serve as a relatively inexpensive geographic information data source. GBF/DIME files include street networks, street addresses, political boundaries, and major hydrographic features. Each street segment is represented as a straight line regardless of its real world shape.

Geographic Information System (GIS)—The generic term used for a system of computer software programs and equipment that is used to acquire, store, manipulate, analyze, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

display spatial data.

Geographic Names Processing System (GNPS)—The DMA system which includes the hardware, software and procedures to maintain the geographic names data required for digital maps and gazetteers.

geographic (geographical)—Signifying basic relationship to the Earth considered as a globe-shaped body. The term **geographic** is applied alike to data based on the geoid and on other spheroids.

geographic coordinates—(JCS) The quantities of latitude and longitude which define the position of a point on the surface of the Earth with respect to the reference spheroid. Also called **astronomic coordinates; gravimetric coordinates; terrestrial coordinates.** See also **coordinates; geodetic coordinates.**

geographic latitude—A general term, applying alike to astronomic latitudes and geodetic latitudes.

geographic limits—The lines having latitude and longitude values bounding the area of a map or chart; that area exclusive of overlap areas. See also **neatlines.**

geographic location—See **geographic position.**

geographic longitude—A general term, applying alike to astronomic and to geodetic longitudes

geographic meridian—A general term, applying alike to an astronomic or geodetic meridian.

geographic name—Also called **place name** and **toponym.** See also **Board of Geographic Names.**

geographic nonlinearity—The error introduced by digitization and subsequent processing to the total error of both absolute and relative positioning of any feature on the graphic.

geographic parallel—A general term, applying alike to an astronomic parallel or a geodetic parallel .

geographic position—The position of a point on the surface of the Earth expressed in terms of latitude and longitude, either geodetic or astronomic.

geographic survey—A general term, not susceptible of defined limitation, covering a wide range of surveys lying between and merging into exploratory surveys on the one hand and basic topographic surveys on the other. Geographic surveys usually cover large areas, are based on coordinated control, and are used to record physical and statistical characteristics of the area surveyed.

geographic vertical—See **vertical.**

geographical area classification system—A logical and orderly geographical division of the world using numbers, letters, and combinations of the same for the designation of areas and subareas.

geographical exploration traverse—A route followed across some parts of the Earth, approximate positions along which are determined by surveying or navigational methods.

geographical mile—The length of 1 minute of arc at the Equator, or 6,087.08 feet (on the Clarke spheroid of 1866).

geographical pole—Either of the two points of intersection of the surface of the Earth with its axis, where all meridians meet.

geoidal contour—A line on the surface of the geoid of constant elevation with reference to the surface of the spheroid of reference. Geoidal contours depend on the surface of reference as well as on the shape of the geoid. The same geoid referred to different surfaces of reference will give different sets of geoidal contours.

geoidal height profile—See **astrogeodetic leveling.**

geoidal height—The distance of the geoid above (positive) or below (negative) the mathematical reference spheroid. Also called **geoid separation; undulation of the geoid.** See also **astrogeodetic undulations.**

geoidal horizon—That circle of the celestial

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

sphere formed by the intersection of the celestial sphere and a plane tangent to the sea level surface of the Earth at the zenith nadir line.

geoidal separation—See **geoidal height.**

geoid—The equipotential surface in the gravity field of the Earth which approximates the undisturbed mean sea level extended continuously through the continents. The geoid is the surface of reference for astronomic observations and for geodetic leveling. See also **compensated geoid; equilibrium theory; equipotential surface; geoidal horizon; gravimetric geoid; isostatic geoid; reference spheroid.**

geokinetics—Local and global motion of the Earth or sea, its measurement, isolation from, and effect upon precision equipment and measuring instruments. Usually applied to the design and test of inertial instruments and systems and stable platforms.

geologic survey—A survey or investigation of the Earth, of the physical changes which the Earth's crust has undergone or is undergoing, and of the causes producing those changes.

Geological Long Range Inclined Asdic (GLORIA)—A low resolution side scan sonar used for reconnaissance surveys of the ocean floor.

geological map—A map showing the structure and composition of the Earth's crust.

geomagnetic coordinates—A system of spherical coordinates based on the best fit of a centered dipole to the actual magnetic field of the Earth.

geomagnetic equator—The terrestrial great circle everywhere 90° from the geomagnetic poles. **Geomagnetic equator** should not be confused with **magnetic equator**, the line connecting all points of zero magnetic dip.

geomagnetic latitude—Angular distance from the geomagnetic equator, measured northward or southward through 90° and labeled "N" or "S" to indicate the direction of measurement. **Geomagnetic latitude** should not be confused with **magnetic latitude. See also dip.**

geomagnetic meridian—The meridional lines of a geomagnetic coordinate system. Not to be confused with **magnetic meridian.**

geomagnetic pole—Either of two antipodal points marking the intersection of the Earth's surface with the extended axis of a powerful bar magnet assumed to be located at the center of the Earth and approximating the source of the actual magnetic field of the Earth. The expression **geomagnetic pole** should not be confused with **magnetic pole**, which relates to the actual magnetic field of the Earth.

geomagnetism—**1.** The magnetic phenomenon, collectively considered, exhibited by the Earth and its atmosphere, and by extension the magnetic phenomena in interplanetary space. **2.** The study of the magnetic field of the Earth. Also called **terrestrial magnetism.**

geometric latitude—See **parametric latitude.**

geometric map projection—See **perspective map projection.**

geometric nonlinearity—Any distortion introduced by digitization and subsequent processing of a mathematically-correct reference system or grid, other than simple rotation, translation, or scale change.

geometric primitive—One of the three basic geometric units of representation: node, edge, and face. Equivalent with geographic primitive.

geometrical dip—The vertical angle, at the eye of an observer, between the horizontal and a straight line tangent to the surface of the Earth. It is larger than dip by the amount of terrestrial refraction.

geometrical horizon—Originally, the celestial horizon; now more commonly the intersection of the celestial sphere and an infinite number of straight lines tangent to the Earth's surface, and radiating from the eye of the observer. If there were no terrestrial refraction, geometrical and visible horizons would coincide.

geomorphic data—A set of model coordinates defining the trace of a geomorphic feature. These consist of ridgelines, valleys, drainage and lakes which are used to refine the elevation data group during the production process.

geomorphic—Pertaining to land forms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

geophysics—The science of the Earth with respect to its structure, composition, and development. Geophysics is a branch of experimental physics dealing with the Earth, including its atmosphere and hydrosphere. It includes the sciences of dynamical geology and physical geography, and makes use of geology, seismology, meteorology, oceanography, magnetism, and other Earth sciences in collecting and interpreting Earth data.

geopositioning—The purpose of photogrammetric geopositioning is two-fold: (1) upgrade metric data associated with source imagery. This consists of measuring tie and control points, and adjusting positions and attitudes of blocks of imagery. The results form the basis for stereoscopic orientation of imagery and supply model information for monoscopic operations; and (2) determine accurately the location of target points. Targeting operations result in ground coordinates and associated error data for selected locations.

geopotential number—The difference between the geopotential on the geoid and the geopotential at a point.

geopotential surface—See **geop**.

geopotential—The gravity potential of the actual Earth. The sum of the gravitational (attraction) potential and the potential of the centrifugal force. A function describing the variation of the geopotential in space. The function whose partial derivative in any direction gives the gravity component in that direction.

geop—An equipotential surface in the gravity field of the Earth. Also called **geopotential surface.**

GEOREF—(JCS) A worldwide position reference system that may be applied to any map or chart graduated in latitude and longitude [with Greenwich as prime meridian] regardless of projection. It is a method of expressing latitude and longitude in a form suitable for rapid reporting and plotting. This term is derived from the words "The World Geographic Reference System."

geosphere—The solid and liquid portions of the Earth; the lithosphere plus the hydrosphere.

giant planets—See **major planets.**

gigabyte (GB)—A unit of memory representing 2^{30} (1,073,741,824) bytes. Commonly, a gigabyte may represent one billion bytes or 1,000 megabytes.

gisement—See **grid declination.**

Global Navigation Chart (GNC)—A 1:5,000,000 scale series of multicolored charts designed for general planning purposes for operations involving long distances or large areas of in-flight navigation in long range, high altitude, high speed aircraft.

Global Positioning System (GPS)— See **Navstar Global Positioning System (GPS).**

globular map projection—A map projection representing a hemisphere, on which the equator and a central geographic meridian are represented by straight lines intersecting at right angles; these lines are divided into equal parts. All meridians, except the central one, are represented by circular arcs connecting points of equal division on the equator with the poles. Excepting the equator, the parallels are circular arcs dividing the central and extreme outer meridians into equal parts. The extreme outer meridian limits the projection and is a full circle .

gnomonic chart—A chart on the gnomonic projection. Also called **great circle chart.**

gnomonic map projection—A perspective map projection on a plane tangent to the surface of a sphere having the point of projection at the center of the sphere. The projection is neither conformal nor equal-area. It is the only projection on which great circles on the sphere are represented as straight lines.

goldenrod paper—A paper, usually a shade of yellow or red, for blocking out nonprinting areas of negatives or film layouts. Also called **masking paper.**

goniometer—An instrument for measuring angles. See also **photogoniometer.**

Goode's interrupted Homolosine projection—An equal-aree projection, based on the Mollweide and sinusoidal projections, using the sinusoidal from the Equator to 40°N and 40°S, and the Mollweide in higher latitudes. The oceans are "interrupted" to allow the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

continents to be recentered on several meridians, so as to attain good overall shape. It is used widely for maps of economic distributions.

gore—1. (surveying) An irregularly shaped tract of land, generally triangular, left between two adjoining surveyed tracts, because of inaccuracies in the boundary surveys or as a remnant of a systematic survey. 2. (globe) A lune-shaped map which may be fitted to the surface of a globe with a negligible amount of distortion.

Government Open Systems Interconnection Protocol (GOSIP)—The exact sequence of bits, characters, and control codes used to transfer data between computers and peripherals through a communications channel.

gradation—The range of tones from the brightest highlights to the deepest shadows.

grade correction—(land surveying) A correction applied to a distance measured on a slope to reduce it to a horizontal distance between the vertical lines through its end points. Also called **correction for inclination of tape; inclination correction; slope correction.**

grade—The rate of slope or degree of inclination. See also **gradient.**

gradient speed—The speed of a photographic material determined on the basis of the exposure corresponding to a particular gradient of the characteristic curve.

gradient tints—See **hypsometric tinting.**

gradienter—An attachment to an engineer's transit with which an angle of inclination is measured in terms of the tangent of the angle rather than in degrees and minutes. It may be used as a telemeter in measuring horizontal distances.

gradient—1. A rate of rise or fall of a quantity against horizontal distance expressed as a ratio, decimal, fraction, percentage, or the tangent of the angle of inclination. Also called **percent of slope; slope.** 2. The rate of increase or decrease of one quantity with respect to another.

gradiometer—An instrument used to measure gravity gradients.

graduation error—Inaccuracy in the graduations of the scale of an instrument

Graf sea gravimeter—A balance-type gravity meter designed for ocean surveys which consists of a mass at the end of a horizontal arm that is supported by a torsion spring rotational axis.

grain direction—The alignment of paper fibers parallel to the movement on the paper machine during manufacture.

grained surface—The roughened or irregular surface of an offset printing plate.

graining—The mechanical roughening or grinding of an abrasive into the surface of a metal press plate to increase the surface area and improve the water receptiveness of the surface.

grain—1. (photography) One of the discrete silver particles resulting from the development of an exposed light-sensitive material. The random distribution of these particles in an area of uniform exposure gives rise to the appearance known as "graininess." 2. (paper) See **grain direction.** 3. (lithography) See **grained surface.**

granularity—The graininess of a developed photographic image, evident particularly on enlargements, that is due either to agglomerations of developed grains or to an overlapping pattern of grains.

Graphic Kernal System (GKS)—An ISO standard for computer graphics programming which contains functions for outputting graphical primitives, controlling the appearance of graphical primitives with attributes, controlling graphical workstations, controlling transformations and coordinate systems, generating and controlling groups of primitives called segments, obtaining graphical input, manipulating groups of device-independent instructions called metafiles, inquiring the capabilities and states of the graphics system, and handling errors.

graphic products—A graphic product may be a paper copy of a map or chart, or the repro-material used to produce the map or chart.

graphic scale—(JCS) A graduated line by

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

means of which distances on the map, chart, or photograph may be measured in terms of ground distances. Also called **bar scale.** See also **scale.**

graphical radial triangulation—A radial triangulation performed by other than analytical means. A radial triangulation is assumed to be made with principal points as radial centers unless the definitive term designates otherwise (as, for example, nadir-point triangulation or nadir-point plot and isocenter triangulation or isocenter plot, and nadir-point slotted-template plot, etc.).

graphical rectification—Any rectification technique employing a graphic method for determining the solution as contrasted with mechanical techniques. See also **paper-strip method.**

graphic—(JCS) Any and all products of the cartographic and photogrammetric art. A graphic may be either a map, chart, mosaic, or even a film strip that was produced using cartographic techniques.

graticule—1. A network of lines representing parallels of latitude and meridians of longitude forming a map projection. See also **fictitious graticule.** 2. A scale at the focal plane of an optical instrument to aid in the measurement of objects. See also **reticle.**

Gravatt leveling rod—A speaking rod, marked with rectangles each 0.01 foot high, the rectangles at the 0.1 of foot being longer and those at the 0.5 being identified by dots.

Gravatt level—A dumpy level with the spirit level mounted on top of a short telescope tube having a large object glass. Later made with wyes.

graver—See **scriber.**

gravimeter (gravity meter)—An accelerometer designed to measure relative differences in the acceleration due to gravity at different locations.

gravimetric coordinates—See **astronomic coordinates.**

gravimetric datum orientation—Adjustment of the ellipsoid of reference for a particular geodetic datum so that the differences between the gravimetric and astrogeodetic deflection components and geoidal undulations are minimized.

gravimetric deflection—A deflection of the vertical determined by methods of gravimetric geodesy.

gravimetric geodesy—The science that utilizes measurements and characteristics of the Earth's gravity field as well as theories regarding this field to deduce the shape of the Earth and in combination with arc measurements, the Earth's size. Also called **physical geodesy.**

gravimetric geoid—An approximation to the geoid as determined from gravity observation.

gravimetric map—A map on which contour lines are used to represent points at which the acceleration of gravity is equal.

gravimetric survey—A survey made to determine the acceleration of gravity at various places on the Earth's surface.

gravimetric undulations—Separations between a gravimetrically determined geoid and a reference ellipsoid of specified flattening.

gravitational constant—See **constant of gravitation.**

gravitational disturbance—See **gravity disturbance.**

gravitational flattening—The ratio of the difference between the polar and equatorial normal gravities to the equatorial normal gravity. Also called **gravity flattening.**

gravitational gradient—The change in the gravity per unit distance.

gravitational harmonics—The spherical harmonics used in approximating the gravitational field of the Earth. See also **gravity field of the Earth; sectorial harmonics; spherical harmonics; tesseral harmonics; zonal harmonics.**

gravitational perturbations— Perturbations caused by body forces due to nonspherical terrestrial effects, lunisolar effect, tides, and the effect of relativity .

gravitational potential—1. The potential associated with the force of gravitation arising

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

from the attraction between mass points, e.g., the Earth's center and a particle in space. 2. At any point, the work needed to remove an object from that point to infinity.

gravitation—The mutual interaction of two masses producing a force between them acting along the line joining their centers of mass. The force is proportional to the product of the two masses divided by the square of the distance between the two centers of mass.

gravity anomaly map—A map showing the positions and magnitudes of gravity anomalies. Also, a map on which contour lines are used to represent points at which the gravity anomalies are equal.

gravity anomaly—The difference between the observed gravity value properly reduced to sea level, and the theoretical gravity obtained from gravity formula. Also called **observed gravity anomaly**. See also **Bouguer anomaly; free-air anomaly; gravity disturbance; Hayford gravity anomalies/Hayford anomalies; isostatic anomaly.**

gravity corer—Any type of corer that achieves bottom penetration solely as a result of gravity.

gravity data—Information concerning that acceleration which attracts bodies and is expressed as observations or in the form of gravity anomaly charts or spherical harmonics for spatial representation of the Earth and other celestial bodies.

gravity disturbance—The difference between the observed gravity and the normal gravity at the same point (the vertical gradient of the disturbing potential) as opposed to gravity anomaly which uses corresponding points on two different surfaces. Because the centrifugal force is the same when both are taken at the same point, it can also be called **gravitational disturbance.**

gravity field of the Earth—The field of force arising from a combination of the mass attraction and rotation of the Earth. The field is normally expressed in terms of point values, mean area values, and/or series expansion for the potential of the field.

gravity flattening—See **gravitational flattening.**

gravity instrument—A device for measuring the acceleration due to gravity (absolute) or gravity differences between two or more points (relative). See also **astatized gravimeter; Brown gravity apparatus; dynamic gravity meter; Graf sea gravimeter; gravimeter; La Coste-Romberg gravimeter; stable gravimeter; stable-type gravimeter; static gravity meter; torsion balance; unstable-type gravimeter.**

gravity network—A network of gravity stations.

gravity reduction—A combination of gravity corrections to obtain reduced gravity on the geoid. See also **Bouguer correction; free-air correction; isostatic correction; terrain correction.**

gravity reference stations—Stations which serve as reference values for a gravity survey, i.e., with respect to which the differences at the other stations are determined in a relative survey. The absolute value of gravity may or may not be known at the reference stations.

gravity station—A station at which observations are made to determine the value of gravity.

gravity—Viewed from a frame of reference fixed in the Earth, acceleration imparted by the Earth to a mass which is rotating with the Earth. Since the Earth is rotating, the acceleration observed as gravity is the resultant of the acceleration of gravitation and the centrifugal acceleration arising from this rotation and the use of an earthbound rotating frame of reference. It is directed normal to sea level and to its geopotential surfaces. See also **absolute gravity; center of gravity; Clairaut's theorem; constant of gravitation; direction of the force of gravity; equatorial gravity value; equipotential surface; formula for theoretical gravity; gravitation; Hayford deflection templets; Hayford effect; Helmert's gravity formula of 1901; Helmert's gravity formula of 1915; intensity of gravity; international gravity formula; isostasy; longitude term gravity formula; normal gravity; observed gravity; reduced gravity; regional gravity; relative gravity; residual gravity; resolution; resolution limit; standard gravity; subgravity;**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

theoretical gravity; virtual gravity.

gray scale—See **step wedge.**

great circle—A circle on the surface of the Earth, the plane of which passes through the center of the Earth. Also called **orthodrome.**

great elliptic arc—An arc defined by a plane which contains the two points and the center of the reference spheroid.

great year—The period of one complete cycle of the equinoxes around the ecliptic, about 25,800 years. Also called **platonic year.**

great-circle bearing—The initial direction of a great circle through two terrestrial points, expressed as angular distance from a reference direction. It is usually measured from 0° at the reference direction clockwise to 360°.

great-circle chart—A chart on which a great circle appears as a straight line; a chart on the gnomonic projection.

great-circle direction—Horizontal direction of a great circle, expressed as angular distance from a reference direction.

great-circle distance—The length of the shorter arc of the great circle joining two points. It is usually expressed in nautical miles.

great-circle line—In land surveying, the line of intersection of the surface of the Earth and the plane of a great circle of the celestial sphere.

great-circle route—(JCS) The route which follows the shortest arc of a great circle between two points.

greatest elongation—The maximum angular distance of a body of the solar system from the Sun, as observed from the Earth. The direction of the body east or west of the Sun is usually specified, as the greatest elongation west.

Greenwich apparent time (GAT)—Local apparent time at the Greenwich meridian.

Greenwich civil time (GCT)—See **Universal Time.**

Greenwich hour angle (GHA)—Angular distance west of the Greenwich celestial meridian; the arc of the celestial equator, or the angle at the celestial equator, or the angle at the celestial pole, between the upper branch of the Greenwich celestial meridian and the hour circle of a point on the celestial sphere, measured westward from the Greenwich celestial meridian through 360°; local hour angle at the Greenwich meridian.

Greenwich interval—An interval based on the Moon's transit of the Greenwich celestial meridian, as distinguished from a local interval based on the Moon's transit of the local celestial meridian.

Greenwich lunar time—Local lunar time at the Greenwich meridian; the arc of the celestial equator or the angle, at the celestial pole, between the lower branch of the Greenwich celestial meridian and the hour circle of the Moon, measured westward from the lower branch of the Greenwich celestial meridian through 24 hours; Greenwich hour angle of the Moon, expressed in time units, plus 12 hours.

Greenwich Mean Time (GMT)—(JCS) Also called **Greenwich civil time; z-time; Zulu time.** See **Universal Time.**

Greenwich meridian—The meridian through Greenwich, England, serving as the reference for Greenwich time, in contrast with local meridians. It is accepted almost universally as the prime meridian, or the origin of measurement of longitude.

Greenwich sidereal date—The number of mean sidereal days that have elapsed on the Greenwich meridian since the beginning of the sidereal day that was in progress at Greenwich mean noon on January 1, 4713 B.C. See also **Greenwich sidereal day number.**

Greenwich sidereal day number—The integral part of the Greenwich sidereal date. It is a means of numbering consecutively successive sidereal days beginning at the instants of upper transit of the mean vernal equinox over the Greenwich meridian. See also **Greenwich sidereal date.**

Greenwich sidereal time (GST)—Local sidereal time at the Greenwich meridian. The arc of the celestial equator, or the angle at the celestial pole, between the upper branch of the Greenwich celestial meridian and the hour circle of the vernal equinox, measured westward from the upper branch of the Greenwich celestial

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

meridian through 24 hours; Greenwich hour angle of the vernal equinox, expressed in time units.

Greenwich time—Time based upon the Greenwich meridian as reference, as contrasted with that based upon a local or zone meridian.

grid amplitude—Amplitude relative to grid east or west. See also **amplitude.**

grid azimuth—The angle in the plane of projection measured clockwise between a straight line and the central meridian of a plane rectangular coordinate system.

grid bearing—(JCS) Bearing measured from grid north.

grid computation—The determination, from a set of tables derived from formulas, of the true shape and dimensions of a grid, for the purpose of constructing such a grid. The grid is mathematically coordinated with its related map projection; they are usually computed concurrently .

grid convergence—(JCS) The horizontal angle at a place between true north and grid north. It is proportional to the longitude difference between the place and the central meridian. See also **convergence.**

grid coordinate system—(JCS) A plane rectangular coordinate system usually based on, and mathematically adjusted to, a map projection in order that geographic positions (latitudes and longitudes) may be readily transformed into plane coordinates and the computations relating to them may be made by the ordinary methods of plane surveying.

grid coordinates—(JCS) Coordinates of a grid coordinate system to which numbers and/or letters are assigned for use in designating a point on a gridded map, photograph, or chart. See also **coordinates.**

grid declination—The angular difference in direction between grid north and true north. It is measured east or west from true north. Also called **declination of grid north; gisement.**

grid direction—Horizontal direction expressed as angular distance from grid north.

grid distance—See **grid length.**

grid equator—A line perpendicular to a prime grid meridian, at the origin.

grid interval—(JCS) The distance represented between the lines of a grid.

grid inverse—The computation of grid length and grid azimuths from grid coordinates.

grid junctions—Those lines delineating the joining of two or more grid systems on a map or chart.

grid latitude—Angular distance from a grid equator. See also **fictitious latitude.**

grid length—The distance between two points obtained by computation from grid coordinates of the points. It differs from the geodetic length by the amount of a small correction based on the scale factor for the line. Also called **grid distance.**

grid line—One of the lines of a grid.

grid longitude—Angular distance between a prime grid meridian and any given grid meridian. See also **fictitious longitude.**

grid magnetic angle—(JCS) Angular difference in direction between grid north and magnetic north. It is measured east or west from grid north. Also called **grivation; grid variation.**

grid meridian—One of the grid lines extending in a grid north-south direction. The reference grid meridian is called **prime grid meridian.** In polar regions the prime grid meridian is usually the 180°—0° geographic meridian. See also **fictitious meridian.**

grid method—(photogrammetry) A method of plotting detail from oblique photographs by superimposing a perspective of a map grid on a photograph and transferring the detail by eye, that is, by using the corresponding lines of the map grid and its perspective as placement guides. See also **perspective grid.**

grid north—(JCS) The northerly or zero direction indicated by the grid datum of directional reference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

grid number—The numerical value of a grid line indicating the distance of that line from the false origin of the grid. See also **grid coordinates.**

grid origin—The point, usually near the center of a grid zone, where a parallel intersects a north-south grid line coincident to a meridian. See also **false origin.**

grid parallel—A line parallel to a grid equator, connecting all points of equal grid latitude. See also **fictitious parallel.**

grid plate—1. (cartography) See **color separation drawing.** 2. (photogrammetry) See **reseau.**

grid prime vertical—The vertical through the grid east and west points of the horizon.

grid rhumb line—A line making the same oblique angle with all grid meridians. Grid parallels and meridians may be considered special cases of the grid rhumb line. See also **fictitious rhumb line.**

grid ticks—(JCS) Small marks on the neatline of a map or chart indicating additional grid reference systems included on that sheet. Grid ticks are sometimes shown on the interior grid lines of some maps for ease of referencing.

grid variation—See **grid magnetic angle.**

grid zone—An arbitrary division of the Earth's surface designated for identification without reference to latitude or longitude.

gridded oblique—An oblique aerial photograph printed with a superimposed grid to assist in the identification of a particular area within the photograph; used chiefly for artillery spotting.

gridded photograph (GP)—A photograph with a precise geographic grid. Used for manual measurement of precise coordinates where elevation is not required. Provides acceptable aircraft inertial navigation system fixpoint derivation as a substitute for survey or other point positioning data.

grid—1. (geodesy) Two sets of parallel lines intersecting at right angles and forming squares; a rectangular Cartesian coordinate system that is superimposed on maps, charts, and other similar representations of the earth's surface in an accurate and consistent manner to permit identification of ground locations with respect to other locations and the computation of direction and distance to other points. 2. A type of Digital Elevation Model (DEM) that is derived from interpolating elevation values from irregularly or regularly spaced points that have X, Y, Z values. 3 A network composed of two families of lines such that a pair of lines, one from each family, intersects in no more than two points. 4. (GIS) Program developed at Harvard in the early 1970s which contained all the standard Boolean operations that today's GIS have. See also **arbitrary grid; atlas grid; British grid reference system; GEOREF; Lambert grid; major grid; military grid; military grid reference system; National grid; overlapping grid; parallactic grid; perspective grid; point-designation grid; polar grid; secondary grid; tangent plane grid system; transverse Mercator grid; Universal Polar Stereographic (UPS) grid; Universal Transverse Mercator (UTM) grid; world polyconic grid, preferred grid, primary grid, and prime grid meridian.**

gripper edge—(JCS)The edge by which paper or other printing material is drawn into the printing machine.

gripper margin—See **gripper edge.**

grivation—See **grid magnetic angle.**

gross error—The result of carelessness or a mistake. May be detected through repetition of the measurements.

gross model—The total overlap area of a pair of aerial photographs. See also **neat model.**

Ground Scale Distance (GSD)—For rectified imagery, GSD is the center-to-center ground distance between adjacent pixels.

ground camera—See **terrestrial camera.**

ground control point—See **control station.**

ground control—(JCS) A system of accurate measurements used to determine the distances and directions or differences in elevation between points on the Earth. See also **common control; control point; field**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

control; traverse.

ground data—See **ground truth.**

ground distance—The great-circle distance between two ground positions, as contrasted with *slant range* the straight-line distance between two points. Also called **ground range.**

ground gained forward (GGF)—(aerial photography) The net gain per photograph in the direction of flight for a specified overlap. The GGF is used to compute the number of exposures in a strip of aerial photography.

ground gained sideways (GGS)— (aerial photography) The net lateral gain per flight for a specified sidelap. The GGS is used to compute the number of flight lines for an area to be photographed.

ground information—See **ground truth.**

ground nadir—(JCS) The point on the ground vertically beneath the perspective center of the camera lens. On a true vertical photograph this coincides with the principal point. Also called **ground plumb point.**

ground parallel—The intersection of the plane of the photograph with the plane of reference of the ground. See also **axis of homology.**

ground photogrammetry—See **terrestrial photogrammetry.**

ground photograph—See **terrestrial photograph.**

ground plane—The horizontal plane passing through the ground nadir of a camera station.

ground plumb point—See **ground nadir.**

ground pyramid—A component of an analytical method for determining the precise degree of photographic tilt, representing a specific spatial configuration from three ground control points (forming a triangle) on the ground to the exposure station of the photograph containing the identical points. When used with the photo pyramid, the ground pyramid permits the exact analytical determination of tilt in the photograph. See also **photo pyramid.**

ground range—See **ground distance.**

ground resolution—The minimum distance which can be detected between two adjacent features, or the minimum size of a feature expressed in size of objects or distances on the ground.

ground return—(JCS) The reflection from the terrain as displayed and/or recorded as an image.

ground speed—(JCS) The horizontal component of the speed of an aircraft relative to the earth's surface.

ground station—A monumented station, established by field survey methods, which is used as a base for ground station equipment for the procurement of shoran or shiran controlled photography or control data.

ground survey—A survey made by ground methods, as distinguished from an aerial survey. A ground survey may or may not include the use of photographs.

ground swing—An error-causing condition in electronic distance measuring which is brought about by the reflection of the microwave beam from the ground or water surface. The reflected beam mixes with the direct beam at the receiving antenna, thereby changing the phase of the direct beam and causing an error in the distance measured. By varying the carrier frequency, the error becomes cyclic, making possible mean instrument readings that are substantially accurate.

ground trace—See **ground parallel.**

ground track—The vertical projection of the flight path onto the surface of the earth.

ground truth—A term coined for data/information obtained from actual ground measurement of surface/subsurface features to aid in the interpretation of remotely sensed data. Also called **ground data; ground information.**

ground-space coordinate system—A scheme by which positions of triangulation stations, control points, and other ground features are related by distance and azimuth or by *x*- and *y*-coordinates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

guard stake—(surveying) A stake driven near a hub, usually sloped with the top of the guard stake over the hub. The guard stake protects, and its markings identify, the hub.

guide meridian—An auxiliary governing line projected north along an astronomic meridian, from points established on the base line or a standard parallel, usually at intervals of 24 miles east or west of the principal meridian, on which township, section, and quarter section corners are established. See also **auxiliary guide meridian; principal meridian.**

guide—A drafting or scribing surface bearing a map image to be traced by drafting or scribing for reproduction. Also called **color separation guide; drafting guide; scribing guide.**

Gunter's chain—A measuring device used in land surveying, composed of 100 metal links fastened together with rings, the length of the chain being 66 feet. Also called **four-pole chain; four-rod chain.** See also **chain.**

gyroazimuth theodolite—See **gyro-theodolite.**

gyrocompass—A compass which functions by virtue of the couples generated in a rotor when the latter's axis is displaced from parallelism with that of the Earth. A gyrocompass is independent of magnetism and will automatically align itself in the celestial meridian. However, it requires a steady source of motive power and is subject to dynamic error under certain conditions. Certain aircraft compasses also use gyroscopes to gain stability, while relying basically on the magnetic meridian; these are to be distinguished from the true gyrocompass.

gyromagnetic compass—(JCS) A directional gyroscope whose azimuth scale is maintained in alignment with the magnetic meridian by a magnetic detector unit.

gyromeridian indicating instrument—See **gyrotheodolite.**

gyroscope—A device consisting of a spinning rotor and associated supporting readouts which makes use of Newton's Law of Rotation to give an indication of the angular velocity of the instrument's case with respect to an inertial reference frame. This instrument is used as the basic sensor in many direction-seeking, direction-keeping, and attitude stabilization systems.

gyroscopic stabilization—Equilibrium in the attitude and/or course of a ship or airborne vehicle maintained by the use of gyroscopes. Also, the maintenance (by the use of gyroscopes) of a camera in a desired attitude within an airborne vehicle.

gyrotheodolite—A theodolite with a gyrocompass attached or built in, whereby a true azimuth reference can be established in any weather, day or night, without the aid of stars, landmarks, or other visible stations. The azimuth obtained from the gyro or inertial theodolite is essentially the astronomic azimuth at the point of observation. This azimuth will differ from the corresponding geodetic azimuth by the amount of the Laplace correction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H

H and D curve—See **characteristic curve.**

hachuring—(JCS) A method of representing relief upon a map or chart by shading in short disconnected lines drawn in the direction of the slopes.

hack chronometer—A chronometer used for visual reference, and not usually for record purposes.

halation—(photography) A spreading of a photographic image beyond its proper boundaries, due especially to reflection from the side of the film or plate support opposite to that on which the emulsion is coated. Particularly noticeable in photographs of bright objects against a darker background.

half mark—See **index mark.**

half model—The stereoscopic model formed by the overlap of two adjacent right- or left-hand exposures of convergent photographs.

half section—Any two quarter sections within a section which have a common boundary; usually identified as the north half, south half, east half, or west half of a particular section.

half tide level—See **mean tide level.**

halftone screen—(JCS) A series of regularly spaced opaque lines on glass, crossing at right angles, producing transparent apertures between intersections. Used in a process camera to break up a solid or continuous tone image into a pattern of small dots. Also called **crossline glass screen.** See also **contact screen; halftone.**

halftone—(JCS) Any photomechanical printing surface or the impression therefrom in which detail and tone values are represented by a series of evenly spaced dots of varying size and shape, varying in direct proportion to the intensity of the tones they represent. See also **halftone screen; middletone.**

Hammer projection—A variation, by E. Hammer in 1892, on the zenithal equal-area (Lambert) projection, made by doubling the horizontal distances along each parallel from the central meridian. This transforms the circular shape of the Lambert into an ellipse, similar in appearance to a Mollweide projection, but with all parallels curved except the Equator, which is a straight line.

hand level—A hand-held instrument for approximate leveling. It consists of a sighting tube with a split field of view; a horizontal cross hair in one-half of the field bisects the image of a spirit level in the other half when the instrument is held level. See also **Abney level.**

hand proof—In offset lithography, a proof of a plate made on a hand proof press where operations are manual for inking, dampening, and taking the impression.

hand templet—A templet made by tracing the radials from the photograph onto a transparent plastic medium. Hand templets are laid out and adjusted by hand to form the radial triangulation.

hand-templet plot—See **hand-templet triangulation.**

hand-templet triangulation—A graphical radial triangulation using any form of hand templet. Also called **hand-templet plot.**

hanging level—A spirit level so mounted that, when in use, its level tube is lower in elevation than its points of support.

Harbor and Approach chart (HA)—A chart which provides detailed portrayal of navigationally useful information supportive of maneuvering in close quarter areas to berth/anchor, channel keeping while entering/exiting port, or navigating in close proximity to dangers associated with close shore/harbor approach areas. Produced from 1:10,000 to 1:100,000 scale.

harbor chart—A nautical chart intended for navigation and anchorage in harbors and smaller waterways.

Hardcopy Exploitation Segment (HE/S)—Segment of DMA's Digital Production System which provides software upgrades for existing DMA photogrammetric equipment and will deliver a new computer for triangulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Outputs include digital elevation data and rectified and orthorectified imagery. See also **Digital Production System; MARK 85.**

hardcopy—Products which are printed on paper, photographic materials or other media, and can be interpreted directly by the human user.

harmonic coefficients—The coefficients of trigonometric terms of an infinite series used to approximate an irregular closed surface. See **also spherical harmonics.**

harmonic component—Any of the simple sinusoidal components into which a periodic quantity may be resolved.

harmonic constants—The amplitude and epochs of the harmonic constituents of the tide or tidal current at any place.

harmonic constituent—See **constituent.**

harmonic expressions—Trigonometric terms of an infinite series used to approximate irregular curves in two or three dimensions.

harmonic function—Any real function that satisfies Laplace's equation.

harmonic motion—The projection of circular motion on a diameter of the circle of such motion.

harmonic—A sinusoidal quantity having a frequency that is an integral multiple of the frequency of a periodic quantity to which it is related. *See also* **compound harmonic motion; gravitational harmonics; sectorial harmonics; simple harmonic motion; spherical harmonics; tesseral harmonics; zonal harmonics.**

Hassler base-line measuring apparatus—An optical base-line measuring apparatus consisting of four rectangular iron bars mounted end to end in a wooden box. Each bar is 2 meters long. The combined length of the apparatus being 8 meters.

Hayford deflection templets—Templets used in connection with studies for the figure of the Earth and isostasy. In obtaining elevation readings from maps in connection with topographic and isostatic reductions, templets of plastic are used. The templets used in connection with deflection of the vertical studies have circles and radial lines drawn upon them, so proportioned with reference to scale of map and azimuth that land elevations and ocean depths within each compartment formed by adjacent arcs and radii can be easily averaged and the effect of the mass therein on a plumb line at the station (center of circles), under various hypotheses, can be computed. See also **Hayford gravity templets.**

Hayford effect—The direct effect on gravity of masses of unit density extending to various distances above and below sea level; it neglects the differences of elevation between the reference spheroid and the geoid.

Hayford gravity anomalies/Hayford anomalies—Isostatic anomalies obtained by computing the isostatic compensation according to the Pratt theory of isostasy as developed by Hayford, using various depths for purposes of comparative analysis.

Hayford gravity templets—Templets used in connection with gravity studies which are similar to Hayford deflection templets except that no account is taken of azimuth, all compartments bounded by a given pair of circles being of the same size and shape. A given templet can be used only on maps of the scale and projection for which it is constructed. See also **Hayford deflection templets.**

Hayford spheroid (ellipsoid)—A reference ellipsoid having the following approximate dimensions: semimajor axis 6,378,388.0 meters; semiminor axis—6,356,909.0 meters; and the flattening or ellipticity 1/297.00.

Hayford-Bowie method of isostatic reduction—A method of computing the effect of topography and isostatic compensation on gravity by which the effect of topography is computed directly and then corrected for the effect of isostatic compensation. The mechanics of this method involve the use of the Hayford gravity templets.

Hayford-Bullard (or Bullard) method of isostatic reduction—A method by which the topographic effect of an infinite slab of density 2.67 and a thickness equal to the elevation of the gravity station is first computed, and then corrected for curvature of the sea level surface and for difference of elevation between the station and the topography.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

haze—The radiance of the atmosphere.

heading—(JCS) (navigation) The direction in which the longitudinal axis of an aircraft or ship is pointed, usually expressed in degrees clockwise from north (true, magnetic, compass, or grid).

height anomaly—The difference between the height of a terrain point above the reference spheroid and the corresponding normal height, measured along the normal plumb line.

height differential—The difference in height between predominant height groupings in a homogeneous surface area.

height displacement—See **relief displacement.**

height finder—A stereoscopic range finder so constructed as to indicate vertical heights rather than slant range. See also **stereometer.**

height of instrument—**1.** (spirit leveling) The height of the line of sight of a leveling instrument above the adopted datum. 2. (stadia surveying) The height of the center of the telescope (horizontal axis) of transit or telescopic alidade above the ground or station mark. 3. (trigonometric leveling) The height of the center of the theodolite (horizontal axis) above the ground or station mark.

height of the tide—The vertical distance from chart datum to the surface water level at any stage of the tide usually measured in feet.

height-of-eye correction—That correction to sextant altitude due to dip of the horizon. Also called **dip correction.**

height—(JCS) The vertical distance of an object, point, or level above the ground or other established reference plane. Height may be indicated as follows: *very low*—below 500 feet but above ground level; *low*—500 to 2,000 feet; *medium*—2,000 to 25,000 feet; *high*—25,000 to 50,000 feet; *very high*—above 50,000 feet. See also **altitude; elevation; ellipsoidal height; geoidal height.**

heliocentric parallax—See **annual parallax.**

heliocentric—Relative to the center of the Sun as origin.

heliotrope—A device used in geodetic surveying for reflecting the Sun's rays to a distant point, to aid in long-distance observations. See also **selenotrope.**

helipad—(JCS) A prepared area designated and used for take off and landing of helicopters. (Includes touchdown or hoverpoint.)

heliport—(JCS) A facility designated for operating, basing, servicing, and maintaining helicopters.

Helmert's gravity formula of 1901—A formula for theoretical gravity developed from the gravity observations available at the time (1901), but not fitted to any preassigned value of the Earth's ellipticity.

Helmert's gravity formula of 1915—A formula for theoretical gravity based on a triaxial ellipsoid and therefore includes a longitude term. See also **longitude term gravity formula.**

hemispherical map—A map of one-half of the Earth's surface, bounded by the Equator, or by meridians.

hidden lines—Line segments obscured from view in a projected image of a three-dimensional object.

hierarchy—(digital) A system of classifying features according to feature type and significance. For example, in some cases features with lower numbers are masked out by higher numbered features occupying the same position.

High Speed Digital Chart (HSDC)-- Vector digital chart of selected and generalized coastal, and harbor and approach chart information in support of navigation and collision-avoidance display systems. HSDC development was coordinated with the National Ocean Service and is produced using Committee on the Exchange of Digital Data format feature encoding. HSDC is no longer supported by DMA.

high altitude—(JCS) Conventionally, an altitude above 10,000 meters (33,000 feet). See also **altitude.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

high density tape (HDT, HDDT)— An analog recording and playback system for storing very high rate data that is not directly compatible with general purpose computer systems. Used by DMA for source imagery under Digital Production System production. Also known as high density digital tape because the source imagery is transferred as digital data.

high rod—See long rod.

high tide—See high water.

high water (HW)—The highest limit of the surface water level reached by the rising tide. High water is caused by the astronomic tide-producing forces and/or the effects of meteorological conditions. Also called **high tide.**

high water full and change (HWF&C)—See **establishment of the port.**

high water interval—*See* **lunitidal interval.**

high water line—The line on the bank or shore to which the waters normally rise at high water. In tidal waters, the high water line is, in strictness, the intersection of the plane of the mean high water with the shore. The high water line is the boundary line between the bed and the bank of a stream.

high water lunitidal interval—See **lunitidal interval.**

high water springs—See **mean high water springs.**

***high-oblique* photograph**—See **oblique air photograph.**

***higher high water* (HHW)**—The higher of two high waters occurring during a tidal day where the tide exhibits mixed characteristics.

higher high water* interval (HHWI)**—The interval of time between the transit (upper or lower) of the Moon over the local or Greenwich meridian and the next higher high water. This expression is use when there is considerable diurnal inequality. See also ***lunitidal interval.

higher low water (HLW)—The higher of two low waters of a tidal day where the tide exhibits mixed characteristics.

higher low water interval (HLWI)—The interval of time between the transit (upper or lower) of the Moon over the local or Greenwich meridian and the next higher low water. This expression is used when there is considerable diurnal inequality. See also ***lunitidal interval.***

highest elevation—That elevation which is the highest point of relief within the area of a map or chart. See also ***critical* elevation.**

high—(JCS) A height between 25,000 and 50,000 feet.

hill plane—The plane containing the positions of three ground marks constituting control points. This may be, but rarely is, a horizontal plane.

hill shading—(JCS) A method of representing relief on a map by depicting the shadows that would be cast by high ground if light were shining from a certain direction, Also called **hillwork.** See also **shaded relief.**

hillwork—See **hill shading.**

historical chart—A chart based on data from previous years to determine probable oceanographic patterns for a specified time. Also called ***first* approximation chart.**

history overlay—A specially prepared matte plastic material which shows the sources of all sounding data used in a bathymetric compilation.

holiday—(JCS) An unintentional omission in imagery coverage of an area. See also **gap.**

hologrammetry—The art or science of interpreting the three-dimensional holographic image and obtaining reliable measurements by means of holography.

hologram—The hologram offers a reconstruction of the external appearance of an object with unique three-dimensional properties. This is accomplished by illuminating the object with coherent light.

***homogeneous* area**—An area which has uniform radar reflecting power at all points.

homogeneous surface area—A grouping

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

of features having the same general surface composition.

homologous images—The images of a single object point that appears on each of two or more overlapping photographs having different perspective centers.

homologous photographs—Two or more overlapping photographs having different camera stations.

homologous rays—The two perspective rays corresponding to a pair of homologous image points.

homologous—The condition where an image of a given object point or series of such points is common to two or more projections having different perspective centers.

homolographic (homalographic) map projection—An equal-area map projection. This term is found in the designations given some particular map projections, such as the Mollweide homalographic projection.

horizon camera—A camera used in conjunction with another aerial camera to photograph the horizon simultaneously with the other photographs. The horizon photographs indicate the tilts of the other photographs.

horizon closure—See **error of closure**, definition 6.

horizon coordinate system—See **horizon system of coordinates.**

horizon photograph—**1.** A photograph of the horizon taken simultaneously with another photograph for the sole purpose of obtaining an indication of the orientation of the other photograph at the time of exposure. **2.** (surveying) A continuous matched set of horizon photographs defining obstructions 360° around a given station.

horizon prism—A prism which can be inserted in the optical path of an instrument, such as a bubble sextant, to permit observation of the visible horizon.

horizon profile—A plot of vertical angles against the horizontal angles taken 360° around a point with annotation. See also **horizon photograph; horizon sweep.**

horizon sweep—(surveying) A preliminary reconnaissance technique where the instrument is pointed initially at the farthest visible known point and recording clockwise angles to tanks, spires, buildings, signals, etc., for purposes of identification and subsequent use. See also **horizon profile.**

horizon system of coordinates—A set of celestial coordinates, usually altitude and azimuth or azimuth angle, based on the celestial horizon as the primary great circle. Also called **horizon coordinate system.**

horizon trace—An imaginary line, in the plane of a photograph, which represents the image of the true horizon; it corresponds to the intersection of the plane of a photograph and the horizontal plane containing the internal perspective center or rear nodal point of the lens. See also **true horizon.**

horizontal angle—Angle in a horizontal plane.

horizontal axis—The axis about which the telescope of a theodolite or transit rotates when moved vertically.

horizontal bridging—See **bridging.**

horizontal circle—A graduated circle affixed to the lower plate of a transit by which horizontal angles can be measured.

horizontal closure error—See **error of closure**, definition 8.

horizontal control datum—A geodetic reference point which is the basis for horizontal control surveys, and of which five quantities are known: latitude, longitude, azimuth of a line from this point, and two constants which are the parameters of the reference ellipsoid. The horizontal control datum may extend over a continent or be limited to a small area. Also called **horizontal datum; horizontal geodetic datum.**

horizontal control point—See **horizontal control station.**

horizontal control station—A station whose position has been accurately determined in *x*- and *y*-grid coordinates, or latitude and longitude. Also called **horizontal control point.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

horizontal control survey net—See **survey net**, definition 1.

horizontal control—A network of stations of known geographic or grid positions referred to a common horizontal datum, which control the horizontal positions of mapped features with respect to parallels and meridians, or northing and easting grid lines shown on the map. Horizontal control includes basic (marked) and supplementary (unmarked) stations.

horizontal coplane—See **basal coplane.**

horizontal datum—See **datum**, definition 2.

horizontal deformation—In relative orientation, the cumulative model warpage affecting the horizontal datum from z-motion error, bridging error, and swing error.

horizontal direction—Observed horizontal angles at a triangulation station reduced to a common initial direction.

horizontal extension—See **extension**, definition 2.

horizontal geodetic datum—See **horizontal control datum.**

horizontal intensity—The intensity of the horizontal component of the magnetic field in the plane of the magnetic meridian.

horizontal line—A line perpendicular to the vertical.

horizontal parallax—**1.** (astronomy) The geocentric parallax of a body on the observer's horizon. This is equal to the angular semidiameter of the Earth as seen from the body. **2.** (photogrammetry) See **absolute stereoscopic parallax.**

horizontal pass point—See **supplemental position.**

horizontal photograph—A photograph taken with the axis of the camera horizontal.

horizontal plane—**1.** A plane perpendicular to the direction of gravity; and plane tangent to the geoid or parallel to such a plane. **2.** (surveying) A plane perpendicular to the plumb line within which, or on which, angles and distances are observed. For any planimetric survey it is assumed that all plumb lines therein are parallel, and all horizontal planes therein are parallel.

horizontal refraction—A natural error in surveying which is the result of the horizontal bending of light rays between a target and an observing instrument. Usually caused by the differences in density of the air along the path of the light rays, resulting from temperature variations. See also **terrestrial refraction.**

horizontal stadia—A method of measuring distances wherein the stadia rod is held in a horizontal position and the stadia hairs of the instrument are vertical during observations.

horizontal taping and plumbing—A method whereby the tape is held horizontally, and the positions of its pertinent graduations are projected to the ground with plumb bobs.

horizontal/vertical bridging—See **bridging.**

horizontal/vertical extension—See **extension** definition 2.

horizontalizing the model—See **leveling**, definition 2.

horizontally controlled photography—Cartographic aerial photography obtained simultaneously with recording of distance measurements between the taking aircraft and each of two or more geodetically positioned ground stations. Shoran or shiran are normally used as the distance measuring equipment. The result is precise relative horizontal positioning of each photograph which has associated recorded distances. This positioning information is used as horizontal control data in the mapping process.

horizon—(JCS) In general, the apparent or visible junction of the earth and sky, as seen from any specific position. Also called the **apparent, visible,** or **local horizon.** A horizontal plane passing through a point of vision or perspective center. [The apparent or visible horizon approximates the true horizon only when the point of vision is very close to sea level.] See also **apparent horizon; attitude indicator; celestial horizon; false horizon; geocentric horizon; geoidal horizon; geometrical horizon; radar horizon; sensible horizon; true horizon.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Horrebow level attachment—A level used in conjunction with a micrometer in a telescope whereby latitude observations by the Horrebow-Talcott method can be made.

Horrebow-Talcott method of latitude determination—See **latitude determination; zenith-telescope method.**

hot spot—(photography) A small area of undesired brilliancy of illumination in the image projected by a printer or a reader.

Hotine Oblique Mercator—A mapping projection developed specifically for LANDSAT imagery, dividing the Earth into five zones of latitude. Within each zone, oblique strips corresponding to individual LANDSAT spacecraft paths are projected onto a plane. The axis of projection corresponds approximately to the path of scene centers.

hour angle difference—See **meridian angle difference.**

hour angle system (of coordinates)—An equatorial system of curvilinear celestial coordinates which has the Equator and the local meridian as primary and secondary reference planes, respectively. The position of a celestial body is given by its hour angle and declination.

hour angle—The hour angle of a celestial body is the time elapsed since its upper transit. It is the angle between the observer's (astronomic) meridian and the declination circle of the body, measured positive westward from the meridian. See also **Greenwich hour angle; local hour angle; sidereal hour angle.**

hour circle—Any great circle on the celestial sphere whose plane is perpendicular to the plane of the celestial equator. Also called **circle of declination; circle of right ascension.** See also **celestial meridian; colures.**

hub—A temporary traverse station marker, usually of wood. The stake is driven flush with the ground with a tack or small nail on top to mark the exact point of reference for angular and linear measurements.

Hull Integrity Test Site Charts (HITS)—Charts which provide detailed bathymetric and environmental data for submarine test dive areas.

Huygen's principle—A general principle applying to all forms of wave motion which states that every point on the instantaneous position of an advancing phase front (wave front) may be regarded as a source of secondary spherical wavelets. The position of the phase front a moment later is then determined as the envelope of all of the secondary wavelets (ad infinitum). This principle is extremely useful in understanding effects due to refraction, reflection, diffraction, and scattering of all types of radiation, including sonic radiation as well as electromagnetic radiation and applying even to ocean wave propagation.

Hydrographic Information Handling System (HIHAN)—HIHAN is a software/hardware system under development by the Detachment Stennis Space Center (NRL) which will automate the preprocessing, merging, integration and preparation of hydrographic data from a variety of sources.

hydrographic chart—(JCS) A nautical chart showing depths of water, nature of bottom, contours of bottom and coastline, and tides and currents in a given sea or sea and land area. Also called **marine map; nautical chart.**

hydrographic datum—The plane of reference of soundings, depth curves, and elevations of foreshore and offshore features. Also called **chart datum.** See also **international low water; low water datum; National Geodetic Vertical datum of 1929; sounding datum; tidal datum.**

hydrographic detail—The features along the shore and the submerged parts of bodies of water. Also called **hydrographic feature.**

hydrographic digital data—Data in discrete numerical form derived from the measurements and description of the physical features of the oceans, seas, lakes, rivers and other waters, and their adjoining coastal areas (with particular reference to navigational usage.)

hydrographic feature—See **hydrographic detail.**

hydrographic reconnaissance—A reconnaissance of an area of water to determine depths, beach gradients, the nature

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

of the bottom, and the location of coral reefs, rocks, shoals, and man-made obstacles.

hydrographic sextant—See **surveying sextant.**

hydrographic sounding—See **sounding,** definitions 1 and 2.

hydrographic survey—A survey made in relation to any considerable body of water, such as a bay, harbor, lake, or river for the purposes of determination of channel depths for navigation, location of rocks, sand bars, lights, and buoys; and in the case of rivers, made for flood control, power development, navigation, water supply, and water storage.

hydrography—**1.** (JCS) The science which deals with the measurements and description of the physical features of the oceans, seas, lakes, rivers, and their adjoining coastal areas, with particular reference to their use for navigational purposes. **2.** That part of topography pertaining to water and drainage features.

hydrology—Hydrology in its broadest extent deals with the properties, laws, and phenomena of water; of its physical, chemical, and physiological relations; of its distribution throughout the habitable Earth; and of the effect of this circulation on human lives and interests.

hydrophone—An electroacoustic transducer that converts sound energy into electrical energy. See also **transducer.**

hydrosphere—That part of the Earth that consists of the oceans, seas, lakes, and rivers; a similar part of any other spatial body if such a body exists. Distinguished from the atmosphere and lithosphere.

hydrostatic equation—In numerical equations, the form assumed by the vertical component of the vector equation of motion when all Coriolis force, Earth curvature, frictional, and vertical acceleration terms are considered negligible compared with those involving the vertical pressure force and the force of gravity.

hydrostatic equilibrium—The state of a fluid whose surfaces of constant pressure and constant mass (or density) coincide and are horizontal throughout. Complete balance exists between the force of gravity and the pressure force. The relation between the pressure and the geometric height is given by the hydrostatic equation.

hygrometric—Relating to the relative humidity or comparative amount of moisture in the atmosphere. Since the atmosphere penetrates the pores or cells of material bodies in varying degrees depending upon the substances of which they are composed, the amount of moisture which it contains will affect the shapes and dimensions of certain instruments and equipment used in surveying and mapping. For this reason it is necessary to select materials which are not sensitive to hygrometric conditions for the construction of leveling rods, planetable sheets, etc., and for the construction and printing of maps.

hygroscopic—The property of materials such as paper and films to absorb or release moisture and, in so doing, to expand or contract.

hyperbolic line of position—A line of position in the shape of a hyperbola, determined by measuring the difference in distance to two fixed points. Loran lines of position are an example.

hyperfocal distance—The distance from the camera lens to the nearest object in focus when the camera lens is focused at infinity.

hyperstereoscopy—(JCS) Stereoscopic viewing in which the relief effect is noticeably exaggerated, caused by the extension of the camera base. Also called **appearance ratio; exaggerated stereo; relief stretching; stereoscopic exaggeration.** See also **vertical exaggeration,** definition 1.

hypsographic detail—The features pertaining to relief or elevation of terrain.

hypsographic map (or chart)—A map or chart showing land or submarine bottom relief in terms of height above, or below, a datum by any method, such as contours, hachures, shading, or tinting. Also called **hypsometric map (or chart).**

hypsography—**1.** The science or art of describing elevations of land surfaces with reference to a datum, usually sea level. **2.** That part of topography dealing with relief or elevation of terrain.

hypsograph—An instrument of the slide rule type used to compute elevations from vertical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

angles and horizontal distances.

hypsometer—An instrument used in determining elevations of points on the Earth's surface in relation to sea level by determining atmospheric pressure through observation of the boiling point (temperature) of water at each point.

hypsometric map (or chart)—See **hypsographic map.**

hypsometric tint scale—A graphic scale in the margin of maps and charts which indicates heights or depths by graduated shades of colors. See also **hypsometric tinting.**

hypsometric tinting—(JCS) A method of showing relief on maps and charts by coloring, in different shades, those parts which lie between selected levels. Also called **altitude tints; color gradients; elevation tints; gradient tints; layer tints.**

hypsometry—The art of determining, by any method, surface elevations on the Earth with reference to sea level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Ice chart—A chart showing prevalence of ice, usually with reference to navigable waterways.

Ice surveillance products—Plotting charts, report forms, and other products used for the collection and dissemination of ice position information.

Iced-bar apparatus—An apparatus for measuring linear distance with great precision and accuracy, and consisting essentially of a steel bar which is maintained at a constant temperature by being surrounded with melting ice. The bar is rectangular in cross section, and is carried in a Y-shaped trough which is filled with melting ice and mounted on a car which moves on a track. Bar lengths are observed with micrometer microscopes mounted on stable supports. Also called **Woodward base-line measuring apparatus.**

Ideal Earth—See **equilibrium theory.**

Ideal sea level—The theoretical sea surface which is everywhere normal to the plumb line. Reference of all depth soundings to this level would make them all comparable.

Identification posts—Posts of wood or other suitable material, appropriately marked and inscribed, and placed near survey stations to aid in their recovery and identification. Also called **supplemental posts for survey monuments.**

Idle pendulum—A working pendulum placed in the receiver in advance of its being used, so that it may assume the same temperature as the dummy pendulum.

Image correlation—The matching of position and physical characteristics between imagery of the same geographic area from different types of sensors, between sensor imagery and a data base, or between two images from the same sensor.

Image degradation—(photometry) The reduction of the inherent optimum imaging potential of individual sensor systems caused by error in sensor operations or processing procedures. Reductions in quality caused by unavoidable factors; i e., atmospherics, snow cover, etc., are not associated with the term.

Image direction—A term used to describe the image orientation of a photographic negative or positive relative to the position of the emulsion. See also **emulsion-to-base; emulsion-to-emulsion; right-reading; wrong-reading.**

Image distortion—Any shift in the position of an image on a photograph which alters the perspective characteristics of the photograph. Causes of image distortion include lens aberration, differential shrinkage of film or paper, and motion of the film or camera.

Image enhancement—Any of several processes that might improve the interpretation quality of an image. Such processes include contrast improvement, greater resolution, special filtering, etc.

Image motion compensation (IMC)—(JCS) Movement intentionally imparted to film at such a rate as to compensate for the forward motion of an air or space vehicle when photographing ground objects.

Image motion compensation distortion—In a panoramic camera system, the displacement of images of ground points from their expected cylindrical position caused by the translation of the lens or negative surface (a motion used to compensate for image motion during exposure time).

Image motion factors—Those factors wherein the image motion varies directly with the aircraft ground speed and lens focal length and inversely with the altitude.

Image motion—The smearing or blurring of imagery on an aerial photograph because of the relative movement of the camera with respect to the ground.

Image plane—See **photograph plane.**

Image point—(photogrammetry) Image on a photograph corresponding to a definite object on the ground.

Image processing—Encompasses all the various operations that can be applied to photographic or image data. These include, but

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

are not limited to, image compression, image restoration, image enhancement, preprocessing, quantization, spatial filtering and other image pattern recognition techniques.

Image ray—Straight line from a ground object, through the camera lens, to the image on the photograph. See also **perspective ray.**

Imagery interpretation key—(JCS) Any diagrams, charts, tables, lists, or sets of examples, etc., which are used to aid imagery interpreters in the rapid identification of objects visible on imagery.

Imagery interpretation—(JCS) The process of location, recognition, identification, and description of objects, activities, and terrain represented on imagery.

Imagery sortie—(JCS) One flight by one aircraft for the purpose of recording air imagery.

Imagery—(JCS) Collectively, the representations of objects reproduced electronically or by optical means on film, electronic display devices, or other media.

Image—1. The permanent record of the likeness of any natural or man-made features, objects, and activities reproduced on photographic materials. This image can be acquired through the sensing of visual or any other segment of the electromagnetic spectrum by sensors, such as thermal infrared, and high resolution radar. See also **erect image; homologous images; inverted image; latent image; real image; reverted image; thermal imagery; virtual image.** 2. A visual representation, as on a radarscope.

Imaging systems—(satellite) Imaging systems carried on satellites can be placed in two basic categories, according to the requirements to be met by their applications: (1) remote sensing for interpretation is characterized by the fact the nature of the object imaged is of primary importance, and the geometry is of minor importance; and (2) satellite photogrammetry is characterized by the geometry which is of primary interest, and the nature of the object is of minor importance.

Impersonal micrometer—See **transit micrometer.**

Import—The process of bringing data or software from one system into another system.

Imposition—Positioning and assembling negatives or positives into printing location on a flat.

Impression—The inked image received by a sheet in a press. Commonly used as a measure of printing production or capacity.

In-and-out station—A recoverable but unoccupied station incorporated into a traverse by recording a fictitious deflection angle of 180° to reverse the azimuth of the course leading into it, so that the next station coincides with the preceding station and the in-and-out station is used as the backsight for continuing the traverse. In the computations it is treated as an ordinary station in the traverse.

Incident nodal point—See **nodal point,** definition 1.

Inclination correction—See **grade correction.**

Inclination of the horizontal axis—The vertical angle between the horizontal axis of a surveying or astronomic instrument and the plane of the horizon.

Inclination—1. The angle which a line or surface makes with the vertical, horizontal, or with another line or surface. 2. The angle between orbit plane and reference plane; for example, the Equator is the reference plane for geocentric, and the ecliptic for heliocentric orbits. 3. See **dip,** definition 2.

Incline sight—A sight made with a surveying instrument at an angle above or below the horizon.

Inclinometer—An instrument for measuring inclination to the horizontal of a ship or aircraft, or of the lines of force of the Earth's magnetic field.

Inclusion—A topological property which provides relationships between objects which are enclosed by or overlap another object. Examples of inclusion are an island surrounded by water, or a town within and a part of a larger administrative area. See also **topology.**

Independent resurvey—A resurvey which is not dependent on the records of the original survey but is intended to supersede them in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

establishing new land boundaries and subdivisions. See also **dependent resurvey.**

Index chart—An outline chart showing the limits and identifying designations of navigational charts, volumes of sailing directions, etc.

Index contour line—(JCS) A contour line accentuated by a heavier line weight to distinguish it from intermediate contours. Index contours are usually shown as every fifth contour with their assigned values, to facilitate reading elevations.

Index correction—**1.** A correction applied to the reading from any graduated measuring device to compensate for a constant error such as would be caused by misplacement of the scale; the reverse of the index error. **2.** (leveling) That correction which must be applied to an observed difference of elevation to eliminate the error introduced into the observations when the zero of the graduations on one or both leveling rods does not coincide exactly with the actual bottom surface of the rod.

Index error—The instrumental error which is constant and attributable to displacement of a vernier or some analogous effect.

Index mark—A real mark, such as a cross or dot, lying in the plane or the object space of a photograph and used singly as a reference mark in certain types of monocular instruments, or as one of a pair to form a floating mark as in certain types of stereoscopes. In stereoscopic map plotting instruments which utilize a stereo pair of index marks, each mark is called a **half mark.** See also **floating mark.**

Index of refraction—See **Snell's law of refraction.**

Index prism—A sextant prism which can be rotated to any angle corresponding to altitudes between established limits. It is the bubble or pendulum sextant counterpart of the index mirror of a marine sextant.

Index to photography—See **photo index,** definition 1.

Indian datum—The Indian datum is accepted as the preferred datum for India and several adjacent countries in Southeast Asia. It is computed on the Everest ellipsoid with its origin at Kalianpur in Central India. Derived in 1830, the Everest ellipsoid is the oldest of the ellipsoids in common use and is much too small. As a result of the latter, the datum cannot be extended too far from the origin or very large geoid separations will occur. For this reason and the fact that the ties between local triangulation in Southeast Asia are typically weak, the Indian datum is probably the least satisfactory of the preferred datums.

Indian spring low water—The approximate mean water level determined from all lower low waters at spring tides.

Indian tide plane—The datum of Indian spring low water.

Indicated corner—A term adopted by the U.S. Geological Survey to designate a corner of the public land surveys whose location cannot be verified by the criteria necessary to class it as a found or existent corner, but which is accepted locally as the correct corner and whose location is perpetuated by such marks as fence line intersections, piles of rocks, and stakes or pipes driven into the ground, which have been recovered by field investigation.

Indicated principal point—See **principal point.**

Indirect effect on the deflections—See **topographic deflection.**

Indirect leveling—See **barometric leveling; thermometric leveling; trigonometric leveling.**

Indirect measurement—Any measurement secured by determining its quantity from its relation to some measured quantity. A technique used in surveying when it is impossible to actually tape a distance across a river or other such obstruction. See also **direct measurement.**

Indirect observation—A measure of a quantity which is a function of the quantity or quantities whose value is desired, such as an observed difference in elevation with a spirit level, used to obtain the elevation of a bench mark.

Indirect photography—Photography in which the camera records an image cast upon a screen or similar display surface by electronic (television, radar, etc.) or other means.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Inequality—A systematic departure from the mean value of a quantity. See also **annual inequality; diurnal inequality; lunar inequality; parallactic inequality; parallax inequality; phase inequality; variational inequality.**

Inertial azimuth—An azimuth which approximates the value which could be obtained from astronomic observations, but which is derived from direct observations along the line of sight with an inertial azimuth measuring device consisting of a north-seeking gyroscope combined with a theodolite. See also **gyrotheodolite.**

Inertial coordinate system—A coordinate system in which the axes do not rotate with respect to the "fixed stars" and in which dynamic behavior can be described using Newton's laws of motion.

Inertial guidance system—A system in which guidance is permitted by means of the measurement and integration of acceleration from within the craft.

Inertial measurement unit (IMU)—(missile guidance) A compact component of an inertial guidance system which has three accelerometers mounted on a gyrostabilized platform. See also **accelerometer.**

Inertial navigation system—A system which is not dependent on man-made electromagnetic signals. Newton's second law of motion is utilized with a system consisting of accelerometers mounted on gyrostabilized platforms, each for measuring longitudinal, lateral, and vertical accelerations. The double integration of all accelerations in three mutually perpendicular directions provides distance traveled (from a known starting point) in three mutually perpendicular directions. Navigation is by a highly refined form of dead reckoning with system position being updated from other navigational references in the more sophisticated systems.

Inertial navigation—The process of measuring a craft's velocity, attitude (in the submarine missile launching applications), and displacement (including changes in altitude in the aircraft application) from a known starting point through sensing the accelerations acting on it in known directions by means of devices that mechanize Newton's laws of motion.

Inertial reference photography—Cartographic aerial photography obtained simultaneously with magnetic tape recorded inertial reference positioning data.

Inferior conjunction—The conjunction of an inferior planet and the Sun when the planet is between the Earth and the Sun.

Inferior planets—The planets with orbits smaller than that of the Earth, i.e., Mercury and Venus.

Inferior transit—See **lower transit.**

Infinity—The point, line, or region beyond measurable limits. An unaltered source of light is regarded as at infinity if it is at such a great distance that rays from it can be considered parallel.

Infrared distance measurement—A distance determined by measuring the phase delay of modulation signals on a light beam (infrared) traveling at a known velocity between a distance meter and reflector.

Infrared film—(JCS) Film carrying an emulsion especially sensitive to "near-infrared." Used to photograph through haze, because of the penetrating power of infrared light, and in camouflage detection to distinguish between living vegetation and dead vegetation or artificial green pigment.

Infrared—Pertaining to or designating the portion of the electromagnetic spectrum with wavelengths just beyond the red end of the visible spectrum, such as radiation emitted by a hot body. Invisible to the eye, infrared rays are detected by their thermal and photographic effects. Their wavelengths are longer than those of visible light and shorter than those of radio waves.

Inherited error—The error in initial values used in a computation; especially the error introduced from the previous steps in a step-by-step integration.

Initial Graphic Exchange Specification (IGES)—An interim standard format developed by the National Bureau of Standards (NIST) for exchanging graphics data between computer systems. IGES is the most widely used exchange format in the world.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Initial monument—A physical structure which marks the location of an initial point in the rectangular system of surveys. *See also* **initial point.**

Initial operating capability (IOC)—The first attainment of the capability to employ effectively a weapon, item of equipment, or system of approved specific characteristics, and which is manned or operated by adequately trained, equipped, and supported operational unit or group.

Initial point—**1.** That point from which any survey is initiated. Also called point of origin. **2.** A point which is established under the rectangular system of surveys and from which is initiated the cadastral survey of the principal meridian and base line that controls the cadastral survey of the public lands within a given area. *See also* base line; initial monument; principal meridian.

Inner orientation—See **interior orientation.**

Inner planets—The four planets nearest the Sun: Mercury, Venus, Earth, and Mars. *See also* **major planets; outer planets; asteroid.**

Inserted grouping—(JCS) (radar) The inclusion of one area of homogeneous surface material in an area of different material.

Inset—(JCS) (cartography) A separate map positioned within the neatline of a larger map. Three forms are recognized: (1) all area geographically outside a sheet but included therein for convenience of publication, usually at the same scale; (2) a portion of the map or chart at an enlarged scale; (3) a smaller scale map or chart of surrounding areas included for location purposes.

Instantaneous field of view (IFOV)—The smallest solid angle resolvable by a scanner when expressed in radians. When expressed in feet, it is the projected area of the detector image on the ground and is a measure of the resolution of a scanner or similar remote sensor with discrete samples.

Instantaneous reading tape—A survey tape on which the foot mark is repeated at each subdivision. Thus, a tape divided into tenths of a foot would have the foot mark imprinted at each tenth of a foot division.

Instrument adjustment—The process of mechanical manipulation of the relation of component parts of an instrument in order to obtain the highest practicable precision and facility in the designed use of the instrument.

Instrument approach chart—An aeronautical chart designed for use under instrument flight conditions, for making instrument approach and letdown to contact flight conditions in the vicinity of an airfield.

Instrument error—A systematic error resulting from imperfections in, or faulty adjustment of, instruments or devices used. Also called **calibration error.**

Instrument parallax—**1.** A change in the apparent position of an object with respect to the reference marks of an instrument which is caused by imperfect adjustment of the instrument. Also called **optical parallax. 2.** Parallax caused by a change in the position of the observer. Also called **personal parallax.**

Instrument phototriangulation—See **stereotriangulation.**

Instrument station—See **setup,** *definition* 1.

Integrated station instrument—An instrument combining horizontal and vertical angles with electronic distance measurement and programmed computer capability in a single piece of hardware.

Integration—(JCS) **1.** A stage in the intelligence cycle in which a pattern is formed through the selection and combination of evaluated information. **2.** (photography) A process by which the average radar picture seen on several scans of the time base may be obtained on a print, or the process by which several photographic images are combined into a single image.

Intensity of gravity—The magnitude with which gravity acts, expressed in suitable units, usually as an acceleration, in gals; as a force, in dynes.

Interactive—A system of allowing two-way electronic communication between the user and the computer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Intercardinal point—Any of the four directions midway between the cardinal points; northeast, southeast, southwest, or northwest.

Interferometer—An apparatus used to produce and measure interference from two or more coherent wave trains from the same source. Used to measure wavelengths, to measure angular width of sources, to determine the angular position of sources (as in satellite tracking), and for other purposes. See also **radio interferometer.**

Interim Terrain Data (ITD)—A DMA digital product consisting of contiguous data sets comprised of attributed and unsymbolized feature information in six thematic files (Surface Configuration, Vegetation, Surface Materials, Surface Drainage, Transportation, and Obstacles) and a seventh file of Digital Terrain Elevation Data (DTED) Level I. The data content is equivalent to the content of either Tactical Terrain Analysis Data Bases (TTADBs) or Planning Terrain Analysis Data Bases (PTADBs). An enhanced transportation file consisting of attributed roads (all-weather through cart track), bridges, and tunnels was added to the data set. See **Planning Terrain Analysis Data Base; Tactical Terrain Analysis Data Base.**

Interior angle traverse—In surveying, a closed traverse wherein distances are measured and only interior angle measurements are used.

Interior orientation—The determining (analytically or in a photogrammetric instrument) of the interior perspective of the photograph as it was at the instant of exposure. Elements of interior orientation are the calibrated focal length, location of the calibrated principal point, and the calibrated lens distortion. Also called **inner orientation.**

Interior perspective center—See **perspective center.**

Interior to a curve—That area lying toward the concave side of a curve and included within the area of the circle of which the curve is a part of the circumference.

Interlocking angle—In tilt analysis of oblique photographs, the angle between the optical axes of the vertical and oblique cameras. The dihedral angle between the planes of the vertical and oblique photographs.

Intermediate contour line—A contour line drawn between index contours. Depending on the contour interval, there are three or four intermediate contours between the index contours.

Intermediate orbit—A central force orbit that is tangent to the real orbit at some point. A fictitious satellite traveling in the intermediate orbit would have the same position, but not the same velocity, as the real satellite at the point of tangency.

Internal data structure—The organization within a system of data and particularly, the reference linkages among data elements. See also **Data structure.**

International Gravity Standardization Net 1971 (IGSN 71)—An adjusted worldwide network of gravity measurements consisting of absolute, pendulum, and gravimeter observations. The IGSN 71 was approved and adopted by the International Union of Geodesy and Geophysics in 1971, and replaces the Potsdam datum as the international gravity standard.

International date line—(JCS) The line coinciding approximately with the antemeridian of Greenwich, modified to avoid certain habitable land. In crossing this line there is a date change of one day. Also called **date line.**

International ellipsoid—A reference ellipsoid defined by Hayford in 1924 with semimajor axis(a) as 6,378,388.0 meters, and flattening of 1/297.

International gravity formula—A development of the formula for theoretical gravity, based on the assumptions that the spheroid of reference is an exact ellipsoid of revolution having the dimensions of the international ellipsoid of reference (Madrid, 1924), rotating about its minor axis once in a sidereal day; that the surface of the ellipsoid is a level surface; and that gravity at the Equator equals 978.049 gals.

International low water (ILW)—A plane of reference below mean sea level; half the range between mean lower low water and mean higher high water multiplied by 1.5.

International system of units (SI)—The practical international metric system of units

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

adopted by the Eleventh General Conference of Weights and Measures in 1960.

Interocular distance—(JCS) The distance between the centers of rotation of the eyeballs of an individual or between the oculars of optical instruments. Also called **eye base; interpupillary distance.**

Interoperability—Consistency between different MC&G software and data sets in terms of accuracy, data structure, feature coding, geo-referencing systems, datums, spatial resolutions, and precision, which allow them to be used on different kinds of equipment systems.

Interpolate—To determine intermediate values between given fixed values. As applied to logical contouring, to interpolate is to ratio vertical distances between given spot elevations.

Interpupillary distance—See **inter-ocular distance.**

Interrogation—Transmission of a radio signal or combination of signals intended to trigger a transponder or group of transponders.

Interrogator-responsor—A radio transmitter and receiver combined to interrogate a transponder and display the resulting replies. Also called **challenger.** See also **interrogator.**

Interrogator—**1.** A radar set or other electronic device that transmits an interrogation. **2.** An interrogator responsor or the transmitting component of an interrogator responsor.

Interrupted map projection—A projection having several standard meridians, each centered over a continent, and with lobate-shaped sections of the projection plotted from each standard meridian. The projection is broken in the ocean areas between the continents, thus reducing the linear scale discrepancies and the overall shape distortion, especially toward the margins, while retaining the equal area property.

Intersected point—See **intersection station.**

Intersection station—An object whose horizontal position is determined by observations from other survey stations, no observations being made at the object itself. Where the object is observed from only two stations, the position is termed a **no-check position,** as there is no proof that such observations are free from blunders. Intersection stations are either objects which would be difficult to occupy with an instrument, or survey signals whose positions can be determined with sufficient accuracy without being occupied. Also called **intersected point.**

Intersection—**1.** (digital mapping) The coexistence of end points at a specific geographic location; the set of all objects common to two or more intersecting sets. **2.** (surveying) The procedure of determining the horizontal position of an unoccupied point (intersection station) by direction observations from two or more known positions. **3.** (photogrammetry) The procedure of determining the horizontal position of a point by intersecting lines of direction obtained photogrammetrically. The lines of direction may be obtained directly from vertical photographs or by graphic or mathematical rectification of tilted photographs. See also **resection.**

Intervalometer—A timing device for automatically operating at specified intervals certain equipment such as a camera shutter for the purpose of obtaining a desired end lap between successive photographs.

Intervisibility test—Any of the several tests used to determine the possible visibility along a sight line in a proposed survey net. Its purpose is to determine the existence of obstructions along a proposed line of sight from which tower and signal requirements may be developed.

Intervisibility—Function which calculates area or line-of-sight which can be "seen" from a specific location or locations. Also termed **viewshed modeling** or **viewshed mapping.**

Invar leveling rod—See **precise leveling rod.**

Invar pendulum—A quarter-meter pendulum made of Invar.

Invar scale—A measuring bar made from Invar. Normally, one side is graduated in the metric system and the other side in the English system.

Invar tape—Any survey tape made of Invar.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Invariable pendulum—A pendulum so designed and equipped with means of support that it can be used in only one position.

Invar—An alloy of nickel and steel having a very low coefficient of thermal expansion. Invar is used in the construction of Jaderin wires (base-line measuring apparatus), subtense bars, precise leveling rods, tapes, and pendulums. See also **Lovar.**

Inventory survey—A survey for the purpose of collecting and correlating engineering data of a particular type, or types, over a given area. An inventory survey may be recorded on a base map.

Inventory—Existing source materials, intermediate and final products identified by type of geographic coverage, and currency.

Inverse chart—See **transverse chart.**

Inverse computation—See **inverse position computation.**

Inverse cylindrical orthomorphic chart—See **transverse Mercator chart.**

Inverse cylindrical orthomorphic map projection—See **transverse Mercator map projection.**

Inverse equator—See **transverse equator.**

Inverse latitude—See **transverse latitude.**

Inverse longitude—See **transverse longitude.**

Inverse Mercator chart—See **transverse Mercator chart.**

Inverse Mercator map projection—See **transverse Mercator map projection.**

Inverse meridian—See **transverse meridian.**

Inverse parallel—See **transverse parallel.**

Inverse position computation—The derivation of the length, and the forward and back azimuths of a line by computation based on the known positions of the ends of the line. Also called **inverse; inverse computation; inverse position problem.**

Inverse position problem—See **inverse position computation.**

Inverse rhumb line—See **transverse rhumb line.**

Inverse—See **inverse position computation; transverse.**

Inversors—(photography) Mechanical devices used to maintain correct conjugate distances and collinearity of negative, lens, and easel planes in autofocusing optical instruments, such as copy cameras and rectifiers. See also **Carpentier inversor; Peaucellier Carpentier inversor; Peaucellier inversor; Pythagorean right angle inversor.**

Inverted image—An image that appears upside down in relation to the object.

Inverted stereo—See **pseudoscopic stereo.**

Inverting telescope—An instrument with the optics so arranged that the light rays entering the objective of the lens meet at the cross hairs and appear inverted when viewed through the eyepiece without altering the orientation of the image. See also **erecting telescope.**

Inverting—See **transit,** definition 3.

Ionosphere—(JCS) That part of the atmosphere, extending from about 70 to 500 kilometers, in which ions and free electrons exist in sufficient quantities to reflect electromagnetic waves. See also **atmosphere.**

Ionospheric correction—The correction made to electromagnetic measurements between satellites and ground stations to compensate for the effect of the ionosphere.

Irradiance—Radiant flux incident per unit area.

Irregular error—See **random error.**

Irregular line—A complex line which cannot be easily described by a mathematical polynomial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Isanomal—A line connecting points of equal variations from a normal value.

Isentropic—Of equal or constant entropy with respect to either space or time.

Islands—(digital mapping) Polygons which are completely enclosed within another polygon.

Isobaric chart—A chart showing isobars. Also called **constant pressure chart.**

Isobar—A line along which the atmospheric pressure is, or is assumed to be, the same or constant.

Isobath—See **depth contour.**

Isocenter plot—See **Isocenter triangulation.**

Isocenter triangulation—Radial triangulation utilizing isocenters as radial centers. Also called **Isocenter plot.**

Isocenter—1. (JCS) The point on a photograph intersected by the bisector of the angle between the plumb line and the photograph perpendicular. 2. The unique point common to the plane of a photograph, its principal plane, and the plane of an assumed truly vertical photograph taken from the same camera station and having an equal principal distance. 3. The point of intersection on a photograph of the principal line and the isometric parallel. The isocenter is significant because it is the center of the radiation for displacements of images due to tilt.

Isochrone—A line on a chart connecting all points having the same time of occurrence of a particular phenomenon or of a particular value of a quantity.

Isoclinal—A line drawn on a map or chart joining points of equal magnetic dip. Also called **isoclinic line.**

Isoclinic chart—A chart of which the chief feature is a system of isoclinic lines, each for a different value of the magnetic inclination.

Isoclinic line—See **Isoclinal.**

Isodiff—One of a series of lines on a map or chart connecting points of equal correction or difference in datum, especially useful in readjustment of surveys from one datum to another. See also **Isolat; Isolong.**

Isodynamic line—A line connecting points of equal magnitude of any force.

Isogal—A contour line of equal gravity values on the surface of the Earth.

Isogonal—A line drawn on a map or chart joining points of equal magnetic declination for a given time. Also called **isogonic line.** See also **agonic line.**

Isogonic chart—A chart of which the chief feature is a system of isogonic lines, each for a different value of the magnetic declination.

Isogonic line—A line drawn on a map or chart joining points of equal magnetic declination for a given time. The line connecting points of zero declination is the agonic line. Lines connecting points of equal annual change are isopors, and are depicted on Magnetic Variation charts for the current 5-year epoch.

Isogram—See **Isopleth.**

Isogriv chart—A chart with lines connecting points of equal grivation.

Isogriv—A line on a map or chart which joins points of equal angular difference between grid north and magnetic north.

Isolat—An isodiff connecting points of equal latitude corrections.

Isoline—1. A line representing the intersection of the plane of a vertical photograph with the plane of an overlapping oblique photograph. If the vertical photograph were tilt free, the isoline would be the isometric parallel of the oblique photograph. 2. A line along which values are, or are assumed to be, constant.

Isolong—An isodiff connecting points of equal longitude corrections.

Isomagnetic chart—A chart showing the configuration of the Earth's magnetic field by isogonic, isoclinic, or isodynamic lines.

Isometric (conformal) latitude—An auxiliary latitude used in the conformal mapping of the spheroid on a sphere. By transforming

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

geographic latitudes on the spheroid into isometric latitudes on a sphere, a conformal map projection (the Mercator) may be calculated, using spherical formulas, for the plotting of geographic data.

Isometric parallel—The intersecting line between the plane of a photograph and a horizontal plane having an equal perpendicular distance from the same perspective center.

Isoperimetric curve—A line on a map projection along which there is no variation from exact scale. There are two isoperimetric curves passing through every point on an equal-area map projection. This characteristic gives that class of projections some preference for engineering maps.

Isopleth—A line of equal or constant value of a given quantity, with respect to either space or time. Also called **isogram.**

Isoporic chart—A chart with lines connecting points of equal magnetic annual change.

Isopor—A line found on magnetic charts showing points of equal annual change. Also called **magnetic isoporic line.**

Isopycnic—A line connecting points of equal density, particularly of ocean water and atmosphere.

Isoradial—A radial from the isocenter.

Isostasy—A condition of approximate equilibrium in the outer part of the Earth, such that the gravitational effect of masses extending above the surface of the geoid in continental areas is approximately counterbalanced by a deficiency of density in the material beneath those masses, while the effect of deficiency of density in ocean waters is counterbalanced by an excess of density in the material under the oceans. See also **depth of isostatic compensation; Hayford-Bowie method of isostatic reduction; Hayford-Bullard (or Bullard) method of isostatic reduction; isostatic adjustment; isostatic compensation; isostatic correction; Pratt-Hayford theory of isostasy.**

Isostatic adjustment—The natural process by which the crust of the Earth adjusts to restore or maintain its state of equilibrium. See also **isostasy.**

Isostatic anomaly—The difference between an observed value of gravity and a theoretical value at the point of observation which has been corrected for elevation of the station above the geoid, and for the effect of topography over the whole Earth, and for its isostatic compensation.

Isostatic compensation—The departure from normal density of material in the lower part of a column of the Earth's crust which balances (compensates) land masses (topography) above sea level and deficiency of mass in ocean waters, and which produces the condition of approximate equilibrium of Earth's crust. See also **isostasy; topographic deflection.**

Isostatic correction—The adjustment made to values of gravity or to deflections of the vertical observed at a point to take account of the assumed mass deficiency under topographic features for which a topographic correction is also made.

Isostatic geoid—An ideal geoid derived from the spheroid of reference by the application of computed values of the deflection of the vertical which depend upon the topography and isostatic compensation.

Isostere—A line connecting points of equal atmospheric density.

Isotimic—Pertaining to a quality which has equal value in space at a particular time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

J

Jacob's staff—A single staff or pole used for mounting a surveyor's compass or other instrument. Sometimes used in place of a tripod.

Jaderin wires (base apparatus)—An apparatus used for base line measurement. It consists of separate steel and brass wires, extended under constant tension over reference tripods in the line of the base. The coefficients of expansion and lengths at a certain temperature of the two wires having been found, the temperatures of the wires themselves may be deduced from the difference of the measurement of the same distance by the two wires. With these temperatures known, the length of the base may be accurately obtained.

Jet Navigation Chart (JNC)— A 1:2,000,000 scale, coordinated series of multicolored charts, designed to satisfy long range navigation of high-altitude, high-speed aircraft.

joining—Connecting two or more separately digitized or stored maps. The junction between two such maps is sometimes visible as a result of imperfections or inconsistencies in the data.

Joint Operations Graphic (JOG)—The standard 1:250,000 scale Department of Defense cartographic product which may be produced in any of the following three versions to meet the validated unified and specified commands and military departments area requirements: the JOG/G (Series 1501) is designed to meet ground use requirements; JOG/A (Series 1501 Air) is designed to meet air use requirements; and JOG/R (Series 1501 Radar) is the Air Target Material version in support of radar/ intelligence planning and operations requirements.

Julian calendar—The calendar established by Julius Caesar in 46 B.C. It is based upon the assumption that the true length of the tropical year was exactly 365.25 mean solar days.

Julian day—The number of each day, as reckoned consecutively since the beginning of the present Julian period on January 1, 4713 B.C. The Julian day number denotes the number of days that has elapsed at Greenwich noon on the day designated, since this epoch day. See also **modified Julian day.**

junction bench mark—A bench mark selected as the common meeting point for lines of levels or links of levels.

junction detail—A sketch or working diagram showing the details of the various levelings at a junction.

junction figure—A triangulation figure in which three or more triangulation arcs meet, or two or more arcs intersect.

junction—1.(digital) The point (node) where two or more line segments join together. 2.(leveling) The place where two or more lines of levels are connected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

K

K-factor—See **base altitude ratio.**

Kalman filtering—The recursive minimum variance estimation of an unbiased stochastic variable. An a priori estimate and covariance are linearly combined with new data to form an updated estimate and covariance.

Kepler's laws—The three laws governing the motions of planets in their orbits: (1) the orbits of the planets are ellipses, with the Sun at a common focus; (2) as a planet moves in its orbit, the line adjoining the planet and Sun sweeps over equal areas in equal intervals of times; (3) the squares of the periods of revolution of any two planets are proportional to the cubes of their mean distances from the Sun. Also called **Kepler's planetary laws.**

Kepler's planetary laws—See **Kepler's laws.**

key flat—The principal or master layout or flat used as a positioning guide for stripping up other flats. Also called **layout guide.** See also **flat,** definition 1; **layout.**

key—A data element in a data base used by the application software to locate a specific data record or group of records.

kilobyte (KB)—A unit of memory representing 2^{10} (1,024) bytes. Commonly, a kilobyte may represent 1000 bytes. A K is used as a suffix when describing memory size, as in 640K.

kiss plate—A press plate used to make an addition or correction to a previously printed sheet. Also called **touch plate.**

Kilmsch-Variomat compilation method (relief)—A process by which reproduction material of large-scale sheets is photographically filtered to retain only index contours. The index contours are then used as the relief compilation for a medium-scale sheet.

knowledge based system (KBS)—A computer system that embodies knowledge, including inexact, heuristic, and subjective knowledge. The results of knowledge engineering.

Krasovsky spheroid (ellipsoid)—A reference ellipsoid having the following approximate dimensions: semimajor axis 6,378,245.0 meters; flattening or ellipticity 1/298.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

L

length equation—A condition equation which expresses the relationship between the fixed lengths of two lines which are connected by triangulation.

length of degree—The length of a degree of latitude measured along a meridian of longitude. The length varies somewhat with the degrees of latitude. Those near the pole are longer and those near the Equator are shorter. The length also varies with different selections of spheroids.

level control—A series of bench marks or other points of known elevation, established throughout a project.

level correction—That correction which is applied to an observed difference of elevation to correct for the error introduced by the fact that the line of sight through the leveling instrument is not absolutely horizontal when the bubble is centered in its vial. See also **level constant**.

level line—1. A line on a level surface; therefore a curved line. 2. A line over which leveling operations are accomplished. See also ***duplicate level line***; ***line of levels***; **multiple level line**; **simultaneous level line**; **spur line of levels**.

level net—See **survey net**, definition 2.

level rod—See **leveling rod**.

level surface—See **equipotential surface**.

level trier—An apparatus for use in measuring the angular value of the divisions of a spirit level.

leveling error of closure—See **error of closure**, definition 4.

leveling instrument—An instrument used for determining differences of elevations between points. See also **Abney level**; ***dumpy level***; **Egault level**; **engineer's level**; **first-order level**; **Fischer level**; **Gravatt level**; **hand level**; ***hanging level***; **latitude level**; **Lenoir level**; **locator's hand level**; **military level**; **pendulum level**; **plate level**; **precise level**; ***reversible level***; **self-leveling level**; **spirit level**; **Stampfer level**; **striding level**; **telescope level**; **tilting level**; ***Troughton level***; ***U.S. Geological Survey level***; **wye (Y) level**.

leveling rod—A straight rod or bar, designed for use in measuring a vertical distance between a point on the ground and the line of collimation of a leveling instrument which has been adjusted to a horizontal position. Also called **level rod**; **rod**. See also **Barlow leveling rod**; **Boston leveling rod**; **double-target leveling rod**; ***foot-meter rod***; **Gravatt leveling rod**; **Invar leveling rod**; **long rod**; **Mollitor precise leveling rod**; **New York leveling rod**; ***Pemberton leveling rod***; **Philadelphia leveling rod**; **precise leveling rod**; **range rod**; **self-reading leveling rod**; ***short rod***; **single-target leveling rod**; **Stephenson leveling rod**; **tape rod**; **target leveling rod**; ***U.S. Engineer precise leveling rod***; **U.S. Geological Survey precise leveling rod**.

leveling the model—See **leveling**, definition 2.

line map—*See **planimetric map**.*

line of apsides—The major axis of an elliptical orbit extended indefinitely in both directions. Also called **apse line**.

line of collimation—(optics) The line through the second nodal point of the objective lens of a telescope and the center of the reticle. Also called **aiming line**; ***line of sight***; ***pointing line***; ***sight line***.

line of constant scale—Any line on a photograph which is parallel to the true horizon or to the isometric parallel. Also called **line of equal scale**.

line of equal scale—See **line of**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

constant scale.

line of force—A line indicating the direction in which a force acts, as in a magnetic field.

local magnetic disturbance—See **local magnetic anomaly.**

labeling—(digital) The process of assigning attributes to polygons.

Laborde map projection—Similar to the transverse Mercator projection, except that the Laborde projects a spheroid rather than a sphere onto a plane. This conformal projection is best suited for regions which are elongated in a direction which is at a considerable angle to the meridian. [USAF Special Flight Charts are based on this projection.]

LaCoste-Romberg gravimeter—A long-period spring suspended cantilevered weight system adapted to the measurement of gravity differences.

ladder grid numbers—Those grid numbers which identify the grid lines within the neatline.

Lambert azimuthal equal-area map projection—See **Lambert zenithal equal-area map projection.**

Lambert azimuthal polar map projection—A Lambert equal-area map projection with the pole of projection at the pole of the sphere, and the radii of the circles which represent the geographic parallels corresponding to the chords of those parallels.

Lambert bearing—A bearing as measured on a Lambert conformal chart or plotting sheet. This approximates a great-circle bearing.

Lambert central equivalent map projection upon the plane of the meridian—An azimuthal map projection having the pole of the projection on the Equator. Also called **Lambert equal area meridional map projection.**

Lambert conformal chart—A chart on the Lambert conformal projection.

Lambert conformal conic map projection—A conformal map projection of the so-called conical type, on which all geographic meridians are represented by straight lines which meet in a common point outside the limits of the map, and the geographic parallels are represented by a series of arcs of circles having this common point for a center. Meridians and parallels intersect at right angles, and angles on the Earth are correctly represented on the projection. This projection may have one standard parallel along which the scale is held exact; or there may be two such standard parallels, both maintaining exact scale. At any point on the map, the scale is the same in every direction. It changes along the meridians and is constant along each parallel. Where there are two standard parallels, the scale between those parallels is too small; beyond them, too large. Also called **Lambert conformal map projection.**

Lambert conformal map projection—See **Lambert conformal conic map projection.**

Lambert equal-area meridional map projection—See **Lambert central equivalent map projection upon the plane of the meridian.**

Lambert grid—An informal designation for a coordinate system based on a Lambert conformal map projection.

Lambert zenithal equal-area map projection—An azimuthal map projection having the pole of the projection at the center of the area mapped. The azimuths of great circles radiating from this center (pole) are truly represented on the map: equal distances on those great circles are represented by equal linear distances on the map, but the scale along those great circle lines so varies with distance from the pole of the projection, that an equal-area projection is produced. Also called **Lambert azimuthal equal-area map projection.**

laminate—**1.** The process of preserving a map sheet or other graphic by sandwiching between two sheets of clear synthetic material (polyethylene polyester plastic). The laminating equipment uses heat and pressure but no adhesive. **2.** (relief model) See **plastic block.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

land boundary—A line of demarcation between two parcels of land.

land effect—See **coastal refraction.**

land mass simulator plate—*See* **factored transparency.**

land survey—The process of determining boundaries and areas of tracts of land. The term **cadastral survey** is sometimes used to designate a land survey, but in this country its use should be restricted to the surveys of public lands of the United States. Also called **boundary survey; property survey.** See also **cadastral survey.**

land-line adjustment—Positioning the public land lines on a topographic map to indicate their true, theoretical, or approximate location relative to the adjacent terrain and culture.

landmark—**1.** An object of enough interest or prominence in relation to its surroundings to make it outstanding or to make it useful in determining a location or a direction. 2. Any monument, material mark, or fixed object used to designate the location of a land boundary on the ground.

LANDSAT—The generic name for a series of earth observing satellites, first launched in 1972 by NASA. LANDSAT serves as platforms for several sensors, including the Return Beam Vidicon (RBV), the Multispectral Scanner (MSS), and the Thematic Mapper (TM). Originally named ERTS, for Earth Resources Technology Satellite.

landscape map—A topographic map made to a relatively large scale and showing all details. Such maps are required by architects and landscape gardeners for use in planning buildings to fit the natural topographic features and for landscaping parks, playgrounds, and private estates. These are generally maps of small areas, and scales vary from 1 inch = 20 feet to 1 foot = 50 feet, depending on the amount of detail.

Laplace azimuth mark—An astronomic azimuth mark at a Laplace Station.

Laplace azimuth—A geodetic azimuth derived from an astronomic azimuth by use of the Laplace equation.

Laplace condition—The Laplace condition, expressed by the Laplace equation, arises from the fact that a deflection of the vertical in the plane of the prime vertical will give a difference between astronomic and geodetic longitude and between astronomic and geodetic azimuth; or, conversely, that the observed differences between astronomic and geodetic values of the longitude and of the azimuth may both be used to determine the deflection in the plane of the prime vertical.

Laplace control—Control and correction of astronomic azimuths through observations of the deflection of the plumb line in the prime vertical (comparison of astronomic and geodetic longitude).

Laplace equation—**1.** The equation which expresses the relationship between astronomic and geodetic azimuths in terms of astronomic and geodetic longitudes and geodetic latitude. 2. (potential) A partial differential equation of the second-order which is satisfied by the Newtonian potential of every finite body at all exterior points.

Laplace station—A triangulation or traverse station at which a Laplace azimuth is determined. At a Laplace station both astronomic longitude and astronomic azimuth are determined.

large-scale map—(JCS) A map having a scale of 1:75,000 or larger. See also **map.**

laser terrain profile recorder—An electronic instrument that emits a continuous wave laser beam from an aircraft to measure vertical distances between the aircraft and the Earth's surface.

laser—A device producing coherent energy beams in the spectrum of light or near-light frequencies. A laser-equipped Geodimeter makes it possible to measure greater distances.

latent image—The invisible image produced in radiation-sensitive materials which

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

becomes visible upon processing.

lateral chromatic aberration—An aberration which affects the sharpness of images off the lens axis because different colors undergo different magnifications.

lateral gain—(JCS) The amount of new ground covered laterally by successive photographic runs over an area.

lateral magnification—The ratio of a length in the image, perpendicular to the lens axis, to a corresponding length in the object.

lateral refraction—The horizontal component of the refraction of light through the atmosphere.

lateral shift—The offset of the position of the peak of an anomaly with the mass of magnetization (or gravitation).

lateral tilt—See **roll**, definition 2.

latitude band—(JCS) Any latitudinal strip designated by accepted units of linear or angular measurement, which circumscribes the Earth. Also called **band; latitudinal band.**

latitude correction—The amount of the adjustment of observed gravity values to an arbitrarily chosen base latitude. Also, correction to latitude in a traverse course.

latitude determination, zenith-telescope method—A precise method of determining astronomic latitude by measuring the difference of the meridional zenith distances of two stars of known declination, one north and the other south of zenith. Also called **Horrebow-Talcott method of latitude determination.**

latitude difference—(plane surveying) Length of the projection of a traverse course onto a meridian. Also called **northing; southing.**

latitude equation—A condition equation which expresses the relationship between the fixed latitudes of two points which are connected by triangulation or traverse.

latitude factor—The change in latitude along a celestial line of position per one minute change in longitude.

latitude level—A sensitive spirit level attached to the telescope of an instrument employed for observing astronomic latitude, in such manner that when the telescope is clamped in position, the level measures, in a vertical plane, variations in the direction of the line of collimation.

latitude of the line—See **difference of latitude**, definition 2.

latitude—1. (general) A linear or angular distance measured north or south of the Equator on a sphere or spheroid. 2. (plane surveying) The perpendicular distance in a horizontal plane of a pointfrom an east-west axis of reference. See also **difference of latitude**, definition 2. 3. (on a sphere) The angle at the center of a sphere between the plane of the Equator and the line to the point on the surface of the sphere. 4. (traverse) The north-south component of a traverse course. 5. (photography) The ability of an emulsion to record a range of brightness values. See also **argument of latitude; assumed latitude; astronomic latitude; authalic (equal-area) latitude; celestial latitude; circle of latitude; difference of latitude; fictitious latitude; galactic latitude; geocentric latitude; geodetic latitude; geographic latitude; geomagnetic latitude; grid latitude; isometric (conformal) latitude; latitude correction; middle latitude; oblique latitude; parallel; parametric latitude; rectifying latitude; terrestrial latitude; transverse latitude; variation of latitude.**

latitudinal band—See **latitude band.**

latitudinal curve—This term denotes an easterly and westerly property line adjusted to the same mean bearing from each monument to the next one in regular order, as distinguished from the long chord or great circle that would connect the initial and terminal points.

lattice—(JCS) A network of intersecting positional lines printed on a map or chart from which a fix may be obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

law of propagation of error—The probable (standard) error of the sum of two or more quantities is equal to the square root of the sum of the squares of their probable (standard) errors.

law of universal gravitation—See **constant of gravitation.**

laydown—Often used to designate a mosaic temporarily assembled from uncropped prints.

layer tints—See **hypsometric tinting.**

layered data—Thematically separated data. The geographic equivalent of third normal form.

layer—(digital mapping) A logical separation of data base information according to theme. Many geographical information systems and CAD/CAM systems allow the user to choose and work on a single thematic layer or any combination of thematic layers at a time.

layout guide—See **key flat.**

layout—The planned positioning of reproduction material to fit the requirements and limitations of lithographic plates, paper, and finishing. Also called lay. See also **flat**, definition 1.

lay—1. See **layout.** 2. To assemble a photomosaic. Often referred to by the method used in assembly, such as **wet lay, staple lay, etc.**

lead line—A long, graduated chain or line at the end of which is attached a lead weight, used to measure depths of water. The lead line is usually used when making soundings by hand in water less than 25 fathoms deep.

leap second—The step adjustment made to UTC to compensate for the approximately 1 second that the time transmitted by UTC signals gains on UT1 or UT2 each year. Normally, UTC is decreased by exactly 1 second (i.e., the leap second) at 24^h on the last day of December and/or June. See also **UT0; UT1; UT2; Coordinated Universal Time.**

leap-frog method—A rapid means of obtaining elevations of stations along a route between two base stations, or to obtain a closed loop of altimeter elevations. The system uses four barometers operating in pairs. One pair of barometers remains at the base station while the other pair is advanced to the first station at which time barometer and weather conditions are read and recorded simultaneously. The original base station pair are advanced to the second station and the process repeated. This method does not produce reliable elevations. The two-base method is considered better.

least count—(micrometer or vernier) The finest reading that can be made directly (without estimation) from a vernier or micrometer.

least squares—The theory of least squares states that in a set of redundant observations, the most probable adjusted values of the observations are obtained when the sum of the squares of the adjustments to the observations to obtain mathematical compatibility is a minimum. A least squares adjustment imposes this condition on a set of redundant and incompatible observations so that the set of equations may be solved uniquely for these most probable values. See also **adjustment of observations.**

left bank—That bank of a stream or river on the left of the observer when he is facing in the direction of flow, or downstream.

Legendre polynomial—A special case of the associated Legendre function in which the function becomes a polynomial.

Legendre's theorem—A mathematical theorem stating that the lengths of the sides of a spherical triangle (very short by comparison with the radius of the sphere) are equal to the lengths of the corresponding sides of a plane triangle in which the plane angles are derived by reducing each of the spherical angles by approximately one-third of the spherical excess.

Lehmann's method—See **triangle-of-error method.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

length correction—(taping) The difference between the nominal length of a tape and its effective length under conditions of standardization. The standard length of a tape is usually expressed by a number of whole units (the nominal length) plus or minus a small distance which is the length correction defined above.

Lenoir level—An instrument which has the telescope passing through steel blocks, one near each end, whose upper and lower faces are plane and closely parallel; the lower faces rest upon a brass circle; the upper faces support a spirit level, which is reversed in leveling the instrument.

lens axis—See **optical axis.**

lens calibration—See **camera calibration.**

lens component—See **lens element.**

lens distortion—(JCS) Image displacement caused by lens irregularities and aberrations.

lens element—One lens of a complex lens system. In a photographic lens, the terms **front element** and **rear element** are often used. Also called **lens component.**

lens speed—See **relative aperture.**

lensatic compass—A type of compass equipped with a lens which permits the observer to read the far side of the movable dial.

lens—A disk of optical glass, or plastic, or a combination of two or more such disks, by which rays of light may be made to converge or to diverge. Such disks have two surfaces, which may both be spherical, one plane and one spherical, or various other combinations (cylindrical, paraboloid, or hyperboloid). See also **achromatic lens; anastigmatic lens; aplanatic lens; apochromatic lens; aspherical lens; astigmatizer; coated lens; compensating lens; convertible lens; cylindrical lens; eyepiece; Fresnel lens; Metrogon lens; narrow-angle lens; negative lens; normal-angle lens; objective lens; positive lens; process lens; spherical lens; superwide-angle lens; thick lens; thin lens; wide-angle lens.**

Level 0 topology—A topological surface which contains a set of entity points and edges. Edges contain only coordinate and attribute information.

Level 1 topology—A topological surface which consists of a set of entity points and edges that meet at nodes. Edges contain start node, end node, next edge, and previous edge information.

Level 2 topology—A topological surface which consists of a set of edges and entity points, when projected on a planar surface, the edges meet only at nodes. Edges contain start node, end node, and next and previous edge information. Edges meet only at nodes.

Level 3 topology—A topological surface which is partitioned by a set of mutually exclusive and collectively exhaustive faces. Edges contain left face, right face, start and end node, and next and previous edge information. Edges meet only at nodes.

level constant—The amount by which the actual line of sight through a leveling instrument (when the bubble is centered in its vial) departs from the truly horizontal line through the center of the instrument, computed in millimeters per millimeter of stadia interval. When leveling rods graduated in yards instead of meters are used, the level constant, C, would be expressed in milliyards per milliyard of stadia interval. Also called **C-constant.**

leveling—**1.** (surveying) The operations of measuring vertical distances, directly or indirectly, to determine elevations. See also **astrogeodetic leveling; astrogravimetric leveling; barometric leveling; differential leveling; direct leveling; first-order leveling; flying levels; geodetic leveling; profile leveling; reciprocal leveling; second-order leveling; spirit leveling; stadia trigonometric leveling; thermometric leveling; third-order leveling; three-wire leveling; trigonometric leveling; vertical angulation; water leveling.**
2. (photogrammetry) In absolute

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

orientation, the operation of bringing the model datum parallel to a reference plane, usually the tabletop of the stereoplotting instrument. Also called **horizontalizing the model; leveling the model.** *See* also **aerolevelling; orientation,** definition 7.

level—A partition within an MC&G data subgroup that is an aggregation of data elements distinguishable by data characteristics (e.g, resolution, scale, accuracy).

library attribute—(digital mapping) Properties of a library. These properties are contained in the Library Header Table, the Library Attribute Table, and other tables related to them.

library negative mold—A negative mold which has been extended to a size compatible with the printed plastic map and forming equipment, and which is kept in file for subsequent castings.

library—(digital mapping) A collection of coverages contained within a specified spatial extent, all of which share a single coordinate system and scale and have a common thematic definition.

libration—A real or apparent oscillatory motion, particularly the apparent oscillation of the Moon, which results in more than half of the Moon's surface being revealed to an observer on the Earth, even though the same side of the Moon is always toward the Earth because the Moon's periods of rotation and revolution are the same.

lift—*See* **selection overlay.**

limb—**1.** The graduated curved part of an instrument for measuring angles, as that part of a marine sextant carrying the altitude scale, or arc. **2.** The circular outer edge of a celestial body. See also **lower limb; upper limb.**

limit of reliable photo coverage—A label placed along a dashed line separating reliable photo compilation from map compilation on a chart. The label is always placed on the photo compilation side of the limit line.

line copy—Any copy suitable for reproduction without using a screen. Copy composed of lines as distinguished from continuous tone copy.

line of levels—A continuous series of measured differences of elevation. The individual measured differences may be single observations in the case of single-run leveling or the means of repeated observations in the case of double-run leveling.

line of nodes—The straight line connecting the two points of intersection of the ecliptic with the orbit of a planet, planetoid or comet; or the line of intersection of the planes of the orbit of a satellite and the equator of its primary.

line of position—A line indicating a series of possible positions, determined by observation or measurement.

line of sight—**1.** The straight line between two points. This line is in the direction of a great circle, but does not follow the curvature of the Earth. **2.** The line extending from an instrument along which distant objects are seen, when viewed with a telescope or other sighting device. Also called **aiming line. 3.** (optics) See **line of collimation.**

line of soundings—A series of soundings obtained by a vessel underway, usually at regular intervals.

line pattern—A photographic negative containing parallel lines of equal-sized widths, which are equally spaced. Line patterns are used for printing tones of a color or to present a pattern of coverage for a chart feature. See also **area pattern screen.**

line rod—See **range rod.**

line thinning (generalization)—**1.** (vector) Reducing the quantity of coordinates necessary to portray the basic shape at a feature by using a series of rules. **2.** (raster) Process whereby a linear feature is represented in a grid by a continuous series of cells, each of which touches along its sides and corners, no more than two other cells belonging to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

feature.

line tree—A tree intersected by a surveyed line, reported in the field notes of the survey, and marked with two hacks or notches cut on each of the sides facing the line. Also called **sight tree.**

line-route map—A map or overlay for signal communication operations that shows the actual routes and types of construction of wire circuits in the field. It also gives the locations of switchboards and telegraph stations.

lineage—Information about the data source, particularly the original scale and accuracy.

lineal convergency—The length by which meridians approach one another when extended from one parallel to another.

linear building frontage—In air photographic interpretation, the side elevation of structures of homogeneous area.

linear distortion—The failure of a lens to reproduce accurately to scale all distances in the object.

linear error of closure—The straight-line distance by which a traverse fails to close.

linear error—A one-dimensional error (such as an error in elevation) defined by the normal distribution function.

linear feature—A feature that is portrayed by a line that does not represent an area. Also called line feature.

linear magnification—The ratio of a linear quantity in the image to a corresponding linear quantity in the object. It may be lateral magnification or longitudinal magnification.

linear parallax—See **absolute stereoscopic parallax.**

linearization—The process of redefining a set of nonlinear equations to a set of approximate linear equations to facilitate solution computations.

lines of communications (LOC)— (JCS) All the routes, land, water, and air, which connect an operating military force with a base of operations, and along which supplies and reinforcements move.

lines on a spheroid—Any direct line between two positions on a spheroid, represented by two points on the Earth. Such a line may be one of mathematical definition, or it may result from a direct survey between the points on the Earth. See also **curve of alignment; geodesic line; normal section line.**

lining pole—See **range rod.**

link and node—A data structure characterized by data in which one or more links are related to one or more features. The beginning and end points of each link are identified by node identifiers either explicitly or implicitly.

link of levels—See **link,** definition 1.

link—**1.** (leveling) A line, a part of a line, or a combination of lines or parts of lines of levels, which, taken as a unit, make a continuous piece of leveling directly from one junction bench mark to another junction bench mark without passing through or over any other junction bench marks. Also called **link of levels. 2.** A unit of linear measure, one one-hundredth of a chain, and equivalent to 7.92 inches. See also **chain. 3.** A line between two consecutive nodes.

liquid hand compass—A type of hand held compass wherein the compass card is damped through the action of a liquid.

List of Lights—The DMA publication which identifies and describes lights and fog signals in foreign waters of the world. This information is also accessible on the Navigation Information Network (NAVINFONET).

list of directions—A listing of objects observed at a triangulation station, together with the horizontal directions in terms of arc of the circle, referred to one of the objects observed as a zero initial.

list—See ***x*-tilt.**

litho copy—See **lithographic copy.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

lithographic copy—A graphic reproduced by the lithographic process. Also called **litho copy**.

lithographic drafting—See **tusching**.

lithography—A planographic method of printing based on the chemical repulsion between grease and water to separate the printing from nonprinting areas. See also **offset lithography; photolithography.**

lithosphere—The solid part of the Earth or other spatial body. Distinguished from the atmosphere and the hydrosphere.

local adjustment—See **station adjustment.**

local apparent time—The apparent solar time for the meridian of the observer.

local astronomic time—Mean time reckoned from the upper branch of the local meridian.

local attraction—See **local magnetic anomaly.**

local chart—A large-scale aeronautical chart designed for contact flight in a congested area.

local civil time—See **local mean time.**

local coordinate system—A right-handed rectangular coordinate system of which the z-axis coincides with the plumb line through the origin .

local datum—The point of reference of the geodetic control used exclusively in a small area. Usually identified by a proper name.

local horizon—See **apparent horizon.**

local hour angle—Angular distance west of the local celestial meridian; the arc of the celestial equator, or the angle at the celestial pole, between the upper branch of the local celestial meridian and the hour circle of a point on the celestial sphere, measured westward from the local celestial meridian through 360°.

local lunar time—The arc of the celestial equator, or the angle at the celestial pole, between the lower branch of the local celestial meridian and the hour circle of the Moon, measured westward from the lower branch of the local celestial meridian through 24 hours; local hour angle of the Moon, expressed in time units, plus 12 hours. See also **Greenwich lunar time.**

local magnetic anomaly—Abnormal or irregular variation of the Earth's magnetic field extending over a relatively small area, due to local magnetic influences. Also called **anomalous magnetic variation; local attraction; local magnetic disturbance; magnetic anomaly.**

local mean time—**1.** (JCS) The time interval elapsed since the mean sun's transit of the observer's antemeridian. **2.** The arc of the celestial equator, or the angle at the celestial pole, between the lower branch of the local celestial meridian and the hour circle of the mean sun, measured westward from the lower branch of the local celestial meridian through 24 hours; local hour angle of the mean sun, expressed in time units, plus 12 hours. Called **local civil time** in United States terminology from 1925 through 1952. See also **local astronomic time; Universal Time.**

local meridian—The meridian through any particular place or observer, serving as the reference for local time. Also called **reference meridian.**

local sidereal time—The local hour angle of the vernal equinox, expressed in time units. Local sidereal time at the Greenwich meridian is called **Greenwich sidereal time.**

local time—**1.** Time based upon the local meridian as reference, as contrasted with that based upon a time zone meridian, or the meridian of Greenwich. **2.** Any time kept locally.

local vertical—The direction of the acceleration of gravity, as opposed to the normal to a reference surface.

location survey—The establishment on the ground of points and lines in positions

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

which have been previously determined by computation or by graphical methods, or by description obtained from data supplied by documents of record, such as deeds, maps, or other sources.

locator's hand level—A hand held type of level used to measure approximate differences in elevation .

locking angle—In tilt analysis of oblique photographs, the complement of the interlocking angle. The depression angle of the oblique when the tilt of the vertical photograph is zero.

logical consistency—(digital mapping) The correctness of the relationships encoded in the data structure of the data set.

logical contouring—A procedure, based on the facts that contours are spaced equally along a uniform slope, which permits the sketching of contours from field notes with considerable accuracy and without the need of running a level line for every contour. Contour lines are interpolated by spacing them proportionately between spot elevations established at every point where there is a change in slope.

long chord—(route surveying) On a simple curve, the chord, or straight line, that extends from the point of curvature to the point of tangency; on a compound curve, the chord that extends from the point of compound curvature to the point of curvature or to the point of tangency. In a description of a circular land boundary, the length and bearing of the long chord is an important factor.

long line azimuth (LOLA) surveys— A measurement by use of photorecording theodolites and airborne strobe lights of long azimuth lines not visible between ground stations.

long rod—A level rod, usually a Philadelphia rod, permitting readings of 13 feet when fully extended. Also called **high rod**. See also **short rod.**

long-period constituent—A tide or tidal current constituent with a period that is independent of the rotation of the Earth but which depends upon the orbital movement of the Moon or of the Earth. The period is usually longer than a day and in general a half-month or larger.

long-period perturbations—Periodic perturbations in the orbit of a planet or satellite which require more than one orbital period to execute one complete periodic variation.

long-range chart—See **long-range navigation chart.**

long-range navigation chart—Any one of a series of small scale, 1:3,000,000 or smaller, aeronautical charts designed for long flights using dead reckoning and celestial navigation as the principal means of navigation. Also called **long-range chart.**

longitude difference—See **departure,** definition 1.

longitude equation—A condition equation which expresses the relationship between the fixed longitudes of two points which are connected by triangulation or traverse.

longitude factor—The change in longitude along a celestial line of position per one minute change in latitude.

longitude of the Moon's nodes—The angular distances along the ecliptic of the Moon's nodes from the vernal equinox; the nodes have a retrograde motion, and complete a cycle of 360° in approximately 19 years.

longitude signal—A sign indicating a time event, observable at different stations, and used in comparing local times of those stations, and determining the difference of their longitudes.

longitude term gravity formula—An additional term in the formula for theoretical gravity which expresses the variation with longitude due to a triaxial ellipsoid of reference. See also **Helmert's gravity formula of 1915.**

longitude—A linear or angular distance measured east or west from a reference meridian (usually Greenwich) on a sphere or spheroid. See also **assumed longitude; astronomic longitude; celestial longitude; circle of**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

longitude; difference of longitude; fictitious longitude; galactic longitude; geodetic longitude; geographic longitude; grid longitude; meridian; oblique longitude; terrestrial longitude; transverse longitude.

longitudinal chromatic aberration—An aberration which affects the sharpness of all parts of an image because different colors come to a focus at different distances from the lens.

longitudinal magnification—The ratio of a length in the image, parallel to the axis, to a corresponding length in the object.

longitudinal separation—Time separation.

longitudinal tilt—See **pitch**, *definition 2.*

look angles—The elevation and azimuth at which a particular satellite is predicted to be found at a specified time. See also **alerts.**

loop closure—(leveling) The difference between the rod sum on the run out and the rod sum on the run back.

loop error of closure—See **error of closure**, definition 5.

loop traverse—A closed traverse that starts and ends at the same station. The traverse provides neither inherent validation of starting position and azimuth, nor validation against systematic distance error. See also **connecting traverse.**

Lorac—A trade name for a hyperbolic radio location system. [This term is derived from the words "long-range accuracy."]

loran chart—A plotting chart on which loran ground wave line of positions and sky wave correction values have been printed, for use in loran navigation.

loran lines—Lines of constant time difference between signals from a master and a slave loran station.

loran tables—Publications containing tabular data for constructing loran hyperbolic lines of position.

LORAN-C Secondary Phase Correction Table—Tabulated secondary phase corrections for LORAN-C in a given service area for each appropriate transmitter pair or station. These corrections account for overland radio wave phase retardation.

loran-C—A long-range radio navigation position fixing system using a combination of time difference of reception and phase difference of signals from two stations to provide a line of position.

loran—(JCS) A long-range radio navigation position fixing system using the time difference of reception of pulse type transmissions from two or more fixed stations. [This term is derived from the words "long-range navigation."]

lorhumb line—A line along which the rates of change of the values of two families of hyperbolae are constants.

lorop photography—A general term referring to any photographs taken with a long focal length (in excess of 100 inches) camera with a narrow-angle lens. [The term is derived from the words "long-range oblique photography."]

lost corner—A corner whose position cannot be determined, beyond reasonable doubt, either from traces of the original marks or from acceptable evidence or testimony that bears on the original position, and whose location can be restored only by reference to one or more interdependent corners.

Lovar tape—A newer version of the Invar tape used in surveying operations. Lovar tape possesses properties and cost factors between that of the less accurate steel tape and the more accurate Invar tape.

Lovar—A steel alloy having a low coefficient of expansion used in construction of precise Lovar tapes. See also **Invar.**

low tide—See **low water.**

low water (LW)—The lowest limit of the surface water level reached by the lowering tide. Low water is caused by the astronomic tide-producing forces and/or the effects of meteorological conditions. Also called **low tide.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

low water datum—An approximation of the plane of mean low water, adopted as a standard datum plane for a limited area, and retained for an indefinite period, even though it might differ slightly from a better determination of mean low water from later observations.

low water full and change—The average interval of time between the transit (upper or lower) of the full or new Moon and the next low water.

low water inequality—See **diurnal inequality.**

low water interval—See **lunitidal interval.**

low water line—The line defined by the boundary of a body of water at its lowest stage (elevation). In tidal waters, the low water line is, strictly, the intersection of the plane of mean low water with the shore.

low water lunitidal interval—See **lunitidal interval.**

low water springs datum—An approximation of the plane of mean low water springs, used as a datum in local areas, and retained for an indefinite period, even though it might differ slightly from a better determination of mean low water springs from later observations.

low water springs—See **mean low water springs.**

low-oblique photograph—See **oblique air photograph.**

lower branch—That half of a meridian or celestial meridian from pole to pole which passes through the antipode or nadir of a place.

lower culmination—See **culmination.**

lower high water (LHW)—The lower of two high waters of any tidal day where the tide exhibits mixed characteristics.

lower high water interval (LHWI)—The interval of time between the transit (upper or lower) of the Moon over the local or Greenwich meridian and the next lower high water. This expression is used when there is considerable diurnal inequality. See also **lunitidal interval.**

lower limb—That half of the outer edge of a celestial body having the least altitude, in contrast with the upper limb, that half having the greatest altitude.

lower low water (LLW)—The lower of two low waters of any tidal day where the tide exhibits mixed characteristics.

lower low water datum—An approximation of the plane of mean lower low water, adopted as a standard datum plane for a limited area, and retained for an indefinite period, even though it might differ slightly from a better determination of mean lower low water from later observations. Used in engineering design of harbor facilities and dredging, when there is a material difference between mean lower low and mean low datums.

lower low water interval (LLWI)—The interval of time between the transit (upper or lower) of the Moon over the local or Greenwich meridian and the next lower low water. This expression is used when there is considerable diurnal inequality. See also **lunitidal interval.**

lower motion—(surveying) The rotation of the lower plate of a repeating instrument.

lower transit—Transit of the lower branch of the celestial meridian. Also called **inferior transit.**

lowest low water springs—A plane of reference approximating the mean lowest low water during syzygy (spring tides).

lowest low water—A plane of reference whose distance below mean sea level corresponds with the mean level of lowest low water of any normal tide. Also called **lowest normal low water.**

lowest normal low water—See **lowest low water.**

loxodrome—A line on the surface of the Earth cutting all meridians at the same angle. Also called a **rhumb line.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

loxodromic curve—See **rhumb line.**

lunar celestial equator—A great circle on the celestial sphere in the plane of the Moon's equator, i.e., in a plane perpendicular to the Moon's axis of rotation.

lunar chart—A chart showing the surface of the Moon.

lunar cycle—Any cycle related to the Moon, particularly the Callippic cycle or the Metonic cycle. See also **saros.**

lunar day—The interval between two successive upper transits of the Moon over a local meridian. The period of the mean lunar day, approximately 24.84 solar hours, is derived from the rotation of the Earth on its axis relative to the movement of the Moon about the Earth. Also called **tidal day.**

lunar declination—Angular distance of the Moon expressed in degrees north or south of the celestial equator; it is indicated as positive when north, and negative when south of the equator. Also called **declination of the Moon.**

lunar distance—The angle between the line of sight toward the Moon and the line of sight toward another celestial body at the point of an observer on the Earth.

lunar earthside chart—A chart showing that portion of the Moon's surface visible from the Earth.

lunar eclipse—The phenomenon observed when the Moon enters the shadow of the Earth. A lunar eclipse is partial if only part of its surface is obscured; and total if the entire surface is obscured.

lunar equation—A factor used to reduce observations of celestial bodies to the barycenter of the Earth-Moon system.

lunar farside chart—A chart showing that portion of the Moon's surface not visible from the Earth.

lunar inequality—**1.** Variation in the Moon's motion in its orbit, due to attraction by other bodies of the solar system. See also **evection; perturbation. 2.** A minute fluctuation of a magnetic needle from its mean position, caused by the Moon.

lunar interval—The difference in time between the transit of the Moon over the Greenwich meridian and a local meridian. The lunar interval equals the difference between the Greenwich and local intervals of a tide or current phase.

lunar month—See **synodical month.**

lunar node—A node of the Moon's orbit.

lunar parallax—The horizontal parallax or the geocentric parallax of the Moon.

lunar satellite—A man-made satellite that makes one or more revolutions about the Moon.

lunar tide—That part of the tide caused solely by the tide-producing forces of the Moon as distinguished from that part caused by the forces of the Sun.

lunar time—**1.** Time based upon the rotation of the Earth relative to the Moon. **2.** Time on the Moon. See also **Greenwich lunar time; local lunar time.**

lunation—See **synodical month.**

lune—That part of the surface of a sphere bounded by halves of two great circles.

lunicentric—See **selenocentric.**

lunisolar effect—Gravitational effects caused by the attraction of the Moon and of the Sun.

lunisolar perturbation—Perturbations of artificial satellite orbits due to the attractions of the Sun and the Moon. The most important effects are secular variations in the mean anomaly, in the right ascension of the ascending node, and in the argument of perigee. All other orbital elements, except the major semiaxis, undergo long periodic changes.

lunisolar precession—That component of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

general precession caused by the combined effect of the Sun and Moon on the equatorial perturbance of the Earth, producing a westward motion of the *equinoxes along the ecliptic.*

lunitidal interval—The interval between the Moon's transit (upper or lower) over the local or Greenwich meridian and the following high or low water. The average of all high water intervals for all phases of the Moon, the mean high water lunitidal interval, is abbreviated to high water interval. Similarly, the mean low water lunitidal interval is abbreviated to low water interval. The high water or low water interval is described as local or Greenwich according to whether the reference is to the transit over the local or the Greenwich meridian. Also called **establishment.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

M

MAD Operational Effectiveness Charts (MOE)—Detailed information on geomagnetic field intensities in ocean areas. Used in planning tactical AntiSubmarine Warfare (ASW) operations. Magnetic anomalies are color coded to assist in setting sensitivity on Magnetic Anomaly Detection (MAD) equipment.

MAD Planning Charts (MADPC)—Information on geomagnetic total-field intensities for selected ocean areas. Used as a planning base for AntiSubmarine Warfare (ASW) operations. Magnetic anomalies are color coded to assist in determining the effectiveness of Magnetic Anomaly Detection (MAD) equipment for a given area.

magazine—(aerial camera) A component in the aerial camera system. It serves to hold the exposed and unexposed film and includes the film drive mechanism and film flattening device (platen).

magenta contact screen—A contact film screen composed of magenta dyed dots of variable density used for making halftone negatives.

magnetic amplitude—Amplitude relative to magnetic east or west. See also **amplitude.**

magnetic annual change—The amount of magnetic secular change undergone in 1 year. Also called **annual change; annual magnetic change; annual rate; annual rate of change.** See also **magnetic secular change.**

magnetic annual variation—The small regular fluctuation in the Earth's magnetism, having a period of 1 year. Also called **annual magnetic variation.**

magnetic anomaly—See **local magnetic anomaly.**

magnetic azimuth—At the point of observation, the angle between the vertical plane through the observed object and the vertical plane in which a freely suspended, symmetrically magnetized needle, influenced by no transient artificial magnetic disturbance, will come to rest. Magnetic azimuth is generally reckoned from magnetic north (0°) clockwise through 360°. Such an azimuth should be marked as being magnetic, and the date of its applicability should be given. Magnetic azimuths are frequently measured with two Wild T-O theodolites in connection with compass rose surveys.

magnetic bearing—**1.** *(navigation)* Bearing relative to magnetic north; compass bearing corrected for deviation. **2.** (surveying) Bearing relative to magnetic north or south, and stated in an east or west direction. The compasses included with the engineer transit and the 1-minute theodolite can be used to read magnetic bearings. Also called **compass bearing.**

magnetic chart—A special purpose map depicting the distribution of one of the magnetic elements, as by isogonic lines, or of its secular change.

magnetic compass—A compass depending upon the attraction of the magnetism of the Earth for its directive force.

magnetic daily variation—The transient change in the Earth's magnetic field associated with the apparent daily motions of the Sun and Moon. In most places the solar daily variation follows approximately a consistent pattern, although with appreciable and unpredictable changes in form and amplitude.

magnetic declination—(JCS) The angle between the magnetic and geographical meridians at any place, expressed in degrees east or west to indicate the direction of magnetic north from true north. In nautical and aeronautical navigation the term **magnetic variation** is used instead of **magnetic declination,** and the angle is termed **variation of the compass** or **magnetic variation.** Magnetic declination is not otherwise synonymous with magnetic variation, which refers to regular or irregular change with time of the magnetic declination, dip, or intensity. [Because of local attraction, the magnetic declination of two close points may differ by several degrees.] See also **magnetic variation.**

magnetic deviation—See **deviation.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

magnetic dip circle—An instrument for measuring the magnetic dip by the use of a needle and a graduated vertical circle.

magnetic dip needle—1. A dip circle or the needle thereof. Also called **dip needle**. 2. A needle arranged to disclose an intense local anomaly of the magnetic dip and useful in the recovery of lost iron survey monuments.

magnetic dip—See **dip**, definition 2.

magnetic direction—Horizontal direction expressed as angular distance from magnetic north.

magnetic disturbance—1. Irregular, large amplitude, rapid time changes of the Earth's magnetic field which occur at approximately the same time all over the Earth. Also called **magnetic storm**. 2. Sometimes used to describe spatial changes in the Earth's magnetic field. See also **local magnetic anomaly**.

magnetic diurnal variation—1. The daily variation. 2. The simple harmonic component of the daily variation having a period of 24 hours.

magnetic elements—The declination, the horizontal intensity, the vertical intensity, the total intensity, the inclination or dip, the strength of the force toward geographic north, and the strength of the force toward geographic east.

magnetic equator—(JCS) A line drawn on a map or chart connecting all points at which the magnetic inclination (dip) is zero for a specified epoch. Also called **aclinic line; dip equator**. See also **geo-magnetic equator**.

magnetic field intensity—The magnetic force exerted on an imaginary unit magnetic pole placed at any specified point of space. It is a vector quantity. Its direction is taken as the direction toward which a north magnetic pole would tend to move under the influence of the field. Also called **magnetic field; magnetic field strength; magnetic force; magnetic intensity**.

magnetic field strength—See **magnetic field intensity** .

magnetic field—See **magnetic field intensity**.

magnetic force—See **magnetic field intensity**.

magnetic inclination—See **dip**, definition 2.

magnetic intensity—See **magnetic field intensity**.

magnetic isoporic line—See **isopor**.

magnetic latitude—See **dip**, definition 2.

magnetic lines of force—Imaginary lines so drawn in a region containing a magnetic field to be everywhere tangent to the magnetic field intensity vector if in vacuum or nonmagnetic material, or parallel to the magnetic induction vector if in a magnetic medium.

magnetic lunar daily variation—A periodic variation of the Earth's magnetic field that is in phase with the transit of the Moon.

magnetic meridian—At any point, the direction of the horizontal component of the Earth's magnetic field. Not to be confused with **geomagnetic meridian**.

magnetic moment—The quantity obtained by multiplying the distance between two magnetic poles by the average strength of the poles.

magnetic north—(JCS) The direction indicated by the north-seeking pole of a freely suspended magnetic needle, influenced only by the Earth's magnetic field. See also **compass north**.

magnetic observation—Measurement of any of the magnetic elements.

magnetic pole—Either of the two places on the surface of the Earth where the magnetic dip is 90°. Not to be confused with **geomagnetic pole**. Also called **dip pole**.

magnetic prime vertical—The vertical circle through the magnetic east and west points of the horizon.

magnetic secular change—Increase or decrease of intensity and/or change of direction of the Earth's magnetic field over a period of many years [usually given as average gammas per year for intensity values and minutes per

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

year for directional values].

magnetic solar daily variation—A periodic variation of the Earth's magnetic field that is in phase with solar (local) time.

magnetic station—A monumented station at which a series of magnetic observations have been made. It usually consists of a bronze marker set in stone or concrete at which, in addition to latitude and longitude, the magnetic value is indicated.

magnetic storm—See **magnetic disturbance.**

magnetic survey—A survey conducted to measure the strength and/or direction of the Earth's magnetic field at specific points on or near the surface of the Earth.

magnetic tape—Ferrous-coated tape. Selective polarization of the surface permits the sequential storage of digital data. Commonly, magnetic tape refers to reel-to-reel tape as opposed to tape enclosed within cartridges.

magnetic variation chart—A chart depicting isogonic lines, or lines of equal magnetic declination, for a given time.

magnetic variation—(JCS)**1.** In navigation, at a given place and time, the horizontal angle between the true north and magnetic north measured east or west according to whether magnetic north lies east or west of true north. Also called **variation.** See also **magnetic declination. 2.** In cartography, the annual change in direction of the horizontal component of the earth's magnetic field.

magnetism—The ability to attract magnetic material, notably iron and steel. Also called **terrestrial magnetism.** See also **blue magnetism; geomagnetism; horizontal intensity; red magnetism; vertical intensity.**

magneto-optical disc—A mass storage media which consists of a transparent rigid substrate, which is coated with a recording layer composed of rare earth and transition metals. Writing is achieved by using a laser to raise the temperature of a small area of the recording layer. This temporarily lowers the coercivity (the strength of the magnetic state), allowing the direction of the magnetic field to be reversed by a small external magnetic field applied by the drive's magnetic head. Erasing is accomplished by the same process, except the applied magnetic field is reversed in direction. Reading of magneto-optical material can be performed either magnetically or optically.

magnetometer survey—A survey wherein the Earth's magnetic field is mapped by the use of a magnetometer. See also **magnetic survey.**

magnetometer—An instrument used in the study of geomagnetism for measuring a magnetic element. See also **flux-gate magnetometer; flux meter; nuclear precession magnetometer; optical pumping magnetometer; theodolite/magnetometer; variometer.**

magnification—(optics) The ratio of the size of an object to the size of its magnified image. Also called **power of a lens.** See also **angular magnification; diopter; lateral magnification; linear magnification; longitudinal magnification; unidimensional magnification.**

magnifying power—The ratio of the apparent length of a linear dimension as seen through an optical instrument, and by the unaided eye. Thus, an instrument with a magnifying power of three makes an object appear three times as high and three times as wide. Also called **diameter.** See also **diopter.**

magnitude—**1.** Relative brightness of a celestial body. Also called **stellar magnitude. 2.** The intensity of a short-period magnetic fluctuation, usually expressed in milligausses or gammas. **3.** Relating to amount, size, or greatness.

main scheme station—A station through which the main computations and adjustments of the survey data are carried and serve for the continued extension of the survey. Also called **primary station; principal station.**

maintainability—The capacity for making updates in an efficient and cost-effective manner to a data base or software.

major axis—The longest diameter of an ellipse or ellipsoid.

major datum—See **preferred datum.**

major grid—The primary grid or grids on a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

map or chart. See also **overlapping grid; secondary grid.**

major planets—The four largest planets; Jupiter, Saturn, Uranus, and Neptune. Also called **giant planets.** See also **asteroid; inner planets; outer planets.**

make line—An accurately scaled line denoting the size to which original copy is to be enlarged or reduced. Also called **make size.** See also **scale of reproduction.**

make ready—The adjustment of feeder, grippers, side guide, pressure between plate and blanket cylinder, impression plate, and ink fountain prior to a press run.

make size—See **make line.**

man-made features—See **culture.**

maneuvering board—A polar coordinate plotting sheet devised to facilitate solution of problems involving relative movement.

manually digitized—The process of converting an analog map or other graphic overlay into numeric format with the use of a digitizing table/tablet and manually tracing the input with a cursor.

manually encoded—The process of data input, including tabular and attribute information, into a computer-compatible format through the use of direct operator interaction with collection and/or storage system peripherals.

manuscript—The original drawing of a map as compiled or constructed frown various data, such as ground surveys and photograph. See also **multiuse manuscript.**

Map and Chart Data Interchange Format (MACDIF)—A project undertaken by the Province of Ontario, Canada, and the national governments of Canada and the USA to develop standards for format, structure, and coding of digital map and chart information. MACDIF provides a coding scheme for encoding a digital description of map or chart data.

map accuracy specifications—Specifications which set up standards to which the finished map must adhere. See also **U.S. National Map Accuracy Standards.**

map accuracy standards—See **U.S. National Map Accuracy Standards.**

map adjustment—An adjustment of the horizontal position of maps to control points or to a specific grid plotted on the map projection at compilation scale.

map catalog—A publication giving both graphical and word descriptions of all maps, charts, and related products issued by a producing agency. It contains information such as the title, scale, edition date, edition number, price, and classification of all publications issued.

map chart—See **combat chart.**

map controlled—Utilization of a map, rather than geodetic or photogrammetric data, for purposes of positioning map detail.

map exchange agreement—An approved agreement between a United States mapping organization and a foreign mapping organization to furnish each other specified mapping, charting, and geodetic data as published, or on a request basis. See also **cooperative mapping agreement.**

map index—(JCS) Graphic key primarily designed to give the relationship between sheets of a series, their coverage, availability, and further information on the series. See also **map.**

map matching guidance—The guidance of a rocket or aerodynamic vehicle by means of a radarscope film previously obtained by a reconnaissance flight over the terrain of the route or from a radar simulation system, and used to direct the vehicle by aligning itself with radar echoes received during flight from the terrain below. See also **stellar map matching.**

map matching—The simultaneous electronic or mechanical optical scanning of an observed map image obtained from a space vehicle, and a reference map image, while the reference map image is being oriented and scaled until a close comparison between the two is found. An inspection of the scale and orientation of the reference map indicates the position of the vehicle. See also **pulse Doppler map matching.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

map nadir—Point on a map or manuscript vertically beneath the perspective center of the camera lens at the instant of exposure.

map of standard format—A map with dimensions, layout, lettering, and symbolization in accordance with the specifications for the series.

map parallel—*See axis of homology.*

map point—A supplemental control point whose horizontal position can be obtained by scaling the coordinates from a map or chart on which the point can be identified.

map projection—An orderly system of lines on a plane representing a corresponding system of imaginary lines on an adopted terrestrial datum surface. A map projection may be derived by geometrical construction or by mathematical analysis. See also **Altoff equal-area map projection; Albers conical equal-map projection; *aphylactic map* projection; authalic map projection; azimuthal equidistant map projection; azimuthal map projection; Bonne map projection; Cassini map projection; Cassini-Soldner map projection; conformal map projection; conic map *projection*; *conic map projection* with two standard parallels; cylindrical equal-area map projection; cylindrical equal-spaced map projection; cylindrical map projection; doubly azimuthal map projection; equal-area map projection; equatorial map projection; globular map projection; gnomonic map projection; Goode's interrupted homolosine projection; Hammer projection; homalographic map projection; interrupted map projection; Laborde map projection; Lambert azimuthal polar map projection; Lambert central equivalent map projection upon the plane of the meridian; Lambert conformal conic map projection; Lambert zenithal equal-area map projection; *Mercator map projection*; meridional orthographic map projection; modified Lambert conformal map projection; modified polyconic map projection; Mollweide homalographic map projection; nonperspective azimuthal map projection; oblique map projection; *oblique Mercator map* projection; orthembadic map projection; *orthographic map projection*; perspective map projection; perspective map projection upon a tangent cylinder; polar map projection; polar orthographic map projection; polar stereographic map projection; polyconic map projection; polyhedric projection; rectangular map projection; rectangular polyconic map projection; simple conic map projection; sinusoidal map projection; skewed map projection; stereographic horizon map projection; stereographic map projection; stereographic meridional map projection; *transverse map projection*; transverse Mercator map projection; transverse polyconic map projection; Werner map projection.**

map reference code—(JCS) A code used primarily for encoding grid coordinates and other information pertaining to maps. This code may be used for other purposes when the encryption of numerals is required.

map revision—See **revision.**

map scale—See **scale**, definition 1.

map series—(JCS) A group of maps or charts usually having the same scale and cartographic specifications, and with each sheet appropriately identified by producing agency as belonging to the same series. Also called **series.**

map sheet—*(JCS) An individual map or chart, either complete in itself or part of a series.* Also called **series.**

map substitute—A hasty reproduction of aerial photographs, photomaps, or mosaics, or of provisional maps, or any other product used in place of a map, when the precise requirements of a map cannot be met.

map test—The accuracy of topographic mapping can be tested by running traverse and level lines across selected areas of any map sheet, and comparing geographic positions of map features with those determined by traverse and comparing interpolated elevations of points from the map with those determined by the level line.

map-controlled mosaic—A technique of constructing mosaics by using topographic maps as the basis for control and orientation purposes. The method may be used in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

preparing both controlled and semicontrolled mosaics although its use is preferred with the latter.

MapGrafix—A vector, Macintosh-based geographic information system developed by ComGrafix, Inc.

MapInfo—A vector, personal computer based geographic information system developed by MapInfo Corporation.

mapping camera—A camera specially designed for oceans. The production of photographs to be used in mapping. The modifier **mapping or surveying** indicates that the camera is equipped with mechanisms to maintain and to indicate the interior orientation of the photographs with sufficient accuracy for mapping purposes. A mapping camera may be an aerial mapping camera or terrestrial mapping camera. Also called **surveying camera.**

mapping photography—Aerial photography obtained by precisely calibrated mapping cameras and conforming to mapping specifications, as distinguished from aerial photography for other purposes. Also called **aerial cartographic photography; cartographic photography; charting photography; survey photography.**

mapping, charting, and geodesy (MC&G)—MC&G comprises the collection, transformation, generation, dissemination, and storing of geodetic, geomagnetic, gravimetric, aeronautical, topographic, hydrographic, cultural, and toponymic data. These data may be used for military planning, training, and operations including aeronautical, nautical, and land navigation, as well as for weapon orientation and target positioning. MC&G also includes the evaluation of topographic, hydrographic, or aeronautical features for their effect on military operations or intelligence. The data may be presented in the form of topographic, planimetric, relief, or thematic maps and graphics; nautical and aeronautical charts and publications; and in simulated, photographic, digital, or computerized formats.

***map*—1.** (JCS) A graphic representation, usually on a plane surface and at an established scale of natural and artificial features on the surface of a part or the whole of the Earth or other planetary body. The features are positioned relative to a coordinate reference system. **2.** To prepare a map or engage in a mapping operation. See also **administrative map; base map; battle map; boundary map; cadastral map; choreographic map; compiled map; contour map; controlled map; county map; distribution map; domestic map; engineering map; experimental map; flight map; flood control map; fluorescent map; forestry map; general map; general-purpose map; geological map; gravimetric map; gravity anomaly map; hemispherical map; hypsographic map (or chart); landscape map; large-scale map; line-route map; medium-scale map; morphographic map; native map; operation map; orthophotomap; orthophotomap; outline map; photocontour map; photorevised map; photocontrol index map; photogrammetric map; photomap; physiographic pictorial map; pictogram; pictomap; planetable map; planimetric base map; planimetric map; planning map; plastic relief map; port plan; provisional map; quadrangle; radar intelligence map; radar map; reconnaissance map; red light readable map; relief model; road map; route map; shaded-relief map; situation map; sketch map; small-scale map; source map; special-purpose map; special job-cover map; standard accuracy map; standard content map; state base map; strategic map; Tactical Commanders' Terrain Analysis; tactical map; topical map; topographic map; Topographic Map of the United States; traffic circulation map; wall map; weather map.**

March equinox—See **vernal equinox.**

margin data—See **marginal data.**

margin information—See **marginal data.**

marginal data—(JCS) All explanatory information given in the margin of a map or chart which clarifies, defines, illustrates, and/or supplements the graphic portion of the sheet. Also called **border data; border information; margin information.**

margin—(JCS) In cartography, the area of a map or chart lying outside the border.

marine map—See **hydrographic chart.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

marine sextant—A sextant designed primarily for marine navigation.

marine survey—See **oceanographic survey.**

MARK 85— MARK 85 is the first of two development phases of the Defense Mapping Agency's Digital Production System (DPS). Its mission was to improve hardcopy production methods, generate improved production management and data base management capabilities, and provide initial softcopy production capability. The six segments include: Hardcopy Exploitation (HE/S), Source Acquisition (SA/S), Data Integration (DI/S), Feature Extraction (FE/S), Universal Rectifier (UR/S), and Digital Comparator (DC/S). See also **Digital Production System; MARK 90.**

MARK 90—MARK 90 is the second of two development phases of the Defense Mapping Agency's Digital Production System (DPS). Its mission is to provide DMA with end-to-end softcopy production capabilities. MARK 90 incorporates the Data Integration, Source Acquisition, Hardcopy Exploitation, and the Universal Rectifier Segments of the MARK 85 phase with new Production Management, Data Services, Source Preparation, Data Extraction, and Product Generation Segments. See also ***Digital Production System; MARK 85.***

mark—**1.** A call used when simultaneous observations are being made, to indicate to the second person the moment a reading is to be made, as when the time of a celestial observation is to be noted; or the moment a reading is a prescribed value. **2.** (surveying) A definite object, such as an imprinted metal disk, used to designate a survey point and sometimes refers to the entire survey monument. **Mark** is used with a qualifying term such as **station, reference,** or **bench.** See also **bench mark; reference mark; station mark; witness mark. 3.** (photogrammetry) See **floating mark; index mark.**

Marsden chart—A chart showing the distribution of meteorological data, especially over the oceans.

masking paper—See **goldenrod paper.**

masking—A means of controlling plastic expansion locally during forming of plastic relief maps to obtain more accurate register of preprinted line work to the landforms of the mold. By masking, differential heating is achieved. Also called **screening.**

mask—**1.** In photomechanical processing, to block out an area by means of actinically opaque material, to prevent exposure in the part blocked out. Also, the covering material itself when so applied. **2.** A clear stable base plastic, coated with an opaque stratum which can be peeled off between photographically etched outline images, thus producing an open-window negative of the desired area. This process of masking is often identified by the trade name of the material used. **3.** A continuous tone positive or negative made from an original negative or positive for the purpose of altering the image produced from the original. Used to alter contrast, correct color portrayal, or produce pictotone or pictoline images.

mass attraction vertical—The normal to any surface of constant geopotential. On the Earth this vertical is a function only of the distribution of the mass and is unaffected by forces resulting from the motions of the Earth; e.g., the direction of a plumb bob on a nonrotating Earth.

master film positive—A positive made from an original negative for the purpose of making additional negatives.

master glass negative—See **calibration plate.**

master model—The developed original terrain model which bears, in miniature, the same spatial relationships as the actual ground it represents. Also called **original model.**

master plot—(JCS) A portion of a map or overlay on which are drawn the outlines of the areas covered by an air photographic sortie. Latitude and longitude, and map and sortie information are shown.

master print—(mosaicking) A photograph which is representative of the mosaic area used as a guide during the developing process to insure the tonal match of subsequent prints.

master projection—The originally computed and constructed map projection from which copies are made; one such projection serves as the master for copies circling the globe within the same set of standard parallels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

master station—That station in a given system of transmitting stations that controls the transmissions of the other stations (the slave stations) and maintains the time relationship between the pulses of the stations. In satellite surveys, positions can be upgraded considerably by translocation. See also **translocation.**

match lines—A series of grease pencil lines drawn on a photograph, radiating from the torn edges of the print onto the adjacent areas to serve as a registration guide when laying the individual print in the mosaic.

match strip—See **tie strip**, definition 1.

matching—The act by which detail or information on the edge, or overlap area, of a map or chart is compared, adjusted, and corrected to agree with the existing overlapping chart. Also called **edge matching.**

matrix data—Data which consists of a matrix of evenly spaced rows and columns of data points. The position within the rows and columns represents the geographic position, while the data point is the value of some spatial variable at that position., An example is DTED, which provides elevations at nominal 100 meter horizontal intervals.

matte print—Print made on photographic paper with a dull finish which is more suitable for pencil or ink annotations than a glossy print.

maximum elevation figure—(JCS) A figure, shown in each quadrangle bounded by ticked graticule lines on aeronautical charts, which represents the height in thousands and hundreds of feet, above mean sea level, of the highest known natural or man-made feature in that quadrangle, plus suitable factors to allow for inaccuracy and incompleteness of the topographical heighting information.

MC&G utility software environment (MUSE)—Software programs which perform general purpose operations on Mapping, Charting & Geodetic (MC&G) data that are applicable to a broad range of systems and users rather than a single, special purpose. Some examples include datum and projection transformations, coordinate conversions, and symbol libraries.

mean anomaly—See **anomaly**, definition 3.

mean center of Moon—**1.** A central point for a lunar coordinate system. **2.** The point on the lunar surface intersected by the lunar radius that is directed toward the Earth's center when the Moon is at the mean ascending node and when the node coincides with the mean perigee or mean apogee.

mean chart—Any chart on which isopleths of the mean value of a given oceanographic element are drawn. Also called **mean map.**

mean deviation—See **average deviation.**

mean distance—See **semimajor axis.**

mean diurnal high water inequality (DHQ)—Half the average difference between the heights of the two high waters of each tidal day over a 19-year period, or a computed equivalent period, obtained by subtracting the mean of all high waters from the mean of the higher high waters.

mean diurnal low water inequality (DLQ)—Half the average difference between the heights of the two low waters of each tidal day over a 19-year period, or a computed equivalent period, obtained by subtracting the mean of the lower low waters from the mean of all low waters.

mean equinox of date—See **mean equinox.**

mean equinox—A fictitious equinox whose position is that of the vernal equinox at a particular date with the effect of nutation removed. Also called **mean equinox of date.**

mean error—The algebraic mean of all errors derived by comparing observed or computed quantities with all fixed values of common dimensional parameters in an integral observation system or computation model. A mean error differing significantly from zero indicates a bias in the system or model that should be removed.

mean free-air anomaly—The representative free-air gravity value for a given geographic area; e.g., 1' x 1' mean, 10' x 10' mean, etc.

mean ground elevation—Average elevation

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

of the terrain above mean sea level of an area to be photographed.

mean high water (MHW)—The average height of all the high waters recorded over a 19-year period, or a computed equivalent period.

mean high water lunitidal interval—See **lunitidal interval.**

mean high water springs (MHWS)— The average height of all high waters recorded during syzygy over a 19-year period, or a computed equivalent period. Also called **high water springs.**

mean higher high water (MHHW)—The average height of all the daily higher high waters recorded over a 19-year period or a computed equivalent period. It is usually associated with a tide exhibiting mixed characteristics.

mean higher high water springs (MHHWS)—The average height of all higher high waters recorded during syzygy over a 19-year period, or a computed equivalent period.

mean low water (MLW)—The average height of all low waters recorded over a 19-year period, or a computed equivalent period.

mean low water lunitidal interval—See **lunitidal interval.**

mean low water springs (MLWS)—The average height of all low waters recorded during syzygy over a 19-year period, or a computed equivalent period.

mean lower low water (MLLW)—The average height of all the lower low waters recorded over a 19-year period, or a computed equivalent period. It is usually associated with a tide exhibiting mixed characteristics.

mean lower low water springs (MLLWS)—The average height of all lower low waters recorded during syzygy over a 19-year period, or a computed equivalent period.

mean map—See **mean chart.**

mean of the errors—The average value of a set of errors.

mean place—See **mean position.**

mean position—The position of a star corrected for secular variations including proper motion, but uncorrected for short term variations. Also called **mean place.**

mean range (Mn)—The difference in height between mean high water and mean low water, measured in feet or meters.

mean refraction—The refraction effect on vertical angles given usually in the plane of a vertical circle for average conditions of temperature and barometric pressure.

mean river level—The average height of the surface of a river at any point for all stages of the tide over a 19-year period, or a computed equivalent period, usually determined from hourly height readings. Unusual variations of river level due to discharge or runoff may be excluded in computation.

mean sea level (MSL)—(JCS) The average height of the surface of the sea for all stages of the tide, used as a reference for elevations. [Usually determined by averaging height readings observed hourly over a minimum period of 19 years.] Also called **sea level datum.**

mean sidereal time—Sidereal time adjusted for nutation, to eliminate slight irregularities in the rate.

mean solar day—The interval of time from a transit of the mean sun across a given meridian to its next successive transit across the same meridian.

mean solar time—Time measured by the diurnal motion of a fictitious body, termed the **mean sun**, which is supposed to move uniformly in the celestial equator, completing the circuit in 1 tropical year. The mean sun may be considered as moving in the celestial equator and having a right ascension equal to the mean celestial longitude of the true sun. Also called **mean time.**

mean sphere depth—The uniform depth to which the water would cover the Earth if the solid surfaces were smoothed off and were parallel to the surface of the geoid.

mean square error—The quantity whose square is equal to the sum of the squares of the individual errors divided by the number of those

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

errors.

mean sun—See **fictitious sun.**

mean tide level (MTL)—The reference plane midway between mean high water and mean low water. Also called **half tide level; ordinary tide level.**

mean time—See **mean solar time.**

meander corner—A corner established at the intersection of standard, township, or section lines with the meander line near banks of navigable streams or any meanderable body of water.

meander line—A traverse of the margin of a permanent natural body of water, along the locus of the bank or shoreline at the elevation of mean or ordinary high water, upon which bank or shoreline a riparian right may be predicated.

measured angle—An angle as read directly from an instrumental observation and without any application of corrections for local conditions. A measured angle which has been corrected for local conditions only at the point of observation is considered as an observed angle.

mechanical arm templet—See **spider templet.**

mechanical-templet plot—See **mechanical-templet triangulation.**

mechanical-templet triangulation—A graphical radial triangulation using slotted, spider, or any form of mechanical templet. Also called **mechanical-templet plot.**

mechanical-templet—Any templet which is manipulated and adjusted mechanically in laying out a radial triangulation.

medium-scale map—(JCS) A map having a scale larger than 1:600,000 and smaller than 1:75,000. *See also* **map.**

megabyte (MB)—A unit of memory. One megabyte equals 1,048,576 bytes or approximately one million bytes.

memorial—A durable article deposited in the ground at the position of a corner to perpetuate that position should the monument be removed or destroyed. The memorial is usually deposited at the base of the monument and may consist of anything durable such as glass or stoneware, a marked stone, charred stake, or a quantity of charcoal.

Mendenhall pendulum—An invariable pendulum, one-quarter meter in length, with a vibration period of one-half second, composed of a lenticular-shaped bob on a thin stem, swung in an airtight case from which the air has been largely exhausted.

mensuration—**1.** The act, process, or art of measuring. **2.** That branch of mathematics dealing with the determination of length, area, or volume.

Mercator bearing—See **rhumb bearing.**

Mercator chart—A chart on the Mercator projection. This is the chart commonly used for marine navigation. Also called **equatorial cylindrical orthomorphic chart.**

Mercator direction—Horizontal direction of a rhumb line, expressed as angular distance from a reference direction. Also called **rhumb direction.**

Mercator equal-area map projection—See **sinusoidal map projection.**

Mercator map projection—A conformal map projection of the cylindrical type. The Equator is represented by a straight line true to scale; the geographic meridians are represented by parallel straight lines perpendicular to the line representing the Equator; they are spaced according to their distance apart at the Equator. The geographic parallels are represented by a second system of straight lines perpendicular to the family of lines representing the meridians and therefore parallel with the Equator. Conformality is achieved by mathematical analysis, the spacing of the parallels being increased with increasing distance from the Equator to conform with the expanding scale along the parallels resulting from the meridians being represented by parallel lines. Also called **equatorial cylindrical orthomorphic map projection.**

Mercator track—See **rhumb line.**

Mercury datum—A worldwide geodetic system derived from an analysis of data from astrogeodetic, gravimetric, and satellite sources. Results of this analysis provided a best-fitting

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

world ellipsoid used in tracking Project Mercury manned space missions and as the reference datum for the electronic navigation systems—Omega, loran, and loran C. See also **Fischer ellipsoid of 1960.**

mercury barometer—A barometer in which atmospheric pressure is balanced against the weight of a column of mercury. See also **aneroid barometer; cistern barometer; siphon barometer.**

merge—To take two or more maps or data sets and combine them together into a single, coherent map or data base without redundant information.

merging—The process of joining two or more overlapping data sets.

meridian altitude—The altitude of a celestial body when it is on the celestial meridian of the observer.

meridian angle difference—The difference between two meridian angles, particularly between the meridian angle of a celestial body and the value used as an argument for entering a table. Also called **hour angle difference.**

meridian angle—Angular distance east or west of the local celestial meridian; the arc of the celestial equator, or the angle at the celestial pole, between the upper branch of the local celestial meridian and the hour circle of a celestial body, measured eastward or westward from the local celestial meridian through 180°, and labeled "E" or "W" to indicate the direction of measurement.

meridian distance—1. (astronomy) The hour angle of a celestial body when close to but not exactly on the astronomic meridian. 2. (plane surveying) The perpendicular distance in a horizontal plane of a point from a meridian of reference. The difference of the meridian distances of the ends of a line is called the **departure of the line.**

meridian extension—That portion of a meridian shown above the top construction line of a projection.

meridian line—(plane surveying) The line of intersection of the plane of the celestial meridian and the plane of the horizon. It is a horizontal direction used in surveying. Its astronomic azimuth is 0° or 180°.

meridian observation—Measurement of the altitude of a celestial body on the celestial meridian of the observer, or the altitude so measured.

meridian passage—See **meridian transit.**

meridian telescope—A portable instrument so designed that it can be used as an astronomic transit, or quickly converted for use as a zenith telescope.

meridian transit—The passage of a celestial body across a celestial meridian. Also called **meridian passage.** See also **culmination; transit,** *definition* 1.

meridian—A north-south reference line, particularly a great circle through the geographical poles of the Earth, from which longitudes and azimuths are determined; or a plane, normal to the geoid or spheroid, defining such a line. See also **astronomic meridian; auxiliary guide meridian; celestial meridian; central meridian; convergence of meridians; double meridian distance; ecliptic meridian; fictitious meridian; geodetic meridian; geographic meridian; geomagnetic meridian; Greenwich meridian; grid meridian; guide meridian; gyro meridian; local meridian; magnetic meridian; meridional difference; meridional part; meridional plane; oblique meridian; photograph meridian; prime fictitious meridian; prime grid meridian; prime meridian; prime oblique meridian; prime transverse meridian; principal meridian; standard meridian; table of meridional parts; time meridian; transverse meridian; true meridian.**

meridional difference—The difference between the meridional parts of any two given parallels. This difference is found by subtraction if the two parallels are on the same side of the Equator, and by addition if on opposite sides.

meridional distance—The distance between latitude lines as determined from the mid-latitude of a map projection.

meridional interval—The value of the distance between meridians of a projection at chart scale.

meridional offsets—Small distances applied

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

to the lengths of meridians in order to create the curves of the top and bottom latitudes of a projection.

meridional orthographic map projection—A map projection having the plane of the projection parallel to the plane of some selected meridian; the geographic parallels and the central meridian are straight lines, the outer meridian is a full circle, and the other meridians are arcs of ellipses.

meridional part—The length of the arc of a meridian between the Equator and a given parallel on a Mercator chart, expressed in units of 1' of longitude at the Equator.

meridional plane—Any plane containing the polar axis of the Earth. See also **astronomic meridian plane; geodetic meridian plane.**

meta data base—**1.** Information about information, e.g. a list or catalog of directories. **2.** Information about data, e.g. a directory of data storage locations. See also **Data base** and **Data dictionary**.

metadata—Information about information: more specifically, information about the meaning of other data.

meteorological chart—Any chart showing meteorological (weather) information.

meter rod—See **precise leveling rod.**

metes-and-bounds survey—A method of describing the boundaries of tracts of land by giving the bearing and length of each successive line. Much of the land in the nonpublic land States has been surveyed and described by this method. This method is also used in the surveys of the public lands to define the boundaries of irregular tracts, such as claims, grants, and reservations, which are non-conformable to the rectangular system of subdivision.

method of repetitions—The determination of the angle between two marks by accumulating, on the horizontal circle of a repeating theodolite, the sum of a series of measurements of the horizontal angle between the two marks.

Metonic cycle—A period of approximately 19 years, or a computed equivalent, during which all phase relationships between Moon, Sun, and Earth occur, and after the lapse of which the phases of the Moon return to a particular date in the calendar year. During any cycle, new and full Moon will recur on approximately the same day in the calendar year.

metric camera—A specially constructed and calibrated camera used to obtain geometrically accurate photographs for use in photogrammetric instruments.

metric mapping support data (MMSD)—Camera model parameters which allow proper geographic positioning of features seen on imagery based on sensor position and pointing angle relative to the earth's geoid.

metric photography—The recording of events by means of photographs, either singly or sequentially, together with appropriate coordinates, to form the basis for accurate measurements.

Metrogon lens—The trade name of a wide-angle lens for aerial cameras used in mapping, charting, and reconnaissance photography. See also **trimetrogon camera.**

microfeatures—Features of relief, drainage, and landforms which can be identified on photographs, but are too small to appear on maps

micrometer method—The determination of the astronomic azimuth of a line by measuring indirectly with an ocular micrometer attached to a theodolite or transit the horizontal angle between a selected star at its elongation and a suitable ground mark (light) placed close to the vertical plane which passes through the star, and applying that angle to the azimuth of the star computed for the epoch of the observation.

micrometer—An auxiliary device to provide measurement of very small angles or dimensions by an instrument such as a telescope. See also **filar micrometer; ocular micrometer; transit micrometer.**

mid-latitude—**1.** See **middle latitude. 2.** (cartography) The one parallel that is at the same scale as indicated on a Mercator projection.

middle latitude—One-half of the arithmetical sum of the latitudes of two places on the same side of the Equator. Middle latitude is labeled

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

"N" or "S" to indicate whether it is north or south of the Equator. Also called **mid-latitude.**

middle ordinate—*The distance from the* middle point of a chord to the middle point of the corresponding circular arc.

middle point (MP)—That point on a circular curve which is equidistant from the two ends of the curve.

middletone—In halftone, any neutral tone intermediate between the highlights and shadows of an original and the resulting reproduction. Also, the tones in a reproduction between the highlights and the shadows.

mileage chart—A chart showing distances between various points.

military city map—See **city products.**

military geography—(JCS) The specialized field of geography dealing with natural and man-made physical features that may affect the planning and conduct of military operations.

military grid reference system (MGRS)—1. The alphanumeric position reporting system used by U.S. Military. 2.(JCS) A system which uses a standard-scaled grid square, based on a point of origin on a map projection of the Earth's surface in an accurate and consistent manner to permit either position referencing or the computation of direction and distance between grid positions. See also **military grid.**

military grid—(JCS) Two sets of parallel lines intersecting at right angles and forming squares; the grid is superimposed on maps, charts, and other similar representations of the Earth's surface in an accurate and consistent manner to permit identification of ground locations with respect to other locations and the computation of direction and distance to other points. See *also* ***military grid reference system.***

military installation map (MIM)—DMA maps constructed from one or more standard series maps and overprinted with information such as fire zones, training areas and range facilities. MIMs are constructed only for major camps, posts, and areas which are used extensively for training and research. Produced at 1:50,000 scale.

military level—A compact ruggedized version of the dumpy level developed specifically for military use. It is primarily used for third-order leveling, but has a second-order capability.

military standard **(MIL-STD)**—In accordance with the basic policy of the Department of Defense, Defense Standardization and Specification Program, military standards are issued for the comprehensive presentation of engineering practices (including test methods), procedures, processes, codes, safety requirements, symbols, abbreviations, nomenclatures, type designations and characteristics for standard equipments or items, either singly or in families. Military standards are also used to cover overall characteristics of families of end items or major components. These characteristics include, as applicable, envelope dimensions, performance ratings, primary structural features, and data required for interchangeability of components.

millemap—A quantitative distribution map on which there are 1,000 dots, representing the quantity depicted; each dot therefore represents 1/1,000 of the total, and is located as accurately as possible according to the available evidence.

milligal—A unit of acceleration equal to 1/1,000 of a gal, or 1/1,000 centimeter per second per second. This unit is used in gravity measurements, being approximately one-millionth of the average gravity at the Earth's surface. Such measurement includes the component of centrifugal acceleration in the direction of the gravitational acceleration.

milligauss—Unit of magnetic force equal to 0.001 gauss (oersted) or 100 gammas.

mine survey—A survey to determine the positions and dimensions of underground passages of a mine; also, of the natural and artificial features (surface and underground) relating to the mine. The data include both horizontal and vertical positions, lengths, directions, and slopes of tunnels; topographic and geologic characteristics of the particular vicinity; ownership of the land and of the mine.

mineral survey—A survey made to mark the legal boundaries of mineral deposits or ore-bearing formations on the public domain, where the boundaries are to be determined by lines other than the normal subdivision of the public lands.

Minimally Redundant Topology (MINI-TOPO)—The internal digital spatial data

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

structure of the DMA Digital Production System. MINI-TOPO consists of two overlapping data structures: cartographic and spatial. The cartographic structure contains all attribute information, relationships between features and feature components, and ties to the underlying spatial structure level. The spatial structure associates the cartographic features to their exact geographic placement. Also known as the Mapping, Charting, & Geodesy Feature Data Exchange Structure (MCGFDES).

minimum bounding rectangle (MBR)—The smallest possible rectangle, with North-South and East-West sides, which encloses an area of interest.

minor axis—The shortest diameter of an ellipse or ellipsoid.

minor control plot—See **radial triangulation.**

minor control—See **photogrammetric control.**

minor planets—See **asteroid.**

minus angle—See **angle of depression.**

minus declination—See **declination,** definition 3.

minus sight—See **foresight,** definition 3.

minute of standard length—The length of 1 minute of longitude at the Equator. The length is variable, depending on the dimensions of the particular ellipsoid (spheroid) used as a reference surface.

mirror image—See **reverted image.**

miscellaneous chart—A chart other than a regular navigational chart; a special chart.

mismatch—A condition which occurs when detail is displaced and perfect matching cannot be achieved.

missile launch site data card— A standardized form containing launcher geodetic information which has been produced on the current world geodetic system. Used in conjunction with the missile target data card.

missile target data—Precise geodetic target positioning data required to support strategic and tactical weapon systems. See also **point position data.**

missing triangle—(pendulum) A triangle which represents the failure of the two sides of a knife edge to reach a perfect intersection in a geometric line.

mistake—See **blunder; error.**

mock-up—See **style sheet.**

model coordinates—(photogrammetry) The space coordinates of any point imaged in a stereoscopic model which define its position with reference to the air base or to the instrument axes.

model datum—**1.** (photogrammetry) That surface in a stereoscopic model conceived as having been reconstructed as part of the model representing the sea level datum of nature. Often modified to designate the type of photography used, such as convergent model datum, and transverse model datum. **2.** For relief maps or models the datum may or may not be sea level but is consistent within a relief map series.

model marriage—The rejoining of sections of a model, after the carving operation, to the original neatline limits of a relief model.

model scale—(photogrammetry) The relationship which exists between a distance measured in a stereoscopic model and the corresponding distance on the Earth.

modeling—The development of the model surface by the application of modeling clay between the step edges of the step cast. A preparatory step in producing relief models.

model—See **airborne landing model; assault landing model; assault models; flat model; gross model; half model; master model; neat model; perspective spatial model; relief model; stereoscopic image; stereoscopic model; strategic planing model; tactical planning models; terrain model; warped model.**

modified facsimile chart—A nautical chart published by a foreign hydrographic agency and reproduced by DMA. Specific authority to reproduce is contained in a bilateral agreement,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

which will also identify any qualifications to the permission to reproduce.

modified Julian day—An abbreviated form of the Julian day which requires fewer digits and translates the beginning of each day from Greenwich noon to Greenwich midnight; obtained by subtracting 2400000.5 from Julian days.

modified Lambert conformal chart—A chart on the modified Lambert conformal map projection. Also called **Ney's chart.**

modified Lambert conformal map projection—A modification of the Lambert conformal projection for use in polar regions, one of the standard parallels being at latitude 89°59'58" and the other at latitude 71° or 74°, and the parallels being expanded slightly to form complete concentric circles. Also called **Ney's map projection.**

modified polyconic map projection—A map projection obtained from the regular polyconic projection by so altering the scale along the central meridian that the scale is exact along two standard meridians, one on either side of the central meridian and equidistant therefrom. Also called **rectangular polyconic map projection.**

modulation error—In electronic distance measuring equipment, the difference in modulating frequencies obtained from crystals, between the actual frequencies of the crystals and the frequencies required for a correct distance measurement

modulation—A variation of some characteristic of a radio wave, called the **carrier wave**, in accordance with instantaneous values of another wave called the **modulating wave**. These variations can be amplitude, frequency, phase, or pulse.

moiré—An interference pattern resulting from the overlaying or overprinting of halftones or tints whose screen angles are not sufficiently separated to make the pattern inconspicuous or to preclude a pattern accuracy.

mold alterations—The slight modification of the landforms of the mold, often necessary in local areas in order to obtain proper register due to unequal stretch required to accommodate the landforms on a plastic relief map.

molded aerial photograph—A vertical aerial photograph, usually annotated with military symbols, which has been formed to show terrain configuration.

Molitor precise leveling rod—A speaking rod of T-shaped cross section, with graduation marks shaped as triangles and rectangles, the smallest division being two millimeters. Read by estimation to single millimeters. Equipped with thermometer and circular level.

Mollweide homalographic map projection—An equal-area map projection showing the Equator and geographic parallels as straight lines, and the geographic meridians as elliptical arcs, with the exception of the central meridian, represented by a straight line, and the meridian 90° from the center, shown as a full circle.

moment of inertia—The quantity obtained by multiplying the mass of each small part of a body by the square of its distance from an axis, and adding all the results.

momentum—Quantity of motion. Linear momentum is the quantity obtained by multiplying the mass of a body by its linear speed. Angular momentum is the quantity obtained by multiplying the moment of inertia of a body by its angular speed.

moment—A tendency to cause rotation about a point or axis, as of a control surface about its hinge or of an airplane about its center of gravity; the measure of this tendency is equal to the product of the force and the perpendicular distance between the point of axis of rotation and the line of action of the force.

monochromator—A dispersive device for isolation of narrow portions of the spectrum.

monochrome—A single hue or color.

monocomparator—A precision instrument, consisting of a measuring system, a viewing system, and a readout system designed for the measurement of image coordinates on a single photograph.

monoscopic (mono) imaging—A single image taken of the target.

monoscopic revision—Extraction of image

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

identifiable feature data from a single image. No effort is made to compensate for relief displacement or tilt. Existing center line data (CLD) feature and elevation data are registered with the imagery. New CLD features may be added and existing CLD features may be deleted or modified spatially or attributionally.

month—The period of the revolution of the Moon around the Earth. The month is designated as sidereal, tropical, anomalistic, nodical or dracontic, or synodical, according to whether the revolution is relative to the stars, the vernal equinox, the perigee, the ascending node, or the Sun. See also **anomalistic month; calendar month; nodical month; sidereal month; synodical month; tropical month.**

monumented bench mark—See **permanent bench mark.**

monument—**1.** A structure used or erected to mark the position of a survey station; permanence is implied. See also **artificial monument; natural monument. 2.** A physical structure, such as an iron post, marked stone, or tree in place, which marks the location of a corner point established by a cadastral survey. Objects, to be ranked as monuments, should have certain physical properties such as visibility, durability, and stability, and they must define location without resorting to measurements. **Monument** and **corner** are not synonymous, although the two terms are often used largely in the same sense. See also **corner.**

Moon position camera method—A means of determining geodetic position, that is unaffected by deflection of the vertical, by photographing the Moon against a star background.

morphographic map—A small-scale map showing physiographic features by means of standardized pictorial symbols, based on the appearance such features would have if viewed obliquely from the air.

mosaicking board—A smooth-surfaced material, usually tempered Masonite, to which the mosaic is fastened with a suitable adhesive.

mosaic—**1.** (JCS) (photogrammetry) An assembly of overlapping aerial photographs which have been matched to form a continuous photographic representation of a portion of the Earth's surface. Also called **aerial mosaic.** See also **controlled mosaic; map-controlled mosaic; orthophotomosaic; scale-ratio mosaic; semicontrolled mosaic; strip mosaic; uncontrolled mosaic. 2.** (cartography) See **panel base.**

most probable value—That value of a quantity which is mathematically determined from a series of observations and is more nearly free from the effects of errors than any other value that might be derived from the same series of observations. Derivation of the most probable value is made after blunders and systematic errors have been removed from the data.

moving average—See **consecutive mean.**

multiplex—A name applied to anaglyphic double-projection stereoplotters with the following characteristics: (1) the stereomodel is projected from diapositives reduced from aerial photograph negatives; (2) the projection system illuminates the entire diapositive format area; and (3) the stereomodel is measured and drawn by observation of a floating mark.

multiband photography—A remote sensing system which produces more than one image of a single area in which each image shows a different wavelength band of the electromagnetic spectrum.

multibeam survey—A hydrographic survey conducted with an array of several or many transducers mounted on the sounding vessel, designed to take simultaneous soundings on a wide swath of the ocean bottom.

multicolor—Two or more colors. Also called **polychrome.** See also **process color printing.**

multinational data base—A standardized data base operated and maintained under the control of two or more participating member nations.

multiple level line—Two or more single lines of spirit leveling run between the same terminal points, but along different routes.

multiple-camera assembly—An assembly of two or more cameras mounted to maintain a fixed angle between their respective optical axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

multiple-lens camera—A camera with two or more lenses, with the axes of the lenses systematically arranged at fixed angles in order to cover a wide field by simultaneous exposures in all chambers.

multiple-lens photograph—A photo- graph made with multiple-lens camera.

multiple-stage rectification—A technique employing standard equipment to rectify oblique photos by applying a series of projections to effect the desired projective transformation.

multiplex control—See **photogrammetric control.**

multiplex triangulation—See **stereo-triangulation.**

multispectral sensing—Employment of one or more sensors to obtain imagery from different portions (bands) of the electromagnetic spectrum.

multispectral—Remote sensing in two or more spectral bands, such as visible and infrared See also **infrared; remote sensing.**

multiuse manuscript (MUM)—A manuscript compilation that, as a minimum, establishes the contours, spot elevations and includes the horizontal position of the significant planimetric features. It is suitable for use in completing a topographic map, or an aeronautical or nautical chart; and the integrity of its horizontal and vertical accuracy is retained in all end products made from it.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

N

nadir point—See **photograph nadir.**

nadir radial—A radial from the nadir point.

nadir-point plot—See **nadir-point triangulation.**

nadir-point triangulation—Radial triangulation in which nadir points are utilized as radial centers. Also called **nadir-point plot.**

nadir—(JCS) That point on the celestial sphere directly beneath the observer and directly opposite the zenith. See also **ground nadir; map nadir; photograph nadir.**

nanotesla—(geomagnetism) A unit of magnetic field intensity generally used in describing the Earth's magnetic field. It is defined as 10^{-9} tesla = 1 gamma. See also **gamma.**

narrow-angle lens—A lens whose focal *length is equal approximately to twice the* diagonal of the format.

National Geodetic Vertical datum of 1929—Known as "sea level datum of 1929" prior to September 1973, this datum was established by constraining the combined interconnected United States and Canadian networks of first-order *leveling, as it existed in* 1929.

national data base—A data base designed, operated and maintained by any nation for purposes specific to that nation.

national map accuracy standards—See **U.S. National Map Accuracy Standards.**

native map—A map of any country produced by that country's governmental or private agencies.

natural detail—The features on the Earth, such as streams, lakes, forests, and mountains; exclusive of the works of man. Also called **natural feature.** See also ***culture; hydrographic detail;*** ***hypsographic detail.***

natural error—Errors arising from variations in temperature, humidity, wind, gravity, *refraction, and magnetic* declination.

natural feature—See **natural detail.**

natural monument—A natural feature, such as a stream, boulder, tree, etc., which serves to mark the location of a survey station or land corner. See also **monument.**

nautical broadcasts—Radio navigational warning broadcasts to shipping on urgent dangers to navigation, advanced information on changing navigational conditions, and special warnings disseminated by official U.S. Government proclamations affecting shipping. This information is also accessible on the *Navigation Information System* (NAVINFONET).

nautical chart—See **hydrographic chart.**

nautical mile—Typically, a measure of distance equal to 1 minute of arc of a great circle. See also **International Nautical *Mile.***

Naval Operating Area Chart (OPAREA)—Nautical charts produced and overprinted with prescribed OPAREA boundaries, submarine transit lanes, aeronautical and other information for fleet exercise areas.

Naval Range Charts—Detailed bathymetry, grid, and range boundaries; information on range acoustics; and seismic instrumentation.

Navigation Information Network (NAVINFONET)—The DMA data base which contains the information to produce the following publications: Notice to *Mariners, Summary of Corrections,* List of Lights and Fog Signals. It also provides users access to up-to-date marine information via commercial telecommunications services.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

navigation chart—See **aeronautical chart; hydrographic chart.**

navigation sight—An auxiliary device used in the taking of aerial photography to show not only the vertical field of view but also the path ahead and behind the aircraft.

Navigational Filmstrips (NFS)—Filmstrips depicting cartographic data as photographed from lithographic copy among the JNC and TPC series for A-7 aircraft, ONC and JOG series for HH-53 aircraft, and all 4 chart series for Remote Map Reader (RMR) equipped (F-14E). Used in aircraft equipped with a moving map display system to provide in-cockpit display of aircraft horizontal position relative to the chart data.

navigational planets—The four planets commonly used for celestial navigation: Venus, Mars, Jupiter, and Saturn.

navigational triangle—The spherical triangle solved in computing altitude and azimuth or great circle sailing problems. The celestial triangle is formed on the celestial sphere by the great circles connecting the elevated pole, zenith of the assumed position of the observer, and a celestial body. The terrestrial triangle is formed on a spherical Earth by the great circles connecting the pole and two places on Earth, either the assumed position of the observer and geographic position of the body for celestial observations, or the points of departure and destination for great circle sailing problems. The expression **navigational triangle** applies to either the celestial or terrestrial triangle used for solving navigation problems.

NAVINFONET—See **Navigation Information Network.**

NAVSTAR Global Positioning System (GPS)—A navigation and positioning system, consisting of 24 satellites, with which the three-dimensional position and the velocity of a user at a point on or near the Earth can be determined in real time. Its reference system is WGS 84. The user's receiver will require tracking of a minimum of four of the satellites from any location at any time to establish position and velocity.

Navy Navigation Satellite System (NNSS)—A set of five or six satellites in polar orbit with which three-dimensional position of a user can be determined. See also **broadcast ephemeris; Doppler navigation; precise ephemeris.**

near-certainty error (3.5 sigma, 3.5σ)—The 99.78 percent error interval based on the bivariate normal distribution function. See also **circular near-certainty error; circular error probable.**

neat model—The portion of the gross overlap of a pair of photographs that is actually utilized in photogrammetric procedures. Generally, the neat model approximates a rectangle whose width equals the air base and whose length equals the width between flights. See also **gross model.**

neatlines—(JCS) The lines that bound the body of a map, usually parallels and meridians (but may be conventional or arbitrary grid lines). Also called **sheet lines.**

negative altitude—Angular distance below the horizon.

negative component in color mixture—A component that is mixed with the sample light in order to desaturate it sufficiently to obtain a match with a mixture of the other two components.

negative corrections—Changes made directly on a negative or a scribed surface. See also **negative engraving.**

negative deflection angle—See **deflection angle,** definition 1.

negative engraving—The operation of making corrections and additions to negatives. This term should not be applied to the process of scribing on coated plastics.

negative forming—In relief model making, forming into a negative mold.

negative lens—A lens diverging a beam of parallel light rays, with no real focus being

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

obtained. Also called **concave lens; diverging lens.**

negative mold—The cast resulting from casting over a master relief model.

negative scribing—See **scribing.**

negative titling—See **titling.**

negative—**1.** In black and white photography an image on film, plate, or paper in which the normal tones of the subject are reversed. In color photography, an image on film, plate, or paper, in which colors appear as their complements. **2.** In cartographic scribing, a scribed sheet is essentially a manually produced negative. See also **duplicate negative; original negative.**

network analysis—Analytical techniques concerned with the relationships between locations on a network, such as the calculation of optimal routes through road networks, flow capacities of network systems, or the best location for facilities along networks.

net—See **survey net.**

neutral data—Digital Geographic Information that is intended to support numerous applications but must be tailored for a specific application.

neutral filter—A filter that reduces the intensity of light reaching the film or plate without affecting the tonal rendition of colors in the original scene.

New York leveling rod—A two-piece rod with movable target. For heights greater than 6 1/2 feet, the target is clamped at 6 1/2 feet and raised by extending the rod. Graduated to hundredths of a foot and read by vernier to thousandths.

new chart—A chart constructed to satisfy the needs of navigation in a particular area. It is laid out in conformity with a broad scheme to meet future needs in the adjacent areas.

new edition—Contains changes of such importance to map or chart users that all previous printings are made obsolete.

new survey—See **resurvey.**

Newton's laws—**1.** (gravitation) Every particle of matter in the universe attracts every other particle with a force proportional to the product of their masses and inversely as the square of the distance between them. **2.** (motion) (1) Every body continues in its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by a force impressed upon it. (2) The rate of change of momentum is proportional to the force impressed, and takes the direction of the straight line in which the force acts. (3) To every action there is an equal and opposite reaction; or, the mutual actions of two bodies are always equal and oppositely directed.

Newton's rings—An interference effect arising from close, but not quite perfect, contact between two surfaces, manifested by irregular concentric rings of color.

Newtonian constant of gravitation—See **constant of gravitation.**

newton—The newton is the metric (SI) unit of force. A force of 1 newton (N) acting on a mass of 1 kilogram imparts an acceleration of 1 meter per second per second. .One newton equals 1 kilogram per meter per second per second. See also **dyne.**

night effect—(JCS) An effect mainly caused by variations in the state of polarization of reflected waves, which sometimes result in errors in direction finding bearings. The effect is most frequent at nightfall.

no-check position—See **intersection station.**

nocturnal arc—See **astronomic arc.**

nodal line—In a tide area, the line about which the tide oscillates and where there is little or no rise and fall of the tide.

nodal plane—A plane perpendicular to the optical axis at a nodal point.

nodal point of emergence—See **nodal point,** definition 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

nodal point of incidence—See **nodal point**, definition 1.

nodal point—1. (optics) One of two points on the optical axis of a lens, or system of lenses, such that a ray emergent from the second point is parallel to the ray incident at the first. This first nodal point is also referred to as the **front nodal point, incident nodal point, or nodal point of incidence**; and the second point as the **rear nodal point, emergent nodal point, or nodal point of emergence.** Also called **node.** 2. (astronomy) See **node**, definition 1. 3. (hydrography) See **amphidromic point.**

node cycle—The time required for the regression of the Moon's nodes to complete a circuit of 360° of longitude; a period of approximately 18.6 years.

node—1. (astronomy) One of the two points of intersection of the orbit of a planet, planetoid, or comet with the ecliptic, or of the orbit of a satellite with the equatorial plane of the orbit of its primary. Also called **nodal point.** See also **ascending node; descending node; ecliptic node; equatorial node; line of nodes; longitude of the Moon's nodes; lunar node; regression of the nodes.** 2. (optics) See **nodal point**, definition 1. 3. A unique point representing the location of a feature, or the location of an intersection or connection of some number of features. Alternatively, a node is any isolated point, or the junction of any two or more edges, or the end points of an edge.

nodical month—The interval of time between two successive passages of the Moon through the same node of its orbit, approximately 27 days. Also called **draconic month.**

nodical period—The interval between two successive passages of a satellite or planet through the ascending node of its orbit.

noise level—The magnitude of random errors in a particular type of measurement.

nominal focal length—(JCS) An approximate value of the focal length, rounded off to some standard figure, used for the classification of lenses, mirrors, or cameras.

nominal orbit—The true or ideal orbit upon which a space vehicle is expected to travel.

nomogram—A diagram showing, to scale, the relationship between several variables in such manner that the value of one which corresponds to known values of the others can be determined graphically.

Non-Submarine Contact (NSC) Lists—Lists which contain a tabulated list of wrecks and other obstructions which might be construed as submarines by sonar.

nonautomatic rectifier—Any rectifier which requires computation of the elements of rectification, each of which must be manually set on its corresponding circle or scale on the rectifier.

nongravitational perturbations—Perturbations caused by surface forces due to mechanical drag of the atmosphere (in case of low flying satellites), electromagnetism, and radiation pressure.

nonmonumented bench mark—See **temporary bench mark.**

nonperspective azimuthal map projection—A projection not based on perspective lines of sight from a single point of view.

nonselective filter—A filter for which transmittance is substantially independent of wavelength.

nontilting-lens rectifier—A class of rectifier wherein the lens is constrained to move in the direction of its fixed axis.

nontilting-negative-plane rectifier—A class of rectifier which contains a nontilting negative carrier. In this class of rectifiers, the negative carrier plane remains horizontal.

normal distribution function—A mathematical function describing the behavior of one-dimensional random errors.

normal equation—One of a set of simultaneous equations derived from

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

observation, condition, or correlate equations, and expressing a condition for a least squares adjustment. In a least squares adjustment, values obtained from the solution of normal equations (either directly or through the correlate equations) are applied to the observation or condition equations to obtain the desired corrections.

normal gravity field—A mathematically derived gravity field used in geodesy to closely approximate the Earth's actual gravity field.

normal gravity—A reference gravity field that is mathematically defined for a normalized earth which has gravitational symmetry. It is commonly taken as the field of a rotating level ellipsoid but may be arbitrarily defined.

normal orbit—The orbit of a satellite considered with no disturbing effects present due to other celestial bodies, or to some physical phenomena. Also called **unperturbed orbit.**

normal section azimuth—The angle between the geodetic meridian of the observer and the plane containing the ellipsoidal normal of the observer, and measured clockwise from the north in a plane perpendicular to the ellipsoidal normal of the observer.

normal section line—A line on the surface of the ellipsoid connecting two points on that surface, and traced by a plane containing the normal at one point and passing through the other point.

normal tension—(taping) The tension to be applied to a tape to compensate for the shortening effect of sag in order to bring the tape to standard length. That pull at which the tension correction and sag exactly balance each other.

normal water level—The most prevalent water level in a watercourse, reservoir, lake, or pond, generally defined by a shoreline of permanent land-type vegetation. Along large bodies of water, wave action may retard vegetation beyond the normal shoreline.

normal-angle lens—A lens having an angle of coverage from 60° to 75°. A lens whose focal length is equal approximately to the diagonal of the format.

normal—1. A straight line perpendicular to a surface or to another line. 2. A condition of being perpendicular to a surface or line. 3. In geodesy, the straight line perpendicular to the surface of the reference ellipsoid. 4. The average, regular, or expected value of a quantity.

North American datum of 1927 (NAD 27)—The datum which defined the geodetic positions in the United States and adjoining countries was derived from a readjustment of the geodetic data as available up to 1927. It is based on the Clarke 1866 ellipsoid. See also **horizontal datum.**

North American datum of 1983 (NAD 83)—The new datum, which replaces NAD 27, is geocentric and based on GRS 80 parameters. Its coverage includes Alaska, Canada, CONUS, and Central American countries from Mexico to Panama. Total number of redefined horizontal stations exceeds 250K.

North Star—See **Polaris.**

north declination—See **declination,** definition 3.

north geographical pole—The geographical pole in the Northern Hemisphere, at latitude 90° N.

north geomagnetic pole—The geomagnetic pole in the Northern Hemisphere.

north magnetic pole—The magnetic pole in the Northern Hemisphere.

north point—See **celestial meridian.**

north polar circle—See **Arctic Circle.**

northbound node—See **ascending node.**

northing—1. (JCS) Northward increasing grid values on a map. See also **false northing.** 2. (plane surveying) See **latitude difference.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

north—The primary reference direction relative to the Earth. See also **compass north; grid north; magnetic north; true north.**

Notice to Mariners—Specific narrative and graphic correction data for all DMA, USCG and NOS nautical charts, publications and new information required to navigate. This information is published weekly. It is also accessible on the Network (NAVINFONET) as incorporated in the summary of corrections.

numerical map—See **digital map.**

nutation in right ascension—See **equation of the equinox.**

nutation—**1.** The oscillation of the axis of any rotating body, as a gyroscope rotor. **2.** (astronomy) Irregularities in the precessional motion of the equinoxes because of varying positions of the Moon and, to a lesser extent, of other celestial bodies with respect to the ecliptic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

O

object oriented—A design and programming methodology that seeks to mimic the way we form models of the world, and embodies three main properties, e.g., encapsulation, inheritance, and polymorphism.

objective lens—In telescopes and microscopes, the optical component which receives light from the object and forms the first or primary image. In a camera, the image formed by the objective lens is the final image. In a telescope or microscope used visually, the image formed by the objective lens is magnified by the eyepiece.

oblate ellipsoid of rotation—An ellipsoid of rotation, the shorter axis of which is the axis of rotation.

oblique air photograph—(JCS) An air photograph taken with the camera axis directed between the horizontal and vertical planes. Commonly referred to as an **oblique**: (1) **high oblique**— one in which the apparent horizon appears; and (2) **low oblique**—one in which the apparent horizon does not appear.

oblique ascension—The arc of the celestial equator, or the angle at the celestial pole, between the hour circle of the vernal equinox and the hour circle through the intersection of the celestial equator and the eastern horizon at the instant a point on the oblique sphere rises, measured eastward from the hour circle of the vernal equinox through 24 hours.

oblique chart—A chart on an oblique map projection.

oblique coordinates—Magnitudes defining a point relative to two intersecting nonperpendicular lines, called **axes.**

oblique cylindrical orthomorphic map projection—See **oblique Mercator projection.**

oblique equator—A great circle the plane of which is perpendicular to the axis of an oblique projection.

oblique graticule—A fictitious graticule based upon an oblique map projection.

oblique latitude—Angular distance from an oblique equator. See also **fictitious latitude.**

oblique longitude—Angular distance between a prime oblique meridian and any given oblique meridian. See also **fictitious longitude.**

oblique map projection—A map projection with an axis inclined at an angle between 0° and 90°.

oblique Mercator projection—A conformal cylindrical map projection in which points on the surface of a sphere or ellipsoid, such as the Earth, are conceived as developed by Mercator principles on a cylindrical tangent along an oblique great circle. Also called **oblique cylindrical orthomorphic projection.**

oblique meridian—A great circle perpendicular to an oblique equator. The reference oblique meridian is called **prime oblique meridian.** See also **fictitious meridian.**

oblique parallel—A circle or line parallel to an oblique equator, connecting all points of equal oblique latitude. See also **fictitious parallel.**

oblique plotting instrument—An instrument for plotting from oblique photographs.

oblique pole—One of the two points 90° from an oblique equator.

oblique rhumb line—**1.** A line making the same oblique angle with all fictitious meridians of an oblique Mercator projection. Oblique parallels and meridians may be considered special cases of the oblique rhumb line. **2.** Any rhumb line, real or fictitious, making an oblique angle with its meridians. In this sense the expression is used to distinguish such a rhumb line from parallels and meridians, real or fictitious, which may be included in the expression **rhumb line.** See also **fictitious rhumb line.**

oblique sketchmaster—A type of sketchmaster in which oblique photographs are utilized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

oblique sphere—The celestial sphere as it appears to an observer between the Equator and the pole, where celestial bodies appear to rise obliquely to the horizon.

obliquity of the ecliptic—The acute angle between the plane of the ecliptic (the plane of the Earth's orbit) and the plane of the celestial equator.

obliterated corner—An obliterated corner is one at whose point there are no remaining traces of the monument or its accessories, but whose location has been perpetuated, or the point for which may be recovered beyond reasonable doubt, by the acts and testimony of the intersected landowners, competent surveyors, or other qualified local authorities, or witnesses, or by some acceptable record evidence.

observation equation—An adjustment equation wherein variables representing corrections to approximations of the unknown parameters, as well as variables representing adjustments to the observations, remain in the equations. A least squares adjustment employing this type of equations is said to be by the observation equations method, or variation of parameters method, as opposed to the condition equations method.

observation—See **observed value.**

observed altitude—Corrected sextant altitude; angular distance to the center of a celestial body above the horizon, corrected for instrumental errors, personal error, dip, refraction, and semidiameter and parallax if necessary. See also **true altitude.**

observed angle—An angle obtained by direct instrumental observation. A measured angle which has been corrected for local conditions only at the point of observation, is considered an observed angle.

observed gravity anomaly—See **gravity anomaly.**

observed gravity—The value of gravity at a station as determined from a gravity meter, a pendulum, or an instrument timing free falling bodies. The gravity obtained is either relative or absolute according to the apparatus used to make the measurements.

observed value—A value of a quantity that is obtained by instrumental measurement of the quantity. The term **observed value** is often applied to the value of a quantity derived from instrumental measurement after corrections have been applied for systematic errors by some method of adjustment.

obsolete chart—A chart which does not contain the latest navigational information.

occultation—**1.** (astronomy) The disappearance of a celestial body behind another body of larger apparent size. When the Moon passes between the observer and a star, the star is said to be occulted. **2.** (surveying) Name applied to a geodetic survey technique which employs the principle of occultation where repeated observations are made on an unknown position, accurately timed with similar observations at another unknown station, and mathematically reducing these data to determine the exact geodetic position of the unknown stations. See also **star occultation method.**

occupy—(surveying) To set a surveying instrument over a point for the purpose of making observations.

oceanographic station—An observation point in ocean from which oceanographic observations are taken.

oceanographic survey—A study or examination of conditions in the ocean or any part of it, with reference to animal or plant life, chemical elements present, temperature gradients, etc. Also called **marine survey.**

oceanography—**1.** The study of the sea, embracing and integrating all knowledge pertaining to the sea and its physical boundaries, the chemistry and physics of sea water, and marine biology.

oceanology—See **oceanography.**

octant—A type of sextant having a range of 90° and an arc of 45°.

ocular micrometer—A filar micrometer so placed that its wire moves in the principal focal plane of a telescope Also called **eyepiece micrometer.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

odograph—A mechanical instrument containing a distance measuring element which is moved or turned by an amount proportional to the actual distance traveled; a compass element which provides a fixed reference direction; and an integrator which provides for the resolution of the direction of motion into components and for the summation or integration of the distance components.

off soundings—Any area where the depth of water cannot be measured by a sounding lead, generally considered to be beyond the 100-fathom line. Opposite of **on soundings.**

off-line—Transmission of information between a computer and a peripheral unit before or after, but not during processing, in contrast to on-line processing.

office computations—Computations based on field measurements, including all calculations relative to the reduction of field survey notes to graphic form for any type of survey or for the continuation of field work.

offset line—A supplementary line close to and roughly parallel with a main line, which is referred by measured offsets. Where the line for such data are desired is in such position that it is difficult to measure over it, the required data are obtained by running an offset line in a convenient location and measuring offsets from it to salient points on the other line.

offset lithography—An indirect method of printing whereby the ink image is transferred from the press plate to an intermediate surface of a rubber blanket, and from that to the paper or other stock. Also called **offset; offset printing.** See also **lithography; photolithography.**

offset press—A press which contains an extra cylinder, rubber covered, upon which the image is printed first and then reprinted or "offset," from this cylinder onto the paper.

offset printing—See **offset lithography.**

offset—1. (cartography) In projection construction, that small distance added to the length of the meridians on each side of the central meridian in order to determine the top latitude of the constructed chart. 2. (surveying) A short line perpendicular to a surveyed line, measured to a line or point for which data are desired, thus locating the second line or point with reference to the first or surveyed line. An offset is also a jog in a survey or other line, the line having approximately the same direction both before and after passing the jog. Offsets are measured from a surveyed line, or lines to the edges of an irregular-shaped body of water, or to any irregular line which it is desired to locate. 3. (printing) See **offset lithography.**

OK sheet—The first press impression from each color, or color combination, approved for accuracy of register and color.

Omega—A long-range hyperbolic navigation system designed to provide worldwide coverage for navigation.

omnidirectional radar prediction—A radar prediction which is intended to be valid from any direction of approach. The Radar Significance Analysis Code on the Series 200 Air Target Chart is an example of omnidirectional radar prediction. Each coded area represents an analysis of relative radar intensity from all directions.

omnigain radar prediction—A radar prediction containing some information about all radar responsive features within the predicted area. This is accomplished by predicting all significant radar returns in relative intensities based on the predicted probability of the return remaining on the radarscope at decreased gain. Generally, the more intense the return appears on the prediction, the more likely it will remain on the radarscope as the gain is decreased.

on soundings—Any area where the depth of water can be measured by a sounding lead, generally considered to be within the 100-fathom line. Opposite of **off soundings.**

on-line—Transmission of information between a computer and a terminal or display device while processing is occurring, in contrast to off-line processing.

one-projector method—See **one-swing method.**

one-swing method—The technique employed in relative orientation for clearing *y*-parallax by maintaining one projector of a pair in a fixed position and making all adjustments with the second projector in relationship to the first. Also called **one-projector method; single-projector method. *y*-swing method;**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

one-to-one (1:1) copy—See **contact size.**

opacity—See **density,** definition 1.

opaque—1. Not transmitting light. 2. Not transmitting the particular wavelengths (which may or may not be visible) which affect given photosensitive materials. Thus, a substance may be opaque to some colors and not to others. It may be visually transparent, yet actinically opaque. 3. A material applied to areas of a negative to make it opaque in those areas. 4. To apply a material or blockout.

open end traverse—See **open traverse.**

open system (OS)—1. Any system that interacts with its environment through input and output. An open system is able to adapt or adjust to changes in the environment in order to continue its existence. 2. In computers, an open system has detailed published operating instructions, hardware and software descriptions, and interface diagrams enabling users and third party vendors to add to or modify the system to adapt it to the user's specific needs. See also **GOSIP.**

open traverse—A survey traverse which begins from a station of known or adopted position, but does not end upon a known or adopted station. Also called **open end traverse.**

open window process—(cartography) A method of preparing color separation negatives or positives by peeling an opaque stratum from its base in the desired areas. It is normally used for preparing large areas covered by vegetation or open water. See also **mask,** definition 2.

operation map—(JCS) A map showing the location and strength of friendly forces involved in an operation. It may indicate predicted movement and location of enemy forces.

Operational Navigation Chart (ONC)—The standard worldwide small-scale (1:1,000,000) aeronautical chart series. It contains cartographic data with an aeronautical overprint depicting obstructions, aerodromes, etc., designed for medium altitude high-speed visual and radar navigation. Also used for mission planning/analysis and intelligence briefings.

operational grid—A grid in current operational use. Generally this would be the preferred grid but could be a previously prescribed grid.

operational libraries—A DMA approved, selective data file consisting of extra copies of originals, duplicate copies, computer printed catalogs, etc., obtained from any designated DoD library or from other nationally designated libraries within the non-DoD agencies, and maintained in a DoD MC&G agency for its direct use in accomplishing assigned production missions. General reference publications such as dictionaries, glossaries, atlases, periodicals, etc., are excluded from controls applied to operational libraries.

operational system—A system that has been developed, verified, and is being operated under the auspices of a unified or specified command.

opposition—1. The situation of two celestial bodies having either celestial longitudes or sidereal hour angles differing by 180°. The term is usually used only in relation to the position of a planet or moon from the Sun. 2. The situation of two periodic quantities differing by half a cycle.

optical axis—(JCS) In a lens element, the straight line which passes through the centers of curvature of the lens surfaces. In an optical system, the line formed by the coinciding principal axes of the series of optical elements. Also called **axis of lens; lens axis; principal axis.**

optical base-line measuring apparatus—A base apparatus composed of bars whose lengths are defined by distances between lines at or near their ends, which are observed by suitably mounted and adjusted microscopes. In using any optical baseline measuring apparatus, the positions of the bars are controlled by microscopes on stable support, whose reticle lines may be brought into coincidence with the fiducial marks on the bars, either by adjusting a bar or a microscope.

optical center—The point of intersection of lines which represent within the lens those rays whose emergent directions are parallel to their respective incident directions. This point lies on the optical axis. An oblique ray, even if it passes through this point, undergoes a longitudinal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

displacement increasing with the thickness of the lens.

optical correlation—The process of electronically relating a stored photographic film chip of a geographic area with a realtime optical image acquired by photographic or television sensors. It is used to provide positioning information to correct or check air navigation and guidance systems.

optical density—A common logarithm of reciprocal of transmittance.

optical flat—A surface, usually of glass, ground and polished plane within a fraction of a wavelength of light. An optical element or glass blank with an optical flat is used to test the flatness of other surfaces. Also called **flat; optical plane.**

optical parallax—See **instrument parallax**, definition 1.

optical path—The path followed by a ray of light through an optical system.

optical plane—See **optical flat.**

optical plummet—See **vertical collimator.**

optical rectification—The process of projecting the image of a tilted aerial photograph onto a horizontal reference plane to eliminate the image displacements caused by tilt of the aerial camera at the time of exposure.

optical square—A small hand instrument used in setting off a right angle. One form of optical square uses two plane mirrors placed at an angle of 45° to each other. In use, one object is sighted direct, and another object is so placed that its twice-reflected image appears directly in line with the first object. The lines to the point of observation from the two observed objects will then meet in a right angle. In another form of optical square, a single plane mirror is so placed that it makes an angle of 45° with a sighting line; one object is sighted direct, and the other so placed that its reflected image is seen also in the sighting line.

optical system—All the parts of a compound lens and accessory optical parts which are designed to contribute to the formation of an image on a photographic emulsion, or of a visual image, or of an image on a projection screen.

optical vernier—A microscope with vernier lines ruled on a glass slide placed in the focal plane common to the objective and the eyepiece, where it is compared with the image of the graduated circle.

optical wedge—See **wedge.**

optical-mechanical scanner—A system utilizing a rotating mirror and a detector in conjunction with lenses and prisms to record reflected and/or emitted electromagnetic energy in a scanning mode along the flight path.

optical-projection instruments—A class of instruments which provide projected images of photographic prints or other opaque material superimposed on a map or map manuscript. Often used for transferring detail from near-vertical photographs or other source material.

optimum ground elevation—(photogrammetry) The elevation of an assumed horizontal surface in the area photographed that would be projected at the optimum distance in the plotting instrument.

orbital altitude—The mean altitude above the surface of the parent body of the orbit of a satellite.

orbital elements—A set of six parameters defining the orbit of a body attracted by a central force.

orbital inclination—The direction that the path of an orbiting body takes. In the case of an Earth satellite, this path may be defined by the angle of inclination of the path to the Equator.

orbital mode—A method for determining the position of an unknown station position when the unknown position cannot be viewed simultaneously with known positions. The arc of the satellite orbit is extrapolated from the ephemeris of the satellite determined by the known stations which permits the determination of the position of the unknown station dependent completely on the satellite's orbital parameters.

orbital motion—Continuous motion in a closed path about and as a direct result of a source of gravitational attraction.

orbital path—One of the tracks on a primary

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

body's surface traced by a satellite that orbits about it several times in a direction other than normal to the primary body's axis of rotation. Each track is displaced in a direction opposite and by an amount equal to the degrees of rotation between each satellite orbit.

orbital period—The interval between successive passages of a satellite through the same point in its orbit. Also called **period of satellite.**

orbital plane—The plane of the ellipse defined by a central force orbit.

orbital velocity—The velocity of an Earth satellite or other orbiting body at any given point in its orbit.

orbit—The path of a body or particle under the influence of a gravitational or other force. For example, the orbit of a celestial body or satellite is its path relative to another body around which it revolves. The term orbit is commonly used to designate a closed path. See also **central force orbit; intermediate orbit; nominal orbit; normal orbit; osculating orbit; perturbed orbit; polar orbit; stationary orbit; two-body orbit.**

ordinates—In a system of rectangular or oblique coordinates, the linear distance of a point measured from the horizontal or *x*-axis, and parallel to the *y*-axis. Also called **total latitudes; *y*-coordinate.**

orientation inset—See **inset.**

orientation point—A picture point selected in areas common to vertical photographs and their corresponding obliques which serves to establish the relationship between the vertical and the oblique. Two such points are usually selected in each vertical photograph and transferred to the matching oblique photo.

orientation—1. The act of establishing correct relationship in direction with reference to the points of the compass. 2. The state of being in correct relationship in direction with reference to the points of the compass. 3. A map is in orientation when the map symbols are parallel with their corresponding ground features. 4. A surveyor's transit is in orientation if the horizontal circle reads 0° when the line of collimation is parallel to the direction it had at an earlier (initial) position of the instrument, or to a standard line of reference. If the line of reference is a meridian, the circle will show azimuths referred to that meridian. 5. A photograph is in orientation when it correctly presents the perspective view of the ground or when images on the photograph appear in the same direction from the point of observation as do the corresponding map symbols. 6. Photogrammetric orientation is the recreation of natural terrain features at a miniature scale by the optical projection of overlapping photographs. The model is formed when all corresponding light rays from the two projectors intersect in space. See also **absolute orientation; aeroleveling; astrogeodetic datum orientation; basal orientation; empirical orientation; exterior orientation; gravimetric datum orientation; interior orientation; preliminary orientation; relative orientation; single astronomic station datum orientation.**

original copy—The photographs, artwork, scribed material, typed matter, and/or other materials to be processed for reproduction. Also called **original** .

original model—See **master model.**

original negative—That negative developed from the film which was in a camera magazine at the instant of exposure. Synonymous with a first-generation photographic product.

original survey—See **survey.**

original—See **original copy.**

origin—The reference position from which angles, coordinates or distances are reckoned. See also **false origin; grid origin.**

orthochromatic—(photography) 1. Of, pertaining to, or producing tonal values (of light or shade) in a photograph, corresponding to the tones of nature. 2. Designating an emulsion sensitive to blue and green light, but not to red.

orthodrome—See **great circle.**

orthogonal map projection—See **orthographic map projection.**

orthogonal—At right angles; rectangularly; meeting, crossing, or lying at right angles.

orthographic chart—A chart on the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

orthographic projection.

orthographic map projection—A perspective azimuthal projection in which the projecting lines, emanating from a point at infinity, are perpendicular to a tangent plane. This projection is used chiefly in navigational astronomy for interconverting coordinates of the celestial equator and horizon systems. Also called **orthogonal map projection.**

orthometric correction—A systematic correction which must be applied to measured difference observed through leveling.

orthometric elevation—An elevation to which the orthometric correction has been applied.

orthomorphic chart—A chart on which very small shapes are correctly represented.

orthomorphic map projection—See **conformal map projection.**

orthophotograph—A photographic copy, prepared from a perspective photograph, in which the displacements of images due to tilt and relief have been removed.

orthophotomap—A photomap made from an assembly of orthophotographs. It may incorporate special cartographic treatment, photographic edge enhancement, color separation, or a combination of these.

orthophotomosaic—An assembly of orthophotographs forming a uniform scale mosaic.

orthophotoscope—A photomechanical device, used for producing orthophotographs.

orthopictomap—A pictomap made from an orthophotomap base.

orthorectification—The process of removing image displacement caused by tilt and terrain relief. Tilt, however, is not relevant in radar images.

orthostereoscopy—A condition wherein the horizontal and vertical distances in a stereoscopic model appear to be at the same scale.

oscillation—A double motion, one in each direction, of a pendulum. An oscillation is composed of two successive vibrations.

osculating elements—The elements that define an osculating orbit. See also **osculating orbit.**

osculating ellipse—An ellipse that is tangent at a point (called the **epoch of osculation**) to a real orbit.

osculating orbit—The ellipse that a satellite would follow after a specific time "t" (the epoch of osculation) if all forces other than central inverse square forces ceased to act from time "t" on. An osculating orbit is tangent to the real perturbed orbit and has the same velocity at the point of tangency.

other grids—Grids other than the Universal Transverse Mercator (UTM) and Universal Polar Stereographic (UPS), such as Ceylon Belt, India Zone IIA, West Malaysian RSO (Metric) Grid, etc.

outer orientation—See **exterior orientation.**

outer planet—The planets with orbits larger than that of Mars; i.e., Jupiter, Saturn, Uranus, Neptune, and Pluto.

outlier—A measurement which does not fit the remainder of measurements of the same quantity, where the reason for the discrepancy cannot be assessed.

outline map—(JCS) A map which represents just sufficient geographic information to permit the correlation of additional data placed upon it.

overcharging—Applying excessive additional information (aeronautical or navigational) to a map or chart resulting in clutter.

overhang—(aerial photography) The additional exposures beyond the boundary of an area to be photographed, usually two exposures at the ends of each strip to assure complete stereoscopic coverage.

overlapping grid—A major grid from a neighboring area primarily intended to facilitate military surveying and fire-control. See also **major grid; secondary grid.**

overlapping mean—See **consecutive**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

mean.

overlapping pair—(photogrammetry) Two photographs taken at different exposure stations in such a manner that a portion of one photograph shows the same terrain as shown on a portion of the other photograph. This term covers the general case and does not imply that the photographs were taken for stereoscopic examination. See also **stereoscopic pair.**

overlap—1. (JCS) In photography, the amount by which one photograph includes the same area covered by another, customarily expressed as a percentage. The overlap between successive air photographs on a flight line is called **forward overlap** or **forward lap**. Also called **end lap**. The overlap between photographs in adjacent parallel light lines is called **side overlap** or **side lap**. 2. (JCS) In cartography, that portion of a map or chart which overlaps the area covered by another of the same series. 3. An area included within two surveys of record, which by record are described as having one or more common boundary lines with no inclusion of identical parts.

overlay—1. (JCS) A printing or drawing on a transparent or semitransparent medium at the same scale as a map, chart, etc., to show details not appearing, or requiring special emphasis, on the original. 2. (digital) A data layer, usually dealing with only one aspect of related information, which is used to supplement the data base, digital, image overlays, e.g., overlaying a raster base map with a vector shoreline map. Overlays are registered to the base by a common coordinate system. 3. (lithography) Additional data, or a pattern, printed after the other features, so as to "overlay" them. See also **correction overlay; history overlay; radarscope overlays; selection overlay.**

overprint—1. (JCS) Information printed or stamped upon a map or chart, in addition to that originally printed, to show data of importance or special use. Also called **surprint.** 2. A feature of a composite map image incidentally printed so as to interfere with another feature.

overrun control—(JCS) Equipment enabling a camera to continue operating for a predetermined number of frames or seconds after normal cutoff.

oversheet—A transparency or a print of a map compilation used for recording supplemental information.

oversize chart—A chart whose neatlines have been extended slightly, thereby increasing the sheet size to include a small land area in order to avoid publishing a separate graphic of that area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

P

panchromatic—(photography) Sensitive to light of all colors, as a film or plate emulsion.

pancratic system—A variable power optical system. Also called **zoom system.**

panel base—(cartography) The completed assembly of pieces of film positives onto a grid or projection which is used as a base for compilation. Also called **film mosaic; panel.**

paneling—1. (cartography) Cutting a film positive of a map, in which some distortion is involved, into several pieces and cementing them in place, on a projection constructed on a stable-base medium, in such a way that the error is distributed in small amounts throughout the area rather than being localized. 2. (surveying) The placement of panels on a control station to facilitate station identification on aerial photography.

panel—1. (cartography) See **panel base.** 2. (photogrammetry) An element of a target used for control station identification on aerial photography. Panels are made of cloth, plastics, plywood, or Masonite, and are positioned in a symmetrical pattern centered on the station. See also **target.**

panoramic camera—A camera which takes a partial or complete panorama of the terrain. Some designs utilize a lens which revolves about an axis perpendicular to the optical axis; in other designs, the camera itself is revolved by clockwork to obtain a panoramic field of view. See also **frame camera.**

panoramic distortion—The displacement of ground points from their expected perspective positions, caused by the cylindrical shape of the negative film surface and the scanning action of the lens in a panoramic camera system.

panoramic photograph—Photography obtained from a panoramic camera.

pantograph—An instrument for copying maps, drawings, or other graphics at a predetermined scale. Pantographs capable of adjustment for several scales are known as fixed ratio pantographs. See also **two-dimensional pantograph.**

paper-strip method—(rectification) A graphical method of making a point-by-point rectification based on the invariance of the cross ratio. A modification of this technique permits map detail to be revised from an oblique aerial photograph based on the projectivity of straight lines.

parallactic aberration—See **differential aberration.**

parallactic angle—1. (astronomy) The angle between a body's hour circle and its vertical circle. Also called **position angle.** 2. (photogrammetry) Also called **angular parallax.** See **angle of convergence.**

parallactic error—An error caused by personal or instrument parallax.

parallactic grid—(photogrammetry) A uniform pattern of rectangular lines drawn or engraved on some transparent material, usually glass, and placed either over the photographs of a stereoscopic pair or in the optical system of a stereoscope, in order to provide a continuous floating mark system.

parallactic inequality—A secondary effect in solar perturbations in the Moon's longitude due to the ellipticity of the Earth's orbit.

parallax age—See **age of parallax inequality.**

parallax bar—See **stereometer.**

parallax difference—The difference in the absolute stereoscopic parallaxes of two points imaged on a pair of photographs. Customarily used in determination of the difference in elevation of objects.

parallax in altitude—Geocentric parallax at any altitude. The expression is used to distinguish the parallax at the given altitude from the horizontal parallax when the body is in the horizon.

parallax inequality—The variation in the range of tide or in the speed of tidal currents because of the continual change in the distance of the Moon from the Earth. The range of tide

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

and speed of tidal currents tend to increase as the Moon approaches perigee and to decrease as it approaches apogee.

parallax—1. (JCS) In photography, the apparent displacement of the position of an object in relation to a reference point due to a change in the point of observation. 2. The apparent displacement between objects on the Earth's surface due to their difference in elevation. Also called **angular parallax; want of correspondence.** See also **absolute stereoscopic parallax; age of parallax inequality; annual parallax; equatorial horizontal parallax; false parallax; geocentric parallax; horizontal parallax; instrument parallax; lunar parallax; residual parallax; solar parallax; y-parallax.**

parallel of altitude—A circle of the celestial sphere parallel to the horizon connecting all points of equal altitude. Also called **almucantar; altitude circle; circle of equal altitude.**

parallel of declination—A circle of the celestial sphere parallel to the celestial equator. Also called **celestial parallel; circle of equal declination.**

parallel of latitude—See **circle of longitude.**

parallel plate—An optical disk with optically flat, parallel surfaces; used especially in optical micrometers. Also called **plane plate.** See also **optical flat.**

parallel sphere—The celestial sphere as it appears to an observer at the pole, where celestial bodies appear to move parallel to the horizon.

parallel—A circle on the surface of the Earth, parallel to the plane of the Equator and connecting all points of equal latitude, or a circle parallel to the primary great circle of a sphere or spheroid; also, a closed curve approximating such a circle. Also called **inverse parallel.** See also **astronomic parallel; ecliptic parallel; fictitious parallel; geodetic parallel; geographic parallel; grid parallel; ground parallel; isometric parallel; oblique parallel; photograph parallel; principal parallel; standard parallel; transverse parallel.**

parameterize—The act or process of describing a geometric situation with a mathematical model containing adjustable constants.

parameter—In general, any quantity of a problem that is not an independent variable. More specifically the term is often used to distinguish from dependent variables quantities which may be assigned arbitrary values for purposes of the problem at hand.

parametric equations—A set of equations in which the independent variables or coordinates are each expressed in terms of a parameter.

parametric latitude—The angle at the center of a sphere which is tangent to the ellipsoid along the geodetic equator, between the plane of the equator and the radius to the point intersected on the sphere by a straight line perpendicular to the plane of the Equator and passing through the point on the ellipsoid whose parametric latitude is defined. Parametric latitude is an auxiliary latitude used in problems of geodesy and cartography.

paraxial ray—A ray whose path lies very near the axis of a lens and which intersects the lens surface at a point very close to its vertex and at nearly normal incidence.

partial tide—See **constituent.**

pass point—A point whose horizontal and/or vertical position is determined from photographs by photogrammetric methods and which is intended for use in the absolute orientation of a model. Also called **photogrammetric point.** See also **annex point; supplemental elevation; supplemental position.**

passive satellite—A satellite which contains no power sources to augment output power; a satellite which is a passive reflector. See also **active satellite.**

pass—1. A single circuit of the Earth by a satellite. See also **orbit.** 2. The period of time a satellite is within telemetry range of a data acquisition station. 3. (mensuration) One complete set of pointings or measurements on a specific plate, reseau, or other media containing photographic imagery.

patch—(digital) A small area of information spliced into a data base or software in order to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

update, complete, or densify the data content.

path—The projection of the orbital plane of the satellite on the Earth's surface; the locus of the satellite subpoint.

PC ARC/INFO—A vector data, relational geographic information system for a personal computer. Produced by Environmental Systems Research Institute as a low-end GIS.

PC-1000 camera—A trade name for a geodetic stellar camera having a focal length of 1,000 mm.

Peaucellier inversor—A class of inversor providing a mechanical solution for the linear and angular elements of rectification. Also called **scissors inversor.**

Peaucellier-Carpentier inversor—A modified Carpentier inversor coupled to the linkage system of a Peaucellier inversor to provide a mechanical means of solving the linear and angular elements of rectification.

peel—(negative engraving) A technique of removing the opaque stratum from its supporting base. Peeling between etched outline images produces a negative; peeling outside of the etched outline images produces a positive. See also **mask**, definition 2.

peepsight alidade—A type of alidade consisting of a peep sight mounted on a straightedge.

peepsight compass—The sights of a compass formed by standards with slits for a sighting medium rather than a telescope.

peg adjustment—A method of adjusting a leveling instrument of the dumpy level type, to make the line of collimation parallel with the axis of the spirit level, and employing two stable marks (pegs) the length of one instrument sight apart. Also called **11/10 peg adjustment.**

peg test—A method of testing the collimation adjustment of a leveling instrument.

Pemberton leveling rod—A speaking rod marked with alternate rows of circular and diamond-shaped dots, running diagonally across the rod. Read to hundredths of a foot.

pendulum alidade—A telescopic alidade in which a pendulum device replaces the conventional bubble for establishing a horizontal reference line from which vertical angles may be measured.

pendulum astrolabe—An astronomic instrument using a constant altitude for position determination. Its distinctive feature is a mirror suspended on top of a pendulum to form the artificial horizon.

pendulum level—A leveling instrument in which the line of sight is automatically maintained horizontal by means of a built-in pendulum device. Also called **automatic level.**

pendulum—1. In general, a body so suspended as to swing freely to and fro under the influence of gravity and momentum. 2. A vertical bar so supported from below by a stiff spring as to vibrate to and fro under the combined action of gravity and the restoring force of the spring. See also **compound pendulum; dummy pendulum; free-swinging pendulum; idle pendulum; Invar pendulum; invariable pendulum; Mendenhall pendulum; quartz pendulum; receiver; relative pendulum; reversible pendulum; simple pendulum; working pendulum.**

percent of enlargement/reduction—The factor by which an original is to be enlarged or reduced in reproduction. A 50 percent linear enlargement of a 4- by 5-inch original would be 6-inches by 7 1/2-inches, while a 50 percent reduction of the same original would be 2-inches by 2 1/2-inches. See also **scale of reproduction.**

percent of slope—See **gradient.**

periapsis—See **pericenter.**

periastron—That point of the orbit of one member of a double star system at which the stars are nearest together. Opposite of **apastron.**

pericenter—In an elliptical orbit the point in the orbit which is the nearest distance from the focus where the attracting mass is located. The pericenter is at one end of the major axis of the orbital ellipse. Opposite of **apoapsis; apocenter; apofocus.** Also called **periapsis; perifocus.**

Source: https://assist.dla.mil Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

pericynthion—See **perilune.**

perifocus—See **pericenter.**

perigee-to-perigee period—See **anomalistic period.**

perigee—The point at which a satellite orbit is the least distance from the center of the gravitational field of the earth. Opposite of **apogee.**

perihelion—The point in the elliptical orbit of a planet which is the nearest to the Sun, when the Sun is the center of attraction. Opposite of **aphelion.**

perilune—The point of closest approach of an orbiting body to the Moon. Opposite of **aplune; apocynthion.** Also called **pericynthion.**

period of satellite—See **orbital period.**

periodic errors—In a complete set of observations there corresponds to every individual error another error which is necessarily more or less equal and opposite. In a limited series the cancellation may not be quite exact, but the error of the mean of *n* observations may be expected to be $1/n$ of that of a single measure, or less.

periodic perturbations—Perturbations to the orbit of a planet or satellite which change direction in regular or periodic fashion in time, such that the average effect over a long period of time is zero.

periodic terms—In the mathematical expression of an orbit, terms which vary with time in both magnitude and direction.

period—**1.** The interval needed to complete a cycle. **2.** The interval between passages at a fixed point of a given phase of a simple harmonic wave; the reciprocal of frequency. See also **anomalistic period; nodical period; orbital period; sidereal period; synodic period.**

permanent bench mark (PBM)—A bench mark of as nearly permanent character as it is practicable to establish. Usually designated simply as a bench mark or BM. A permanent bench mark is intended to maintain its elevation with reference to an adopted datum without change over a long period of time. Also called **monumented bench mark.**

perpendicular equation—(traverse) A condition equation to reduce to zero the algebraic sum of the projections of the separate lines of a traverse upon perpendiculars to a fixed line with which the traverse forms a closed figure.

perpendicular—A perpendicular line, plane, etc. A distinction is sometimes made between perpendicular and normal, the former applying to a line at right angles to a straight line or plane, and the latter referring to a line at right angles to a curve or curved surface.

personal equation—The time interval between the sensory perception of a phenomenon and the motor reaction thereto. A personal equation may be either positive or negative, as an observer may anticipate the occurrence of an event, or wait until he actually sees it occur before making a record. This is a systematic error, treated as the constant type.

personal error—An error caused by an individual's personal habits, his inability to perceive or measure dimensional values exactly, or by his tendency to react mentally and physically in a uniform manner under similar conditions. It may be a systematic error, if it occurs regularly or a blunder if it occurs once. A certain amount of minor personal error can be included with random errors.

personal parallax—See **instrument parallax,** definition 2.

perspective axis—See **axis of homology.**

perspective center—The point of origin or termination of bundles of perspective rays. The two such points usually associated with a survey photograph are the interior perspective center and the exterior perspective center. In a perfect lens-camera system, perspective rays from the interior perspective center to the photographic images enclose the same angles as do the corresponding rays from the exterior perspective center to the objects photographed. In a lens having distortion, this is true only for a particular zone of the photograph. In a perfectly adjusted lens-camera system, the exterior and interior perspective centers correspond, respectively, to the front and rear nodal points of the camera lens. Also called **center of projection.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

perspective chart—A chart on a perspective projection.

perspective grid—(JCS) A network of lines, drawn or superimposed on a photograph, to represent the perspective of a systematic network of lines on the ground or datum plane. Also called **Canadian grid**. See also **grid method**.

perspective map projection—A map projection produced by straight lines radiating from a selected point and passing through points on the earth's surface to the projection surface. Also called **geometric map projection**.

perspective plane—Any plane containing the perspective center. The intersection of a perspective plane and the ground will always appear as a straight line on an aerial photograph.

perspective projection—The projection of points by straight lines drawn through them from some given point to an intersection with the plane of projection. Unless otherwise indicated, the point of projection is understood to be at a finite distance from the plane of projection.

perspective ray—A line joining a perspective center and a point object. See also **image ray**.

perspective spatial model—Optical reconstruction of an area of terrain showing depth by viewing a pair of aerial photographs through a stereoscope.

perspective view—A three-dimensional representation generated with a reference to a specific viewer location on or above the surface portrayed.

perspective—The appearance to the eye of objects in respect to their relative distance and position.

perspectivity—The correspondence between the points, lines, or planes of two geometric configurations in perspective. Usually referred to as linear perspectivity because the true perspective center must be recoverable before angular perspectivity can be included.

perturbation—In celestial mechanics, differences of the actual orbit from a central force orbit, arising from some external force such as a third body attracting the other two; a resisting medium (atmosphere); failure of the parent body to act as a point mass, and so forth. See also **gravitational perturbations; long period perturbations; lunisolar perturbations; nongravitational perturbations; periodic perturbations; secular perturbations; short period perturbations; terrestrial perturbations.**

perturbed orbit—The orbit of a satellite differing from its normal orbit due to various disturbing effects such as nonsymmetrical gravitational effects, atmospheric drag, radiation pressure, and so forth. See also **perturbation.**

perturbing factors (forces)—In celestial mechanics, any force that acts on the orbiting body to change its orbit from a central force orbit.

phase age—See **age of phase inequality.**

phase angle—1. The phase difference of two periodically recurring phenomena of the same frequency, expressed in angular measure. 2. The angle at a celestial body between the Sun and Earth.

phase inequality—Variations in the tide or tidal currents associated with changes in the phase of the Moon. At new and full Moon (springs) the tide-producing forces of the Sun and Moon act in conjunction, resulting in greater than average tide and tidal currents. At first and last quarters of the Moon (neaps) the tide-producing forces oppose each other, resulting in smaller than average tide and tidal currents.

phase—1. (general) Of a periodic quantity, for a particular value of the independent variable, the fractional part of a period through which the independent variable has advanced, measured from an arbitrary reference. 2. (surveying) The apparent displacement of an object or signal caused by one side being more strongly illuminated than the other. The resultant error in pointing is similar to the error caused by observing an eccentric signal. 3. (astronomy) A stage in a cycle of recurring aspects, caused by a systematic variation of the illumination of an object. The Moon passes through its phases, new Moon to full Moon and back to new Moon, as its position relative to the Sun and Earth changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Philadelphia leveling rod—A two-piece target rod, with graduation marks so styled that it may also be used as a speaking rod. For heights greater than 7 feet the target is clamped at 7 feet, and raised by extending the rod. As a target rod, it is read by vernier to thousandths of a foot; as a speaking rod, to half-hundredths of a foot.

photo altitude—Height of an aircraft above the mean elevation of the terrain to be photographed.

photo index—1. An index map made by assembling the individual aerial photographs into their proper relative positions and copying the assembly photographically at a reduced scale. Also called **index to photography; photo plot; plot map. 2.** See **sortie plot.**

photo plot—See **photo index**, *definition* 1.

photo pyramid—A component of an analytical method of precise determination of photographic tilt which represents a specific spatial configuration formed by three control points of known position on the photograph (forming a triangle) and the exposure station. When used with the ground pyramid, it permits the exact position of the exposure station to be determined and, by analytical techniques, the exact tilt of the photograph. See also ground pyramid.

photo revision—The process of making changes on a map based upon information obtained from a study of aerial photographs.

photo scale—See **scale**, definition 1.

photoalidade—A photogrammetric instrument having a telescopic alidade, a plateholder, and a hinged ruling arm mounted on a tripod frame. It is used for plotting lines of direction and measuring vertical angles to selected features appearing on oblique and terrestrial photographs.

photoangulator—See **angulator.**

photobase—The distance between the principal points of two adjacent prints of a series of vertical aerial photographs. It is usually measured on one print after transferring the principal point of the other print. See also **base line**, definition 2.

photocompose—To mechanically impose one or more images by step-and-repeat exposures in predetermined positions on a press plate or negative by means of a photocomposing machine.

photocontour map—Essentially, a topographic map upon which the planimetric detail is depicted photographically in its correct position. It is usually prepared from convergent photography although conventional vertical photography can be used.

photocontour process—A process developed to combine, in a photocontour map, that information normally portrayed on a topographic drawing and an aerial photograph. The system usually is composed of three elements: (1) a conventional stereoplotter for contouring; (2) a rectifier for tilt rectification of the aerial photographs; and (3) a zone printer to eliminate relief displacement. It is designed to utilize convergent photographs although normal vertical photographs can be utilized as well.

photocontrol base—See **control base.**

photocontrol diagram—Any selected base map or photo index on which proposed ground control networks, to include proposed positions for pass points, are delineated. See also **photocontrol index map.**

photocontrol index map—Any selected base map or photo index on which ground control and photo identified ground points are depicted and identified. See also **photocontrol diagram.**

photocontrol point—See **picture control point.**

photogoniometer—An instrument for measuring angles from the true perspective center to points on a photograph.

photogrammetric camera—A general term applicable to any camera used in any of the several branches of photogrammetry.

photogrammetric compilation—See **compilation**, *definition* 2.

photogrammetric control point—A horizontal control point which has been established by photogrammetric triangulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

photogrammetric control—(JCS) Control established by photogrammetric methods as distinguished from control established by ground methods. Also called **minor control; multiplex control.**

photogrammetric map—A topographic map produced from aerial photographs and geodetic control data by means of photogrammetric instruments. Also called **stereometric map; stereotopographic map.**

photogrammetric point—See **pass point.**

photogrammetric pyramid—An analytical method for the precise determination of photographic tilt, consisting of a ground pyramid and a photo pyramid, which represent a spatial configuration formed by three control points of known position on the photograph (forming a triangle) and the exposure station. See also **ground pyramid; photo pyramid.**

photogrammetric rectification—See **rectification.**

photogrammetric survey—A survey utilizing either terrestrial or aerial photographs.

photogrammetric triangulation—See **phototriangulation.**

photogrammetrically digitized—Digitized from aerial photographs and geodetic control data by means of photogrammetric instruments, providing three-dimensional coordinates.

photogrammetry—**1.** (JCS) The science or art of obtaining reliable measurements from photographic images. **2.** The preparation of charts and maps from aerial photographs using stereoscopic equipment and methods. See also **aerial photogrammetry; analytical photogrammetry; stereophotogrammetry; terrestrial photogrammetry.**

photograph center—The center of a photograph as indicated by the images of the fiducial mark(s) of the camera. In a perfectly adjusted camera, the photograph center and the principal point are identical.

photograph coordinates—A system of coordinates, either rectangular or polar, describing the position of a point on a photograph.

photograph meridian—The image on a photograph of any horizontal line in the object space which is parallel to the principal plane. Since all such lines meet at infinity, the image of the meeting point is at the intersection of the principal line and the horizon trace and all photograph meridians pass through that point.

photograph nadir—The point at which a vertical line through the perspective center of the camera lens pierces the plane of the photograph. Also called **nadir point; photographic plumb point; plumb point.**

photograph parallel—The image on a photograph of any horizontal line in the object space which is perpendicular to the principal plane. All photograph parallels are perpendicular to the principal line.

photograph perpendicular—The perpendicular from the interior perspective center to the plane of the photograph.

photograph plane—The plane in the camera in which the plate or film is held. It is not exactly the primary focal plane of the lens, but is a plane placed so as to secure the best balance of sharp focus on all parts of the plate or film. Also called **image plane.**

photograph plumb point—See **photograph nadir.**

photographic coverage—(JCS) The extent to which an area is covered by photography from one mission or a series of missions or in a period of time. Covarage in this sense conveys the ideal of availability of photography and is not a synonym for the word **photography.**

photographic datum—The effective datum for each photograph. It is a horizontal plane at the average elevation of the terrain, on which distances measured will be at the average scale of the photograph.

photographic exposure—The time of exposure multiplied by irradiance or illuminance.

photographic interpretation—The examination of photographic images for the purpose of identifying objects and deducing their significance. Also called **photointerpretation.** See also **imagery interpretation.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

photographic reading—(JCS) The simple recognition of natural or man-made features from photographs not involving imagery interpretation techniques.

photographic reduction—The production of a negative, diapositive, or print at a scale smaller than the original.

photographic scale—(JCS) The ratio of a distance measured on a photograph or mosaic to the corresponding distance on the ground, classified as follows: very large-scale—1:4,999 and larger; large-scale—1:5,000 to 1:9,999; medium-scale—1:10,000 to 1:24,999; small-scale—1:25,000 to 1:49,999; very small-scale—1:50,000 and smaller. See also **scale.**

photographic survey—A survey accomplished from either aerial photographs or terrestrial photographs, or from a combination of both.

photographic zenith tube (PZT)—The most precise instrument for meridian observations. No corrections are required for level, azimuth, collimation, or flexure. Each observation gives a measure of both the time and the latitude.

photography—The art or process of producing images on sensitized material through the action of light. The term **photography** is sometimes incorrectly used in place of the term **photographs.** See also **analytical photography; composite air photography; continuous strip photography; control point photography; convergent photography; cross-flight photography; direct photography; fan camera photography; horizontally controlled photography; indirect photography; inertial reference photography; iorop photography; mapping photography; metric photography; multiband photography; positional camera photography; process photography; radar photography; radarscope photography; reconnaissance photography; shoran controlled photography; split vertical photography; supplemental photography; terrain profile photography; tricamera photography.**

photograph—A general term for a positive or negative picture made with a camera on sensitized material, or prints from such a camera original. See also **aerial photograph; annotated photo-graph; equivalent vertical photograph; homologous photo-graphs; horizon photograph; horizontal photograph; molded aerial photograph; multiple-lens photograph; oblique air photo-graph; orthophotograph; pano-ramic photograph; pinpoint photo-graph; terrestrial photograph; vertical photograph; wing photo-graph.**

photoidentification—(surveying) The detection, identification, and marking of ground survey stations on aerial photographs. Positive identification and location is required if survey data are to be used to control photogrammetric compilation. Also called **control-station identification.**

photointerpretation key—Reference materials designed to facilitate rapid and accurate identification and the determination of the significance of objects or conditions from an analysis of their photo images.

photointerpretation—See **photographic interpretation.**

photointerpretometer—A device, used in conjunction with a pocket stereoscope, for making vertical and horizontal measurements.

photolithography—A lithographic process in which photographic products are used to produce an image on the printing surface. See also **lithography; offset lithography.**

photomap backup—A photomap printed on the back of a line map of the same area and at the same scale.

photomapping—The process of making maps or charts from various types of photographs, with reference to other source maps, charts, or surveys.

photomap—(JCS) A reproduction of a photograph or photomosaic upon which the grid lines, marginal data, contours, place names, boundaries, and other data may be added.

photomechanical—Pertaining to or designating any reproduction process by a combination of photographic and mechanical operations.

photorevised map—A topographic or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

planimetric map which has been revised by photoplanimetric methods.

photosphere—The intensely bright portion of the Sun visible to the unaided eye.

phototheodolite—A ground surveying instrument combining a survey camera and a transit; used for measuring the angular orientation of the camera at the moment of exposure. Also called **camera transit.**

phototopography—The science of surveying in which the detail is plotted entirely from photographs taken at suitable ground stations. See also **terrestrial photogrammetry.**

phototriangulation—The process for the extension of horizontal and/or vertical control whereby the measurements of angles and/or distances on overlapping photographs are related into a spatial solution using the perspective principles of the photographs. Generally, this process involves using aerial photographs and is called **aerotriangulation, aerial triangulation,** or **photogrammetric triangulation.** See also **analytical nadir-point triangulation; analytical phototriangulation; analytical radial triangulation; Arundel method; bridging; cantilever extension; direct radial triangulation; extension of control; graphical radial triangulation; hand-templet triangulation; isocenter triangulation; mechanical templet triangulation; nadir-point triangulation; radial triangulation; slotted-templet triangulation; spider-templet triangulation; stereotemplet triangulation; stereotriangulation; strip radial triangulation; templet method.**

phototrig traverse—A vertical-angle traverse employing phototrig methods; a procedure for determining elevations trigonometrically, wherein horizontal distances are determined photogrammetrically and vertical angles are either measured instrumentally in the field, or are obtained from measurements on terrestrial photographs.

phototypesetter—A type setting unit comprising two separate and independent units, the keyboard unit and the photographic unit. Composition is accomplished at the keyboard unit, essentially an electric typewriter, which produces a typewritten proof copy and a perforated tape. The tape is then fed at any convenient time thereafter to the photographic unit which produces a right-reading film positive suitable for stickup work.

physical characteristic—(target) The visible material aspects of a target or installation, including, but not limited to, dimensions, structural materials, predominant height, configuration, and orientation of its various components such as buildings, structures, runways, and associated facilities and services.

physical geodesy—See **gravimetric geodesy.**

physiographic pictorial map—A map with relief depicted by the systematic application of a standardized set of conventional pictorial symbols, based on the simplified appearance of the physical features they represent, as viewed obliquely from the air at an angle of about 45°.

piano-wire tape—Piano wire used instead of a metallic ribbon tape when it is advisable to control hydrography by precise traverse rather than by a weak extension of triangulation.

pictochrome process—The process employed to produce pictomaps. Consists of three tonal separations photographically extracted from a photomosaic, blockout masks, drafted symbols, and names data.

pictogram—A map of distributions, especially commodities in which small pictorial representative symbols (e.g., sacks, bricks, barrels) are located over the area of production.

pictoline process—A photographic masking process utilizing a rotating vacuum frame to produce an edge-enhanced line image from a continuous tone image.

pictomap—(JCS) A topographic map in which the photographic imagery of a standard mosaic has been converted into interpretable colors and symbols by means of a pictomap process. See also **pictochrome process.**

pictorial symbolization—(JCS) The use of symbols which convey the visual character of the features they represent.

pictotone process—A photo- lithographic method from which film for reproduction and transfer to printing plates is derived for the printing of monochrome photomaps and pictomaps. The process provides a random

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

granular-like effect which visibly sharpens the definition of features and separation of tones, and, in many instances, is superior to halftone printing.

picture control points—Supplementary horizontal and vertical control points that are required for the immediate control of mapping operations in a given area. These points are established by field survey parties in specific locations and are precisely identified on the aerial photographs for the project. Also called **photocontrol point; picture point.**

picture element (pixel)—**See pixel.**

picture plane—A plane upon which can be projected a system of lines or rays from an object to form an image or picture. In perspective drawing, the system of rays is understood to converge to a single point. In photogrammetry, the photograph is the picture plane.

picture point—See **control point; picture control points.**

pilot chart—Special charts, covering the oceans of the world for each month of the year, issued on one sheet for 3 months at a time on a quarterly basis. They show meteorologic, oceanographic, and hydrographic data for use in conjunction with conventional charts. Timely articles of professional interest to the seafarer are published on the backs.

pilot sheet—A sample of a new series, made as a trial in anticipation of a map series, to disclose the problems which occur in the various stages of compilation, drafting, and reproduction. It is later used as a guide in developing the series. Also called **prototype.** See also **experimental map.**

pilot's trace—(JCS) A rough overlay to a map made by the pilot of a photographic reconnaissance aircraft during or immediately after a sortie. It shows the location, direction, number, and order of photographic runs made, together with the camera(s) used on each run.

pilot—See **sailing directions.**

pinholes—Tiny clear spots on negative images caused by dust, air bubbles, or undissolved chemicals.

pinpoint photograph—(JCS) A single photograph or a stereopair of a specific object or target.

pinpoint target—(JCS) In artillery and naval gunfire support, a target less than 50 meters in diameter. See also **area target; precise installation position.**

pin—(surveying) A metal pin used for marking taped measurements on the ground. A set consists of 11 pins. Also called **chaining pin; surveyor's arrow; taping arrow; taping pin.** See also **turning point pin.**

pitch—1. (JCS) The rotation of an aircraft or ship about its lateral axis. 2. (JCS) In air photography, the camera rotation about the transverse axis of the aircraft. 3. (photogrammetry) A rotation of the camera, or of the photograph coordinate system, about either the photograph *y*-axis or the exterior *y*-axis. In some photogrammetric instruments and in analytical applications, the symbol phi (ϕ) may be used. Also called **longitudinal tilt; tip** (which is an obsolete term); ***y*-tilt.**

pixel—A picture element, smallest unit of information in a grid cell map or scanner image. Also known as PEL.

place name—See **toponym.**

place—See **position,** definition 2.

plan position indicator (PPI)—1. (JCS) A cathode-ray tube on which radar returns are so displayed as to bear the same relationship to the transmitter as the objects giving rise to them. 2. A cathode-ray indicator in which a signal appears on a radial line. Distance is indicated radially and bearing as an angle.

plan range—(JCS) In air photographic reconnaissance, the horizontal distance from the point below the aircraft to an object on the ground.

plane coordinates—See **plane rectangular coordinates.**

plane curve—See **plane elliptical arc.**

plane elliptical arc—Any part of the line formed by the intersection of a plane and an ellipsoid. Also called **plane curve.**

plane parallel plate—See **parallel plate.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

plane polar coordinates—A system of polar coordinates in which the points all lie in one plane. In the terminology of analytical geometry, the distance from the origin to the point is the magnitude of the radius vector and the polar distance is the vectorial angle.

plane rectangular coordinates—A system of coordinates in a horizontal plane, used to describe the positions of points with respect to an origin. Also called **plane coordinates.**

plane survey—A survey in which the surface of the Earth is considered a plane. For small areas, precise results may be obtained with plane-surveying methods, but the accuracy and precision of such results will decrease as the area surveyed increases in size.

planetable map—A map compiled by planetable methods. The term includes maps made by complete field mapping on a base projection and field contouring on a planimetric base map.

planetable traverse—A graphical traverse accomplished by planetable methods.

planetable—A field device for plotting the lines of a survey directly from observations. It consists essentially of a drawing board mounted on a tripod, with a leveling device designed as part of the board and tripod. See also **alidade; Philadelphia leveling rod; stadia.**

planetary aberration—The angular displacement of the geometric direction between the object and the observer at the instant of light emission, from the geometric direction at the instant of observation.

planetary configurations—Apparent positions of the planets relative to each other and to other bodies of the solar system, as seen from the Earth.

planetary geometry—**1.** Mathematical treatment of the shape and figure of a planet. **2.** Mathematical treatment of relationships between two or more planets and/or their orbits.

planetary precession—That component of general precession caused by the effect of other planets on the equatorial protuberance of the Earth, producing an eastward motion of the equinoxes along the ecliptic.

planetoid—See **asteroid.**

planet—A celestial body of the solar system revolving around the Sun in a nearly circular orbit, or a similar body revolving around a star. See also **asteroid; inferior planets; inner planets; major planets; navigational planets; outer planets; principal planets; superior planets; terrestrial planet.**

plane—See **astronomic meridian plane; basal plane; collimation plane; epipolar plane; focal plane; geodetic meridian plane; ground plane; hill plane; horizontal plane; meridional plane; nodal plane; orbital plane; perspective plane; photograph plane; picture plane; principal plane; tangent plane; vertical plane.**

planimeter—A mechanical integrator for measuring the area of a plane surface. See also **polar planimeter.**

planimetric map—(JCS) A map representing only the horizontal position of features. Also called **line map.** See also **topographic map.**

planimetric-base map—A map prepared from aerial photographs by photogrammetric methods, as a guide or base for contouring.

planimetry—**1.** The science of measuring plane surfaces; horizontal measurements. **2.** Parts of a map which represent everything except relief; that is, works of man, and natural features such as woods and water.

planisphere—A representation, on a plane, of the celestial sphere, especially one on a polar projection, with means provided for making certain measurements such as altitude and azimuth. Also, a map representation, on a plane, of the Earth's sphere.

planispheric astrolabe—An astrolabe consisting of a full graduated circle with a centrally mounted alidade and accessory adjustable plates on which are engraved stereographic projections of the heavens and of the sphere for local latitudes.

Planning Terrain Analysis Data Base (PTADB)—A 1:250,000 scale hardcopy, geographic information system consisting of a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

set of selected single thematic terrain information overlays.

planning chart—A chart designed specifically for planning flight operations.

planning map—Small-scale military map used for general planning purposes.

plastic block—The block of bonded cellulose acetate sheets, each sheet equal in thickness to the contour interval at the scale of the relief model, from which the terrain base is cut. Also called **laminate.**

plastic relief map—A topographic map printed on plastic and molded into a three-dimensional form. The plastic medium is generally formed by heat and vacuum over a terrain model to achieve the three-dimensional representation.

plate coordinates—The *x*- and *y*-coordinates of control points appearing on a photographic plate.

plate level—A spirit level attached to the plate of a surveying instrument for leveling the graduated circle or, indirectly, making the vertical axis truly vertical.

plate reduction—Scaling of control point images on a stellar plate.

plate—**1.** (lithography) A thin metal, plastic, or paper sheet, that carries the printing image and whose surface is treated to make only the image areas ink receptive. Also called **press plate. See also color plate; combination plate. 2.** (photography) A transparent medium, usually glass, coated with a photographic emulsion. See also **diapositive; stellar plate.**

platform—The vehicle that holds a sensor. It is usually a satellite, but may be a plane or a helicopter. Sensors can be mounted on tripods for certain uses, such as examining electromagnetic radiation from various types of vegetation.

Platonic year—See **great year.**

plat—A diagram drawn to scale showing land boundaries and subdivisions, together with all data essential to the description and identification of the several units shown thereon, and including one or more certificates indicating due approval. A plat differs from a map in that it does not necessarily show additional cultural, drainage, and relief features. See also **cadastral map.**

plot map—See **photo index.**

plotting chart—**1.** (JCS) A chart designed for the graphical processes of navigation. **2.** A chart designed primarily for plotting and dead reckoning or lines of position from celestial observations or radio aids. Relief, culture, and drainage are shown as necessary.

plotting scale—The relationship of the size of the compilation to the size of the ground area it represents.

plot—(JCS) **1.** A map, chart, or graph representing data of any sort. **2.** To represent on a diagram or chart the position or course of a target in terms of angles and distances from known positions; locate a position on a map or chart. **3.** The visual display of a single geographical location of an airborne object at a particular instant of time. **4.** A portion of a map or overlay on which are drawn the outlines of the areas covered by one or more photographs. See also **master plot.**

plumb bob—A conical device, usually of brass and suspended by a cord, by means of which a point can be projected vertically into space over relatively short distances.

plumb line—**1.** The line of force in the geopotential field. The continuous curve to which the direction of gravity is everywhere tangential. **2.** A cord with a plumb bob at one end for determining the direction of gravity.

plumb point—See **photograph nadir.**

plunge—See **transit,** definition 3.

plus angle—See **angle of elevation.**

plus declination—See **declination,** definition 3.

plus distance—Fractional part of 100 feet used in designating the location of a point on a survey line as "4 + 47.2," meaning 47.2 feet beyond Station No. 4 or 447.2 feet from the initial point, measured along a specified line. See also **plus station.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

plus point—An intermediate point on a traverse course located by a plus distance from the beginning of the course.

plus sight—See **backsight.**

plus station—An intermediate point on a traverse, not at an even tape length distance from the initial point. See also **plus distance; taping station.**

Point Positioning Data Base (PPDB)—Sets of geodetically controlled photographs and accompanying data that enable trained personnel using appropriate hardware and software to derive precise coordinates for any feature identifiable within the PPDB.

point anomaly—The value of the gravity anomaly at a specific location as observed or predicted.

point base—A manuscript which contains radial centers, picture points, pass points, control points, and tie points from the photographs used in the radial triangulation method.

point feature—An object whose location can be described by a single set of coordinates.

point marker—A device used for identifying points on diapositives by either marking a small hole in the emulsion or marking a small ring around the detail point itself Also called **snap marker. See also point-transfer device.**

point of certainty—In a simple two-point intersection problem, that point where the two intersecting rays cross and the point is confirmed by the intersection of a third or check ray passing through the same point.

point of compound curvature (PCC)—The point on a line survey where a circular curve of one radius is tangent to a circular curve of a different radius, both curves lying on the same side of their common tangent.

point of contact—Any level surface along a terrain profile recorder (TPR) flight line that can be flown over both before and after the changing or adjustment of a TPR positional camera magazine, a chart roll, or a recording pen.

point of curvature (PC)—The point in a line survey where a tangent ends and a circular curve begins. See also **point of tangency.**

point of cusp—The point of tangency of two curves, the direction of the extension of said curves being of opposite sign; such as the vertex of a Y of a railroad track or a point on the edge of a convex-concave lens. May also be applied to the point of tangency of a straight line and a curve where the direction of extension of the line and curve are of opposite sign.

point of inflection—The point at which a reversal of direction of curvature takes place.

point of intersection (PI)—The point where the two tangents of a circular curve meet. Also called **vertex of curve.**

point of origin—See **initial point.**

point of reverse curvature (PRC)—The point of tangency common to two curves, the curves lying on the opposite side of the common tangent.

point of symmetry—The point in the focal plane of a camera about which all lens distortions are symmetrical. If the lens were perfectly mounted, the point of symmetry would coincide with the principal point.

point of tangency (PT)—The point in a line survey where a circular curve ends and a tangent begins. The point of tangency and point of curve are both points of tangency, their different designations being determined by the direction of progress along the line; the point of curvature is reached first.

point of vertical curve (PVC)—The point of change from a line of uniform slope to a vertical curve.

point of vertical intersection (PVI)—The point of intersection of two lines, each having different uniform slopes.

point of vertical tangent (PVT)—The point of change from a vertical curve to a line of uniform slope.

point position data (PPD)—The collective result of an analytical triangulation effort that provides evaluated geodetic positions of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

photoidentifiable ground points or reseau intersections. These positions are the result of an evaluated adjustment of the points to a specific mathematical surface and are expressed in terms of latitude, longitude, elevation, and positional accuracy for each point.

point positioning—(surveying) The process of establishing independent survey position. See also **short arc; short arc geodetic adjustment; translocation.**

point position—(Doppler) The geocentric or geodetic position of a point determined from satellite tracking data by a Doppler receiver and the satellite(s) ephemerides.

point the instrument—Turning the survey instrument to where the cross hairs (vertical, horizontal, or both) are accurately aligned with the target.

point-designation grid—(JCS) A system of lines having no relation to the actual scale or orientation, drawn on a map, chart, or air (aerial) photograph, dividing it into squares so that points can be more readily located.

point-in-polygon—A geographic information system (GIS) routine which assesses whether a point falls within a polygon.

point-matching method—(rectification) The technique of utilizing an autofocus rectifier for tilt removal by the manual matching of projected image points to those plotted in their correct horizontal position on a film templet.

point-transfer device—A stereoscopic instrument used to make corresponding image points on overlapping photographs. Also called **transcriber.** See also **point marker.**

pointer—(digital) The address of a record or other data groupings contained in another record.

pointing accuracy—The exactness, in surveying or photogrammetry, with which the line of sight or floating mark can be directed toward a target or image point.

pointing errors—Errors which reflect the accuracy with which the floating mark of a stereoplotting system can be located on a sharp model point. These errors generally follow a more or less random distribution but show a systematic trend with progressive working time on the instrument due to eye fatigue and its effect on stereoscopic perception.

pointing line—See **line of collimation.**

pointing—**1.** (mensuration) Placing the reticle or index mark of a precision measuring instrument, such as a comparator, within the symmetrical center or center of gravity of a point being measured to determine its position relative to the position of other points in some system of coordinates. **2.** (stereocompilation) A general term applied to the movement of the tracing table of a stereoplotting instrument to specific control and/or picture points on the datum during orientation of a stereomodel. **3.** See **line of sight,** definition 2.

point—A position on a reference system determined by a survey. See also **amphidromic point; angle point; annex point; antisolar point; astrogravimetric points; cardinal points; check point; control point; datum point; detail points; distant points; fix; image point; initial point; intercardinal point; map point; middle point; nodal point; orientation point; pass point; plus point; principal point; sublunar point; subsatellite point; subsolar point; substellar point; tie point; turning point; wing point; witness point.**

polar axis—The primary axis of direction in a system of polar or spherical coordinates.

polar bearing—In a system of polar or spherical coordinates, the angle formed by the intersection of the reference meridional plane and the meridional plane containing the point.

polar chart—**1.** A chart of polar areas. **2.** A chart on a polar projection. The projections most used for polar charts are the gnomonic, stereographic, azimuthal equidistant, transverse Mercator, and modified Lambert conformal.

polar circle—Either the Arctic Circle (north polar circle) or the Antarctic Circle (south polar circle).

polar coordinates—(JCS) **1.** Coordinates derived from the distance and angular measurements from a fixed point (pole). **2.** In artillery and naval gunfire support, the direction, distance, and vertical correction from the observer/spotter position to the target.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

polar diameter—The diameter of the Earth between the poles.

polar distance—Angular distance from a celestial pole; the arc of an hour circle between a celestial pole, usually the elevated pole, and a point on the celestial sphere, measured from the celestial pole through 180°. See also **codeclination.**

polar grid—A grid system utilized for aerial navigation in the polar regions. It consists of a rectangular grid with *x*- and *y*-axes aligned with the 0°—180° and the 90°E—90°W meridians respectively. When plotted on a transverse Mercator map projection of the polar regions, it represents a system of transverse meridians and parallels whose poles are at the intersections of the Equator and the 0°—180° meridian.

polar map projection—A map projection centered on a pole.

polar motion—See **variation of the pole.**

polar orbit—An Earth satellite orbit that has an inclination of about 90° and, hence, passes over the Earth's poles.

polar orthographic map projection—A map projection having the plane of the projection perpendicular to the axis of rotation of the Earth (parallel with the plane of the Equator); in this projection, the geographic parallels are full circles, true to scale, and the geographic meridians are straight lines.

polar planimeter—An instrument used in measuring areas from a drawing. The instrument rotates about a pole, hence its name.

polar radius—The radius of the Earth measured along its axis of rotation.

polar satellite—Any satellite that passes over the north and south poles of the Earth; i.e., one that has an inclination of about 90° with respect to the Earth's Equator.

polar stereographic map projection— A stereographic projection having the center of the projection located at a pole of the sphere.

Polaris correction—A correction to be applied to the observed altitude of Polaris to obtain the latitude.

Polaris—The second-magnitude star, Alpha, in the constellation Ursa Minor (Little Dipper). Also called **North Star; polestar.**

polarization filter—Any of the manufactured plastic fibers which plane polarizes ordinary light when it passes through the filter. Usually identified by a trade name.

polarization—(optics) The act or process of modifying light in such a way that the vibrations are restricted to a single plane. According to the wave theory, ordinary (unpolarized) light vibrates in all planes perpendicular to the direction of propagation. On passing through or contacting a polarizing medium (such as Polaroid or a Kerr cell) ordinary light becomes plane polarized, that is, its vibrations are limited to a single plane.

polastrodial—A mechanical counter for determining the azimuth and altitude of Polaris at any time.

pole of the Milky Way—The pole in the galactic system of coordinates.

polestar—See **Polaris.**

pole—1. Either of the two points of intersection of the surface of a sphere or spheroid and its axis. 2. The origin of a system of polar coordinates. See also **average terrestrial pole; celestial pole; depressed pole; ecliptic pole; elevated pole; fictitious pole; galactic pole; geomagnetic pole; magnetic pole; north geographical pole; north geomagnetic pole; north magnetic pole; oblique pole; south geographical pole; south geomagnetic pole; south magnetic pole; terrestrial pole; transverse pole.**

polhody—A chart depicting the motion of the terrestrial pole as a function of time. See also **variation of the poles.**

polyart—A plastic based printing substrate with excellent wet strength.

polychrome—See **multicolor.**

polyconic chart—A chart on the polyconic map projection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

polyconic map projection—A *map* projection having the central geographic *meridian represented* by a straight line, along which the spacing for lines representing the geographic parallels is proportional to the distances between the parallels; the parallels are represented by arcs of circles which are not concentric, but whose centers lie on the line representing the central meridian, and whose *radii are determined* by the lengths of the elements of cones which are tangent along the parallels. All meridians except the central ones are curved. The projection is neither conformal nor equal area, but it has been widely used for maps of *small areas because of the ease with* which it can be constructed.

polygonization—The process of connecting together linear feature information to form polygons.

polygon—Thematically common areas composed of contiguous faces.

polyhedric projection—A projection used for a large-scale topographic map whereby a small quadrangle on the spheroid is projected onto a plane trapezoid. Scale is made true either on the central meridian or along the sides.

Porro prism—A *prism* that deviates the axis 180° and inverts the image in the plane in which the reflection takes place. It may be described as two right-angle prisms cemented together.

Porro-Koppe principle—The *principle* applied in some photogrammetric instruments to *eliminate the effect of camera*-lens distortion. The photographic positive or negative is observed through a lens or optical system identical in distortion characteristics to the camera objective which made the original exposure. *In effect, this method of observation* is a reverse use of the camera, with the focal plane becoming the object which is imaged at infinity by parallel bundles of rays emerging from the lens. The chief ray of each bundle assumes its correct direction, and the cone of rays is identical to that whose vertex was the incident node of the *camera lens at the instant* of exposure. The parallel bundles may be observed by means of a telescopic system focused at infinity and made rotatable about the incident node of the lens. This method of eliminating lens distortion is *utilized in* photogrammetric instruments of both the *monoscopic type, such as* the photogoniometer, and the stereoscopic type used for stereoplotting.

port plan—A special-purpose large-scale map of a port area showing piers, railroad extensions, repair facilities, pilot office, customhouse, and other *applicable nonnavigational features.*

portable automatic tide gage—A small automatic tide gage, designed for use where a short series of observations is necessary for the reduction of soundings to a common datum.

position angle—**See parallactic angle,** definition 1.

position plotting sheet—A blank chart, usually on the Mercator projection, showing only the graticule and a compass rose, so that the chart can be used for any longitude. See also **universal plotting sheet.**

positional accuracy—*(cartography)* A *term* used in evaluating the overall reliability of the positions of cartographic features on a map or chart relative to their true position, or to an established standard.

positional camera photography—*Photography obtained with a camera* aligned with the TPR radar beam, used for correlation and transfer of recorded vertical data to the cartographic photography.

positional error—(cartography) The amount by which a cartographic feature fails to agree *with its true position.*

positioning camera—A camera used for correlation purposes in the airborne profile recorder system. It is mounted on the radar antenna and records the area *illuminated* by the radar beam.

position—1. Data which define the location of a point with respect to a reference system. 2. The place occupied by a point on the surface of the Earth or in space. Also called **place.** 3. The coordinates which define the location of a point on the geoid or spheroid. 4. A prescribed *setting (reading)* of the horizontal circle of a direction theodolite which is to be used for the observation on the initial station of a series of stations which are to be observed. Also called **circle position.** See also **adjusted position; apparent position; astrometric position; astronomic position; celestial fix; celestial line of position; circle of**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

position; convergent position; electronic line of position; field position; fix; geocentric station position; geodetic position; geographic position; line of position; mean position; point position; point positioning; precise installation position; preliminary position; relative position; supplemental position; transverse position; true position.

positive altitude—Angular distance above the horizon.

positive deflection angle—See **deflection angle.**

positive forming—In relief model making, forming over a positive mold.

positive lens—A lens that converges a beam of parallel light rays to a point focus. Also called **converging lens; convex lens.**

positive mold—The cast pulled from a negative mold when making a relief model.

positive—In photography, an image on film, plate or paper having approximately the same total rendition of light and shade as the original subject.

potential disturbance—See **disturbing potential,** definition 1.

potential of disturbing masses—See **disturbing potential,** definition 1.

potential of random masses—See **disturbing potential,** definition 1.

potential—A scalar function, the gradient of which results in a vector field. Use of the scalar function simplifies investigation and description of the phenomenon considered. Used extensively for magnetic, gravitational, and gravity field investigations. In celestial mechanics and geodesy, the negative of the potential, sometimes called the **force function,** is usually employed. See also **disturbing potential; gravitational potential.**

power of a lens—See **diopter; magnification.**

power of telescope—(surveying) The magnification of a telescope when focused at infinity.

Pratt-Hayford theory of isostasy—A theory of isostatic compensation which assumes that every topographic excess or defect of mass is compensated by an equal and opposite defect or excess, evenly distributed immediately below it between ground level or sea bottom level and a fixed depth, called the **depth of compensation,** commonly 113.7 km. Also called **fermenting dough theory.** See also **Airy theory of isostasy.**

Pre-Ship Readiness Review (PSR)—A review under the verification phase of DMA's modernization program (DPS). PSR examines segment factory testing, planned testing activities, and production center readiness to determine when to ship DPS equipment from the contractor to the production centers.

precession in declination—The component of general precession along a celestial meridian, amounting to about 20.0"of arc per year.

precession in right ascension—The component of general precession along the celestial equator, amounting to about 46.1" of arc per year.

precession of the equinoxes—The conical motion of the Earth's axis about the vertical to the plane of the ecliptic, caused by the attractive force of the Sun, Moon, and other planets on the equatorial protuberance of the Earth. See also **general precession.**

precession—Change in the direction of the axis of rotation of a spinning body, as a gyroscope, when acted upon by a torque. The direction of motion of the axis is such that it causes the direction of spin of the gyroscope to tend to coincide with that of the impressed torque. See also **drift; general precession; planetary precession; topple; topple axis; total drift.**

Precise Bathymetric Navigation Zone Charts (PBNZCs)—Charts depicting highly detailed underwater topography in the form of bathymetric curves. Detailed multibeam/swath bathymetry allow submarines to precisely fix their positions from bottom contour matching. Also called **Bathymetric Recovery Area Charts (BRACs).**

precise ephemeris—Coordinates and velocity of an artificial satellite, computed for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

uniform time intervals from data acquired from a worldwide tracking network. The ephemeris is computed from observations taken from many stations spaced worldwide and adjusted together by least squares methods for maximum accuracy. See **broadcast ephemeris; ephemeris; Navy Navigation Satellite System.**

precise installation position (PIP)—Geodetic coordinates of installation reference points reflecting their maximum possible refinement by utilizing all intelligence sources and optimum computer techniques of analytical area adjustment.

precise leveling rod—A rod used for precise leveling. The graduations are on an invar ribbon which is maintained under constant tension and which, for all practical purposes, eliminates the need for correcting for changes in length. These rods are usually graduated in whole and fractional meters. The back side of the rod is graduated in feet and tenths of feet. Also called **invar leveling rod; meter rod; Molitar precise leveling rod; National Geodetic Survey first-order leveling rod.**

precise level—An instrument designed specifically for obtaining precise results by direct leveling techniques. It is essentially the same as an engineer's level except that it contains micrometer screws for more precise leveling of the instrument and contains a prism arrangement whereby the level bubble can be observed simultaneously with the rod reading.

precise radar significant location (PRSL)—The horizontal and vertical values derived for a selected ground feature that is radar significant. The return may be either positive or negative.

Precision Automatic Photogrammetric Intervalometer System (PAPI)—Automatic intervalometer utilizing radar in determining interval for desired aerial photography forward lap.

precision altimeter—A sensitive aneroid barometer. In surveying, it will produce results accurate to within a meter only when the two-base method is carefully applied. See also **two-base method.**

precision camera—A relative term used to designate any camera capable of giving high resolution and dimensional results of a high order of accuracy.

precision depth recorder—A device which records sounding information on electrosensitive paper for depths up to 6,000 fathoms. It provides a trigger to the sonar and performs a time measuring function.

precision—The closeness with which repeated measurements made under similar conditions are grouped together. NOTE: Precision is affected only by the random errors in the measuring process. Accuracy is affected by precision and unknown and/or systematic errors.

predominant height—(JCS) In air reconnaissance, the height of 51 percent or more of the structures within an area of similar surface material.

preferred datum—A geodetic datum selected as a base for consolidation of local independent datums within a geographical area. Also called **major datum.**

preferred grid—1. The grid designated by DoD for production of new maps, charts, and digital geographic data; and shown on the "Index to Preferred Grids, Datums, and Ellipsoids Specified for New Mapping." 2. The grid preferred by DMA for production of new maps and charts.

Preliminary Design Review (PDR)—A review conducted prior to commencing the detailed design process to assure that the approach is feasible and sound from a design, development, test, and activation point of view.

preliminary edition—See **provisional edition.**

preliminary elevation—An elevation arrived at in the office after the index, level, rod, and temperature corrections have been applied to the observed differences of elevation and new elevations have been computed.

preliminary orientation—An initial, rough orientation of projectors prior to accomplishing relative orientation of a stereomodel. It is the approximate leveling and scaling of the instrument frame and projectors, based on the best estimate of what their ultimate orientations are assumed to be.

preliminary position—In the adjustment of triangulation, the term preliminary is applied to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

geographic positions derived from selected observations for use in forming latitude and longitude condition equations.

preliminary survey—The collection of survey data on which to base studies for a proposed project. See **also reconnaissance survey.**

preliminary triangle—In the adjustment of triangulation, the term **preliminary** is applied to triangles derived from selected observations for use in forming latitude and longitude condition equations.

preliminary—Not of the desired accuracy and precision, and adopted for temporary use with the proviso of later being superseded.

preprocessing—Preliminary transformation of raw data required to facilitate further cartographic processing.

prepunch register system—A method in which a system of precisely located holes are punched in the margins of map or chart materials (such as films, vinyls, etc.) prior to their actual use. Exact register of materials can be accomplished by placing register studs (small plastic or metal pins) through the holes, thereby assuring exact register of detail. See also **register marks.**

prescribed grid—The grid that is locally prescribed by the country of origin of military commander.

Presentation Manager—A graphical user interface developed by IBM for use with the OS/2 operating system.

press plate—See **plate.**

press proof—A lithographed impression taken from among the first copies run on the press and used for checking purposes. Also called **press pull.**

press pull—See **press proof.**

pressure altimeter—See **barometric altimeter.**

pressure altitude—(JCS) An atmospheric pressure, expressed in terms of altitude which corresponds to that pressure in the standard atmosphere. See also **altitude.**

pressure gage—A tide gage that is operated at the bottom of the body of water being gaged and which records tidal height changes by the difference in pressure due to the rise and fall of the tide.

primary bench mark—A bench mark close to a tide station to which the tide staff and tidal datum originally were referenced.

primary body—The celestial body or central force field about which a satellite or other body orbits, or from which it is escaping, or towards which it is falling. Also called **primary.**

primary circle—See **primary great circle.**

primary compilation—A specially prepared matte plastic material used to depict sounding data corrected to true depths in bathymetric compilations.

primary great circle—A great circle used as the origin of measurement of a coordinate; particularly such a circle 90° from the poles of a system of spherical coordinates, as the Equator. Also called **fundamental circle; primary circle.**

primary grid—The major, or principal grid on map or chart.

primary key—A key that uniquely identifies a row of data in a data base.

primary row—A single row of attribute data associated with a simple feature.

primary station—See **main scheme station.**

primary tide station—A place at which continuous tide observations are made over a number of years to obtain basic tidal data for the locality.

primary—See **primary body.**

prime fictitious meridian—The reference meridian (real or fictitious) used as the origin for measurement of fictitious longitude.

prime grid meridian—The reference meridian of a grid. In polar regions it is usually the 180°—0° geographic meridian, used as the origin for measuring grid longitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

prime inverse meridian—See **prime transverse meridian.**

prime meridian—The meridian of longitude 0°, used as the origin for measurement of longitude. The meridian of Greenwich, England, is almost universally used for this purpose.

prime oblique meridian—The reference *fictitious meridian of an* oblique graticule.

prime transverse meridian—The reference meridian of a transverse graticule. Also called ***prime inverse meridian.***

prime vertical circle—*The vertical circle* through the east and west points of the horizon. It may be true, magnetic, compass, or grid, depending upon which east or west points are involved. Also called **prime vertical.**

prime vertical plane—*The plane* perpendicular to the meridian plane (astronomic or geodetic) containing the normal. The intersections of the astronomic prime vertical plane with the horizon are the east and west points.--Also called **east point.**

prime vertical—See **prime vertical circle.**

primitive—(digital mapping) The lowest level of attribution.

principal axis—See **optical axis.**

principal distance—**1.** *The* perpendicular distance from the internal perspective center to the plane of a particular finished negative or print. This distance is equal to the calibrated focal length corrected for both the enlargement or reduction ratio and the film or paper shrinkage or expension. It maintains the same perspective angles at the internal perspective center to points on the finished negative or print, as existed in the taking camera at the moment of exposure. This is a geometrical property of each particular finished negative or print. Also called **effective focal length. 2.** (multiplex) The perpendicular distance from the internal perspective center of the projector lens to the plane of the emulsion side of the diapositive.

principal focus—See **focus.**

principal line—The trace of the principal plane upon a photograph (e.g., the line through the principal point and the nadir point).

principal meridian—**1.** The meridian extended from an initial point, upon which regular quarter-quarter section, section, and township corners have been or are to be established. See also **auxiliary guide meridian; guide meridian. 2.** (photogrammetry) See **principal point.**

principal parallel—(JCS) On an oblique photograph, a line parallel to the true horizon and passing through the principal point.

principal planets—The larger bodies revolving about the Sun in nearly circular orbits. The known principal planets, in order of their distance from the Sun, are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, and Pluto.

principal plane—**1.** (JCS) (photogrammetry) A vertical plane which contains the principal point of an oblique photograph, the perspective center of the lens, and the ground nadir. **2.** (optics) A plane through a principal point and perpendicular to the optical axis. See also **axis of homology; axis of tilt; ground parallel; ground trace; horizon trace; isometric parallel; line of constant scale; line of equal scale; photograph meridian; photograph parallel; principal line; principal meridian; principal parallel: vanishing line; vanishing point.**

principal point—**1.** (JCS) (photogrammetry) The foot of the perpendicular to the photo plane through the perspective center. Generally determined by intersection of the lines joining opposite collimating or fiducial marks. [If the fiducial marks are not visible on the photograph, the principal point may be found by drawing diagonals between opposite corners or by measuring one-half the distance along each side of the photograph and connecting these marks in the same manner as fiducial marks, or by reseau marks.] Also called **indicated principal point.** See also **photograph center. 2.** (calibrated) The center of radial lens distortion, usually given as *x*- and *y*- distances from indicated principal point.

principal scale—(JCS) In cartography, the scale of a reduced or generating globe representing the sphere or spheroid defined by the fractional relation of their respective radii. Also called **nominal scale.** See also **scale.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

principal station—See **main scheme station.**

principal vertical circle—The vertical circle through the north and south points of the horizon, coinciding with the celestial meridian.

principal vertical—(JCS) On an oblique photograph, a line perpendicular to the true horizon and passing through the principal point.

principal-distance error—In a stereoplotting system, an instrument error resulting from improper calibration of the aerial camera, diapositive printer, or projector. The error is of little importance in a flat surface model but the effects are increased in proportion to the relief in the model.

principal-point assumption—The assumption with respect to approximately vertical photographs that radial directions are correct if measured from the principal point.

principal-point error—A personal error in which the principal points in a stereoplotting system are displaced in such a manner that they have unequal *x* components with a resultant error in vertical scale. Such errors are usually introduced into the system by either improper orientation of the diapositive plate in the printer, in the projector, or both.

principal-point radial—A radial from the principal point of a photograph.

principal-point triangulation—See **radial triangulation.**

print—A photographic copy made by projection or contact printing from a negative or transparency. See also **contact print; diapositive; enlargement; matte print; photographic reduction; projection print; ratio print; rectified print; transformed print.**

prismatic astrolabe—An astrolabe consisting of a telescope in a horizontal position, with a prism and artificial horizon attached at its objective end, used for determining astronomic positions.

prismatic compass—A small magnetic compass held in the hand when in use and equipped with peepsights and glass prism so arranged that the magnetic bearing or azimuth of a line can be read at the same time that the line is sighted over.

prismatic error—That error due to lack of parallelism of the two faces of an optical element, such as a mirror or a shade glass.

prism—A transparent body bounded in part by two plane faces that are not parallel; used to deviate or disperse a beam of light. See also **dove prism; horizon prism; index prism; Porro prism; reflecting prism; refracting prism; retrodirective prism; rhomboidal prism; right-angle prism; roof prism; wedge.**

Probabilistic Vertical Obstruction Data (PVOD)—A file which combines residual density functions with discrete vertical obstruction data extracted from Vertical Obstruction Data (VOD), Digital Feature Analysis Data (DFAD), the DMA Digital Vertical Obstruction File (DVOF), Powerline Data, and intelligence databases. PVOD provides cruise missile and aircraft mission planners vertical obstruction information which can be used to compute probability of mission success, set mean clearance heights, and plan mission routes. PVOD is the replacement product for VOD.

probability interval—See **confidence interval.**

probable error—**1.** In a measured value, it is the most probable value of the resultant error in the measurement. This is a plus or minus quantity that may be larger than the resultant error or smaller than the resultant error, and its probability of being larger is equal to its probability of being smaller. **2.** The 50 percent error interval based on the bivariate normal distribution function.

process camera—See **copy camera.**

process color printing—(lithography) A technique for the reproduction of a subject or chart in full color rendition, by combining tones of the primary colors, (yellow, magenta, cyan) and black. See also **process plates.**

process lens—A lens for photochemical copying, enlarging, or projection purposes, free from aberrations, usually of low aperture and of symmetrical construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

process photography—Line and halftone photography in which the resulting negatives and positives are subsequently used in the preparation of press plates.

process plates—Two or more color press plates combined to produce other colors and shades. See also **color plate; combination plate; process color printing.**

Product Generation Segment (PG/S)—A segment of DMA's DPS which will provide the capability to generate both digital products and reproducible materials for hard copy maps and charts. By utilizing digitizing capabilities and graphic workstations, the segment will provide for collection of MC&G data from existing hard copy graphics including maps, charts, and manuscripts. During product generation, the segment will extract appropriate data from the MARK 90 MC&G database to generate required products. In producing color separates for graphic products, PG/S will convert the digital data to graphic symbols, generalize and displace graphic symbols, perform color separation, and generate the reproducible copy. See also **MARK 90; Digital Production System.**

product data—Data that is either collected in a specific product form or data which is derived from neutral data for a specific application.

Production Management Segment (PM/S)—Segment of DMA's DPS which will incorporate the MARK 85 Data Integration Segment to provide a total production planning and management capability using the MARK 90 production processes. This segment will perform requirements, definition, long-range planning, program development, production scheduling, and status reporting. See also **MARK 90; Digital Production System.**

profile leveling—The determination of elevations of points at short measured intervals along a definitely located line, such as the center line of a highway.

profile—1. A vertical section of the surface of the ground, or of underlying strata, or both, along any fixed line. 2. Elevation of the terrain along some definite line. Elevations are measured at a sufficient number of points to enable defining the configuration of the ground surface.

Programmers Hierarchical Interactive Graphics System (PHIGS)—An ANSI standard which provides a set of functions for modification of graphics data and the relationships between the graphical data; and the definition, display, and modification of geometrically related objects, and two and three dimensional graphical data. See also **Graphic Kernal System.**

progress sketch—A map or sketch showing work accomplished. In triangulation and traverse surveys, each point established is shown on the progress sketch, and also lines observed over and base lines measured. In a leveling survey, the progress sketch shows the route followed and the towns passed through, but not necessarily the locations of the bench marks.

progressive motion—Motion in an orbit in the usual orbital direction of celestial bodies within a given system. Specifically, motion of a satellite in the same direction to the direction of the primary. Opposite of **retrograde motion.**

progressive proofs—A series of color prints that show the individually separated color printings of a job and their progressive combinations as each color is overprinted.

projected map display (PMD)—An inflight navigation aid which uses a continuous color or black-and-white sprocketed filmstrip containing chart imagery, projected on a display device and driven by the aircraft's computer. The aircraft's position is displayed with correlated chart image to show location and direction actually being flown.

projection computation—The determination, from a set of tables derived from formulas, of the true shape and dimensions of a map projection, for the purpose of constructing such a projection. See also **grid computation.**

projection distance—The distance from the external node of a projection lens to the plane onto which the image is projected.

projection generation—The transformation of a spherical surface, such as the earth, to a plane surface. Regardless of the transformation method, there is always some distortion in the plane view.

projection printer—An optical device for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

enlarging or reducing the image of a negative or positive transparency by projecting it onto a sensitized surface.

projection print—(JCS) An enlarged or reduced photographic print made by projection of the image of a negative or a transparency onto a sensitized surface.

projection tables—Data made available in tabular form for determining a definite relationship which exists between any grid intersection and any adjacent intersection of latitude and longitude lines on the map projection.

projection ticks and crosses—Ticks perpendicular to and inside the neatline of a map placed to indicate points through which parallels and meridians would pass if they had been extended. Small cross marks indicate where the lines intersect within the map.

projection transformation—The process of changing from one map projection to another map projection. Transformations may involve changes in equations, ellipsoids, scale factors, cone constants, standard lines, standard parallels, origins, central meridians, etc.

projection—1. (geometry) The extension of lines or planes to intersect a given surface; the transfer of a point from one surface to a corresponding position on another surface by graphical or analytical methods. See also map **projection.** 2. (photography) The process of placing a negative or positive photograph in a projecting camera and reproducing the image on a screen or on a sensitized photographic medium. 3. (surveying) The extension of a line beyond the points which determine its character and position. The transfer of a series of survey lines to a single theoretical line by a series of lines perpendicular to the theoretical line. In surveying a traverse, a series of measured short lines may be projected onto a single long line, connecting two main survey stations, and the long line is then treated as a measured line of the traverse. See also **prolongation.**

projector station—The position of a projector unit of a stereoplotter when absolute orientation has been accomplished. This position recreates the conditions existing at the corresponding camera station at the instant of exposure.

projector—An optical instrument which throws the image of a negative or print upon a screen or other viewing surface, usually at a larger scale. See also **reflecting projector.**

prolate ellipsoid of rotation—See **prolate spheroid.**

prolate spheroid—An ellipsoid of rotation along the major axis, which is along the axis of earth rotation. Also called **prolate ellipsoid of rotation.**

prolongation—In surveying, a line is prolonged when the last segment of the surveyed line is extended in the same direction as the segment itself. A prolongation of a curve under such a definition of extension would be a line tangent to the curve at the point of extension, although the term frequently is used to mean a continuation along the curvature of the curve.

proofing—The operation of pulling proofs of plates for proofreading, revising, approval, and other purposes prior to production printing.

proof—A trial print, produced by any method, for examination or editing, to be marked for necessary corrections or approval. See also **color composite; color proof; color proof process; composite; final composite; galley proof; hand proof; OK sheet; press proof; progressive proofs; proofing.**

proper motion—That component of the space motion of a celestial body perpendicular to the line of sight, resulting in the change of a star's apparent position relative to a coordinate system such as right ascension and declination. This change is expressed as a velocity, such as seconds of arc per century.

property map—See **cadastral map.**

property survey—See **land survey.**

proportionate measurement—A measurement that applies an even distribution of a determined excess or deficiency of measurement, ascertained by retracement of an established line, to provide concordant relations between all parts. See also **double proportionate measurement; single proportionate measurement.**

prototype—See **experimental map; pilot sheet.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

provisional edition—A map or chart printed and distributed for temporary use with the proviso that it will later be superseded. Also called **preliminary edition.**

provisional map—Any nonofficial map, photo, or other material which is used as a map. It may vary from a highly accurate captured enemy map, which has not been sanctioned for use, to a hastily made drawing or sketch. It is usually a hastily made line map based on aerial photographs, used as a map substitute.

proximity analysis—Analytical technique used to determine the relationship between a specified point and its neighbors.

proximity search—An analytical procedure to identify occurrences of predefined data elements in the neighborhood of a selected point.

proximity—A measure of closeness to a specified point as defined by a user.

pseudoscopic stereo—A three-dimensional impression of relief which is the reverse of that actually existing when the positions of a stereo pair of photographs are interchanged. Also called **inverted stereo; reverse stereo.** See also **false stereo.**

publication scale—See **reproduction scale.**

Pulkovo 1942 datum—The geodetic datum defined at Pulkovo Observatory, U.S.S.R. using the Krasovsky ellipsoid.

pull up—See **selection overlay.**

pulse Doppler map matching (PDMM)—An image matching concept employing a pulse Doppler mapping technique to locate three preselected unique edges within the terrain scene viewed prior to reentry of the vehicle. Edges are defined by differences in radar reflectance. Range and range rate data are obtained from a small elemental area of the terrain illuminated by the antenna beam spot during a fix action. A range/range rate map of the spot area is then correlated against a reference map of the particular edge to provide a range and velocity update to the onboard vehicle inertial measurement unit (IMU). Three such successive fix correlations are executed to enable a trilateration solution of the vehicle's position relative to the target scene by the sensor's computer. Therefore, the IMU can calculate a corrective maneuver prior to impact.

pygmy meter—A small cup type current meter for use in low velocity measurements in shallow streams. This meter is used in conjunction with wading rods only.

Pythagorean right-angle inversor—A simple device which provides a mechanical solution for linear and angular elements of rectification, thus permitting any enlarger to be made autofocusing provided negative, lens, and easel planes are parallel.

PZS triangle—See **astronomic triangle.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Q

quad corner areas—A user-defined measure relating to the portion of a quadrangle mapping sheet where it joins two or more other sheets.

quadrangle map—See **quadrangle.**

quadrangle report—A brief history of the mapping of a specific quadrangle. It accompanies the mapping material through each phase of production, and is filed with the map material. The narrative summary for each operational phase stresses conditions that may affect later phases.

quadrangle—A rectangular, or nearly so, area covered by a map or plat, usually bounded by given meridians of longitude and parallels of latitude. Also called **quad; quadrangle map.** See also **standard quadrangle.**

quadrant—1. (mathematics) A sector having an arc of 90°. 2. (surveying) A surveying or astronomic instrument composed of a graduated arc about 90° in length (180° in range), equipped with a sighting device. The quadrant may be considered a form of sector. Some survey quadrants combine both surveying and astronomic functions.

quadrature—1. The position in the phase cycle when the two principal tide producing bodies (Moon and Sun) are nearly at a right angle to the Earth; the Moon is then in quadrature in its first quarter or last quarter. 2. The situation of two periodic quantities differing by a quarter of a cycle.

quad—Abbreviated form of quadrangle or quadrilateral.

quality control—The process of taking steps to ensure the quality of data or operations in keeping with standards set for the system.

quarter section corner—A corner at an extremity of a boundary of a quarter section. Written as 1/4 section corner, not as one-fourth section corner.

quarter section—One-fourth of a section, containing 160 acres more or less.

quarter-quarter section corner—A corner at an extremity of a boundary of a quarter-quarter section; midpoint between or 20 chains from the controlling corners on the section or township boundaries. Written as 1/16 section corner. Also called **sixteenth section corner.**

quartz pendulum—A pendulum of fused quartz used for determining the acceleration of gravity. Quartz is employed in the construction because its thermal expansion coefficient is only one-fourth that of Invar.

query language—A set of commands used for data manipulation in data base management systems.

quick-look plot—A "draft" plot done very fast, which may not adhere to the positional accuracy or symbolization of the final plot.

quintant—A sextant having a range of 144°, or an arc of 72°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

R

R4D—(Ready for Distribution) A term referring to printed MC&G products ready to be distributed to the customer.

R4R—(Ready for Reproduction) A term referring to reproduction material ready to be printed.

radar altimeter—An instrument used for determining aircraft flying height above terrain by measurement of time intervals between emission and return of electromagnetic pulses.

radar altitude—The altitude of an aircraft or spacecraft as determined by a radar altimeter; thus, the actual distance from the nearest terrain feature.

radar chart—A chart intended primarily for use with radar, or one suitable for this purpose.

radar clutter—(JCS) Unwantad signals, echoes, or images on the face of the display tube which interfere with observation of desired signals.

radar correlation—The process of electronically relating realtime radar images with stored digital data on the radar reflectance and position of terrain and features on the Earth's surface. It is used to provide positioning information to correct or check air navigation and guidance systems.

radar coverage—(JCS) The limits within which objects can be detected by one or more radar stations.

radar fix point (RFP)—The most significant radarscope ground feature for a given geographic area. The feature may be radar reflective or completely void of reflectivity to show contrast with the surrounding area; e.g., land/water or show/no show. Positioning data established for RFPs are used in offset aiming procedures or for enroute and final update of navigation systems.

radar horizon—(JCS) The locus of points at which the rays from a radar antenna become tangential to the earth's surface. On the open sea this locus is horizontal but on land it varies according to the topographical features of the terrain.

radar intelligence item—A feature which is radar significant but which cannot be identified exactly at the moment of its appearance as homogeneous.

radar intelligence map (RIM)—An intermediate element in the process of light-optical radar simulation and in the production of analytical predictions.

radar map—A map produced through the application of radar techniques.

radar photography—A combination of the photographic process and radar techniques. Electrical impulses are sent out in predetermined directions and the reflected or returned rays are utilized to present images on cathode-ray tubes. Photographs are then taken of the information displayed on the tubes.

radar prediction categories—In the broadest sense, radar prediction is separated into two major categories, experience prediction and analytical prediction.

radar prediction formats—Radar predictions appear in a wide variety of formats, generally indicative of intended application. The four most common formats are: spot predictions; strip predictions; radar intelligence maps (RIM); and Series 200 Air Target Charts.

radar prediction types—Each major radar prediction category is divided into three types of predictions: single heading predictions; omnidirectional predictions; and omnigain predictions.

radar reconnaissance—(JCS) Reconnaissance by means of radar to obtain information, on enemy activity and to determine the nature of terrain.

radar reflectivity plate—A scaled, three-dimensional model of a target area constructed of radar reflective materials on a transparent plastic plate, used in a radar trainer to simulate the radar returns of that area. Also called **radar simulation plate; radar trainer plate.**

radar reflector—A device capable of or intended for reflecting radar signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

radar return analysis—Those items of the radar significance analysis code (RSAC), the special area (SA) information, and the radar significant powerline (RSPL) information which has been developed from an analysis of cartographic, photographic, and intelligence sources.

radar return code (RRC)—An omnidirectional radar prediction based on the decibel radar prediction system and depicted in a color code on certain air target charts.

radar shadow—A condition in which radar signals do not reach a region because of an intervening obstruction.

radar significance analysis code (RSAC)—The unique radar intensity categories of built-up areas and other radar reflective objects and structures based on surface material/height factors, and depicted by a system of color coding.

radar significant power line (RSPL)—A power transmission line which, because of its unique physical characteristics and/or voltage capacity, is known to possess radar reflective qualities, and is therefore distinctively displayed on a target graphic.

radar simulation plate—See **radar reflectivity plate.**

radar target—An object which reflects a sufficient amount of a radar signal to produce an echo signal on the radar screen.

radar trainer plate—See **radar reflectivity plate.**

Radar, Synthetic Aperture (SAR)—A radar in which a synthetically long apparent or effective aperture is constructed by integrating multiple returns from the same ground cell, taking advantage of the Doppler effect to produce a phase history film or tape that may be optically or digitally processed to reproduce an image.

radargrammetry—That branch of photogrammetry which utilizes radar images.

radarscope overlays—(JCS) Transparent overlays for placing on the radarscope for comparison and identification of radar returns.

radarscope photography—(JCS) A film record of the returns shown by a radar screen.

radar—(JCS) Radio detection and ranging equipment that determines the distance and usually the direction of objects by transmission and return of electromagnetic energy.

radial assumption—In an aerial photograph containing both tilt displacement and relief displacement, neither the nadir point nor the isocenter is the theoretically correct radial center. The photographic nadir point should be used as the radial center if relief is the major consideration and the isocenter should be used if tilt is the major consideration.

radial center—The selected point on a photograph from which radials (directions) to various image points are drawn or measured; that is, the origin of radials. The radial center is either the principal point, the nadir point, the isocenter, or a substitute center. Also called **center of radiation; center point.**

radial distortion—Linear displacement of image points radially to or from the center of the image field, caused by the fact that objects at different angular distances from the lens axis undergo different magnifications.

radial line intersection—That point at which two or more radial lines cross or intersect.

radial line plotter—See **radial plotter.**

radial line—(surveying) A radius line of a circular curve to a designated point in the curve; if the line is extended beyond the convex side of the curve, it is a prolongation of the radial line.

radial plotter—A device whereby two overlapping photographs are viewed stereoscopically, and the planimetric details in their common area can than be transferred to a map or base sheet through a mechanical linkage utilizing the radial line principle. Also called **radial line plotter.**

radial plot—See **radial triangulation.**

radial secator—See **templet cutter.**

radial triangulation—The aerotriangulation procedure, either graphical or analytical, in which directions from the radial center, or approximate

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

radial center, of each overlapping photograph are used for horizontal control extension by the successive intersection and resection of these direction lines. A radial triangulation also is correctly called a **radial plot** or a **minor control plot**. If made by analytical methods, it is called an **analytical radial triangulation**. A radial triangulation is assumed to be graphical unless prefixed by the word **analytical**. It is also assumed to be based on the principal point unless prefixed by definitive terms such as **isocenter** or **nadir point**.

radial—(photogrammetry) A line or direction from the radial center to any point on a photograph. The radial center is assumed to be the principal point, unless otherwise designated (e.g., nadir radial). See also **isoradial; nadir radial; principal-point radial.**

radiant energy—The energy of any type of electromagnetic radiation. See also **radiation**, definition 2.

radiation—**1.** (surveying) The process of locating points by a knowledge of their direction and distance from a known point. The directions may be azimuths or bearings read from a theodolite or graphical directions determined by alidade and planetable. The distances may be taped or measured by stadia. **2.** The process by which electromagnetic energy is propegated through free space by virtue of joint undulatory variations in the electric and magnetic fields in space.

radio acoustic ranging—A means of determining distance by a combination of radio and sound; the radio being used to determine the instant of transmission or reception of the sound, and the distance being determined by the time of transit of sound, usually in water.

radio beacon—(JCS) A radio transmitter which emits a distinctive, or characteristic, signal used for the determination of bearings, courses, or location.

radio direction finding—(JCS) Radio location in which only the direction of a station is determined by means of its emissions.

radio facility chart—**See enroute chart.**

radio fix—(JCS) The location of a ship or aircraft by determining the direction of radio signals coming to the ship or aircraft from two or more sending stations, the locations of which are known.

radio interferometer—An interferometer operating at radio frequencies; used in radio astronomy and in satellite tracking.

radio navigation—(JCS) Radio location intended for the determination of position or direction or for obstruction warning in navigation.

radio range findings—(JCS) Radio location in which the distance of an object is determined by means of its radio emissions, whether independent, reflected or retransmitted on the same or other wavelength.

radio range station—(JCS) A radio navigation land station in the aeronautical radio navigation service providing radio equisignal zones. In certain instances a radio range station may be placed on board a ship.

radiometric camera calibration—The calibration of a camera for spectral recording characteristics.

radiometric linearity—The gray levels are in linear proportion to the light intensities within a color band.

radiometric nonlinearity—The analog to digital convarsion system that provides signal to noise (S/N) ratios of the sensors, where the S/N is calculated by the difference of the sensor's average dark signal value divided by the root mean square dark noise value. Intermediate intensities will be linear representations from average white reference to the average dark reference. Intermediate intensities will be represented using a linear tonal transfer curve for each color channel. Error introduced during the digitization process which causes the gray scale values for a color component (RGB) to not be in linear proportion to the source intensities for that component.

radius vector—The line (distance) and direction connecting the origin with the point whose position is being defined. See also **polar coordinates.**

random error—Random errors are those not classified as blunders, systematic errors, or periodic errors. They are numerous, individually small, and each is as likely to be positive as negative. Also called **accidental error; casual error.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

random line—A trial line, directed as closely as possible toward a fixed terminal point which is invisible from the initial point. The error of closure permits the computation of a correction to the initial azimuth of the random line; it also permits the computation of offsets from the random line to establish points on the true line.

random microlenticular screen (RML)—A high density halftone screen with no specific screen angle or pattern used to produce high resolution printing from continuous tone images.

random traverse—A survey traverse run from one survey station to another station which cannot be seen from the first station in order to determine their relative positions.

Range Only Correlation System (ROCS)—An all-weather terminal guidance system that uses ranging information to determine its position by comparing two radar images. The images are taken 90° apart and compared with a reference (or prediction) of contrast edge information (bright spots) in the range return signal. The checkpoint is a geographic reference location, selected prior to the mission, by which the vehicle can determine course correction.

range finder—An instrument, using the *parallax principle*, for finding the distance from a place of observation to points at which no instruments are placed.

range line—A boundary of a township surveyed in a north-south direction. See also **township lines.**

range marker—(JCS) A single calibration blip fed on to the time base of a radial display. The rotation of the time base shows the single blips as a circle on the plan position indicator scope. It may be used to measure range.

range pile—Any pile serving as a guide for marine surveying.

range pole—See **range rod.**

range rate data—Information gathered by an instrument that measures the rate of change in the distance (range) to a moving object.

range resolution—(JCS) The ability of the radar equipment to separate two reflecting objects on a similar bearing, but at different ranges from the antenna. The ability is determined primarily by the pulse length in use.

range rod—A slender wood or metal rod, 6 to 8 feet long, with a pointed metal shoe, usually painted in contrasting colors (red and white), alternately, at 1-foot intervals. It is frequently used as a sighting signal at the ends of traverse courses. Also called **line rod; lining pole; range pole; ranging pole; sight rod.**

range signal—A buoy, rod, flag, or other similar object used to mark and identify range points when taking soundings during a hydrographic survey.

range—**1.** (JCS) The distance between any given point and an object or target. **2.** The difference between the maximum and minimum of a given set of quantities. See also **distance. 3.** Two or more objects in line. Such objects are said to be in range. **4.** Any series of contiguous townships situated north and south of each other; also sections similarly situated within a township. **5.** The well-defined lines or courses whose positions are known and are used in determining soundings in a hydrographic survey.

ranging data—Information gathered by an instrument that measures the distance (range) to the object in question.

ranging pole—See **range rod.**

ranging-in—See **wiggling-in.**

Raster Product Standard(s) (RPS)—A suite of user system oriented standards encompassing raster data structure digitizing conventions, storage formats and tiling schemes.

raster data structure—A method of representing MC&G data characterized by a matrix of evenly spaced rows and columns of data points. These data points (called "pixels" in image and scanned map data) typically represent some value at that point, while the position within the columns and rows determines the geographic position. Raster data structures are typically used to record scanned maps and charts (MC&G graphic data), image data, or gridded data.

raster format—See **raster data structure.**

raster lines—The lines of scan of the CRT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

beam across the CRT face. See also **raster.**

raster map product (RMP)—A generic term for raster scanned map images not found to be in accordance with the DMA ADRG Specification or production program; excludes TEC's ADRG-S (raster separates).

raster to vector—The process of converting an image made up of cells into one described by lines and polygons.

raster—1. Pattern of horizontal, parallel lines of varying intensity forming an image (usually) on a cathode ray tube screen. 2. A regular grid of cells covering an area.

rate station—See **drift station.**

ratio print—A print in which the scale has been changed from that of the negative by photographic enlargement or reduction.

ratiograph—See **ratiometer.**

ratiometer—An instrument used to help solve the mathematical relationship of a photograph to a mosaic. It determines scale ratios from which, through mathematical formulas, a rectified print can be made on a properly calibrated rectifying printer.

rational horizon—See **celestial horizon.**

rationalization method—A technique of relative orientation which takes into consideration the limiting factors of the equipment being used, the nature and variations of tilt and crab angles at successive camera stations, and providing approximate projector adjustments based on these data.

ray of light—The geometrical concept of a single element of light propagated in a straight line and of infinitesimal cross section; used in analytically tracing the path of light through an optical system. See also **beam of light.**

ray tracing—(optics) A trigonometric calculation of the path of a light ray through an optical system.

Raydist—The trade name of an electronic distance measuring system. A nonline-of-sight system capable of simultaneous multiparty, range-range operation; it gives continuous range information from two base stations operating simultaneously with one or more aircraft and surface vessels.

real image—An image actually produced and capable of being shown on a surface, as in a camera.

realtime math model—A set of equations that defines image coordinates as functions of three-dimensional object coordinates for a region of imagery.

rear element—See **lens element.**

rear nodal point—See **nodal point.**

recall—(digital) A procedure which restores the previous entry.

recast—To change a map from one horizontal datum to another by appropriately changing the geographic values of the map graticule.

receiver—1. (pendulum) A heavy-cast metal box within which the pendulum is suspended and some auxiliary equipment placed when making observations for the intensity of gravity. 2. (satellite surveying) The equipment necessary to receive signals broadcast from the Navy Navigation Satellite System, including an antenna, preamplifier, processor, oscillator, output device, and power system.

reciprocal bearing—See **back bearing,** definition 1.

reciprocal leveling—Trigonometric leveling wherein vertical angles have been observed at both ends of the line to eliminate errors.

reciprocal observations—Observations taken backward and forward such as vertical angles at both termini of a line for trigonometric leveling.

reciprocal vertical angle—A vertical angle measured over a line at both ends in trigonometric leveling to eliminate (at least partly) the effects of curvature and refraction. Reciprocal observations must be made as simultaneously as practicable to obviate error caused by changing refractive conditions.

reclassify—(digital) A procedure to change the classification of existing data.

recognition—In photointerpretation, the act of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

discovering the true identity of an object.

recompilation—The process of producing a map or chart that is essentially a new item and which replaces a previously published item. Normally, recompilation of a map or chart involves significant change to the horizontal position of features, revision of vertical values, improvement in planimetric or navigational data, or any combination of these factors.

reconnaissance map—The plotted results of a reconnaissance survey and data obtained from other sources.

reconnaissance photography—(JCS) Photography taken to obtain information on the results of bombing, or on enemy movements, concentrations, activities and forces. The primary purposes do not include making maps, charts, or mosaics.

reconnaissance sketch—A drawing which resembles a reconnaissance map but is lacking in some map element.

reconnaissance survey—A preliminary survey, usually executed rapidly and at relatively low cost. The information obtained is recorded, to some extent, in the form of a reconnaissance map or sketch.

reconnaissance—(JCS) A mission undertaken to obtain, by visual observation or other detection methods, information about the activities and resources of an enemy or potential enemy; or to secure data concerning the meteorological, hydrographic, or geographic characteristics of a particular area. See also **aerial reconnaissance; hydrographic reconnaissance; radar reconnaissance; triangulation reconnaissance.**

reconstructed imagery—Imagery on film derived from processed digital image data by means of film printer.

recording statoscope—A statoscope equipped with a recording camera whose shutter is synchronized with that of the aerial camera and the image of the statoscope is recorded on each individual frame.

recovered control—See **recover.**

recovery of station—See **recover.**

recover—(surveying) To visit a survey station, identify its mark as authentic and in its original location, and verify or revise its description. The term is usually modified to indicate the type or nature of the recovery, such as recovered bench mark, or a recovered triangulation station.

rectagraver—A scribing instrument which rests on the scribing surface during the operation and only the cutter arm moves to scribe each symbol.

rectangular chart—A chart on the rectangular projection.

rectangular coordinate plotter—See **coordinatograph.**

rectangular coordinates—Coordinates on any system in which the axes of reference intersect at right angles. See also **Cartesian coordinates.**

rectangular map projection—A cylindrical map projection with uniform spacing of the parallels.

rectangular polyconic map projection—A modified polyconic map projection having a line representing a standard parallel divided to exact scale, through whose division points pass the lines representing the geographic meridians, intersecting the lines which represent the geographic parallels in right angles.

rectangular space coordinates—The perpendicular distances of a point from places defined by each pair of a set of three axes which are mutually perpendicular to each other at a common point of origin. In photogrammetry, space coordinates are also called **survey coordinates,** and are the *x*-coordinates and *y*-coordinates which define the horizontal position of a point on a ground system, and the *z*-coordinate, which is the elevation of the point with reference to the ground system. Also called **air coordinates.**

rectangular surveys—A system of surveys in which an area is divided by a base line intersected at right angles by a principal meridian, the intersection termed the initial point from which the partitions are subdivided into equal size townships containing 36 sections of land each.

rectification—(JCS) In photogrammetry, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

process of projecting a tilted or oblique photograph onto a horizontal reference plane. [Although the process is applied principally to aerial photographs, it may also be applied to the correction of map deformation.] See also **analytical orientation; empirical orientation; graphical rectification; multiple-stage rectification; optical rectification; paper-strip method; point-matching method; transformation.**

rectified altitude—See **apparent altitude.**

rectified print—A photograph in which tilt displacement has been removed from the original negative, and which has been brought to a desired scale.

rectifier—A specially designed projection printer whose geometry is variable in order to eliminate tilt from an aerial negative. There are two basic types; those in which the optical axis of the rectifier lens is the common reference or base direction of the instrument, and those in which the line between the principle point of the negative and the rectifier lens is the common reference. Also called **automatic rectifier; nonautomatic rectifier; nontilting- lens rectifier; nontilting-negative- plane rectifier; rectifying camera; tilting-lens rectifier.** See also **autofocus rectifier; transforming printer.**

rectifying camera—See **rectifier.**

rectifying latitude—The latitude on a sphere such that a great circle on it has the same length as a meridian on the spheroid and such that all lengths along a meridian from the Equator are exactly equal to the corresponding lengths on the spheroid. Rectifying latitude is an auxiliary latitude used in problems of geodesy and cartography.

rectifying printer—See **rectifier.**

rectilinear coordinates—See **rectangular coordinates.**

rectobilque plotter—See **angulator.**

rectoplanigraph—An instrument utilizing a vertical photograph mounted in a vertical position, and used in the preparation of planimetric maps.

red light readable map—A map printed with special inks which can be read under conditions requiring special lighting; e.g., in a tank or aircraft during nighttime operations.

red magnetism—The magnetism of the north-seeking end of a freely suspended magnet. This is the magnetism of the Earth's south magnetic pole. See also **blue magnetism.**

reduced gravity—Observed gravity that has been reduced to the geoid or to some other reference surface by one of the gravity reductions.

reduced latitude—See **parametric latitude.**

reduction factor—See **scale of reproduction.**

reduction printer—See **diapositive printer.**

reduction to ellipsoid—The correction subtracted from or added to the measured horizontal length of a line at average topographic elevation to reduce it to the corresponding length on the pertinent ellipsoid. The height used in this reduction will differ by the amount of the geoid height from the height used in reduction to sea level.

reduction to sea level—A reduction applied to a measured horizontal length on the Earth's surface to reduce it to the surface of the local sea level datum.

reduction to the meridian—The process of applying a correction to an altitude observed when a body is near the celestial meridian of the observer, to find the altitude at meridian transit. The altitude at the time of such an observation is called an **exmeridian altitude.**

reduction-to-center—1. The amount which must be applied to a direction observed at an eccentric station or to an eccentric signal, to reduce such direction to what it would be if there were no such eccentricity. 2. (astronomy) One of the values used in finding the equation of time.

reference datum—A general term applied to any datum, plane, or surface used as a reference or base from which other quantities can be measured.

reference direction—A direction used as a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

basis for comparison of other directions.

reference ellipsoid—See **reference spheroid.**

reference frame—See **coordinates.**

reference grid—See **grid.**

reference level—See **datum level.**

reference line—Any line which can serve as a reference or base for the measurement of other quantities. Also called **datum line.**

reference mark—A permanent supplementary mark close to a survey station to which it is related by an accurately measured distance and direction, and/or a difference in elevation.

reference meridian—See **local meridian.**

reference monument—An iron post or rock cap accessory used where the point for a corner monument is such that, for practical purposes, a permanent corner monument cannot be established, or if monumented, a full complement of bearing trees or bearing objects are not obtainable.

reference plane—See **datum level.**

reference point—See **datum point.**

reference signal—In telemetry, the signal against which data carrying signals are compared to measure differences in time, phase, frequency, or other values or quantities.

reference spheroid—A theoretical figure whose dimensions closely approach the dimensions of the geoid; the exact dimensions are determined by various considerations of the section of the Earth's surface concerned. Also called **reference ellipsoid; spheroid of reference.** See also **World Geodetic System.**

reference station—A place where tide or tidal current constants have been determined from observations, and which are used as a standard for the comparison of simultaneous observations at a subordinate station. Also, a place for which independent daily predictions are obtained for other locations by means of differences or factors. Also called **standard port; standard station.**

referencing—The process of measuring the horizontal distances and directions from a survey station to nearby landmarks, reference marks, and other objects which can be used in the recovery of the station.

reflected ray—A ray extending outward from a point of reflection.

reflecting prism—A prism that deviates a light beam by internal reflection. Practically all prisms used in photogrammetric instruments are of this type.

reflecting projector—An instrument which is used to project the image of photographs, maps, or other graphics onto a copying table. The scale of the projected image can be varied by raising or lowering the projector or, in some models, the copy board. These latter models also allow the tilting of the copy board in *x*- and *y*-directions in order to compensate for tip and tilt distortion in aerial photographs.

reflection—The return or change in the direction of travel of particles or radiant energy which impinges on a surface but does not enter the substance providing the reflecting surface. See also **diffraction; diffuse reflection; refraction; specular reflection.**

reflector constant—The amount that a distance measurement must be reduced when using glass reflectors because the velocity of light is slower in glass than it is in air. The constant will also include the distance difference between the reflector housing plumbing point and the effective reflecting plane of the prism.

reflight—Another flight over the same course to secure photographs to fill in for those missing or defective.

refracted ray—A ray extending onward from the point of refraction.

refracting prism—A prism that deviates a beam of light by refraction. The angular deviation is a function of the wavelength of light; therefore, if the beam is composed of white light, the prism will spread the beam into a spectrum. Refracting prisms can be used in optical instruments only for small deviations. See also **wedge.**

refraction angle—That portion of an

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

observed zenith distance, which is due to the effect of atmospheric refraction.

refraction displacement—Displacement of images radially outward from the photograph nadir because of atmospheric refraction. It is assumed that the refraction is symmetrical about the nadir direction.

refraction line—A line of sight to a survey signal which becomes visible only by the effect of atmospheric refraction.

refraction—The change in direction of motion of a ray of radiant energy as it passes obliquely from one medium into another in which the speed of propagation is different. See also **angle of incidence; angle of refraction; astronomic refraction; atmospheric refraction; coastal refraction; coefficient of refraction; electronic refraction; horizontal refraction; lateral refraction; mean refraction; refracted ray; refraction angle; refraction line; Snell's law of refraction; terrestrial refraction.**

regional gradient—See **regional gravity.**

regional gravity—In gravity prospecting, contributions to the observed anomalies due to density irregularities at much greater depths than those of the possible structures, the location of which was the purpose of the survey. Also called **regional gradient.**

register hole punch—See **prepunch register system.**

register marks—(JCS) Designated marks, such as small crosses, circles, or other patterns applied to original copy prior to reproduction to facilitate registration of plates and to indicate the relative positions of successive impressions. Also called **corner marks; corner ticks; register ticks; registration ticks; ticks.**

register studs—See **prepunch register system.**

register ticks—See **register marks.**

register trials—The test runs necessary to obtain the proper combination of the degree of partial vacuum and the length of the heating cycle required for individual models in forming a plastic relief map.

register—(JCS) The correct position of one component of a composite map image in relation to the other components, at each stage of production.

registration ticks—See **register marks.**

regression of the nodes—Precessional motion in a direction opposite to the direction of revolution of a set of nodes. See also **precession.**

regular error—See **systematic error.**

relational data base—A data base that consists of at least one table, which, in turn, is made up of rows and columns. Tables in the data base are related to one another through the value in at least one column that is common to two or more tables.

relative accuracy—1. (general) An evaluation of the random errors in determining the positional orientation (e.g., distance, azimuth) of one point or feature with respect to another. 2. (chart, feature to graticule) An evaluation of the random errors in chart features with respect to the graticule excluding any error in the graticule or the datum defined by the graticule. 3. (chart, feature to feature) An accuracy evaluation based on random errors in determining the positional accuracy of one point feature with respect to another feature.

relative aperture—(JCS) The ratio of the equivalent focal length to the diameter of the entrance pupil of photographic lens, expressed as f:4.5, etc. Also called **aperture ratio; aperture stop; diaphragm stop; f-number; lens speed; speed of lens; stop; stop numbers.**

relative coordinate system—Any coordinate system which is moving with respect to an inertial coordinate system.

relative deflection—See **astrogeodetic deflection.**

relative direction—Horizontal direction expressed as angular distance from a heading.

relative distance—Distance relative to a specified reference point, usually one in motion.

relative error of closure—The value

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

obtained by dividing the total error of closure by the total length of the traverse, commonly expressed by a fraction having a numerator equal to unity, e.g., 1/1,540. It is used for determining the degree of accuracy of a survey.

relative gravity—Gravity determined from gravity difference measurements (e.g., gravimeter, relative pendulum) between the observer and a reference station. The value obtained is relative with respect to the reference station.

relative motion—See **apparent motion.**

relative movement—Motion of one object or body relative to another. The expression is usually used when describing relative movements other than that of a celestial body. See also **apparent motion; direction of relative movement.**

relative orientation—The reconstruction (analytically or in a photogrammetric instrument) of the same geometric conditions between a pair of photographs that existed when the photographs were taken. In the instrument, this is achieved by a systematic procedure of rotational and translational movements of the projectors. Also called **clearing *y*-parallax.** See also **absolute orientation; vertical deformation.**

relative pendulum—A device for measuring relative gravity through the difference in the period of a pendulum at two stations.

relative position—The location of a point or feature with respect to other points or features, either fixed or moving.

relative relief—The relation of the altitudes of the highest and lowest points of land in any area. The difference between the highest and lowest points is the amplitude of relative relief. Various types of maps have been devised to show this, usually depending on gridding the area on a map, finding a value for the amplitude in each grid square, and producing an isopleth or dot map to depict the distribution of these values.

relative setting—In tilt analysis of oblique photography, the dihedral angle between the two planes passing through the principal point of the opposite obliques, the principal point of the vertical photograph, and the common exposure station. This angle is measured on the vertical photograph as the angle between the two isolines, or as the deflection angle between the perpendiculars from the principal point of the vertical photograph to the two isolines.

relative swing—In the tilt analysis of oblique photographs, the angle of rotation of the oblique camera about its own axis with respect to the plane of the vertical photograph, measured on the oblique photograph by the angle between the isoline and a line joining the fore and aft fiducial marks.

relative tilt—The angular relationship between two overlapping vertical photographs with no reference to an established datum.

releasability code letter—A code letter prefix to the chart identification number which limits the releasability of a particular chart to specific users.

reliability diagram—A diagram included on some MC&G products depicting horizontal and vertical accuracies and date(s) of information

relief displacement—Displacement radial from the nadir point of a photograph caused by differences in elevation of the corresponding ground objects. Also called **height displacement; relief distortion.**

relief distortion—See **relief displacement.**

relief model—A general category which denotes any three-dimensional representation of an object or geographic area, modeled in any size or medium. See also **plastic relief map; terrain model.**

relief stretching—See **hyperstereoscopy.**

relief—(JCS) Inequalities of elevation and the configuration of land features on the surface of the Earth which may be represented on maps or charts by contours, hypsometric tints, shading, spot elevations, or hachures.

Relocatable Target Assessment Data (RTAD)—DMA digital data sets of attributed and nonsymbolized feature information segregated into thematic files portraying roads, railroads, vegetation, surface drainage, and limited surface materials which can be combined into three separate levels of data. All three levels use standard DTED Level 1 to determine

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

elevation, slope, and intervisibility and are intended to be used with multispectral or electro-optical image products. RTAD is used to assess potential target locations.

remote sensing—The measurement or acquisition of information of some property of an object or phenomenon by a recording device that is not in physical or intimate contact with the object or phenomenon under study. Sometimes restricted to the practice of data collection in the wavelengths from ultraviolet to radio regions.

remote station—See **slave station.**

repeatability—A measure of the variation in the accuracy of an instrument when tests are made over the same line(s) at different times of the year, with different operators, and with different but equivalent instruments, all using the same procedures. See also **external error.**

repeating instrument—See **repeating theodolite.**

repeating theodolite—A theodolite so designed that successive measures of an angle may be accumulated on the graduated circle, and a final reading of the circle made which represents the sum of the repetitions. Also called **double-center theodolite; *repeating* instrument.**

repetition of angles—The accumulation of a series of measures of the same angle on the horizontal circle of a repeating theodolite or surveyor's transit.

replace—A procedure to replace all existing occurrences of a specified data element with a new data element.

representative fraction (RF)—(JCS) The scale of a map or chart expressed as a fraction or ratio. [Relates unit distance on the map to distance measured in the same unit on the ground.] Also called ***fractional scale*; natural scale.**

representative pattern—(cartography) 1. An accurate portrayal of the surface of the Earth in the area being compiled. 2. The selection and portrayal of the most prominent of a dense group of similar features.

reprint—The process of using existing reproducible without change to print additional quantities of a product.

reproducible—Any copy capable of being used as a master-to-be. May be either a negative or positive transparency.

reproduction material—(JCS) Material, generally in the form of positive or negative copies on film or glass for each color plate, from which a map or chart may be reproduced without redrafting. Also called ***repromat.***

reproduction positive mold—The positive mold which has been drilled through with vacuum holes, and over which the plastic relief map is formed.

reproduction ratio—See **scale of *reproduction.***

reproduction scale—The scale at which a map or chart is published or is to be published. Also called **publication scale.**

reproduction—1. The summation of all the processes involved in printing copies from an original drawing. 2. A printed copy of an original drawing made by any of the processes of reproduction.

repromat—See **reproduction material.**

Repsold base-line measuring apparatus—An optical base-line measuring apparatus, composed of a steel bar approximately 4 meters long, whose exact length at any temperature is known, and whose temperature is determined by means of a metallic thermometer composed of the steel measuring bar and a similar bar of zinc, the two being fastened together at their middle points.

resampling—(digital mapping) Technique for transforming a raster image to a particular scale and projection. Common resampling techniques include zero-order (nearest-neighbor), first order (bilinear interpolation) and cubic convolution.

rescale—An adjustment of values or parameters representing magnitudes or intensity so that the data reflects an aspect more suited to the user.

reseau—1. A glass plate on which is etched an accurately ruled grid. Sometimes used as a focal-plane plate to provide a means of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

calibrating film distortion; used also for calibrating plotting instruments. Also called **grid plate.** 2. Intersecting orthogonal lines superimposed over photo imagery.

resection station—A station located by resection methods.

resection—1. The graphical or analytical determination of a position, as the intersection of at least three lines of known direction to corresponding points of known position. 2. (surveying) The determination of the horizontal position of a survey station by observed directions from the station to points of known positions. Also, the line drawn through the plotted location of a station to the occupied station. 3. (photogrammetry) The determination of the position and/or attitude of a camera, or the photograph taken with that camera, with respect to the exterior coordinate system. Also called **three-point method.** See also **intersection,** definition 2.

residual deviation—Deviation of a magnetic compass after adjustment or compensation.

residual error—The difference between any value of a quantity in a series of observations, corrected for known systematic errors, and the value of the quantity obtained from the combination or adjustment of that series. Frequently used as the difference between an observed value and the mean of all observed values of a statistically valid set. See also **error; residual.** The latter term is generally used in referring to actual values in a specific computation.

residual gravity—In gravity prospecting, the portion of a gravity effect remaining after removal of some type of regional gravity, usually the relatively small or local anomaly components of the total or observed gravity field.

residual parallax—Small amounts of *y*-parallax which may remain in a model after relative orientation is accomplished.

residual—A general term denoting a quantity remaining after some other quantity has been subtracted. It occurs in a variety of particular contexts. For example, if the true value of a variable is subtracted from an observed value then the difference may be called a **residual;** it is also frequently called an **error.** Similarly, if a mathematical model is fitted to data, the values by which the observations differ from the model values are called **residuals.**

resolution in bearing—The minimum detectable separation of objects at the same range and the same elevation, expressed in terms of the horizontal angular distance between such objects.

resolution in elevation—The minimum detectable separation of objects at the same range and same bearing, expressed in terms of the vertical angular distance.

resolution in range—The minimum detectable separation of objects in the same line of sight, expressed in terms of the distance between them.

resolution limit—In gravity and magnetic prospecting, the separation of two disturbing bodies at which some obvious indication, in a measured quantity, of the presence of two separate bodies ceases to be visible.

resolution—1. For raster applications, resolution is the number or pixels per unit distance. 2 (JCS) A measurement of the smallest detail which can be distinguished by a sensor system under specific conditions. 3. The minimum distance between two adjacent features, or the minimum size of a feature, which can be detected by a remote sensory system. 4. In gravity or magnetic prospecting, the indication in some measured quantity, such as the vertical component of gravity, of the presence of two or more close but separate disturbing bodies.

resolving power target—A test chart used for the evaluation of photographic, optical, and electrooptical systems. The design usually consists of ruled lines, squares, or circles varying in size according to a specified geometric progression.

resolving power—A mathematical expression of definition in an imaging system, usually stated as the maximum number of lines per millimeter that can be seen as separate lines in the image.

responder—In general, an instrument that indicates reception of an electric or electromagnetic signal. See also ***transponder.***

responsor—A radio receiver which receives the reply from a transponder and produces an output suitable for feeding to a display system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

restitution—(JCS) The process of determining the true planimetric position of objects whose images appear on photographs. [Restitution corrects for distortion resulting from both tilt and relief displacement.]

restoration—The recovery of one or more lines or corner positions, or both, of a prior survey; the replacement of one or more lost corners or obliterated monuments by approved methods, including the substantial renewal of one or more monuments, as required for the purpose of a survey.

resultant error—The error in any measurement that is the difference between the measured value and the true value for a quantity. Also called **true error.**

resurvey—A retracing on the ground of the lines of an earlier survey, in which all points of the earlier survey that are recovered are held fixed and used as a control. If too few points of the earlier survey are recovered to satisfy the control requirements of the resurvey, a new survey may be made. A resurvey is related directly to an original survey, though several resurveys may interpose between them. See also **dependent resurvey; independent resurvey; retracement.**

reticle (reticule)—1. (surveying) A system of wires, hairs, threads, etched lines, or the like, placed normal to the axis of a telescope at its principal focus, by means of which the telescope is sighted on a star, or target, or by means of which appropriate readings are made on some scale, such as a leveling or stadia rod. 2. (optics) A mark, such as a cross or system of lines, lying in the image plane of a viewing apparatus and used singly as a reference mark in certain types of monocular instruments or as one of a pair to form a floating mark, as in certain types of stereoscopic instruments. See also **floating mark; graticule; index mark; parallactic grid.**

retouching—Corrective treatment of a plate, negative, positive, or copy by means of brush, pencil, pen, airbrush, or other method.

retracement—A term applied to a survey that is made for the purpose of verifying the direction and length of lines, and identifying the monuments and other marks of an established prior survey. See also **resurvey.**

retrodirective prism—A prism consisting of a solid glass element having three mutually perpendicular reflecting surfaces and a fourth surface oblique to the three reflecting surfaces. A light beam entering through the oblique surface is reflected on each of the three other surfaces and turned 180° to be returned along the same airpath which it traveled to the retrodirective prism.

retrograde motion—1. Motion in an orbit opposite to the usual orbital direction of celestial bodies within a given system. Specifically, of a satellite, motion in a direction opposite to the direction of rotation of the primary. 2. The apparent motion of a planet westward among the stars. Also called **retrogression.** Opposite of **progressive motion.**

retrograde vernier—A vernier scale which has spaces or divisions slightly longer than those of the primary scale. The numbers on the vernier scale run in the opposite direction from those on the primary scale.

retrogression—See **retrograde motion.**

reverse stereo—See **pseudoscopic stereo.**

reversible level—A spirit level having a bubble tube with the inner surface ground barrel shaped so that the tangent lines to curves on the upper and lower sides are parallel when exactly opposite, permitting the level to be used in either the erect or inverted positions.

reversible pendulum—A pendulum so designed and equipped with means of support that it may be used with either end up or down.

reversing in azimuth and altitude—See **double centering.**

reverted image—(optics) An image in which detail is in reverse order, from left to right, compared to the corresponding detail of the object. The order of detail from top to bottom remains unchanged. A mirror image.

revision—The process of updating a product to reflect current information. Typically, revision of a map or chart does not require significant changes to the horizontal position of features or vertical data values; rather, improvement in planimetric data is provided. Normally, publications are revised, not recompiled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

revolution—1. The turning of a body about an exterior point or axis. The correct distinction between revolution and rotation is given in the statement "the Earth revolves around the Sun, and rotates on its axis." 2. (surveying) A turning of an instrument or part of an instrument.

rhomboidal prism—A prism that displaces the axis of the beam of light only laterally.

rhumb bearing—The direction of a rhumb line through two terrestrial points, expressed as angular distance from a reference direction. It is usually measured from 0° at the reference direction clockwise to 360°. Also called **Mercator bearing.**

rhumb direction—See **Mercator direction.**

rhumb line distance—Distance along a rhumb line, usually expressed in nautical miles.

rhumb line—A line on the surface of the Earth cutting all meridians at the same angle. [A loxodrome or loxodromic curve spiraling toward the poles in a constant true direction. Parallels and meridians, which also maintain constant true directions, may be considered special cases of the rhumb line. A rhumb line is a straight line on a Mercator projection.] Also called **equiangular spiral; loxodrome; loxodromic curve; Mercator track.**

ridge line—A graphic representation of major ridges used to give more definition to the topographic character of an area for the determination of low altitude radar predictions. They are shown only in areas of rise gradient to depict those places in which the elevated terrain forms a sufficient background to partially screen vision at low altitude.

right ascension system—An equatorial system of curvilinear celestial coordinates which has the Equator as the primary reference plane and the perpendicular hour circle through the vernal equinox as the secondary reference plane. The direction to a body is given by its right ascension and declination.

right ascension—The angular distance measured eastward on the Equator from the vernal equinox to the hour circle through the celestial body, from 0 to 24 hours.

right bank—That bank of a stream or river on the right of the observer when he is facing in the direction of flow, or downstream.

right sphere—The celestial sphere as it appears to an observer at the Equator, where celestial bodies appear to rise vertically above the horizon.

right-angle prism—A prism that turns a beam of light through a right angle. It inverts (turns upside down) or reverts (turns right for left) according to the orientation of the prism.

right-reading—A descriptive term for an image which, when viewed through the base, reads the same as the original. Other terms sometimes used to identify image direction, such as **normal reading, natural readings,** etc., are not recommended because of possible confusion in negative-positive relationship.

rigid tripod engraver—A scribing instrument with three points of contact surface, to absorb the normal imbalance of the operator's hand pressure.

ring—(digital mapping) An edge which is a member of a different closed loop of edges.

rise gradient—A color-coded omnidirectional graphic representation of those terrain slopes which are predicted to be low altitude radar significant.

rise time—The time at which a satellite's broadcast can be picked up by a suitably equipped observer, as taken from an alert. Set time and time of closest approach are also given. See also **alerts; rise.**

rise—1. To cross the visible horizon while ascending. 2. (satellite surveying) To cross the observer's horizon while ascending; detectable by broadcast data received.

rising tide—See **flood tide.**

river crossing—(leveling) Carrying a line of levels across a stream or other body of water, when no suitable bridge is available and the width of the body of water is greater than the maximum allowable length of sight for the leveling, requires a special series of observations which taken collectively is known as a river crossing.

riverine area—(JCS) An inland or coastal area

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

comprising both land and water, characterized by limited land lines of communication, with extensive water surface and/or inland waterways that provide natural routes for surface transportation and communications.

road map—A medium-or small-scale special-purpose map, generally showing only planimetric detail, with emphasis upon the road network and related data. Its main purpose is to furnish pertinent road information for tactical and administrative troop movement.

road net—(JCS) The system of roads available within a particular locality or area.

roamer—(JCS) Grids constructed to common map scales used for determination of map coordinates.

rod correction—(leveling) That correction which is applied to an observed difference of elevation to correct for the error introduced when the leveling rods are not actually of the length indicated by the graduations.

rod float—A small cylindrical tube of any material, closed at the bottom and weighted with shot until it floats in an upright position with about 2 to 6 inches projecting above the water surface. Current velocities are determined by direct observations.

rod level—An accessory for use with a leveling rod or a stadia rod to assure a vertical position of the rod prior to instrument reading.

rod sum—(leveling) The algebraic total of plus and minus sights in a given level line.

rod—See **leveling rod.**

roll—(JCS) **1.** The rotation of an aircraft or ship about its longitudinal axis. **2.** In air photography, the camera rotation about the longitudinal access of the aircraft. Also called **tilt.** See also **tilt angle.**

romanization—**1.** The process of recording in roman script either the sounds of a language or the graphic symbols of a nonroman writing system. **2.** An item of a language which has undergone this process. See also **transcription; transliteration.**

roof prism—A type of prism in which the image is reverted by a roof, that is, two surfaces inclined at 90° to each other.

root mean square error—See **standard error.**

roots of mountain theory—See **Airy theory of isotasy.**

rotating prism camera—A class of panoramic camera in which a double dove prism is rotated while the lens system remains fixed. This configuration can achieve a scan of 180° or more.

rotating prism—See **dove prism.**

rotational movement—(photogrammetry) The systematic rotation of projectors or projector assemblies. When applied to the projector body within the gimbal inner ring, the movement is about the *z*-axis and is called **swing.** Rotation of the inner ring is about an *x*-axis (secondary axis) and is called *x*-**tilt.** Rotation of the outer ring is about a *y*-axis (primary axis) and is called *y*-**tilt.**

rotation—**1.** (astronomy) A turning of a body about a self-contained axis; as the daily rotation of the Earth. See also

route chart—**1.** A chart showing routes between various places, usually with distances indicated. **2.** An aeronautical chart covering the route between specific terminals, and usually of such scale as to include the entire route on a single chart. Also called **flight chart.**

route map—A map showing roads to be followed and nearby points of military significance.

route survey—Surveys for linear construction such as railroads, highways, and transmission lines which include the layout of lines and grades for these projects.

row—(digital) An ordered collection of fields in a data base pertaining to an entity or record.

rubber blanket—See **offset lithography; offset press.**

rubber sheeting—A rectification process which resamples data to transform (stretch or compress) it from one statistical model to fit another model.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

rubberband—The ability to fix the ends of a line, or selected points along a line, in a static position and then "stretch" the line between the fixed points to establish new geometric shapes.

run of micrometer—See **error of run.**

run-length code—A data compression method for large homogenous regions of raster data. Data is recorded as tuples which designate the quantity of a repeating value and the value itself. Also known as run-length encoding (RLE).

running fix—(JCS) The intersection of two or more position lines, not obtained simultaneously, adjusted to a common time.

running mean—See **consecutive mean.**

running—(leveling) A continuous series of measured differences of elevation, made setup by a setup in one direction along a section of a line of levels, which results in a measurement of the difference of elevation between the bench marks or other points, either temporary or permanent, at the ends of the section.

run—1. (lithography) The number of impressions made on a press for a given sheet. 2. (micrometer) See **error of run.** 3. (JCS) (aerial photography) That part of a flight of one photographic reconnaissance aircraft during which photographs are taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

S

24-hour satellite—See **synchronous satellite.**

sag correction—(taping) The difference between the effective length of a tape, or part of a tape, when supported continuously throughout its length and when supported at a limited number of independent points. Base tapes usually are used with three or five points of support, and hang in curves (catenaries) between adjacent supports. Correction for sag is not required when the method of support in use is the same as was used in the standardization of the tape; only the standardization correction is applied. A base tape may also be used supported throughout or with four points of support, as on a railway rail. Also called **catenary correction.**

sailing chart—A small-scale chart used for offshore sailing between distant coastal ports and for plotting the navigator's position out of sight of land and as he approaches the coast from the open ocean. They show offshore soundings and the most important lights, outer buoys, and natural landmarks which are visible at considerable distances.

sailing directions—A descriptive book for the use of mariners, containing detailed information of coastal waters, harbor facilities, etc. Also called **coast pilot; pilot.**

sample variance—True variance can only be computed from some sample. The term **sample variance** is used when wishing to make reference to a computed variance from a specific sample, as opposed to an assumed or previously estimated variance of some particular class of observations or parameters. In general, if the sample variance of some class of observations in a weighted least square adjustment differs too greatly from the assumed variance used for weighing the observations, the adjustment should be repeated with refined weights based on the sample variance. See **variance.**

sample—A set of observations, used to determine the most probable value of a particular parameter, and/or used to estimate the accuracies of the observations and/ or accuracy of the parameter.

Sanson-Flamsteed map projection—See **sinusoidal map projection.**

saros—The eclipse cycle of about 18 years, almost the same length as 223 synodical months. At the end of each saros, the Sun, Moon, and line of nodes return to approximately the same relative positions and another series of eclipses begins, closely resembling the series just completed. See also **lunar cycle.**

Satellite Geophysics Programs—DMA programs that provide for the operation and maintenance of fixed and mobile satellite tracking systems and the processing of data from geodetic satellites. The data collected by these stations are used in geoid profiles, precise satellite ephemerides, geodetic point positions and related products.

satellite geodesy—The discipline which employs observations of an Earth satellite to extract geodetic information.

satellite surveying—1. (Doppler) The process of positioning one or more points on the Earth's surface by collecting Doppler shift data from passes of Navy navigation satellites. See also **point positioning; short arc; translocation.** 2. (NAVSTAR) The position (by resection) of a point receiving signals from four satellites of the Global Positioning System.

satellite trail—A streak like image of a satellite recorded on a stellar plate by a photographic time exposure.

satellite triangulation stations—Triangulation stations whose angular positions relative to one another are determined by the simultaneous observation of an Earth satellite from two or more of them.

satellite triangulation—The determination of the angular relationships between two or more stations by the simultaneous observation of an Earth satellite from these stations.

satellite—An attendant body, natural or man-made, that revolves about another body, the primary. See also **active satellite; communications satellite; Earth satellite; equatorial satellite; geodetic**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

satellite; lunar satellite; NAVSTAR Global Positioning System; Navy Navigation Satellite System; passive satellite; polar satellite; synchronous satellite.

saturable reactor—See **flux-gate magnetometer.**

scalar—Any physical quantity whose field can be described by a single numerical value at each point in space. A scalar quantity is distinguished from a vector quantity by the fact that a scalar quantity possesses only magnitude, whereas a vector quantity possesses both magnitude and direction.

scale checking—1. The process of determining the scale of an aerial photograph, or more correctly, the altitude above sea level which best fits the print. 2. (stereocompilation) The process of determining the scale of a vertical photograph for points at a specific elevation and the subsequent measurement of direction and distance therefrom.

scale error—A systematic error in the lengths of survey lines usually proportional to the lengths of the lines. See also **instrument error.**

scale factor—A multiplier for reducing a distance obtained from a map by computation or scaling to the actual distance on the datum of the map. Also, in the state coordinate systems, scale factors are applied to geodetic lengths to obtain grid lengths, or to grid lengths to obtain geodetic lengths. Both are lengths on a sea level datum, but the grid lengths are affected by the scale change of the map projection.

scale indicator—A logarithmic scale devised as a rapid and convenient method of determining the natural scale of a map from the divisions marked on the graphic scales, or from the intervals of latitude on a map.

scale of reproduction—The enlargement or reduction ratio of an original to the final copy. This ratio is expressed as a diameter, percent, times (X), or a fraction. Also called **enlargement factor; reduction factor; reproduction ratio. See also contact size; diameter enlargement; make line; percent of enlargement/reduction; times (X) enlargement.**

scale-ratio mosaic—An assembly of photographic prints brought to a common scale by projection printing to scale factors obtained from map distances to allow the best possible fit of contiguous photo detail.

scale—1. (JCS) The ratio or fraction between the distance on a map, chart, or photograph and the corresponding distance on the surface of the Earth. 2. A series of marks or graduations at definite intervals on a measuring device or instrument. 3. Measurement by means of a scale. See also **compilation scale; conversion scale; equivalent scale; graphic scale; Invar scale; model scale; photographic scale; plotting scale; principle scale; representative fraction; reproduction scale; scaling; *x*-scale; *y*-scale; *z*-scale.**

scaling the model—See **scaling,** definition 2.

scaling—1. Alteration of the scale in photogrammetric triangulation to bring the model into agreement with a plot of horizontal control. 2. Fitting a stereoscopic model to a horizontal control plot. A step in absolute orientation. Also called **scaling the model.** 3. Determining the scale of a photograph or graphic. 4. (cartography) See **cartometric scaling.**

scan lines—Successive parallel strips of captured data from a sensor, such as a row of a raster grid, to be printed on a single line of a raster display device; or recorded in a single sweep of the mirror of an optical scanner system, or radar scanning system; or a single addressing of the linear array detector of a pushbroom scanner.

scan positional distortion—In a panoramic camera system, the displacement of images of ground points from their expected cylindrical positions caused by the forward motion of the vehicle as the lens scans.

scanner—1. A device for automatically converting images from maps and photographs, or from part of the real world into digital form. 2. Any device that systematically decomposes a sensed image or scene into pixels and then records some attribute of each pixel.

scatterometry—A method of using radar to measure the variation of radar scattering coefficients. These variations may be used by geoscientists to discriminate between surfaces

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

with different roughness and materials. The scatterometer is distinguished from other radars by its ability to measure amplitude.

scene generation—A two-dimensional display with underlaying terrain data portrayed to appear as if three-dimensional. Normally, an image consists of an oblique view with perspective.

scene matching area correlator (SMAC)—An electrooptical correlation system which uses photographic reference material matched with onboard realtime scenes to achieve correlation. The reference scene is prepared from reconnaissance photographs and is stored in the form of a photographic transparency. The reference is placed around the inner circumference of a drum that rotates at a high angular velocity. The reference then forms a continuous strip which modulates or chops the projected image being focused on it by the optical system. The realtime energy from the ground within the field-of-view is spatially modulated by the transmissivity of the reference transparency.

Scheimpflug condition—The requirement that object, lens, and image planes intersect in a common line for sharp focus in any direct projection system.

schema—The description of the logical structure of an entire data base according to a conceptual model.

Schott base-line measuring apparatus—A contact, compensating base-line measuring apparatus composed of three parallel bars; the middle bar of zinc, the outer bars of steel. One end of each steel bar is free; the other end is fastened to an end of the zinc bar, a different end for each steel bar. The lengths of the bars are so proportioned with respect to their coefficients of thermal expansion that a constant distance is maintained between the free ends of the steel bars.

scintillation—*See* **shimmer.**

scissors inversor—See **Peaucellier inversor.**

screen angle—(photography) The angle which the rows of halftone dots make with the vertical when right-reading. The angle is measured clockwise with 0° at 12 o'clock.

screening—See **masking.**

screen—See **area pattern screen; biangle screen; dot screen; halftone screen; magenta contact screen.**

scribed plate—See **scribed sheet.**

scribed sheet—A scribing surface on which the reproduction scribing has been completed. Also called **scribed plate.**

scriber—An instrument holding a scribing point; used for scribing on coated plastics. Also called **engraver; graver; scribing instrument.** See also **engraver rectagraver; rigid tripod engraver; straight line graver; subdivider; swivel graver; turret graver.**

scribing guide—See **guide.**

scribing instrument—See **scriber.**

scribing points—Needles or blades in various diameters or cross-section shapes ground and sharpened to prescribed dimensions. Used in scribers for scribing on coated plastics.

scribing—(JCS) A method of preparing a map or chart by cutting the lines into a prepared coating. [The process of preparing a negative which can be reproduced by contact exposure. Portions of a photographically opaque coating are removed from a transparent base with specially designed tools.] Also called **negative scribing.**

sea level contour—A contour line delineating points at sea level.

sea level datum of 1929—See **National Geodetic Vertical datum of 1929.**

sea level datum—See **mean sea level.**

sea level variation—Sea level varies from day to day, from month to month, and from year to year. This variation is attributed to meteorological conditions and should not be confused with the lunar tidas.

sea level—The height of the surface of the sea at any time. See also **ideal sea level; mean sea level.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

seamless data base—A data base structured without segmentation or tiling, thereby eliminating the need to begin and terminate chains at boundaries and have topological pointers reference tiles as well as other entities. The result is a data base physically smaller than a tiled one but one in which search times can be quite long.

search and rescue chart—A chart designed primarily for directing and conducting search and rescue operations.

secant conic chart—See **conic chart with two standard parallels.**

secant conic map projection—See **conic map projection with two standard parallels.**

secant method—A method of determining the parallel of latitude for the survey of a base-line or standard parallel by offsets from a great-circle line which cuts the parallel at the first and fifth mile corners of the township boundary. See also **secant.**

secant—1. A line that cuts a geometric curve or surface at two or more points. 2. A trigonometric function of an angle. See also **secant method.**

secator—See **templet cutter.**

second-order bench mark—A bench mark connected to the datum (usually mean sea level) by continuous second-order leveling or by a continuous combination of first-and second-order leveling.

second-order leveling—Spirit leveling which does not attain the quality of first-order leveling but does conform to the current specification for second-order (class I or class II) leveling per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for densification of the National Network and for localized crustal movement and engineering projects.

second-order traverse—A survey traverse which extends between adjusted positions of the first-order or second-order control surveys and which conforms to the current specifications for second-order (class I or class II) traverse per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for densification of the National Network and metropolitan area surveys.

second-order triangulation—Second-order triangulation was at one time known as **secondary triangulation**; changed in 1921 to **primary triangulation**, and in 1925 to **second-order triangulation.** These surveys conform to the current specifications for second-order (class I or class II) triangulation per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for densification of the National Network and metropolitan area surveys.

second-order work—The designation given any survey work of next-to-the-highest order of accuracy and precision.

secondary circle—See **secondary great circle.**

secondary compilation—A specially prepared matte plastic material used to depict uncorrected or discrete soundings in bathymetric compilation.

secondary control point (SCP)—A point photographically identifiable, positioned to a high degree of accuracy using an average of several shiran horizontally controlled photographs. Seven to thirteen controlled photographs, each containing the point, are taken over each of four quadridirectional passes intersecting perpendicularly over the point. Photogrammetric measurements are used to determine the position of the point by relating it to the shiran positional nadir point on each photograph. These positions are mathematically averaged to obtain the most probable coordinates for the point.

secondary great circle—A great circle perpendicular to a primary great circle such as a meridian other than the prime meridian. Also called **secondary; secondary circle.**

secondary grid—Any grid, other than the primary grid, required for combined operations application. Tick marks along the neat lines are the preferred method of portrayal. Such grids should remain on the maps or charts so long as the secondary grid remains in use.

secondary station—An additional triangulation station, usually marked and identified, established to strengthen horizontal map control. Secondary stations are connected

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

to the main scheme stations but are not considered as being part of the main-scheme net. They are often used for providing means for checks and for photogrammetric purposes.

secondary tide station—A tide station which is operated for a short period of time to obtain data for a specific purpose.

secondary—**1. See secondary great circle. 2.** A celestial body revolving around another body, its primary.

SECOR—A phase-comparison electronic long-range distance-measuring system used to determine positions and orbits of satellites or flight vehicles that contain the necessary transponders. This term is an acronym for "sequential collation of range" (now obsolete).

section corner—A corner at the extremity of a section boundary.

sectional chart—A series of aeronautical charts at a 1:500,000 scale covering the entire United States, suitable for contact or visual flying.

section—**1.** The unit of subdivision of a township with boundaries conforming to the rectangular system of surveys, nominally 1 mile square, containing 640 acres. See also **fractional section. 2.** (leveling) That portion of a line of levels which is recorded and abstracted as a unit. See also **fractional section; half section; quarter section.**

sectorial harmonics—The set of spherical harmonics which change from positive to negative as a function of longitude only. See also **tesseral harmonics; zonal harmonics.**

secular aberration—The aberration due to the motion of the center of mass of the solar system in space. Also called **aberration of fixed stars.**

secular perturbations—Perturbations to the orbit of a planet or satellite that continue to act in one direction without limit, in contrast to periodic perturbations which change direction in a regular manner.

secular terms—In the mathematical expression of an orbit, terms which are proportional to time, resulting in secular perturbations.

selection overlay—A tracing of selected map source detail compiled on transparent material; usually described by the name of the features or details depicted, such as contour overlay, vegetation overlay. Also called **lift; pull up; trace.**

selenocentric coordinates—Quantities which express the position of a point with respect to the center of the Moon.

selenocentric—Relating to the center of the Moon; referring to the Moon as a center. Also called **lunicentric.**

selenodesy—(JCS) That branch of applied mathematics which determines, by observation and measurement, the exact positions of points and the figures and areas of large portions of the Moon's surface, or the shape and size of the Moon.

selenodetic—(JCS) Of or pertaining to, or determined by, selenodesy.

selenographic—**1.** Relating to the physical geography of the Moon. **2.** Specifically, referring to positions on the Moon measured in latitude from the Moon's equator and in longitude from a reference meridian.

selenology—That branch of astronomy that deals with the Moon, its magnitude, motion, constitution, and the like.

selenotrope—A device used in geodetic surveying for reflecting the Moon's rays to a distant point, to aid in long-distance observations. See also **heliotrope.**

self-leveling level—A level utilizing the action of gravity in its operation. A prismatic device, called a **compensator**, is an integral part of the instrument which, once the instrument has been roughly leveled, causes the optical system to swing into proper horizontal line of sight and to maintain that position during readings at a given station.

self-reading leveling rod—A rod with graduation marks designed to be read by the observer at the leveling instrument. Also called **speaking rod.**

self-registering gage—Any tide or stream gage which provides a continuous record of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

variation of tide or stream level with the passage of time and which will operate, unattended, for a number of days. Also called **automatic gage.**

semianalytical triangulation—The measurement of *x*-, *y*-, and *z*-model coordinates on an analog instrument and the transformation from model coordinates to grid coordinates by a computational procedure.

semicontrolled mosaic—(JCS) A mosaic composed of corrected or uncorrected prints laid so that major ground features match their geographical coordinates. See also **controlled mosaic; mosaic; uncontrolled mosaic.**

semidiameter correction—A correction due to semidiameter, particularly that sextant altitude correction resulting from observation of the upper or lower limb of a celestial body, rather than the center of that body.

semidiameter—**1.** The radius of a closed figure. **2.** Half the angle at the observer subtended by the visible disk of a celestial body.

semidiurnal constituent—A tidal constituent that has two maximums and two minimums each constituent day.

semidiurnal—Having a period of, occurring in, or related to approximately half a day.

semimajor axis—**1.** One-half the longest diameter of an ellipse. Also called **mean distance. 2.** (geodesy) Equatorial axis of a spheroid or ellipsoid.

semiminor axis—One-half the shortest diameter of an ellipse.

sensibility—(spirit level) See **sensitivity.**

sensible horizon—That circle of the celestial sphere formed by the intersection of the celestial sphere and a plane through any point, such as the eye of an observer, and perpendicular to the zenith-nadir line.

sensitive altimeter—See **barometric altimeter.**

sensitivity—(spirit level) The accuracy and precision which a spirit level is capable of producing. Sensitivity depends on the radius of curvature of its longitudinal section; the longer the radius, the more sensitive the level. Sensitivity is rated by equating the linear length of a division between graduation marks on the level tube and its angular value at the center of curvature of the tube. Also called **sensibility.**

sensitometric curve—See **characteristic curve.**

sensitometry—The measurement of the response of a photosensitive material to the action of light.

Sensor Image Simulator (SIS)—A DMA system built by Goodyear (now Loral) to mimic simulators which use DFAD and DTED. SIS merges DFAD and DTED and assigns radar reflectivity. Also used by DMA as an edit station.

sensor simulation system—A device, such as navigation simulator or trainer, in which specific types of sensor simulator materials are utilized for training purposes. See also **sensor simulator materials.**

sensor simulator materials—Those terrain models or maps, factored transparencies or radar reflectivity plates developed or produced from mapping, charting, geodetic, and/or intelligence data or compilations for use in weapon system or navigation simulators or trainers.

sensor—A technical means to extend man's natural senses. Also a sensing device or equipment which detects and records in the form of imagery, the energy reflected or emitted by environmental areas, features, objects, and events, including natural and cultural features and physical phenomena, as well as man-made features, objects and activities. The energy may be nuclear, electromagnetic, including the visible and invisible portions of the spectrum, chemical, biological, thermal, or mechanical, including sound, blast, and Earth vibration.

September equinox—See **autumnal equinox.**

sequential data structure—A vector data structure in which each feature in a vector data set as a point, line, or polygon, with its coordinates and attributes all in the same record. No spatial relationships are established between features. Also called **spaghetti vector.**

series designation—A descriptive title, a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

number, or a combination of a letter and number, used individually or collectively to identify a group or family of maps, charts, or related publications.

series specifications—See **specifications.**

series—See **coordinated series; map series.**

set forward—See **setup.**

setback—The horizontal distance from the fiducial mark on the front end of a tape or part of tape, which is in use at the time, back to the point on the ground mark or monument to which the particular measure is being made.

setup—**1.** The instrument (transit or level) placed in position and leveled, ready for taking measurements; or a point where an instrument is to be or has been placed. Also called **instrument station. 2.** In base-line measurements, the horizontal distance from the fiducial mark on the front end of a tape or part of tape which is in use at the time, measured in a forward direction to the point on the ground mark or monument to which the particular measure is being made. Also called **set forward.**

set—**1.** The direction toward which the current flows. Usually indicated in degrees true or points of the compass. **2.** A finite or infinite number of objects of any kind, of entities, or of concepts, that have a given property or properties in common. **3.** (surveying) A specified number of observations, as of astronomic azimuth, astronomic longitude.

sexagesimal system—A system of notation by increments of 60; as the division of the circle into 360 degrees, each degree into 60 minutes, and each minute into 60 seconds.

sextant altitude—The altitude of a celestial body as indicated by a sextant or a similar instrument before corrections are applied.

sextant chart—A chart with curves enabling a graphical solution of the three-point problem rather than using a three-arm protractor.

sextant—A double-reflecting instrument for measuring angles, primarily altitudes of celestial bodies. As originally used, the term applied only to instruments having an arc of 60° (and a range of 120°), from which the instrument derived its name. In modern practice the term applies to similar instruments, regardless of range. Also called hydrographic sextant. See also **bubble sextant; marine sextant; octant; quadrant; quintant; surveying sextant.**

shade error—That error of an optical instrument due to refraction in the shade glasses.

shaded relief—(JCS) A cartographic technique that provides an apparent three-dimensional configuration of the terrain on maps and charts by the use of graded shadows that would be cast by high ground if light were shining from the northwest. Shaded relief is usually used in combination with contours. See also **hill shading.**

shaded-relief map—A map on which hypsography is made to appear three-dimensional by the use of graded shadow effects. Generally, the features are shaded as though illuminated from the northwest. A shaded-relief map may also contain contours or hachures in combination with the shading.

shadow factor—(JCS) A multiplication factor derived from the Sun's declination, the latitude of the target, and the time of photography, used in determining the heights of objects from shadow length. Also called **tan alt.**

shadow projector—An optical device developed for checking dimensional accuracy of the various casts of relief models.

shear—Transformation of a rectangle into a parallelogram.

sheet lines—See **neatlines.**

sheet—A single map, either a complete map in one sheet, or belonging to a series.

shift— (JCS) (radar) The ability to move the origin of a radial display away from the center of the cathode-ray tube.

shimmer—An atmospheric effect due to atmospheric turbulences. It may be more critical in photographic observations of celestial objects than refraction. The shimmer makes the image fluctuate rapidly. It averages out in the case of long exposures but is serious in case of flashes. Shimmer affects both right ascension and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

declination in a random manner and, unlike regular refraction, is not zero at the zenith. Also called **scintillation.**

ship-to-shore triangulation—A method of triangulation involving simultaneous observations from three shore stations on a target carried by a ship offshore of the middle station. The middle station must be visible from each of the two end stations.

Ships Inertial Navigation System (SINS)—A precise dead-reckoning system which maintains ships' position and heading through measurements made with gyroscopes and accelerometers.

shiran—An electronic distance-measuring system for measuring distances with geodetic accuracy from an airborne station to each of four ground stations. [This term is an acronym for "S-band high precision short-range electronic navigation. "]

shoot—1. (astronomy; surveying) To make an observation with an instrument. 2. (photography) A slang term used to denote photographing copy, such as a map manuscript, with a copy camera.

shop calibration—Adjustments to precision instruments made in an instrument maintenance shop having a limited amount of specialized testing equipment.

shoran range—The maximum possible operating distance between shoran aircraft and ground stations as limited by flying height, ground-station elevation, terrain, and Earth curvature.

shoran reduction—The computation process of converting from a shoran-distance reading to an equivalent geodetic distance.

shoran trilateration—A method of extending horizontal control in which the sides of appropriate figures are measured by the shoran-line crossing method.

shoran-controlled photography—A method by which the positions of aircraft are determined by distance measurements to two shoran ground stations simultaneously with photographic exposures.

shoran-line crossing—A method of determining distance between two points by flying across the adjoining line.

shoran-wave path—The path taken by the shoran wave as it travels from the mobile (airborne or shipborne) station to the ground station.

shoran—(JCS) A precise short-range electronic navigation system which uses the time of travel of pulse-type transmissions from two or more fixed stations to measure slant-range distance from the stations. Also, in conjunction with a suitable computer, used in precision bombing. [The term is an acronym for the phrase "short-range navigation."]

shoreline effect—See **coastal refraction.**

short arc geodetic adjustment (SAGA)—The least squares adjustment for position, elevation, azimuth, and distance of a number of stations using Doppler satellite observations of the same passes. The satellite positions are permitted to vary. Only portions of satellite arcs are observed. Points along these short arcs are computed for the times of the observations as an intermediate step towards deriving the station positions. See also **point positioning; short arc; short arc network.**

short arc network—A network of positions established by adjustment of simultaneous satellite observations.

short arc reduction method—A computational procedure in which only short arcs of the satellite orbit are employed in order to minimize the effects of secular and long period perturbations.

short arc—A small portion (usually less than half) of the orbital arc traversed by a satellite in making 1 revolution about the Earth.

short distance navigational aids—(JCS) An equipment or system which provides navigational assistance to a range not exceeding 200 statute miles/320 kilometers.

short period perturbations—Periodic perturbations in the orbit of a planet or satellite which execute one complete periodic variation in the time of one orbital period or less.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

short rod—A level rod, usually a Philadelphia rod, permitting readings of 7 feet or less. See also **long rod.**

side can sonar—A system utilizing sonar transducers mounted nearly horizontally and perpendicular to the line of travel, to portray bottom features to the side of the survey vessel.

side equation tests—Side equation tests are a modification of side equations which are helpful to geodetic field parties in checking the accuracy of observations and in locating the points where horizontal-direction observations may be in error.

side equation—A condition equation which expresses the relationship between the various sides in a triangulation figure as they can be derived by computation from one another.

side line—Applied to a strip of land such as a street or right-of-way, it defines the boundaries of that strip; not applied to the ends of a strip.

side overlap—Also called **side lap.** See **overlap,** definition 1.

side shot—A reading or measurement from a survey station to locate a point which is not intended to be used as a base for the extension of the survey. A side shot is usually made for the purpose of determining the position of some object which is to be shown on the map.

side test—In triangulation of a quadrilateral or similar figure, where distances can be computed two different ways, the ratio of the difference between the two computed results to the length of the line.

side-looking airborne radar (SLAR)—(JCS) An airborne radar, viewing at right angles to the axis of the vehicle, which produces a presentation of terrain or moving targets.

side-sight—A sight made with the transit to a point not on the line of traverse. It may be a side shot or an observation to locate an in-and-out station, an azimuth mark, or an intersected point.

sidereal day—The interval of time from a transit of the (true) vernal equinox across the upper branch of a given meridian to its next successive transit across the upper branch of the same meridian. Also called **equinoctial day.**

sidereal focus—The position of the principal focal plane of a lens system. A camera or telescope is in sidereal focus when incident rays from a great distance come to a focus in the plane of the photographic plate or of the reticle. Also called **solar focus.**

sidereal hour angle (SHA)—Angular distance west of the vernal equinox; the arc of the celestial equator, or the angle at the celestial pole, between the hour circle of the vernal equinox and the hour circle of a point on the celestial sphere, measured westward from the hour circle of the vernal equinox through 360°.

sidereal month—The interval of time between two successive passages of the Moon past a fixed star.

sidereal period—**1.** The time taken by a planet or satellite to complete one revolution about its primary and as referred to a fixed star. **2.** Specifically, the interval between two successive returns of an Earth satellite in orbit to the same geocentric right ascension.

sidereal time—Time based upon the rotation of the Earth relative to the vernal equinox.

sidereal year—The period of one apparent revolution of the Earth around the Sun, with respect to the fixed stars, with an accepted value of 365 days, 6 hours, 9 minutes, 9.5 seconds in 1900; and increasing at the rate of 0.0001 second annually.

sidereal—Of or pertaining to the stars. Although sidereal generally refers to the stars and tropical to the vernal equinox, sidereal time and the sidereal day are based upon the position of the vernal equinox relative to the meridian.

sight lamp—A compact, portable, battery-operated electric lamp used as a target for observations on surveys of high precision (usually on first- and second-order geodetic triangulation). The parabolic reflector is mounted in a special case to facilitate pointing and adjusting.

sight line—See **line of collimation.**

sight reduction tables—Tables for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

performing sight reductions, particularly those for determining computed altitude for comparison with the observed altitude of a celestial body to determine the altitude difference for establishing a line of position.

sight reduction—The process of deriving from observation of a celestial body the information needed for establishing a line of position.

sight rod—See **range rod.**

sight tree—See **line tree.**

sight—Observation of the altitude, and sometimes also the azimuth, of a celestial body for a line of position; or the data obtained by such observation.

signature—The characteristics or patterns of physical features that permit objects to be recognized on aerial imagery. A category is said to have a signature only if the characteristic pattern is highly representative of all units of that category.

significant date—The date that represents the best approximation of the date of the product's information; e.g., the date of compilation or the date of the source materials used to revise the product.

simple conic chart—A chart on a simple conic projection.

simple conic map projection—A conic map projection in which the surface of a sphere or spheroid, such as the Earth, is conceived as developed on a tangent cone, which is then spread out to form a plane.

simple feature—See **feature.**

simple harmonic motion—The projection of uniform circular motion on a diameter of the circle of such motion.

simple pendulum—A theoretical concept. A heavy particle suspended from a fixed point by a fine thread which is inextensible and without weight. A simple pendulum cannot be realized in actual work. A simple pendulum is, however, the basis of reductions of observations made with an actual pendulum. Those observations have corrections applied to them to obtain results which would have been produced by an equivalent simple pendulum.

simplification—Smoothing the character of features without destroying their visible shape. Simplification increases as map scale decreases.

Simpson's 1/3 rule—A mathematical expression for determining areas between an irregular boundary and a traverse line where equally spaced offset measurements have been taken.

simultaneous altitudes—Altitudes of two or more celestial bodies observed at the same time.

simultaneous double line—See **simultaneous level line.**

simultaneous level line—A line of spirit leveling composed of two single lines run over the same route, both in the same direction, but using different turning points. Also called **simultaneous double line.**

simultaneous mode—A satellite method for determining the position of an unknown station by the simultaneous 'ranging' from three stations of known position and the unknown station, or simultaneously observing direction from two stations of known position and the unknown station, and mathematically reducing the data to solve for a line or surface of position of the unknown. This technique permits position determination independent of a satellite's orbital parameters.

simultaneous observations—(satellite) Observations of a satellite that are made from two or more distinct points or tracking stations at exactly the same time.

single astronomic station datum orientation—The orientation of a geodetic datum by accepting the astronomically determined coordinates of the origin and the azimuth to one other station without any correction.

single proportionate measurement—A method of proportioning measurements in the restoration of a lost corner whose position is determined with reference to alignment in one direction. Examples of such corners are quarter section corners on the line between two section corners, all corners on standard parallels, and all intermediate positions on any township boundary line. The ordinary field problem

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

consists of distributing the excess of deficiency between two existent corners in such a way that the amount given to each interval shall bear the same proportion to the whole difference as the record length of the interval bears to the whole record distance. After having applied the proportionate difference to the record length of each interval, the sum of the several parts will equal the new measurement of the whole distance.

single-base method—A technique of barometric leveling utilizing two barometers. One barometer is designated as a base and a second, or roving, barometer is used to determine pressures at specific points. Time and pressure are recorded at each position occupied by the roving barometer and time and pressure are recorded every 5 minutes by the base barometer. Data are reduced to elevations by office computations. See also **barometric leveling.**

single-heading radar prediction—A radar prediction made for a single aircraft position or from one specific point in relation to the target. It may be either an experience or an analytical prediction.

single-model instrument—A general class of stereoscopic plotting instruments with a capability for projecting a single stereomodel per setup. This class of plotter is designed for compilation only and is dependent upon supplementary photogrammetric techniques to accomplish necessary stereotriangulation.

single-point transfer instrument—Any instrument used for the transfer of planimetric detail from a single photograph. These instruments are of two general types, reflecting projector and camera lucida.

single-projector method—See **one-swing method.**

single-target leveling rod—Any target rod having graduations on one face only.

sinusoidal map projection—A particular type of the Bonne map projection, employing the Equator as the standard parallel, and showing all geographic parallels as truly spaced parallel straight lines, along which exact scale is preserved. This is an equal-area map projection. Also called **Mercator equal-area map projection.** See also **Sanson-Flamsteed map projection.**

siphon barometer—A mercury barometer consisting of a column of mercury in a glass tube which is bent so as to have two vertical branches, one about one-fourth the length of the other. The end of the longer branch is closed, and the air in it is displaced by the mercury, but the shorter branch is left open, and the mercury is thereby subjected to atmospheric pressure. The difference of the height of the mercury in the two branches is a measure of the atmospheric pressure.

situation map—(JCS) A map showing the tactical or administrative situation at a particular time. See also **map.**

sixteenth section corner—A corner at an extremity of a boundary of a quarter-quarter section; midpoint between or 20 chains from the controlling corners on the section or township boundaries. Written as 1/16 section corner. Also called **quarter-quarter section corner.**

size—1. To coat with any of the various glutinous materials used for filling the pores in the surface of paper, fiber, or of a mosaicking board. 2. To calculate the measurements required in photographing a map to a desired scale.

sizing the litho—An operation performed in order to determine the actual measurement of the original lithographic maps to be used as source for a map revision in order to determine what distortion and changes of dimensions are necessary to fit the old map inside the new projection.

sketch map—A map made from loose, uncontrolled surveys. The information thereon is generally sparse.

sketchmaster—A form of camera lucida that permits superimposition of a rectified virtual image of a photograph over a map manuscript. See also **oblique sketchmaster; universal analog photographic rectification system; universal sketchmaster; vertical sketchmaster.**

skewed map projection—Any standard projection used in map or chart construction which does not conform to a general north-south format with relation to the neatlines of the map or chart.

slant range—(JCS) The line-of-sight distance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

between two points not at the same level relative to a specific datum.

slave station—That station in a given system of stations that is controlled by the master station. Also called **remote station.**

slivers—(digital) Polygons formed when two adjacent polygons do not abut along a single common line and leave a small space between the larger two.

slope angle—The angle between a slope and the horizontal.

slope chaining—See **slope taping.**

slope correction of tape—See **grade correction.**

slope correction—**1.** (hydrographic surveying) The correction applied to soundings erroneously positioned as a result of an echo sounder receiving its initial return from a point upslope from its recorded position. **2.** (land surveying) See **grade correction.**

slope taping—Taping wherein the tape (or chain) is held as required by the slope of the ground, the slope of the tape measured, and the horizontal distance computed. Also called **slope chaining.**

slope—See **gradient.**

slot cutter—See **templet cutter.**

slotted templet—A templet on which the radials are represented as a slot cut in a sheet of cardboard, metal, or other material.

slotted-templet plot—See **slotted-templet triangulation.**

slotted-templet triangulation—A graphical radial triangulation made by the use of slotted templets. Also called **slotted-templet plot.**

Small Wetted Area Twin Hull (SWATH)—A vessel designed for stability by having its buoyant volume mostly underwater, presently being evaluated for use in hydrographic surveying projects.

small circle—A circle on the surface of the Earth, the plane of which does not pass through the Earth's center.

small-scale map—A map having a scale of 1:600,000 or smaller.

smooth sheet—A final plot of field control and hydrographic development such as; soundings, fathom curves, wire drag areas, etc., to be used in chart construction.

smoothing—**1.** A set of procedures for removing short-range, erratic variations from lines, surfaces, or data series. **2.** (image processing) The averaging of densities in adjacent areas to produce more gradual transitions.

snakeslip—See **etch slip.**

snap marker—See **point marker.**

Snell's law of refraction—This law states that the sine of the angle of incidence divided by the sine of the angle of refraction equals a constant termed the index of refraction when one of the media is air. The index of refraction can also be explained as the ratio of the velocity of light in one medium to that in another.

softcopy—Products which are maintained in digital format for use with automated data processing equipment.

solar altitude—Angular distance of the Sun above the horizon.

solar attachment—An auxiliary instrument which may be attached to an engineer's transit, permitting its use as a solar compass.

solar day—**1.** The duration of one rotation of the Earth on its axis, with respect to the Sun. This may be either a mean solar day, or an apparent solar day, as the reference is the mean or apparent sun, respectively. See also **apparent solar day; mean solar day. 2.** The duration of one rotation of the Sun.

solar declination—Angular distance of the Sun expressed in degrees north or south of the celestial equator; it is indicated as "+" when north and "-" when south of the Equator. Also called **declination of the Sun.**

solar eclipse method—A means of determining the angular distance between two observers along the center line of the path of a solar eclipse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

solar eclipse—The obscuration of the light of the Sun by the Moon. A solar eclipse is partial if the Sun is partly obscured; total if the entire surface is obscured; or annular if a thin ring of the Sun's surface appears around the obscuring body.

solar ephemeris—A daily tabulation of astronomic positions of the Sun.

solar focus—See **sidereal focus.**

solar occultation—An occultation of the Sun by the Moon.

solar parallax—The angle subtended by the equatorial radius of the Earth at a distance of one astronomic unit (i.e., the equatorial horizontal parallax of the Sun).

solar radiation pressure—A perturbation of high flying artificial satellites of large diameter. The greater part is directly from the Sun, a minor part is from the Earth, which is usually divided into direct (reflected) and indirect terrestrial (radiated) radiation pressures.

solar time—1. Time based upon the rotation of the Earth relative to the Sun. 2. Time on the Sun.

solar transit—A regular transit to which has been added a solar attachment, which effects the instantaneous mechanical solution of the astronomic triangle (Sun-zenith-pole) and permits the establishment and surveying of the astronomic meridian or astronomic parallel directly by observation.

solar year—*See* ***tropical year.***

solid angle—The integrated angular spread at the vertex of a cone, pyramid, or other solid figure.

solstice—One of two points of the ecliptic farthest from the celestial equator; one of the two points on the celestial sphere occupied by the Sun at maximum declination. Also called **solstitial point.** See also **summer solstice; winter solstice.**

solstitial colure—The hour circle through the solstices.

solstitial point—See **solstice.**

sonar—1. (JCS) A sonic device used primarily for the detection and location of underwater objects. 2. A system for determining distance of an underwater object by measuring the interval of time between transmission of an underwater sonic or ultrasonic signal and return of its echo. [This term is derived form the words "sound navigation and ranging."]

sonic depth finder—See **echo sounder.**

sonic navigation—See **acoustic navigation.**

sortie plot—(JCS) An overlay representing the area on a map covered by imagery taken during one sortie. Also called **photo index.**

sortie—(JCS) In air operations, an operational flight by one aircraft.

Sound Surveillance Systems Charts (SOSUS)—Charts which portray generalized bathymetry and other prescribed system oriented information supporting Anti Submarine Warfare (ASW) surveillance.

sounding datum—The plane to which soundings are referred.

sounding pole—A round, wooden, 15-foot-long pole, used for shoal water soundings. It is graduated in feet and half-feet from the center toward both ends and numbered consecutively from the ends toward the center.

sounding—1. The measured or charted depth of water. 2. A measurement of the depth of water expressed in feet or fathoms and reduced to the tidal datum shown in the chart title. Also called **hydrographic sounding.** See also **depth; depth number.** 3. In geophysics, any penetration of the natural environment for **scientific observation.** See also **echo sounder; echo sounding; *lead line*; *off* soundings; on soundings; wire drag.**

Source Acquisition Segment (SA/S)—Segment of DMA's DPS which provides the capability to manage image library holdings and to maintain the geodetic control data base. See also **Digital Production System.**

Source Preparation Segment (SP/S)—Segment of DMA's DPS which provides the capability to accomplish source assessment,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

source refinement, and geopositioning. Operators perform assessment of imagery and of textual and graphics material to determine their characteristics and utility. This segment also provides for mensuration to support triangulation with the existing Hardcopy Exploitation Segment. See also **Digital Production System; MARK 90; Hardcopy Exploitation Segment.**

source map—The map used for the selection of map or chart detail.

source material—Data of any type required for the production of MC&G products including, but not limited to, ground control, aerial and terrestrial photographs, sketches, maps, and charts; topographic, hydrographic, hypsographic, magnetic, geodetic, oceanographic, and meteorological information; intelligence documents and written reports pertaining to natural and man-made features of the area to be mapped or charted.

south declination—See **declination**, definition 3.

south geographical pole—The geographical pole in the Southern Hemisphere, at latitude 90°S.

south geomagnetic pole—The geomagnetic pole in the Southern Hemisphere.

south magnetic pole—The magnetic pole in the Southern Hemisphere.

south point—See **celestial meridian.**

south polar circle—See **Antarctic Circle.**

southbound node—See **descending node.**

southing—See **latitude difference.**

Space Oblique Mercator—A mapping projection modeled on the dynamics of the LANDSAT spacecraft motion. It incorporates time-dependent values of satellite platform motion and the Earth's motion in a continuous projection of the area viewed in the LANDSAT scene.

space coordinates—(photogrammetry) Any general three-dimensional coordinate system used to define the position of a point in the object space, as distinguished from the image of the point on a photograph.

space motion—Motion of a celestial body through space. See also **proper motion.**

space-polar coordinates—A system of coordinates by which a point on the surface of a sphere is located in space by (1) its distance from a fixed point at the center, called the **pole**; (2) the colatitude or angle between the polar axis (a reference line through the pole) and the radius vector (a straight line connecting the pole and the point); and (3) the longitude or angle between a reference plane through the polar axis and a plane through the radius vector and polar axis.

spacecraft—Devices, manned and unmanned, which are designed to be placed into an orbit about the Earth or into a trajectory to another celestial body.

spading—Removing scribe coating from the base material by use of a wide flat blade.

spaghetti vector—A digital storage format in which all lines and points are unrelated to each other.

Spatial Data Transfer Specification (SDTS)—A specification under the Spatial Data Transfer Standard (SDTS) designed to meet the recognized requirement for easy transfer of spatial data from one spatial data handling system to another, with both systems possibly residing on computer hardware and operating system software of different makes. Under SDTS, the contents (specification of fields and subfields) are clearly separated from the implementation (the encoding of fields and subfields on the media).See **Spatial Data Transfer Standard.**

Spatial Data Transfer Standard (SDTS)—A standard consisting of definitions and references, the Spatial Data Transfer Specification, a quality report on digital cartographic data, a conceptual model for describing cartographic features, and a standard set of defined entities and attributes derived from topographic maps and general nautical charts. See **Spatial Data Transfer Specification.**

spatial analysis—Analytical techniques associated with the study of the location of geographical entities together with their spatial

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

dimensions. Also referred to as quantitative analysis.

spatial data bases—Collections of similar and related spatial data records that are recorded for use by a computer. See also **Geographic Information System**.

spatial data sets—Collections of similar and related spatial data records that are recorded for use by a computer.

spatial data—Data pertaining to the location of geographical entities together with their spatial dimensions. Spatial data are classified as point, line, area or surface.

spatial model—See **stereoscopic model.**

speaking rod—See **self-reading leveling rod.**

special-purpose map—Any map designed primarily to meet specific requirements.

special area (SA) annotation—A structure or group of structures possessing unique physical characteristics, but whose area or linear dimensions do not qualify for application of one of the radar significance analysis codes.

special job-cover map—(JCS) A small-scale map used to record progress on photographic reconnaissance tasks covering very large areas. As each portion of the task is completed, the area covered is outlined on the map.

special meander corner—A corner established at the intersection of a surveyed subdivision-of-section line and a meander line or the intersection of a computed center line of a section and a meander line. In the latter case, the centerline of the section is calculated and surveyed on a theoretical bearing to an intersection with the meander line of a lake (over 50 acres in area) which is located entirely within a section.

special-subject map—See **topical map.**

specific force—The difference between the inertial acceleration and gravitation acting on a body. The physical gravity sensed by accelerometers. All spirit levels are normal to the specific force vector.

specifications—The rules, regulations, symbology, and a comprehensive set of standards which have been established for a particular map or chart series or scale group. Specifications vary with the scale and the purpose of the graphic.

spectral band—A set of adjacent wavelengths in the electromagnetic spectrum with a common characteristic, such as the visible band.

spectrophotometer—A device for the measurement of spectral transmittance, spectral reflectance, or relative spectral emittance.

spectroradiometer—A device for the measurement of spectral distribution of radiant energy.

specular reflection—(optics or microwave theory) The type of reflection characteristic of a highly polished plane surface from which all rays are reflected at an angle equal to the angle of incidence. See also **diffuse reflection.**

speed of lens—See **relative aperture.**

speed—(photography) The response or sensitivity of the material to light, often expressed numerically according to one of several systems (e.g., H and D, DIN, Scheiner, and ASA exposure index). See also **relative aperture.**

sphere—A body or the space bounded by a spherical surface. See also **celestial sphere; oblique sphere; parallel sphere; right sphere; terrestrial sphere.**

spherical aberration—An aberration caused by rays from various zones of a lens coming to focus at different places along the axis. This results in an object point being imaged as a blurred circle.

spherical angle—The angle between two intersecting great circles.

spherical coordinates—A system of polar coordinates in which the origin is the center of a sphere and the points all lie on the surface of the sphere. The polar axis of such a system cuts the sphere at its two poles. In photogrammetry, spherical coordinates are useful in defining the relative orientation of perspective rays or axes and make it possible to state and solve, in simple forms, many related problems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

spherical excess—The amount by which the sum of the three angles of a triangle on a sphere exceeds 180°. In geodetic work, in the computation of triangles, the difference between spherical angles and spheroidal angles is generally neglected; spherical angles are used, and Legendre's theorem is applied to the distribution of the spherical excess. That is, approximately one-third of the spherical excess of a given spherical triangle is subtracted from each angle of the triangle.

spherical harmonics—Trigonometric terms of an infinite series used to approximate a two- or three-dimensional function of locations on or above the Earth.

spherical lens—A lens in which all surfaces are segments of spheres.

spherical triangle—The closed figure formed when any three points on the surface of a sphere are joined by arcs of great circles.

spheroid junction—An accentuated line on a map or chart, separating two or more major grids which are based on different spheroids.

spheroid of reference—See **reference spheroid.**

spheroidal angle—An angle between two curves on a spheroid; measured by the angle between their tangents at the point of intersection.

spheroidal excess—The amount by which the sun of the three angles of a triangle on a spheroid exceeds 180°. See also **spherical excess.**

spheroidal triangle—A triangle on the surface of a spheroid.

spheroid—**1.** (general) Any figure differing slightly from a sphere. **2.** (geodesy) A mathematical figure closely approaching the geoid in form and size and used as a surface of reference for geodetic surveys. In geodesy spheroid and ellipsoid are synonymous terms. See also **Airy spheroid (ellipsoid); Australian National spheroid; Bessel spheroid (ellipsoid); Clarke spheroid (ellipsoid) of 1866; Clarke spheroid (ellipsoid) of 1880; ellipsoid of rotation; equilibrium spheroid; Everest spheroid (ellipsoid); Hayford spheroid (ellipsoid); International spheroid (ellipsoid); Krasovsky spheroid (ellipsoid); oblate spheroid; prolate spheroid; reference spheroid.**

spheropotential surface—See **spherop.**

spherop—An equipotential surface in the normal gravity field of the Earth. Also called **spheropotential surface.**

spider templet—A mechanical templet which is formed by attaching slotted steel arms, representing radials, to a central core. The spider templet can be disassembled and the parts used again. Also called **mechanical arm templet.**

spider-templet plot—See **spider-templet triangulation.**

spider-templet triangulation—A graphical radial triangulation made by the use of spider templets. Also called **spider-templet plot.**

spiral curve—(route surveying) A curve of uniformly varying radius connecting a circular curve and a tangent, or two circular curves whose radii are, respectively, longer and shorter than its own extreme radii. Also called **easement curve; transition curve.**

spiral to spiral (SS)—A common point between two spirals.

spirit level axis—The line tangent to the surface of a spirit level tube (vial) against which the bubble forms, at the center of the graduated scale of the level, and in the plane of the longitudinal axis of the tube (vial) and its center of curvature. Also called **axis of level; axis of the level bubble; bubble axis.**

spirit level wind—Lack of parallelism between the axis of a spirit level vial and the line joining the centers of its supports. When wind (pronounced to rhyme with find) is present, the bubble will respond with a longitudinal movement when the spirit level is rocked on its supports.

spirit leveling—The determination of elevations of points with respect to each other or with respect to a common datum, by use of a leveling rod and an instrument using a spirit level to establish a horizontal line of sight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

spirit level—A closed glass tube (vial) of circular cross section, its center line also forming a circular arc, its interior surface being ground to precise form; it is filled with ether or liquid of low viscosity enough free space being left for the formation of a bubble of air and gas. Also called **bubble level.** See also **chambered spirit level; circular level; hanging level; latitude level; level trier; plate level; reversible level; rod level; striding level; telescope level.**

split cameras—(JCS) An assembly of two cameras disposed at a fixed overlapping angle relative to each other. [Mainly used for reconnaissance purposes.] Also called **split-vertical camera.**

split photography—See **split-vertical photography.**

split-vertical camera—See **split cameras.**

split-vertical photography—(JCS) Photographs taken simultaneously by two cameras mounted at an angle from the vertical, one tilted to the left and one to the right, to obtain a small side overlap. Also called **split photography.**

spoking—(JCS) (radar) Periodic flashes of the rotating time base on a radial display. Sometimes caused by mutual interference.

spot elevation—(JCS) A point on a map or chart whose elevation is noted. [Elevations are shown, wherever practicable, for road forks and intersections, grade crossings, summits of hills, mountains, and mountain passes, water surfaces of lakes and ponds, stream forks, bottom elevations in depressions, and large flat areas.] Also called **spot height.** See also **checked spot elevation; elevation; unchecked spot elevation.**

spot height—See **spot elevation.**

spot prediction—A single heading radar prediction intended to portray, as nearly as possible, a comprehensive analysis of the radarscope at a precise geographic location.

spot size—(JCS) The size of the electron spot on the face of the cathode-ray tube.

spring balance—An accessory of taping and base measuring apparatus which is used in applying proper tension to a tape.

spur line of levels—A line of levels run as a branch from the main line of levels, either for the purpose of determining the elevations of marks not conveniently reached by the main line of levels or to connect with tidal bench marks or other previously established bench marks in obtaining checks on old leveling either at the beginning or end of a line of levels or at intermediate junctions along the new line of levels.

spur traverse—Any short traverse that branches off the established traverse to reach some vantage point or position. Also called **stub traverse.**

stabilized mount—A mount controlled by a gyroscope vertical reference unit designed to maintain a mapping or positional camera or other devices such as TPR antenna in a near vertical orientation independent of aircraft pitch and roll.

stable base—A general term applied to mapping materials possessing a high degree of dimensional stability.

stable gravimeter—A gravimeter having a single weight or spring such that the sensitivity is proportional to the square of its period.

stable-base film—(JCS) A particular type of film having high stability in regard to shrinkage and stretching. [Suitable for aerial mapping photography and map production. Usually referred to by its commercial name.]

stable-type gravimeter—A gravimeter which uses a high order of optical and/or mechanical magnification so that a change in position of a weight or associated property is measured directly.

stadia circle—See **Beaman arc.**

stadia constant—(leveling) The ratio which is multiplied by the stadia interval to obtain the length of a sight in meters. Also, the ratio by which the sum of the stadia intervals of all sights of a run is converted to the length of the run in kilometers.

stadia diagram—A chart or drawing which provides a means for rapid field reduction of stadia readings. Usually it is prepared on cross-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

section paper and drawn to the scale of the survey being performed.

stadia intercept—See **stadia interval.**

stadia interval—(leveling) The length of rod subtended between the top and bottom cross hairs (cross wires) in the leveling instrument as seen projected against the face of the leveling rod. Also called **stadia intercept.**

stadia rod—See **stadia.**

stadia slide rule—The most rapid method of reducing stadia readings is by the use of a slide rule which has, in addition to the ordinary scale of numbers (logarithms of the distances), two scales especially constructed for stadia work, one consisting of values of log $\cos^2 a$ and the other of log 1/2 sin 2 a for different values of a. On some rules, the values of a range from 0°34' to 45°; on others, from 0°03' to 45°. In some forms the horizontal distance is read directly; in others the horizontal correction (1— $\cos^2$) or $\sin^2$ is given. A 10-inch slide rule gives results sufficiently accurate for all ordinary purposes.

stadia traverse—A traverse in which distances are measured by the stadia method.

stadia trigonometric leveling—A technique of extending supplemental vertical control in areas of moderate or low relief. Distances are measured by stadia methods and can be done with planetable, transit, or theodolite. Field work is reduced to usable form by trigonometric computations.

stadia—A graduated rod used in the determination of distance by observing the intercept on the rod subtending a small known angle at the point of observation. In practice, the angle is usually defined by two fixed lines in the reticle of a telescope (transit or telescopic alidade). The term **stadia** is also used in connection with surveys where distances are determined with a stadia, as stadia survey, stadia method, stadia distance, etc.; also used to designate parts of the instrument used, as stadia wires. Also called **stadia rod.** See also **horizontal stadia.**

stadimeter—An instrument for determining the distance to an object of known height by measuring the angle subtended at the observer by the object. The instrument is graduated directly in distance. See also **range finder.**

staff gage—The simplest form of tide or stream gage consisting of a graduated staff securely fastened to a pole or other suitable support. It is so designed that a segment of the staff will be below lowest low water when mounted and the remainder will be above water and positioned for direct observations from shore or some other vantage point.

Stampfer level—A type of leveling instrument having the telescope tube so mounted that it could be moved in a vertical plane about a horizontal axis, involving the use of a striding level and a micrometer screw.

Standard Linear Format (SLF)—A standard format used and modified by DMA for digital cartographic data. SLF uses a chain-node spatial structure to avoid duplication of common boundaries.

standard automatic tide gage—A chronograph used where extended time readings of tidal changes are required. The rise and fall of the tide is communicated by a wire (attached to a float) to a worm screw on the gage, which moves a pen transferring the data to a permanent paper record.

standard corner—A senior corner on a standard parallel or base line.

standard deviation—See **standard error.**

standard error (σ)—The square root of the variance. It implies a 68.27% probability that the parameter of observation has an error the absolute value of which does not exceed σ. Also called **standard deviation.**

standard error of the mean—The standard error(σ) of a most probable value established as the mean of n observations. It is rigorously derived as the standard error (σ) of the n observations divided by the square root of n. It also has an implied probability of 68.27%.

standard grids—The Universal Transverse Mercator (UTM) grid and the Universal Polar Stereographic (UPS) grid.

standard indexing system (SIS)—A system developed for use within the Department of Defense for the indexing of all aerial photography held at national level. Aerial photographic missions are plotted on acetate sheets covering 1° squares of the world at a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

scale of 1:250,000.

standard map—*A map which complies with specific* Map Accuracy Standards and *guidelines.*

standard meridian—**1.** The meridian used for determining standard time. **2.** A meridian of a map projection, along which the scale is as *stated.*

standard of length—A physical representation of a linear unit that is approved by competent authority.

standard parallel—(JCS) A parallel on a map or chart along which the scale is as stated for that map or chart. **2.** A parallel of latitude used in the computation of a map projection.

standard port—See **reference station.**

standard quadrangle—A quadrangle of a specific series, conforming with the systematic *pattern of the series.*

standard station—See **reference station.**

standard survey—A survey which, in scale, *accuracy,* and content, satisfies criteria prescribed for such a survey by competent authority.

standard tension—(taping) That tension or *pull at which a* tape was standardized.

standard time—Mean solar time for a selected meridian adopted for use throughout *a* belt or zone.

standardization—*The comparison of an* instrument or device with a standard to determine the value of the *instrument or device* in terms of an adopted unit.

standard—(JCS) An exact value, a physical entity, or an abstract concept, established and defined by *authority, custom, or common* consent to serve as a reference, model, or rule in measuring quantities or qualities, establishing practices or procedures, or evaluating results. A fixed quantity or quality.

star chart—A chart or map of the celestial sphere showing principal stars which are useful for observations for navigation or field astronomy.

star finder—A device to facilitate the *recognition of stars, particularly for purposes of* navigation and geodetic astronomy.

star trail—A streak-like image of a star recorded on a stellar plate by a photographic *time exposure caused by the rotation of* the Earth.

start node—(digital mapping) The first node of an edge (an edge is traversed from start node to end node).

starting control—Control available for the absolute orientation of the first plate pair along a line of flight for which control is to be extended.

state base map—A base map of the area of a state as the unit used as a base upon which data of a specialized nature are compiled or overprinted.

state coordinate systems—The plane rectangular coordinate systems established by the National Geodetic Survey, one for each state in the United States, for use in defining *positions of* geodetic stations in terms of plane rectangular (*x* and *y*) coordinates. Also called **state system of plane coordinates.**

state system of plane coordinates—See **state coordinate systems.**

state vector—A set of parameters that define the position and *velocity of an object at a* specific time with respect to a reference system.

static gravity meter—A type of gravity instrument in which a linear or angular *displacement is observed or nulled by an* opposing force.

static markings—(photogrammetry) Marks on photographic negatives or other sensor imagery caused by *unwanted discharges* of static electricity.

station adjustment—The adjustment of angle measurements at a triangulation or *traverse station to satisfy* local requirements (such as horizon closure) without regard to observations or conditions at other points. Also *called* **local adjustment.**

station *error*—*See* **deflection of the vertical.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

station mark—A mark on the ground, either a monument of special construction, or a natural or artificial object, which pinpoints the location of a survey station. See also **mark**, definition 2.

station pointer—See **three-arm protractor.**

stationary field—Any natural field of force, as a gravimetric or magnetic field.

stationary orbit—An orbit in which the satellite revolves about the primary at the angular rate at which the primary rotates on its axis. From the primary, the satellite appears to be stationary over a point on the primary. See also **synchronous satellite.**

station—1. (surveying) A definite point on the Earth whose location has been determined by surveying methods. It may or may not be marked on the ground. A station usually is defined by the addition of a term which describes its origin or purpose. Usually marked on the ground by a monument of special construction, or by a natural or artificial structure. 2. (route surveying) Any point whose position is given by its total distance from the starting hub; also, each stake set at 100-foot intervals along a route survey. See also **A-station; air station; astronomic station; auxiliary station; B-station; base station; control station; data acquisition station; drift station; eccentric station; gravity reference stations; gravity station; ground station; horizontal control station; in-and-out station; intersection station; Laplace station; magnetic station; main scheme station; master station; oceanographic station; plus station; primary tide station; projector station; radio range station; reference station; resection station; satellite triangulation stations; secondary station; secondary tide station; setup; slave station; stream gaging station; subordinate station; subsidiary station; supplementary station; taping station; tide station; tracking station; traverse station; triangulation station.**

statoscope—A sensitive form of barometer used in aerial photography for measuring small differences in altitude between successive air stations. Usually recorded automatically on the film at the instant of exposure. See **recording statoscope.**

stellar aberration—The displacement of the observed position from the position where the body was geometrically located at the instant of observation due to the motion of the observing platform.

stellar camera—A camera for photographing the stars.

stellar magnitude—See **magnitude**, definition 1.

stellar map matching—A process during the flight of a vehicle by which a chart of the stars set into the guidance system is automatically matched with the position of the stars observed through telescopes so as to give guidance to the vehicle. See also **map matching guidance.**

stellar parallax—See **annual parallax.**

stellar plate—A precisely ground glass plate coated with a photographic emulsion used for recording satellite images against a stellar background.

step cast—The negative or positive reproduction of the stepped terrain base of a relief model.

step tablet—See **step wedge.**

step wedge—A strip of film or a glass plate whose transparency diminishes in graduated steps from one end to the other; often used to determine the density of a photograph. Also called **gray scale; step tablet.** See also **continuous tone gray scale.**

Stephenson leveling rod—A speaking rod having graduations forming a diagonal scale, with horizontal lines through the tenth-of-foot marks. This rod is read to hundredths of a foot.

steradian—The unit of measure of a solid angle.

stereo oblique plotter—A device which permits continuous plotting of planimetric detail from oblique photographs. Essentially, the device consists of two photoangulators linked under a stereoscope and is provided with plotting arms.

stereo pair—See **stereoscopic pair.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

stereo triplet—A stereogram composed of three photographs, the center photo having a common field of view with the two adjacent photos, arranged in such a manner as to permit complete stereoscopic viewing of the center photograph.

stereocomparagraph—A relatively simple and mobile stereoscopic instrument used for the preparation of topographic maps from photography. Differences in elevation are determined by measuring parallax difference on a stereoscopic pair.

stereocomparator—A stereoscopic instrument for measuring parallax; usually includes a means of measuring photograph coordinates of image points.

stereocompilation—See **compilation**, definition 2.

stereogram—(JCS) A stereoscopic set (pair) of photographs or drawings correctly oriented and mounted (or projected) for stereoscopic viewing. See also **stereo triplet.**

stereographic chart—A chart on the stereographic projection.

stereographic horizon map projection—A stereographic projection having the center of the projection on some selected parallel of latitude other than the Equator.

stereographic map projection—A perspective, conformal map projection on a tangent plane, with the point of projection at the opposite end of the diameter of the sphere from the point of tangency of the plane. Also called **azimuthal orthomorphic map projection.**

stereographic meridional map projection—A stereographic projection having the center of the projection on the Equator.

stereograph—A stereometer with a pencil attachment which is used to plot topographic detail from a properly oriented stereogram.

stereometer—A measuring device containing a micrometer movement by means of which the separation of two index marks can be changed to measure parallax difference on a stereoscopic pair of photographs. Also called **parallax bar.**

stereometric camera—A combination of two cameras mounted with parallel optical axes on a short rigid base, used in terrestrial photogrammetry for taking photographs in stereoscopic pairs.

stereometric map—See **photogrammetric map.**

stereomodel parameters—Numerical data for each image of a complete stereomodel or stereomodel section. The photo coordinates of an object imaged are related to the ground coordinates of the object by a projective relationship defined by a (sensor) mathematical model.

stereomodel—See **stereoscopic model.**

stereophotogrammetry—Use of stereo images, such as overlapping photographs, in the science of photogrammetry. See also **photogrammetry.**

stereoplanigraph—A precise stereoscopic plotting instrument, especially valuable for extension of control, and capable of handling most types of stereoscopic photography, including terrestrial.

stereoscope—(JCS) A binocular optical instrument for helping an observer to view photographs or diagrams, to obtain a three-dimensional mental impression (stereoscopic model). [The design of stereoscopic viewing instruments utilizes lenses, mirrors, and prisms, or a combination thereof.]

stereoscopic base—The distance and direction between complimentary image points on a stereoscopic pair of photographs.

stereoscopic cover—(JCS) Photo- graphs taken with sufficient overlap to permit complete stereoscopic examination.

stereoscopic exaggeration—See **hyperstereoscopy.**

stereoscopic fusion—The mental process which combines two perspective views to give a impression of a three-dimensional model.

stereoscopic image—See **stereoscopic model.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

stereoscopic model—(JCS) The mental impression of an area or object seen as being in three dimensions when viewed stereoscopically on photographs. Also called **spatial model; stereomodel; stereoscopic image.**

stereoscopic pair—(JCS) Two photographs with sufficient overlap of detail to make possible stereoscopic examination of an object or an area common to both. Also called **stereo pair.**

stereoscopic parallax—See **absolute stereoscopic parallax.**

stereoscopic plotting instrument—An instrument for compiling a map or obtaining spatial solutions by observation of stereoscopic models formed by stereoscopic pairs of photographs. See also **double-projection direct-viewing stereoplotter; radial plotter: single-model instrument; stereo oblique plotter; stereocomparagraph; stereoplanigraph.**

stereoscopic principle—The formation of a single, three-dimensional image by binocular vision of two photographic images of the same terrain taken from different exposure stations.

stereoscopic vision—The particular application of binocular vision which enables the observer to obtain the impression of depth, usually by means of two different perspectives of an object (as two photographs taken from different camera stations).

stereoscopic—Of or pertaining to stereoscopy.

stereoscopy—The science which deals with three-dimensional effects and the methods by which they are produced.

stereotemplet triangulation— Aerotriangulation by means of stereotemplets. The method permits scale solutions by area and is not restricted to solutions along flight strips.

stereotemplet—A composite slotted templet adjustable in scale and representative of the horizontal plot of a stereoscopic model. An assembly of stereotemplets provides a means of aerotriangulation for horizontal positions with a stereoscopic plotting instrument not designed for bridging.

stereotopographic map—See **photogrammetric map.**

stereotriangulation—A triangulation procedure that uses a stereoscopic plotting instrument to obtain the successive orientations of the stereoscopic pairs of photographs into a continuous strip. The spatial solution for the extension of horizontal and/or vertical control using these strips (or flight) coordinates may be made by either graphical or computational procedures. Also called **bridging; instrument phototriangulation; multiplex triangulation.** See also **vertical stereotriangulation.**

stereo—1. Contracted or short form of stereoscopic. 2. The orientation of photographs when properly positioned for stereoscopic viewing. Photographs so oriented are said to be "in stereo."

stickup—Adhesive-backed or wax-backed film or paper which map names, symbols, descriptive terms, etc., have been printed, for application in map and chart production.

stilling device—Any device or structure placed in the vicinity of a gage to reduce wave action and afford more accurate reading of the gage.

stipple—A random dot pattern used to depict certain topographic features such as sand.

Stokes' formula—A formula for computing geoid heights from gravity data.

stone bound—A substantial stone post set into the ground with its top approximately flush with the ground surface to mark accurately and permanently the important corners of a land survey.

stop numbers—See **relative aperture.**

stop—See **aperture stop.**

storage tube display—A CRT display on which an image can be stored on the screen for several minutes or longer with a single pass of the electron beam.

straight line graver—A variation of the rigid tripod graver so designed that the scribing point, the vertical vane, and one supporting leg are all

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

directly in line; used with a straightedge for scribing long, straight lines.

strategic map—(JCS) A map of medium scale, or smaller, used for planning of operations, including the movement, concentration, and supply of troops. See also **map.**

strategic planning model—Small-scale terrain models depicting only the general character of the terrain and features of considerable prominence. They generally embrace continental areas, countries, extensive land mass areas, or principal island masses and are most frequently used in high echelon planning activities.

stratosphere—(JCS) The layer of the atmosphere above the troposphere in which the change of temperature with height is relatively small. See also **atmosphere.**

stream gaging station—A point along a stream at which periodic measurements of velocity or discharge are made, and at which daily or continuous records of the stage of height of the water surface above a given datum is obtained.

strength of figure—(triangulation) The comparative precision of computed lengths in a triangulation net as determined by the size of the angles, the number of conditions to be satisfied, and the distribution of base lines and points of fixed position. Strength of figure in triangulation is not based on an absolute scale but rather is an expression of relative strength. Also applicable to the individual geometric figures within a given net.

stretching apparatus—See **tape stretcher.**

striding level—A spirit level so mounted that it can be placed above and parallel with the horizontal axis of a surveying or astronomic instrument, and so supported that it can be used to measure the inclination of the horizontal axis to the plane of the horizon.

strip adjustment—Similar to a block adjustment, but limited to a single strip of photographs.

strip coordinates—The coordinates of any point in a strip, whether on the ground or actually on air station, referred to the origin and axes of the coordinate system of the first overlap.

strip film—A photographic film in which the emulsion membrane can be removed from its temporary base after exposure and processing the membrane is then transferred to a new base. Principally used in correction work. Also called **stripping film.**

strip mosaic—A mosaic consisting of one strip of aerial photographs taken on a single flight.

strip plot—(JCS) A portion of a map or overlay on which a number of photographs taken along a flight line is delineated without defining the outlines of individual prints.

strip prediction—A single heading prediction intended to convey the general nature and pattern of radar returns continuously along a specific flight path.

strip radial plot—See **strip radial triangulation.**

strip radial triangulation—A direct radial triangulation in which the photographs are plotted in flight strips without reference to ground control and the strips are later adjusted together and to the ground control. Also called **strip radial plot.**

strip width—The average dimension, measured normal to the flight line, of a series of neat models in a flight strip. Strip width is generally considered as equal to width between flights.

stripping film—See **strip film.**

stripping—The cutting, attachment, and other operations for assembling cut film sections to produce a flat.

strip—See **flight strip.**

Structured Query Language (SQL)—A user-friendly, non-procedural data language which is a standard for data manipulation in relational database management systems.

stub traverse—See **spur traverse.**

style sheet—A graphic guide for the format and portrayal of grid and marginal information. Also called **mock-up.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

subaqueous reconnaissance survey—A hydrographic survey which is a rapidly executed preliminary survey of a region to provide advance information to meet immediate military needs. Normally made at small scale, it is usually not controlled by triangulation, and may be little more than a sketch with only a few critical soundings shown.

subaqueous running survey—A hydrographic survey of an exploratory nature along an unknown or hostile coast made from shipboard to determine the general form of the coast and the nature of the area.

subastral point—See **substellar point.**

subdivision survey—A type of land survey in which the legal boundaries of an area are located and the area is divided into parcels of lots, streets, right-of-way, and other accessories. All necessary corners or dividing lines are marked or monumented.

subgravity—A condition in which the resultant ambient acceleration is between zero and one g.

sublunar point—The geographical position of the Moon. That point of the Earth at which the Moon is in the zenith at a specified time. See also **subsatellite point.**

submarine relief—Variations in elevation of the ocean floor, or their representation by depth curves, tints, or soundings.

subordinate station—1. One of the places for which tide or tidal current predictions are determined by applying a correction to the predictions of a reference station. 2. A tide or tidal current station at which a short series of observations has been made, which are reduced by comparison with simultaneous observations at a reference station.

subsatellite point—The point at which a line from the satellite perpendicular to the ellipsoid intersects the surface of the Earth. See also **sublunar point.**

subsidiary station—A station established to overcome some local obstacle to the progress of a survey, and not to determine position data for the station point. The term subsidiary station is usually applied to A-stations of a traverse survey. Subsidiary stations usually are temporary in character and not permanently marked. If serving the additional purpose of supplying control for a local survey, such station may be permanently marked and it is then a supplementary station.

subsolar point—The geographical position of the Sun. That point on the Earth at which the Sun is in the zenith at a specified time. See also **substellar point.**

substellar point—The geographical position of a star. That point on the Earth at which the star is in the zenith at a specified time. Also called **subastral point.** See also **subsolar point.**

substitute center—A point which, because of its ease of identification on overlapping photographs, is used instead of the principal point as a radial center.

subsurface float—A hollow cylinder, with its axis held vertical, at a constant depth by the buoyant effect of an indicating surface float; used to determine current velocities in streams or channels having a relatively uniform depth.

subtense bar traverse—A traverse method in which course lengths are measured by use of a subtense bar.

subtense bar—A horizontally held bar of precisely determined length, used to measure distances by observing the angle it subtends at the distance to be measured.

subtense base traverse—A traverse method in which distances are determined by precisely measuring, at one end of the course, the angle subtended by a precisely measured base at the other end of the course and approximately normal to it.

subtense method—A procedure by which distance measurements are obtained by use of a subtense bar.

subtracting tape—A calibrated surveyor's tape with the first foot (or meter) at each end graduated in tenths or hundredths Also called **cut tape.** See also **adding tape.**

Summary of Corrections—A summary of outstanding notices affecting all nautical charts and publications. This information is also accessible on the Navigation Information Network (NAVINFONET).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

summer solstice—1. That point on the ecliptic occupied by the Sun at maximum northerly declination. Also called **first point of Cancer.** 2. That instant at which the Sun reaches the point of maximum northerly declination, about June 21.

sun synchronous—An Earth satellite orbit in which the orbital plane is near polar and the altitude such that the satellite passes over all places on Earth having the same latitude twice daily at the same local time.

Sun-zenith distance—The angle between the zenith and the Sun's disk.

Sun—The luminous celestial body at the center of the solar system, around which the planets, planetoids, and comets revolve. It is an average star. See also **apparent sun; dynamical mean sun; fictitious sun.**

superfeature—Several feature components having the same, or similar, attribution that are joined together to create one feature. An example of a superfeature could be a railroad system consisting of all the railroad tracks, spurs, sidings, terminals, bridges, and culverts within that system.

superior conjunction—The conjunction of a planet and the Sun when the Sun is between the Earth and the other planet.

superior planets—The planets with orbits larger than that of the Earth; Mars, Jupiter, Saturn, Uranus, Neptune, and Pluto.

superior transit—See **upper transit.**

superwide-angle lens—A lens having an angle of coverage greater than 100°. A lens whose focal length is approximately less than one-half the diagonal of the format. Also called **ultrawide-angle lens.**

supplemental control point—A photoimage point for which an elevation or a horizontal position, or both, is to be, or has been determined. See also **control point.**

supplemental control—Points established by subordinate surveys, to relate aerial photographs used in mapping with the system of geodetic control. The points must be positively photoidentified, that is, the points on the ground must be positively correlated with their images on the photographs.

supplemental elevation—A point whose vertical position has been determined by photogrammetric methods and is intended for use in the orientation of other photographs. Also called **vertical pass point.**

supplemental photography— Noncartographic aerial and terrestrial photography that is used to enhance specific characteristics of mapping and charting photographic products. Primarily, supplemental photography is obtained with a reconnaissance camera using a relatively long focal length to provide greater image detail than is available in photographs obtained with mapping cameras.

supplemental plat—A plat prepared entirely from office records designed to show a revised subdivision of one or more sections without change in the section boundaries and without other modification of the subsisting record.

supplemental position—A point whose horizontal position has been determined by photogrammetric methods and is intended for use in the orientation of other photographs. Also called **horizontal pass point.**

supplemental posts for survey monuments—See **identification posts.**

supplemental station—Those stations established only for supplemental vertical control. They normally are not permanently marked (some are merely photoidentified) and accuracy does not have to be of the same order as the horizontal control to which it is tied. Also called **vertical-angle station.**

supplementary bench mark—See **temporary bench mark.**

supplementary contour—A contour line between intermediate contour lines to increase the topographic expression of an area, usually in areas of extremely low relief. Also called **auxiliary contour.**

supplementary instructions—New information, amendments, or changes to specifications or compilation instructions affecting the production of a specific map or chart, or a series of maps or charts.

supplementary station—An auxiliary survey

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

station, established to increase the number of control stations in a given area, or to place a station in a desired location where it is impracticable or unnecessary to establish a principal station. Supplementary stations are permanently marked, and are established with an accuracy and precision somewhat lower than is required for a principal station, since they do not serve as bases from which extensive surveys are run. Also called **secondary station.**

surface anomalies—Irregularities at the Earth's surface, in the weathering zone, or in near surface beds which interfere with geophysical measurements.

surface chart—See **weather map.**

surface corrections—Corrections of geophysical measurements for surface anomalies and ground elevations.

surface float—A device, specially designed or improved, used in hydrographic surveys to determine surface movement of a stream.

surface—A level of spatial measurement [illegible] referring to a three-dimensional defined space, e.g. contours, isolines, bathymetry, etc.

surprint—See **overprint**, definition 1.

survey coordinates—See **rectangular space coordinates.**

survey net—**1.** (horizontal control) Arcs of triangulation, sometimes with lines of traverse, connected to form a system of loops or circuits extending over an area. Also called **horizontal control survey net; traverse net; triangulation net.** See also **triangulation system; trilateration net. 2.** (vertical control) Lines of spirit leveling connected to form a system of loops or circuits extending over an area. Also called **control net; framework of control; level net; net.** See also **area triangulation; U.S. control survey nets.**

survey photography—See **mapping photography.**

survey signal—A natural or artificial object or structure whose horizontal and sometimes vertical position is obtained by surveying methods. Signals are given special designations according to the kind of survey in which they are determined, or which they may later serve.

survey tower—A structure designed for rapid construction and removal to raise the survey instrument and observer above obstructions such as trees and buildings to permit a line-of-sight as required in higher order triangulation, trilateration, or traverse. See also **Bilby steel tower.**

surveying accessories—Those surveying devices which assist in making measurements with a surveying instrument.

surveying altimeter—An aneroid barometer with a dial graduated to read feet or meters of altitude, used to determine approximate differences in elevation between points.

surveying camera—See **mapping camera.**

surveying instruments—Those surveying devices with which measurements are made. See also **electronic distance-measuring equipment; leveling instrument; tachymeter; theodolite; transit.**

surveying sextant—A sextant intended primarily for use in hydrographic surveying. Also called **hydrographic sextant.** See also **marine sextant.**

surveyor's arrow—See **pin.**

surveyor's chain—See **Gunter's chain.**

survey—**1.** The act or operation of making measurements for determining the relative positions of points on, above, or beneath the Earth's surface. **2.** The results of such operations. **3.** An organization for making surveys. See also **aerial survey; Airborne Control system; airborne electronic survey control; area survey; astronomic surveying; boundary survey; cadastral survey; city survey; compass survey; control survey; control survey classification; dependent resurvey; electronic survey; engineering survey; exploratory survey; field inspection; first-order work; geodetic survey; geoelectric survey; geographic survey; geologic survey; gravimetric survey; ground survey; hydrographic survey; independent resurvey; inventory survey; land survey; location survey; magnetic survey; magnetometer survey; metes-and-bounds survey; mine survey;**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

mineral survey; oceanographic survey; photogrammetric survey; photographic survey; plane survey; preliminary survey; reconnaissance survey; rectangular surveys; resurvey; route survey; satellite surveying; second-order work; standard survey; subaqueous reconnaissance survey; subaqueous running survey; subdivision survey; third-order work; topo-graphic survey; town-site survey; transit-and-stadia survey; trilinear surveying.

SWATH survey—See **multibeam survey.** See also **Small Wetted Area Twin Hull.**

sweep bar—A heavy section of steel rail suspended at a predetermined depth by two vertical cables and towed by a vessel for precise determination of navigation obstructions during a hydrographic survey.

swing offset—The perpendicular distance from a point to a transit line found by holding the zero point of a tape at the given point and swinging the tape in an arc until the minimum (horizontal) distance is obtained.

swing-swing method—A technique for clearing *y*-parallax during relative orientation by applying identical swing (or *y*-motion) to both projectors of a pair at the same time. This method has the advantage of affecting *y*-parallax correction without the use of translational motions.

swing—**1.** The rotation of a photograph in its own plane about its camera axis. **2.** On *trimetrogron obliques*, the angle between the principal line and the *y*-axis, or the angle between the isometric parallel and the *x*-axis. See also **relative swing. 3.** The angle at the principal point of a photo measured clockwise from the positive *y*-axis to the principal line at the nadir point. **4.** (*triangulation*) See **eccentric reduction.**

swivel graver—A scribing instrument with a swivel mechanism that permits changes in direction of scribing.

symbolization—The method of portraying topographic features onto a manuscript. The symbols used on the manuscript are either a point (dot), a line, or an area (a delimiting line closing upon itself).

symbolized graphic data (SGD)—A data set which contains attribution for graphic output. Data may include symbolization options such as point, line, area, pattern, orientation, color (hue, value, tone, and intensity), size, shape, and texture.

symbol—A diagram, design, letter, character, or abbreviation placed on maps, charts, and other graphics which by convention, usage, or reference to a legend is understood to stand for or represent a specific characteristic or feature.

synchronous satellite—An Earth satellite moving eastward in an equatorial, circular orbit at an altitude (approximately 35,900 kilometers) such that its period of revolution is exactly equal to (synchronous with) the rotational period of the Earth. Such a satellite will remain fixed over a point on the Earth's Equator. Also called **fixed satellite; 24-hour satellite.** See also **stationary orbit.**

synodic period—The interval of time between any planetary configuration of a celestial body, with respect to the Sun, and the next successive same configuration of that body, as from inferior conjunction to inferior conjunction.

synodical month—The average period of revolution of the Moon about the Earth with respect to the Sun, approximately 29 1/2 days. Also called **lunar month; lunation.**

synoptic chart—See **weather map.**

synthetic aperture radar (SAR)—See **Radar, Synthetic Aperture (SAR).**

System 9—A geographic information system with a feature oriented data base and continuous mapping. Developed by Computervision.

system model—A representation of the information types, directions of flow, production stages, and user interfaces for a system.

system of astronomic constants—An interrelated group of values constituting a model of the Earth and the motions which together with the theory of celestial mechanics serves for the calculation of ephemerides.

systematic error—An error that occurs with the same sign, and often with a similar magnitude, in a number of consecutive or otherwise related observations. For example, when a base is measured with a wrongly

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

calibrated tape, there will be systematic errors. In addition, random errors will occur. Repetition does little or nothing to reduce the ill effect of systematic errors, which are a most undesirable feature of any set of observations. Much of the care in making observations is directed toward eliminating or correcting systematic errors. Also called **regular error.** See also **accumulative error.**

syzygy—A point of the orbit of a planet or satellite at which it is in conjunction or opposition. The term is used chiefly in connection with the Moon, when it refers to the points occupied by the Moon at new and full phase. See also **equinoctial colure; solstitial colure.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T

Table of meridional parts—A table listing lengths of the meridian from the Equator to the various parallels of latitude increased in the proportion required to show lengths along the parallels equal to the corresponding length along the Equator.

Tabular data—Data in row and column format. See also **relational data base.**

Tachymeter (tacheometer or tachometer)—A surveying instrument designed for use in the rapid determination of distance, direction, and difference of elevation from a single observation. There are several forms of these instruments that may be classed as tachymeters: (1) An instrument in which the base line for distance determination is an integral part of the instrument. The term **tachymeter** is usually applied to this group. (2) An instrument equipped with stadia wires or gradienter, the base for distance determination being a graduated rod held at the distant point. See also **autoreducing tachymeter.**

Tachymetry (tachometry)—A surveying method used to quickly determine distance, direction, and relative elevation of a point with respect to the instrument station by a single observation. An example of tachymetry in the United States (where the term is less familiar) is the stadia method.

Tactical Pilotage Chart (TPC)—A 1:500,000 scale, coordinated series of multicolored charts which are produced in selected areas of interest. Designed to satisfy visual and radar navigation of high speed tactical aircraft operating at low altitude. Also used for detailed preflight planning and mission analysis.

Tactical Terrain Analysis Data Base (TTADB)—A set of transparent overlays keyed to 1:50,000 Tactical Line Maps (TLMs) portraying natural and cultural features of military significance. The data base consists of overlays and supporting data for: surface configuration (slope), surface materials (soils), vegetation, surface drainage, transportation, and obstacles. TTADBs provide terrain and man-made information to support the generation of a variety of field user produced synthesized products to include: Lines of Communication, Cross-Country Movement, and Cover & Concealment.

Tactical Terrain Data (TTD)—The basic digital operational terrain data set supporting future land combat. TTD will be a value added database allowing for the addition and update of data by the user. Features and attributes are coded using categories contained in the Feature Attribute Coding System. TTD will contain contents of TTADB thematic overlays, as well as selected features from TLMs and Combat Charts. See also **Tactical Terrain Analysis Data Base.**

tactical map—(JCS) A large-scale map used for tactical and administrative purposes. See also **map.**

tactical planning model—Medium- or large-scale models providing considerable detailed terrain information; generally used for planning operations of a tactical nature.

tan alt—See **shadow factor.**

tangent conical map projection—See **conic map projection.**

tangent distance—The distance from the point of intersection (vertex) of a curve to its point of tangency or point of curvature.

tangent plane grid system—(engineer surveying) A grid system in a tangent plane with origin at the point of tangency. Usually the origin is designated 10,000 N and 10,000 E, or some similar amounts, to keep all coordinates positive. This system never extends for any great distance. See also **plane rectangular coordinates.**

tangent plane—A plane that touches a curved surface of double curvature at one and only one point or that touches a curved surface of single curvature along one or more parallel straight lines which are elements of the surface, without intersecting the surface. In geodetic work, a plane tangent to the spheroid at any point is perpendicular to the normal at that point.

tangent to spiral (TS)—The point at the end of a tangent and the beginning of a spiral.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

tangential distortion—Linear displacement of image points in a direction normal to radial lines from the center of the field.

tangent—(surveying) 1. That part of a traverse or alignment included between the point of tangency of one curve and the point of curvature of the next curve. 2. A great circle line tangent to a parallel of latitude at a township corner. 3. Sometimes applied to a long straight line of a traverse, especially on a route survey, whether or not the termini of the line are points of curve.

tape corrections—Quantities applied to a taped distance to eliminate or reduce errors due to the physical condition of the tape and to the way in which it is used. See also **alignment correction; grade correction; length correction; sag correction; temperature correction,** definition 3; **tension correction.**

tape gage—A device consisting of a tagged or indexed chain, tape, or other line used for measurement or testing. Gage height or measurement is read on a graduated staff or index. Also called **chain gage.**

tape rod—A rod consisting of a frame with rollers at both ends over which an endless, graduated metal tape moves. It is designed to permit direct readings by the instrument man, eliminating all addition and subtraction functions required by other types of rod readings. Also called **automatic rod.**

tape stretcher—A mechanical device which facilitates holding a tape at a prescribed tension and in a prescribed position. Also called **stretching apparatus.**

tape thermometer—A precision thermometer fitted in a specially designed case to clip on and against a metal tape in order to determine temperature corrections for precision base or traverse tape measurements.

tape—(surveying) A ribbon of steel, Invar, specially made cloth, or other suitable material on which graduations are placed for the measurement of lengths or distances. See also **adding tape; base tape; instantaneous reading tape; Invar tape; Lovar tape; plano-wire tape; subtracting tape.**

taping arrow—See **pin.**

taping buck—See **taping stool.**

taping pin—See **pin.**

taping station—The stake marking each interval (one tape length) along a traverse from the initial point along road centerlines and similar survey operations. See also **plus station.**

taping stool—A metal stool used for precise taping operations. Stools are portable and provide a stable elevated table on which the positions of the survey tape ends can be accurately marked. Also called **taping buck.**

taping—The operation of measuring distances on the ground with a tape or chain. Formerly the words **chaining** and **taping** were used synonymously, but the word **taping** is now preferred for all surveys except those of the public land system. For the latter, because of historical and legal reasons, the term **chaining** is preferred.

tare—An abrupt offset in the gravimeter normal reading level.

Target Materials Program (TMP)—1. A DoD program established for the production of all documents specifically supporting installation target analysis and operational planning employment. The DoD overall TMP manager is DIA. 2. The production programs responding to the JCS charter for DMA to produce unique, critical geo-locational information supporting the long-range worldwide requirements of the unified and specified commands, military departments, and allied participants. [The program consists of precision point location products such as the Installation Reference Point Graphic (IRPG), Aim Point Graphic (APG), and grid photos; deployable Point Positioning Data Base (PPDB) in either film, video, or digital form; cartographic Air Target Materials like the Air Target Chart and Jog-Radar; and a variety of other special materials and services.]

target acquisition—1. (JCS) The detection, identification, and location of a target in sufficient detail to permit the effective employment of weapons. See also **target analysis.** 2. The process of optically, manually, mechanically, or electronically orienting a tracking system in direction and range to lock on a target.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

target analysis—(JCS) An examination of potential targets to determine military importance, priority of attack, and weapons required to obtain a desired level of damage or casualties. See also **target acquisition.**

target area survey base—(JCS) A base line used for the locating of targets or other points by the intersection of observations from two stations located at opposite ends of the line.

target complex—(JCS) A geographically integrated series of target concentrations. See also **target.**

target concentration—(JCS) A grouping of geographically proximate targets. See also **target; target complex.**

target dossiers—(JCS) Files of assembled target intelligence about a specific geographic area.

target folders—(JCS) The folders containing target intelligence and related materials prepared for planning and executing action against a specific target.

target intelligence—(JCS) Intelligence which portrays and locates the components of a target or target complex and indicates its vulnerability and relative importance.

target leveling rod—A type of leveling rod, carrying a target, which is moved into position according to signals given by the instrument man; when the target is bisected by the line of collimation of the instrument, it is read and recorded by the rodman. See also **double-target leveling rod; single-target leveling rod.**

target material graphics—See **target materials.**

target materials—(JCS) Graphic, textual, tabular, or other presentations of target intelligence, primarily designed to support operations against designated targets by one or more weapon systems. Target materials are suitable for training, planning, executing, and evaluating such operations. Also called **target material graphics.** See also **Air Target Chart.**

target positioning data—The accurate horizontal and vertical values which define the location of a target or point. See also **precise installation position; precise radar significant location.**

target system component—(JCS) A set of targets belonging to one or more groups of industries and basic utilities required to produce component parts of an end product such as periscopes, or one type of a series of interrelated commodities, such as aviation gasoline.

target system—(JCS) All the targets situated in a particular geographic area and functionally related. See also **target complex.**

target—**1.** (JCS) A geographical area, complex, or installation planned for capture or destruction by military forces. **2.** An object which reflects a sufficient amount of a radiated signal to produce an echo signal on detection equipment. **3.** The distinctive marking or instrumentation of a ground point to aid in its identification on a photograph. In photogrammetry, target designates a material marking so arranged and placed on the ground as to form a distinctive pattern over a geodetic or other control point marker, on a property corner or line, or at the position of an identifying point above an underground facility or feature. A target is also the image pattern on aerial photographs of the actual mark placed on the ground prior to photography. See also **area target; pinpoint target.**

taut-wire apparatus—A 100-meter stranded sounding wire, graduated at 25-meter intervals, used to measure the distances between offshore control buoys during a hydrographic survey.

telemeter—(surveying) An instrument for determining the distance from one point to another. Some such instruments employ a telescope and measure the angle subtended by a short base of known length. See also **electronic telemeter; telemetry.**

telemetry—The science of measuring a quantity or quantities, transmitting the measured value to a distant station, and there interpreting, indicating, or recording the quantities measured.

telescope level—A spirit level attached to a telescope, with its axis parallel to the telescope axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

telescope—An optical instrument used as an aid in viewing or photographing distant objects, particularly celestial objects. See also **achromatic telescope; erecting telescope; inverting telescope; meridian telescope; zenith telescope.**

telescopic alidade—A usual designation for an instrument composed of a telescope mounted on a straightedge ruler, and used with a planetable in topographic surveying.

telescoping—See **transit**, definition 3.

telluroid—A surface near the terrain being the locus of points in which the spheropotential is the same as the geopotential of corresponding points on the terrain. Its distance from the spheroid is the normal height.

Tellurometer—A trade name for a microwave distance-measuring system in which the velocity of a radio wave is used to determine the distance between two instruments operating alternately as master station (interrogator) and remote station (responder).

temperature correction—1. (leveling) That correction which is applied to an observed difference of elevation to correct for the error introduced when the temperature at which the leveling rods are used in the field is different from the temperature at which they were standardized. 2. (pendulum) The quantity that is applied to the period of vibration of a pendulum to allow for the difference in the length of the pendulum at the temperature of observation and its length at some other temperature which has been adopted for purposes of standardization or for combining or comparing corresponding values. 3. (taping) The quantity applied to the nominal length of a tape to allow for a change in its effective length due to its being used at a temperature other than that for which its standard length is given.

templet (template)—1. A pattern or guide, usually constructed of paper, plastic, or metal, used to shape, delimit, or locate an area. 2. A device used in radial triangulation to represent the aerial photograph; the templet provides a record of the directions of radials taken from the photograph. See also **calibration templet; double-model stereotemplet; hand templet; Hayford deflection templets; Hayford gravity templets; mechanical templet; slotted templet; spider templet; stereotemplet.**

templet cutter—A mechanical device for punching center holes and slots in templets. The slots are centered on points transferred from aerial photographs and are radial to the center hole. Also called **secator; slot cutter; radial secator.**

templet laydown—The process of assembling individual slotted templets into a radial control net.

templet method—Any of the various methods utilized in graphical radial triangulation.

templet ratiograph—(photogrammetry) A device for determining the ratio in decimals between two distances. One distance is that between the principal point and another designated point on the aerial photograph. The other is the corresponding distance between the principal point on a templet and the marked center of the stud for the designated point upon completion of the templet laydown. The ratiograph is designed for a specific templet cutter. See also **ratiometer.**

temporary bench mark (TBM)—A bench mark at a junction of sections of a line of levels, at which no permanent bench mark is established. Also called **nonmonumented bench mark; supplementary bench mark.**

tension correction—(taping) The correction applied to the nominal length of a tape to allow for a change in effective length due to its being used at a tension other than that for which its standard length is known.

terabyte (TB)—A unit of memory representing 2^{14} (1,099,511,627,776) bytes. Commonly, one terabyte represents one billion bytes.

Terrain Analysis Products (TAP)—(NATO) A standardized or non-standardized product from the Terrain Analysis System normally presented in graphic form for use in military decision making.

Terrain Analysis System (TAS)— (NATO) A system to meet military requirements for quick and comprehensive information on the terrain in analog and/or digital form. It is a system in which selected data about the terrain are collected and stored in a structured way to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

permit their flexible exploitation in the provision of Terrain Analysis Products.

Terrain Edit Station/Elevation Matrix Processing System (TES/EMPS)—An Intergraph workstation used by DMA to edit terrain products. Gives both plan and perspective views.

Terrain Profile Recorder (TPR)—An electronic instrument that emits a pulsed-type radar signal from an aircraft to the Earth's surface, measuring vertical distances in order to obtain a profile beneath the track of the aircraft. Also called **Airborne Profile Recorder. See also laser terrain profile recorder.**

terrain analysis (TA)— (NATO) The process of collecting, analyzing and evaluating geographic information on the natural and man-made features of the terrain and its interpretation in combination with other factors to provide predictive information and advice about the effect of the terrain on military operations.

terrain contour matching (TERCOM)—The process which updates navigational guidance systems through correlation with gridded terrain elevation data derived from photogrammetric or large scale cartographic sources. See also **terrain correlation.**

terrain correction—A positive correction used in conjunction with other corrections in making gravity reductions. It takes into account actual deviations from level terrain in the area surrounding a station by removing masses above the horizon and filling in mass deficiencies below. Also called **topographic correction.**

terrain correlation—A process used by a vehicle's guidance system in evaluating the elevations of the terrain it is flying over and comparing it with prestored digital terrain elevation data. See also **terrain contour matching.**

terrain emboss—A model-making technique for portraying relief on a chart. A photographic process is used to produce the shaded relief effect from an embossed model.

terrain following—The flight mode by which a vehicle maintains a specified altitude above the Earth's surface.

terrain intelligence—(JCS) Processed information on the military significance of natural and man-made characteristics of an area.

terrain model—A three-dimensional graphic representation of an area, showing the conformation of the ground, modeled to scale and usually handpainted to depict realistically man-made and natural physical features. The vertical scale is usually exaggerated, without severe distortion, to accentuate the aspect of relief.

terrain profile photography— Cartographic photography obtained simultaneously with positional camera photography and recording of data relating to profile elevation information of the terrain along or near the ground track of the aircraft. The terrain profile recorder is normally used as the measuring device.

terrain profiling—Obtaining an elevation profile of the Earth's terrain along or near the ground track of the aircraft by use of a Terrain Profile Recorder.

terrain study—(JCS) An analysis and interpretation of natural and manmade, their effects on military operations, and the effect of weather and climate on these features.

terrain—An area of ground considered as to its extent and topography.

terrestrial camera—A camera designed for use on the ground. Also called **ground camera.**

terrestrial coordinates—See **geographic coordinates.**

terrestrial equator—See **astronomic equator; geodetic equator.**

terrestrial globe—A sphere, on the outer surface of which, by means of symbols and reference lines, the features of the surface of the Earth are shown in relative positions.

terrestrial latitude—Latitude on the Earth; angular distance from the Equator.

terrestrial longitude—Longitude on the Earth; the arc of a parallel, or the angle at the pole, between the prime meridian and the meridian of a point on the Earth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

terrestrial magnetism—See **geomagnetism.**

terrestrial meridian—See **astronomic meridian.**

terrestrial perturbations—The largest gravitational perturbations of artificial satellites which are caused by the fact that the gravity field of the Earth is not spherically symmetrical.

terrestrial photogrammetry— Photogrammetry utilizing terrestrial photographs. Also called **ground photogrammetry.**

terrestrial photograph—A photograph taken by a camera located on the ground. Also called **ground photograph.**

terrestrial planet—A planet that approximates the Earth in size (Mercury, Venus, Mars, and Pluto) and physical makeup.

terrestrial pole—See **geographical pole.**

terrestrial refraction—The refraction by the Earth's atmosphere of light from a terrestrial source. The path of light from a terrestrial source is usually not far from horizontal; it passes through only the lower strata of the atmosphere and suffers refraction throughout its entire length. See also **atmospheric refraction; horizontal refraction; lateral refraction.**

terrestrial sphere—The Earth.

terrestrial triangle—A triangle on the surface of the Earth, especially the navigational triangle.

tesla—(geomagnetism) The electromagnetic unit of magnetic induction. 1 tesla = 10^{-9} nanotesla. See also **gauss.**

Tesselated Spheroid Model-IV (TS)—A coordinate system which divides the world into 5 latitude zones to obtain equal spacing.

tesseral harmonics—The set of all spherical harmonics that are functions of both latitude and longitude. Sectorial harmonics are a special subset of tesseral harmonics.

test chart—See **resolving power target.**

test range support—DMA provided geodetic surveys for the DoD test and training ranges to support weapons systems and weapons systems users research, development, testing and evaluation.

texture—In a photo image, the frequency of change and arrangement of tone.

thematic mapper (TM)—A seven-channel, predominantly 30 meter instantaneous field of view (IFOV) multispectral scanner, designed for monitoring earth resources. TM is used on board LANDSAT satellites.

thematic map—See **topical map.**

thematically separate—The segregation of a feature or a limited number of features within a data base to highlight the spatial characteristics of a subset or the environment. See also **layer.**

theodolite magnetometer—An instrument used in magnetic surveys consisting of a theodolite and a magnetometer modified to fit into a common base, which permits the determination of the true meridian and the magnetic meridian in a single observation.

theodolite—A precision surveying instrument consisting of an alidade with a telescope. It is mounted on an accurately graduated circle and is equipped with necessary levels and reading devices. Sometimes, the alidade carries a graduated vertical circle. See also **cine theodolite; direction instrument theodolite; gyrotheodolite; phototheodolite; repeating theodolite.**

theoretical corner—A term adopted by the U.S. Geological Survey to designate the corners on the map for which no marks are identified on the ground. The locations are determined by adjustment and are indicated on the map only by the intersection of the subdivision lines.

theoretical error—A systematic error arising from natural physical conditions, beyond the control of the observer. See also **external error.**

theoretical gravity—The value of gravity calculated for a particular latitude according to an accepted formula. See also **formula for theoretical gravity.**

theory of anharmonic ratio—A theory principally concerned with the processes of transformation and rectification whereby

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

projectively related figures possess certain metric characteristics which are invariant under projection. Also called **theory of cross ratio.**

theory of cross ratio—See **theory of anharmonic ratio.**

thermal imagery—(JCS) Imagery produced by sensing and recording the thermal energy emitted or reflected from the objects which are imaged.

thermometric leveling—The determination of elevations above sea level from observed values of the boiling point of water. A type of indirect leveling.

thick lens—A term used in geometrical optics to indicate that the thickness of a lens is considered and that all distances are being measured from the nodal points instead of the lens center.

thin lens—A term used in geometrical optics to indicate that the thickness of a lens is ignored and that all distances are measured from the lens center; used for approximate computations.

thinning—The process whereby a linear feature is generalized through the use of a series of rules that reduces the number of data points while maintaining the basic shape of the feature.

third-order leveling—Spirit leveling which does not attain the quality of second-order leveling, but does conform to the current specifications for third-order leveling per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for most general vertical control purposes within a limited area.

third-order traverse—A survey traverse which extends between adjusted positions of other control surveys which conform to the current specifications for third-order (class I or class II) triangulation per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for most general horizontal control purposes within a limited area.

third-order triangulation—Formerly known as tertiary triangulation, these surveys conform to current specifications for third-order (class I or class II) triangulation per "Classification Standards of Accuracy and General Specifications of Geodetic Control Surveys." Recommended for most general horizontal control purposes within a limited area.

third-order work—This is the lowest order of control surveys for which monumentation is authorized.

three sigma(3σ)—The 99.73 percent confidence interval of a distribution. See also **standard error.**

three-arm protractor—A full-circle protractor, equipped with three arms, the fiducial edges (extended) of which pass through the center of the circle. The middle arm is fixed and reads 0° on the graduated circle. The other arms are movable, and their positions on the circle are read with the aid of verniers. The two movable arms are equipped with clamps and may be set at any angle with respect to the fixed arm, within the limits of the instrument. It is used for finding a (ship's) position graphically when the angles between three known fixed points are available. Also called **station pointer.**

three-body problem—That problem in classical celestial mechanics which treats the motion of a small body, usually of negligible mass, relative to and under the gravitational influence of two other finite point masses.

three-dimensional (3-D) data—Volumetric data representing measurements in three dimensions, as angular or linear measures such as phi-lambda-kappa or latitude-longitude-elevation.

three-point method—See **resection.**

three-point problem—The determination of the horizontal position of a point of observation from data comprising two observed horizontal angles between three objects of known position. The problem is solved graphically by the use of a three-arm protractor, and analytically by trigonometrical computation. See also **resection; triangle-of-error method.**

three-wire leveling—A method of leveling applied when the reticle of the level has three lines. The rod is read at each of the three lines and the average is used for the final result with an accuracy as great as if three lines of levels had been run and the results averaged.

ticks—See **register marks.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

tidal bench mark—A bench mark set to reference a tide staff at a tidal station and the elevation of which is determined with relation to the local tidal datum.

tidal constituent—See **constituent.**

tidal correction—A correction applied to gravitational observations to remove the effect of Earth tides on gravimetric observations.

tidal current chart—A chart showing, by arrows and numbers, the average direction and speed of tidal currents at a particular part of the current cycle. A number of such charts, one for each hour of the current cycle, usually are published together.

tidal current—The alternating horizontal movement of water associated with the rise and fall of the tide caused by the astronomic tide-producing forces.

tidal datum plane—See **tidal datum.**

tidal datum—Specific tide levels which are used as surfaces of reference for depth measurements in the sea and as a base for the determination of elevation on land. Many different datums have been used, particularly for leveling operations. Also called **tidal datum plane.**

tidal day—See **lunar day.**

tidal variation of gravity—Periodic deviations from normal of the gravity on Earth and the direction of the plumb line caused by the attraction of the Moon and the Sun's mass.

tide gage—A device for measuring the height of tide. It may be simply a graduated staff in a sheltered location where visual observations can be made at any desired time; or it may consist of an elaborate recording instrument making a continuous graphic record of tide height against time. Such an instrument is usually actuated by a float in a pipe communicating with the sea through a small hole which filters out shorter waves. See also **float gage; nonrecording gage; portable automatic tide gage; pressure gage; self-registering gage; staff gage; standard automatic tide gage.**

tide level—See **mean tide level.**

tide over run—A reprint of a chart or map necessitated by unusual conditions before extensive revisions can be accomplished. Also called **emergency run.**

tide station—A place at which tide observations are made. See also **primary tide station; secondary tide station.**

tide-producing force(s)—The slight local difference between the gravitational attraction of two astronomic bodies and the centrifugal force that holds them apart. These forces are exactly equal and opposite at the center of gravity of either of the bodies, but, since gravitational attraction is inversely proportional to the square of the distance, it varies from point to point on the surface of the bodies. Therefore, gravitational attraction predominates at the surface point nearest to the other body, while centrifugal repulsion predominates at the surface point farthest from the other body. Hence, there are two regions where tide-producing forces are at a maximum, and normally there are two tides each lunar day and solar day.

tidemark—1. A high water mark left by tidal water. 2. The highest point reached by a high tide. 3. A mark placed to indicate the highest point reached by a high tide, or occasionally, any specified state of tide.

tide—The periodic rise and fall of the surface of the ocean resulting from the gravitational attraction of the Moon and Sun acting upon the rotating Earth. See also **age of diurnal inequality; age of parallax inequality; age of phase inequality; amphidromic point; amphidromic region; annual inequality; anomalistic tide cycle; apogean tides; constituent; constituent day; corrected establishment; cotidal hour; degenerate amphidromic system; diurnal constituent; diurnal inequality; ebb tide; establishment of the port; flood tide; harmonic constants; height of the tide; high water; high water line; higher high water; higher high water interval; higher low water; higher low water interval; Indian spring low water; Indian tide plane; international low water; low water; low water full and change; low water line; lower high water; lower high water interval; lower low water; lower low water interval; lowest low water; lowest low water**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

springs; lunar day; lunar tide; lunitidal interval; mean diurnal high water inequality; mean diurnal low water inequality; mean high water; mean high water springs; mean higher high water; mean higher high water springs; mean low water; mean low water springs; mean lower low water; mean lower low water springs; mean range; mean river level; mean sea level; mean tide level; nodal line; parallax inequality; phase inequality; set; semidiurnal constituent; tidal correction; tidal current; tidemark.

tie flight—See **control strip.**

tie point—1. Image points identified on oblique photographs in the overlap area between two or more adjacent strips of photography. They serve to tie the individual sets of photographs into a single flight unit and to tie adjacent flights into a common network. 2. Point of closure of a survey either on itself or on another survey.

tie strip—1. (cartography) An overlay containing all planimetric and relief features in the area along the edge of a map or chart. It is used to insure the matching of these features on adjoining sheets. Also called match strip. 2. (aerial photography) See **control strip.**

tie-in—See **tie.**

tier—Any series of contiguous townships situated east and west of each other; also sections similarly situated within a township.

tie—A survey connection from a point of known position to a point whose position is desired. A tie is made to determine the position of a supplementary point whose position is desired for mapping or reference purposes, or to close a survey on a previously determined point. To "tie-in" is to make such a connection. See also **tie point,** definition 2.

tiled coverage—A coverage that has been physically partitioned into smaller coverages. One or more of these smaller coverages, where all share the same set of feature classes with the same definitions. Each tile defines an independent topology.

tiling scheme—The scheme used to define tile shape and size, and unique tile identification number.

tilt angle—(JCS) The angle between the optical axis of an air camera and the vertical at the time of exposure. See also **angle of depression; tilt.**

tilt circle—In a tilted aerial photograph, a circle passing through the isocenter and having a diameter lying along the principal line. When this diameter is drawn to a convenient linear scale, then any chord through the isocenter gives the component of tilt for that particular direction.

tilt displacement—Displacement radial from the isocenter of the photograph caused by the tilt of the photograph.

tilt slide rule—A device which facilitates the determination for settings on a fixed-lens rectifier when certain tilt factors of an aerial photograph are known.

tilting level—A leveling instrument in which the telescope with its attached bubble tube can be leveled by a fine screw at the eyepiece end of the telescope independently of the vertical axis, thus avoiding the need for careful leveling of the instrument as a whole. This type of level was first designed for precise work, but the principle has come into popular use for ordinary levels.

tilting-lens rectifier—A class of rectifiers in which the principal point is fixed on its axis of swing, and cannot be displaced.

tilt—(JCS) See **roll.** Also called **angle of tilt.** See also **cross tilt; direction of tilt; pitch; relative tilt; *x*-tilt.**

time diagram—A diagram in which the celestial equator appears as a circle, and celestial meridians and hour circles as radial lines; used to facilitate solution of time problems and others involving arcs of the celestial equator or angles at the pole, by indicating relations between various quantities involved. Conventionally, the relationships are given as viewed from a point over the South Pole, westward direction being counterclockwise. Also called **diagram on the plane of the equinoctial.** See also **diagram on the plane of the celestial meridian.**

time distance—Time required for any object to travel between two given points at a given rate of speed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

time gamma curve—See **characteristic curve.**

time meridian—Any meridian used as a reference for reckoning time, particularly a zone or standard meridian.

time zone chart—A small-scale chart of the world designed to show the legal time kept on land.

time zone—An area in all parts of which the same time is kept. In general, each zone is 15° of longitude in width, centered on a meridian whose longitude is generally divisible by 15°.

times (X) enlargement—The multiplication factor by which an original is to be enlarged in reproduction. A two-times (2X) linear enlargement of a 4- by 5-inch original would be 8 by 10 inches. See also **diameter enlargement; scale of reproduction.**

time—The measurable aspect of duration. See also **A1 time; apparent sidereal time; apparent solar time; astrograph mean time; astronomic time; atomic time; civil time; day; ephemeris time; equation of time; Greenwich apparent time; Greenwich lunar time; Greenwich mean time; Greenwich sidereal time; Greenwich time; local apparent time; local astronomic time; local lunar time; local mean time; local sidereal time; local time; lunar time; mean sidereal time; mean solar time; month; rise time; sidereal time; solar time; standard time; Universal Time; UT0 time; UT1 time; UT2 time; WWV time; year; zone time.**

timing correction—A correction applied to the length of a trilateration measurement to compensate for the delay of the radar signal as it passes through the ground transponder unit of an electronic distance-measuring device.

tints—Color gradations used on maps to designate depth or height. See also **hypsometric tinting.**

tipped panoramic distortion—In a panoramic camera system, the displacement of images of ground points from their expected vertical panoramic positions caused by the tipping of the scan axis within the vertical plane of the flight path. This distortion is additive and modifies again the image positions of points already influenced by panoramic distortion, scan positional distortion, and image motion compensation distortion.

tip—**See pitch.**

title block—A space on a nonstandard graphic, such as a mosaic, photograph, or plan devoted to identification, reference, and scale information.

titling (title information)—That information lettered on aerial photographic negatives for identification purposes. Also, the placing of such information on the negatives. Also called **film titling; negative titling.**

Tokyo datum—This datum, with its origin in Tokyo, is defined in terms of the Bessel ellipsoid. It also covers Korea and Okinawa. Note: In the absence of information to confirm that this datum is identical to the Old Tokyo Observatory Datum of 1918, the two datums are to be considered separately.

tolerance—The maximum allowable variation from a standard or from specified conditions.

tone copy—That material in which tones or shades of solid color appear.

tone—(JCS) Each distinguishable shade variation from black to white on imagery.

topical map—A map designed to portray a special subject; e.g., administrative subdivisions, railroads, telecommunications, power lines, navigable waterways. Also called **special-subject map; thematic map.**

topoangulator—An instrument used to measure vertical angles in the principal plane of an oblique photograph.

topocentric coordinates—Coordinates whose origin is on the Earth's surface as distinguished from geocentric coordinates whose origin is at the center of the Earth.

topocentric equatorial coordinates—A coordinate system centered at the observer's position on the surface of the Earth with one coordinate plane parallel to the Equator and one axis parallel to the north polar axis of the Earth.

topocentric horizon—**See apparent**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

horizon.

topocentric—Of measurements or coordinates, referred to the position of the observer on the Earth as the origin.

Topographic Engineering Center (TEC)—Army Corps of Engineers laboratories which support the Army's modern weapons and command and control systems through research and development in mapping, charting, terrain analysis, geodesy, remote sensing, point positioning, surveying and land navigation. TEC also conducts research on environmental effects on military equipment, develops topographic support systems, and produces terrain analysis products. Formerly known as **Engineer Topographic Laboratories.**

Topographic Map of the United States—The recommended designation for the topographic map of the United States being prepared of quadrangle areas in atlas sheet form, chiefly by the U.S. Geological Survey. This map portrays all basic information about location, elevation, and extent of physical and cultural features that are required for preliminary economic and engineering studies, and for incorporation in a base for maps prepared for special purposes.

topographic base film—An aerial photographic film with a dimensionally stable base used primarily for mapping.

topographic correction—See **terrain correction.**

topographic deflection—That part of the deflection of the plumb line which is caused by the gravitational pull exerted by topographic masses. Topographic deflection is not the same as deflection of the plumb line or station error, but is the theoretical effect produced by the resultant gravitational pull of the unevenly distributed topographic masses around the station, no allowance being made for isostatic compensation. Also called **indirect effect on the deflections.**

topographic expression—The effect achieved by shaping and spacing contour lines so that topographic features can be interpreted with ease and fidelity. Good expression is achieved by delineating the contours in appropriate relationship to each other, with due consideration given to the scale and contour interval of the map. Also called **configuration of terrain. See also topography.**

topographic feature—See **topography,** definition 1.

topographic map—(JCS) A map which presents the vertical position of features in measurable form as well as their horizontal positions. See also **map.**

topographic plot—Representation, by means of contour lines, of the ground relief of an area, shown in a stereoscopic model. See also **compilation,** definition 2.

topographic survey—A survey which has for its major purposes the determination of the relief of the surface of the Earth and the location of natural and man-made features thereon.

topographical latitude—See **geodetic latitude.**

topography—1. The configuration of the surface of the Earth, including its relief, the position of its streams, roads, cities, etc. The Earth's natural and physical features collectively. A single feature such as a mountain or valley is termed a **topographic feature.** Topography is subdivided into hypsography (the relief features), hydrography (the water and drainage features), culture (man-made features), and vegetation. 2. The science of delineation of natural and man-made features of a place or region especially in a way to show their positions and elevations. The term includes the scientific and technical fields of surveying, geodesy, geophysics, military geography, photogrammetry, cartography, graphic arts, and related activities to the extent that they are essential to the accomplishment of the military mapping, geodesy, and military geographic intelligence mission. 3. In oceanography, the term is applied to a surface such as the sea bottom or a surface of given characteristics within the water mass.

topologic error checking—The process of ensuring that the logical consistency of the data is intact, that is, all polygons are closed, all arcs are connected to nodes, etc.

topological concepts—Properties of geometric figures that do not change under continuous (smooth) transformation; such as, is connected to, is inside of, is disconnected, has three connected parts (none connected to any other), is a point on the boundary of, is on the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

left of, is a node where five segments meet, etc.

topological data base—A data base that employs the concepts of topology to indicate the relationships of cartographic features, geometrically expressed as points, lines, and polygons. Commonly employed as the basis for data bases to be used in geographic information systems.

topological data structure—A vector data structure which has the same characteristics as the chain-node data structure, but also establishes the topological relationships (connectivity, adjacency, and inclusion) between features in a vector data set.

topological entities—A collection of objects (i.e., nodes, edges and faces) used to define the spatial relationships among the features of the Earth's surface.

topological relationships—How data elements relate to each other within the data base. In particular, how a change to one element affects other elements.

topological structuring—The process of organizing data topologically so that the relationships and reference linkages are specified.

topological verification—The process of verifying the topological relationship between data elements.

Topologically Integrated Geographic and Resource Information System (TIGRIS)—An object oriented geographic information system with a consolidated data structure. Developed by Intergraph Corporation.

topological—Properties of geometric figures as adjacency that are not altered by distortion as long as the surface is not torn.

topology—The way in which geographical elements are linked together, including the spatial relationships existing among geographic features and the methods used within the computer environment to geometrically depict, store, and examine the data within this relational context. Topology includes adjacency, inclusion, and connectivity. See also **adjacency; connectivity; inclusion.**

toponymy—1. The study and treatment of toponyms. 2. A body of toponyms.

toponym—A name applied to a natural or cultural feature. For U.S. Government usage, policies and decisions governing place names on Earth are established by the United States Board on Geographic Names. Also called **geographic name** and **place name. See also descriptive name.**

topple axis—That horizontal axis, perpendicular to the (horizontal) spin axis of a gyroscope, around which topple occurs.

topple—The vertical component of precession or wander, or the algebraic sum of the two.

torsion balance—A device for measuring combinations of the second derivatives of the gravity potential, which are closely related to the horizontal components of the deflection of the vertical. It consists of a bar suspended horizontally by an elastic filament, one end of the bar being subjected to the influence of the attracting force to a greater degree than the other end. The attracting force is balanced and its comparative strength measured by the torsional reaction of the filament.

total departures—See **abscissa.**

total drift—The algebraic sum of drift due to precession and that due to wander. Also called **drift.**

total latitudes—See **ordinates.**

total magnetic intensity—The vector resultant of the intensity of the horizontal and vertical components of the Earth's magnetic field at a specified point.

total station instrument—A survey instrument capable of measuring delta *x*, *y*, and *z* components without significant alterations.

touch plate—See **kiss plate.**

town plan inset—See **Inset.**

town site survey—The marking of lines and corners within one or more regular units of the township subdivision by which the land is divided into blocks, streets, and alleys as a basis for the disposal of title in parcels of land.

township corner—A corner of a township. See also **closing township corner.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

township lines—The township boundaries that run north and south are termed **range lines**; with few exceptions the range lines are run on cardinal and have been intended to be on cardinal. The boundaries running east and west are termed **township lines**. By law, they were intended to be on true parallels of latitude.

township—The unit of survey of the public lands; normally a quadrangle approximately 6 miles on a side with boundaries conforming to meridians and parallels within established limits, containing thirty-six sections, some of which are designed to correct for the convergence of meridians or range lines. See also **fractional township.**

trace—See **selection overlay.**

track adjustment—Adjustment to a ship's track resulting from set and drift of the vessel.

track chart—A chart showing recommended, required, or established tracks, and usually indicating turning points, courses, and distances.

tracking camera—See **ballistic camera.**

tracking station—A ground-based complex set up to track an object moving through the atmosphere or space, by visual, photographic, photoelectric, or electronic methods.

track—(JCS) The actual path of an aircraft above, or a ship on the surface of the Earth. The course is the path which is planned; the track is the path which is actually taken.

traffic separation schemes—Portrayal on nautical charts of schemes aimed at reducing the risk of collision in congested and/or converging areas by separating traffic moving in opposite, or nearly opposite, directions.

traffic-circulation map—(JCS) A map showing traffic routes and the measures for traffic regulation. It indicates the roads for use of certain classes of traffic the location of traffic control stations, and the directions in which traffic may move. Also called **circulation map.** See also **map.**

trajectory—In general, the curve that a body describes in space. An orbit is a trajectory which does not intersect the Earth.

transcriber—See **point-transfer device.**

transcription—1. The process of recording the sounds and/or grammatical elements of a language in terms of a specific writing system. 2. An item of a language which has undergone this process.

transducer—Any device for converting energy from one form to another (electrical, mechanical, or acoustic). In sonar, it usually combines the functions of a hydrophone and a projector.

transfer standard—A set of formats and protocols used to move data from one system to another without loss. May be defined at the logical level, the physical level, or both.

transformation—1. (photogrammetry) The process of projecting a photograph (mathematically, graphically, or photographically) from its plane onto another plane by translation, rotation, and/or scale change. The projection is made onto a plane determined by the angular relations of the camera axes and not necessarily onto a horizontal plane. See also **rectification.** 2. (surveying) The computational process of converting a position from UTM or other grid coordinates to geodetic, and vice versa; from one datum and ellipsoid to another using datum shift constants and ellipsoid parameters. The survey position of a point is frequently given in several different grids or ellipsoids; local datum and Doppler-derived WGS 84 are common requirements.

transformed print—A photographic print made by projection in a transforming printer.

transforming printer—A specially designed projection printer of fixed geometry used for transforming the oblique components of a coupled camera installation, a multiple-lens camera, or a panoramic camera onto a plane perpendicular to the axis of the system. See also **rectifier; universal transforming printer.**

transit instrument—See **transit,** definition 4.

transit line—Any line of a traverse which is projected, either with or without measurement, by the use of a transit or other device. It is not necessarily an actual line of final survey but may be an accessory line. Also called **traverse line.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

transit micrometer contact correction—A quantity applied to the chronograph record of a star transit observed with the aid of a transit micrometer to allow for the time required for the contact spring to cross one-half of the width of a contact strip in the head of the micrometer.

transit micrometer—A form of registering micrometer with its movable wire placed in the local plane of an astronomic transit and at right angles to the direction of motion of the image of a star which is observed at or near culmination. Also called **impersonal micrometer**, because it almost completely eliminates the effect of the personal equation on time observations made with it.

transit rule—A method of balancing a survey. Corrections corresponding to the closing errors in latitude and departure are distributed according to the proportion: latitude and departure of each line of the traverse to the arithmetical sums of the latitudes and departures of the entire traverse. The transit rule is used when it is assumed that the closing errors are due less to the errors in the observed angles than to errors in the measured distances.

transit traverse—A survey traverse in which the angles are measured with an engineer's transit or theodolite and the lengths with a metal tape. A transit traverse is usually executed for the control of local surveys and is of second-order or third-order quality.

transit-and-stadia survey—A survey in which horizontal and vertical directions or angles are observed with a transit and distances are measured by transit and stadia.

transition curve—See **spiral curve.**

transit—1. The apparent passage of a star or other celestial body across a defined line of the celestial sphere, as a meridian, prime vertical, or almucantar. The apparent passage of a star or other celestial body across a line in the reticle of a telescope, or some line of sight. The apparent passage of a smaller celestial body across the disk of a larger celestial body. The transit of a star across the meridian occurs at the moment of its culmination, and the two terms are sometimes used as having identical meanings; such usage is not correct, even where the instrument is in perfect adjustment. At the poles, a star may have no culmination but it will transit the meridians. See also **culmination; lower transit; meridian transit; upper transit.** 2. A surveying instrument composed of a horizontal circle graduated in circular measure and an alidade with a telescope which can be reversed in its supports without being lifted therefrom. See also **theodolite.** 3. The act of reversing the direction of a telescope by rotation around its horizontal axis. Also called **plunge; inverting; telescoping.** 4. An astronomic instrument having a telescope which can be so adjusted in position that the line of sight may be made to define a vertical circle. Also called **astronomic transit.** See also **broken telescope transit; solar transit.**

translational movement—The systematic movement of projector assemblies in line-of-flight directions in a stereoplotting instrument.

translation—1. The process of rendering oral or written text of one language in terms of text of corresponding meaning of another language. See also **romanization transcription; transliteration.** 2. Movement in a straight line without rotation.

transliteration—1. The process of recording the graphic symbols of one writing system in terms of corresponding graphic symbols of a second writing system. 2. An item of a language which has undergone this process.

translocation—The determination of the relative position between two points from simultaneous Doppler satellite observations.

translunar space—As seen from the Earth at any moment, space lying beyond the orbit of the Moon.

translunar trajectory—A trajectory extending outside the Moon's orbit about the Earth.

translunar—Outside the Moon's orbit about the Earth. See also **cislunar.**

transmission—(optics) The ratio of transmitted light to the incident light. If 100 units of light fall upon a translucent material and 10 of them succeed in passing through, then it can be said that the material has 1/10 or 10 percent transmission.

transparency—(JCS) An image fixed on a clear base by means of a photographic, printing, chemical, or other process, especially adaptable for viewing by transmitted light. See also **diapositive.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

transponder—(JCS) A receiver-transmitter which will generate a reply signal upon proper interrogation. See also **responsor.**

transverse axis—The distance between the apsides. It is identical to the semimajor axis for elliptical orbits.

transverse chart—A chart on a transverse projection. Also called **inverse chart.**

transverse cylindrical orthomorphic chart (TCOC)—See **transverse Mercator chart.**

transverse cylindrical orthomorphic map projection—See **transverse Mercator map projection.**

transverse equator—The plane which is perpendicular to the axis of a transverse projection. Also called **inverse equator.**

transverse graticule—A fictitious graticule based upon a transverse projection. See also **fictitious graticule.**

transverse latitude—Angular distance from a transverse equator. Also called **inverse latitude.** See also **fictitious latitude.**

transverse longitude—Angular distance between a prime transverse meridian and any given transverse meridian. Also called **inverse longitude.** See also **fictitious longitude.**

transverse map projection—A map projection in which the projection axis is rotated 90° in azimuth.

transverse Mercator chart—A chart on the transverse Mercator projection. Also called **inverse cylindrical orthomorphic chart; inverse Mercator chart; transverse cylindrical orthomorphic chart.**

transverse Mercator grid—An informal designation for a state coordinate system based on a transverse Mercator map projection. Also called **Gauss-Kruger grid.**

transverse Mercator map projection—A conformal cylindrical map projection, being in principle equivalent to the regular Mercator map projection turned (transversed) 90° in azimuth. In this projection, the central meridian is represented by a straight line, corresponding to the line which represents the Equator on the regular Mercator map projection. Neither the geographic meridians (except the central meridian) nor the geodetic parallels (except the Equator) are represented by straight lines. Also called **inverse cylindrical orthomorphic map projection; inverse Mercator map projection; transverse cylindrical orthomorphic map projection.**

transverse meridian—A great circle perpendicular to a transverse equator. The reference transverse meridian is called **prime transverse meridian.** Also called **inverse meridian.** See also **fictitious meridian.**

transverse model datum—See **model datum,** *definition* 1.

transverse parallel—A circle or line parallel to a transverse equator, connecting all points of equal transverse latitude. Also called **inverse parallel.** See also **fictitious parallel.**

transverse pole—One of the two points 90° from a transverse equator.

transverse polyconic map projection—A polyconic map projection which is turned (transversed) 90° in azimuth by substituting for the central meridian, a great circle perpendicular to the geographic meridian to provide a control axis for the projection, along which axis will lie the centers of the circular arcs representing lines of tangency of cones with the surface of the sphere.

transverse position—A split camera installation so positioned that the plane containing the camera axis is perpendicular to the line of flight.

transverse rhumb line—A line making the same oblique angle with all fictitious meridians of a transverse Mercator projection. Transverse parallels and meridians may be considered special cases of the transverse rhumb line. Also called **inverse rhumb line.** See also **fictitious rhumb line.**

transverse—In cartography, pertaining to or measured on a map projection in which a meridian is used as a fictitious equator. Also called **inverse.**

traverse adjustment—See **balancing a survey.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

traverse angle—Measurement of the horizontal angle from a preceding adjacent station to the following adjacent station.

traverse error of closure—See **error of closure**, definition 8.

traverse line—See **transit line.**

traverse net—See **survey net**, definition 1.

traverse station—A point on a traverse over which an instrument is placed (a setup). Also, on a traverse, a length of 100 feet measured on a given line, either straight, broken, or curved.

traverse tables—Mathematical tables listing the lengths of the sides opposite the oblique angles for each of a series of right-angle plane triangles as functions of the length and azimuth (or bearing) of the hypotenuse.

traverse the instrument—To rotate a survey instrument about its vertical axis; that is, turning the instrument in azimuth.

traverse—(JCS) A method of surveying in which lengths and directions of lines between points on the Earth are obtained by or from field measurements, and used in determining positions of the points. [A survey traverse may determine the relative positions of the points which it connects in series, and if tied to control stations on an adopted datum, the positions may be referred to that datum. Survey traverses are classified and identified in a variety of ways: according to methods used, as astronomic traverse; according to quality of results, as first-order traverse; according to purpose served, as geographical exploration traverse; and according to form, as closed traverse, etc.] See also **angle-to-right traverse; azimuth traverse; closed traverse; connecting traverse; deflection angle traverse; first-order traverse; fourth-order traverse; geographical exploration traverse; interior angle traverse; loop traverse; open traverse; phototrig traverse; planetable traverse; random traverse; second-order traverse; spur traverse; stadia traverse; subtense bar traverse; subtense base traverse; third-order traverse; transit traverse.**

triangle closure—See **error of closure**, definition 7.

triangle error of closure—See **error of closure**, definition 7.

triangle of doubt—In a simple two-point problem, the triangle resulting when the check ray fails to pass through the point of intersection of the two intersecting rays.

triangle of error—The triangle formed when three plotted rays fail to intersect perfectly. The center of the triangle may be considered to be the adjusted position. See also **resection.**

triangle-of-error method—In surveying, a technique for solving the three-point problem graphically by a triangle of error. These methods are generally referred to by name, such as Bessel's method, Coast-Survey method, and Lehmann's method, each of which is based upon its own factors. See also **triangle of error.**

triangle—See **astronomic triangle; celestial triangle; navigational triangle; preliminary triangle; spherical triangle; spheroidal triangle; terrestrial triangle.**

triangulated irregular network (TIN)—A terrain model created from continuously connected triangles derived from the Delauney algorithm. The vertices of the triangles form irregularly spaced elevation posts. Unlike a grid, the TIN allows extra information to be displayed in areas of complex relief without displaying dense or redundant data gathered in areas of simple relief.

triangulation base line—See **base line.**

triangulation net—See **survey net**, definition 1.

triangulation reconnaissance—A preliminary survey to select the location of stations to give the most feasible triangulation scheme.

triangulation signal—A rigid structure erected over or close to a triangulation station and used for supporting an instrument and observer, or target, or instrument and observer and target, in a triangulation survey. Also, any object, natural or artificial, whose position is obtained in a triangulation survey. The term may be applied to a structure whose position is determined by triangulation, but whose primary purpose is to serve later in a hydrographic or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

topographic survey, when it may become known as a **hydrographic or topographic signal.**

triangulation station—(JCS) A point on the Earth whose position is determined by triangulation. Also called **trig point.**

triangulation system—The main scheme, or net, of primary stations and the auxiliary stations. The main scheme is the framework of the system and is tied at several points to previously established triangulation stations of equal or higher order. See also **survey net,** definition 1.

triangulation theodolite—See **direction instrument theodolite.**

triangulation tower—A structure used to elevate the line of sight above intervening obstacles. Usually consists of two independent structures, one within the other; the center structure supports the theodolite and the outer structure supports the observer and the signal. See also **Bilby steel tower.**

triangulation—A method of surveying in which the stations are points on the ground which are located at the vertices of a chain or network of triangles. The angles of the triangles are measured instrumentally and the sides are derived by computation from selected sides which are termed **base lines,** the lengths of which are obtained from direct measurements on the ground. See also **analytical three-point resection radial triangulation; arc triangulation; area triangulation; base net; direct radial triangulation; first-order triangulation; flare triangulation; graphical radial triangulation; hand-templet triangulation; isocenter triangulation; mechanical-templet triangulation; nadir-point triangulation; phototriangulation; radial triangulation; satellite triangulation; second-order triangulation; semianalytical triangulation; ship-to-shore triangulation; slotted-templet triangulation; spider-templet triangulation; third-order triangulation; trilateration.**

triaxial ellipsoid—An ellipsoid having three unequal axes, the shortest being its polar axis, while the two longer ones lie in the plane of its equator.

tricamera photography—(JCS) Photography obtained by simultaneous exposure of three cameras systematically disposed in the air vehicle at fixed overlapping angles relative to each other in order to cover a wide field. See also **fan camera photography.**

trig control—See **field control.**

trig dossier—A detailed record on the triangulation of an area, giving the coordinates of the triangulation stations.

trig list—(JCS) A list published by certain Army units which includes essential information of accurately located survey points. [A publication containing all available positional data and elevations with the respective descriptions of horizontal and/or vertical control points, usually arranged according to the location of the control points within the limits of map sheets of large-scale series.]

trig point—See **triangulation station.**

trigonometric leveling—The determination of differences of elevations from observed vertical angles combined with lengths of lines. A type of indirect leveling.

trilateration net—A network of points whose positions relative to one another are determined by measurement and adjustment of the length of the sides of the triangles formed by these points.

trilateration—A method of surveying wherein the lengths of the triangle sides are measured, usually by electronic methods, and the angles are computed from the measured lengths. See also **triangulation.**

trilinear surveying—The determination of the position of a point of observation by measuring the angles at that point between lines to three points of known position. See also **resection.**

trim marks—Lines placed on original copy to serve as guides in cutting or trimming the printed sheets to their prescribed size.

trim size—(JCS) The size of a map or chart sheet when the excess paper outside the margin has been trimmed off after printing.

trimetrogon camera—A triple camera assembly with one vertical and two fixed-angle obliques whose imagery overlapped the vertical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

and with all three axes lying in a plane perpendicular to the line of flight. Most camera assemblies of this design were referred to as **trimetrogon camera** because of the widespread usage of the Metrogon lens in early tricamera photography.

trimetrogon photography—See **fan camera photography.**

trimming and mounting diagram—A sketch showing how the prints of a transformed multiple-lens photograph should be corrected to obtain, in effect, a photograph made by a single lens. The information is given in the form of distances referred to the fiducial marks on the photograph, and is the result of the calibration test for the particular camera used.

Tropic of Cancer—The northern parallel of declination, approximately 23°27' from the celestial equator, reached by the Sun at its maximum declination, or the corresponding parallel on the Earth.

Tropic of Capricorn—The southern parallel of declination, approximately 23°27' from the celestial equator, reached by the Sun at its maximum declination, or the corresponding parallel on the Earth.

tropical month—The average period of the revolution of the Moon about the Earth with respect to the vernal equinox, approximately 27 1/3 days.

tropical year—The interval of time between two successive passages of the vernal equinox by the Sun. The tropical year is the year of the seasons, and the basis of the conventional calendar year. Also called **astronomic year; equinoctial year; natural year; solar year.**

tropopause—(JCS) The transition zone between the stratosphere and the troposphere. The tropopause normally occurs at an altitude of about 25,000 to 45,000 feet in polar and temperate zones, and at 55,000 feet in the tropics. See also **atmosphere.**

troposphere—(JCS) The lower layers of atmosphere, in which the change of temperature with height is relatively large. It is the region where clouds form, convection is active, and mixing is continuous and more or less complete. See also **atmosphere.**

tropospheric scatter—(JCS) The propagation of radio waves by scattering as a result of irregularities or discontinuities in the physical properties of the troposphere.

trough compass—See **declinatoire.**

Troughton level—An English instrument having the spirit level permanently attached to the top of the telescope tube

true altitude—1. (JCS) The height of an aircraft as measured from mean sea level. 2. The actual altitude of a celestial body above the celestial horizon. Also called ***observed altitude.***

true amplitude—Amplitude relative to true east or west. See also **amplitude.**

true anomaly—See **anomaly,** definition 3.

true azimuth—The horizontal direction of any line measured clockwise from true north.

true bearing—The horizontal angle between the meridian line and a line on the Earth. The term **true bearing** is used in many of the early descriptions of land boundaries in the United States. It is associated with true north, referring to the direction of the north point as determined by astronomic observations. If an astronomically determined bearing is used, however, the term ***astronomic bearing*** is preferred over **true bearing.**

true depression angle—The setting of the oblique cameras in the photographic aircraft with relation to the true horizon. It is defined by a ray from the exposure station through the principal point of the oblique photograph and a ray to the true horizon.

true direction—horizontal direction expressed as angular distance from true north.

true error—See ***resultant error.***

true horizon—(JCS) 1. The boundary of a horizontal plane passing through a point of vision. 2. In photogrammetry, the boundary of a horizontal plane passing through the perspective center of a lens system. See also **horizon trace.**

true line—A line of constant bearing (rhumb

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

line) between two corners of a survey.

true meridian—A term used to distinguish the great circle through the geographical poles from magnetic meridian, compass meridian, or grid meridian.

true north—(JCS) The direction from an observer's position to the geographical North Pole. The north direction of any geographic meridian. [The term was originally applied to **astronomic north** to distinguish it from **magnetic north**.]

true place—See **true position.**

true position—The position of a celestial body after all known corrections including precession and nutation have been made. Also called **true place.**

true prime vertical—The vertical circle through the true east and west points of the horizon, as distinguished from magnetic or grid prime vertical through the magnetic or grid east and west points, respectively.

true sidereal time—See **apparent sidereal time.**

true solar time—See **apparent solar time.**

true sun—See **apparent sun.**

true value—That value of quantity which is completely free from errors. Since the errors to which physical measurements are subject cannot be known exactly, it follows that the true value of a quantity cannot be known with exactness. In survey work, the most probable value is used as best representing the true value of the quantity.

true-to-scale—A condition where map measurements are in exact agreement with the stated map scale. Since all map projections involve some scale change, the scale is not true at all places on a map.

true—1. Related to true north as opposed to magnetic north. 2. Actual, as contrasted with fictitious, as true Sun. 3. Related to a fixed point, either on the Earth or in space, as true wind; in contrast with relative, which is related to a moving point. 4. Corrected, as true altitude. 5. One of two alternate values (True/False) used in Boolean logic.

turning point (TP)—A point on which both a minus sight (foresight) and a plus sight (backsight) are taken on a line of direct levels. Also, in topographic surveys, any point on which the rod is held while the instrument is moved to another station. These turing points are often marked for future use as tie or check points.

turning point pin—A steel pin about one foot long. In leveling operations, the turning point is driven into the ground where it is necessary to establish a point that will be stable in elevation for a short period of time. When it has served its usefulness at one point, it is removed and carried by the rodman for subsequent reuse.

turret graver—A scribing instrument which permits the use of points of several weights interchangeably without the inconvenience of interrupting the scribing to replace points. The turret head is revolved to position the desired point, and scribing is resumed.

tusche—An ink for drafting printing areas on lithographic plates.

tusching—The operation of adding work to the image on a press plate, correcting lines and lettering, and adding solids by means of tusche. Also called **lithographic drafting.**

two-base method—A technique of barometric leveling utilizing three barometers. Two barometers, one high established over a known elevation, and one low established over a known elevation, and a roving barometer operating between the two known positions. High and low barometers are read and recorded, with temperature, every five minutes. The roving barometer and temperature are read and recorded at each station occupied. Data are reduced to position and elevation by office computation. The only barometric leveling method able to consistently produce errors less than ±1 meter. See also **fly-by-method; loapfrog method.**

two-body orbit—The motion of a point mass in the presence of the gravitational attraction, of another point mass, and in the absence of any other forces. This orbit is usually an ellipse, but may be a parabola or hyperbola.

two-body problem—That problem in celestial mechanics which treats of the relative motion or of two-point masses under their mutual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

gravitional attraction.

two-dimensional (2-D) data—Areal data in two dimension, such as northing-easting or latitude-longitude.

two-dimensional pantograph—In relief model making, a machine permitting the cutting, at a predetermined scale, of the three-dimensional terrain base from the flat map contour drawing

two-point problem—A problem in determining the position of a point with the known factor being the length of one line that does not include the point to be located.

two-step enlargement/reduction—A technique of projecting and printing a small image; then copying and projecting it again to the required size. This is often necessary when copy size/copy camera limitations do not permit enlargement or reduction in a single operation.

two-transit method—A method of ship-to-shore triangulation whereby the position of the sounding boat or pole is determined by angle observations from two transits on shore set up over points previously positioned.

type—In printing (typography) a metal block having a raised letter or figure which, when inked, is used to make an impression on paper or other material. Type can also be in the form of negative or positive stripping film. Categories of type include hand-set cold type; hot type, such as Linotype; Monotype (punched tape to metal); phototype (film negative or positive); and Photontype (tape to film). Type is identified by its style and size.

typography—The art of type composition and printing from raised type surfaces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

U

U.S. control survey nets—The two control survey nets being extended over the United States by the National Geodetic Survey for the control of nautical charts and topographic maps, and comprising: (1) The horizontal-control survey net consisting of arcs of first-order and second-order triangulation and lines of first-order and second-order traverse, a few of which have been executed by the United States Geological Survey, The Corps of Engineers, and other organizations. The data derived in this survey are being coordinated and correlated on the North American datum of 1927. The National Geodetic Survey is currently recomputing the horizontal control network to the North American datum of 1983. (2) The vertical control survey net consisting of lines of first-order and second-order spirit leveling which determine the elevations of thousands of bench marks above a common datum, mean sea level. This net includes lines of levels run by the United States Geological Survey, the Corps of Engineers, and other organizations.

U.S. Engineer precise leveling rod— A speaking rod of T-shaped cross section, 12 feet long, graduated in centimeters.

U.S. Geological Survey level—A level of the dumpy type, constructed of stainless steel. It has an internal focusing telescope; the level bubble is centered by the end-coincidence method, effected with the aid of a prism device and Stellite mirror which can be adjusted by the observer.

U.S. Geological Survey precise leveling rod—**1.** A speaking rod graduated in yards and fractions of a yard. It is read for each of three cross wires to the nearest thousandth of a yard. The sum of the three readings is then the mean reading in feet to the nearest thousandth. **2.** A target rod of plus-sign (+) cross section, a little over 12 feet in length. There are two forms of this rod: the single-target rod and the double-target rod.

U.S. National Map Accuracy Standards—**1.** Horizontal accuracy: (1) for maps at publication scales larger than 1:20,000, 90 percent of all well-defined features, with the exception of those unavoidably displaced by exaggerated symbolization, will be located within 1/30 inch (0.85 mm) of their geographic positions as referred to the map projection; (2) for maps at publication scales of 1:20,000 or smaller, 1/50 inch (0.50 mm). **2.** Vertical accuracy: 90 percent of all contours and elevations interpolated from contours will be accurate within one-half of the basic contour interval. Discrepancies in the accuracy of contours and elevations beyond this tolerance may be decreased by assuming a horizontal displacement within 1/50 inch (0.50 mm). Also called **map accuracy standards; national map accuracy standards.**

ultrawide-angle lens—See **superwide-angle lens.**

unchecked spot elevation—Elevation determined by unchecked field survey methods, such as side shots on stadia lines, unchecked vertical angles, and barometric leveling. Also an elevation determined by repeated photogrammetric reading.

uncontrolled mosaic—(JCS) A mosaic composed of uncorrected prints, the details of which have been matched from print to print without ground control or other orientation. Accurate measurement and direction cannot be accomplished. See also **controlled mosaic.**

underground mark—A surveying mark set and plumbed below the center of a surface mark and separated therefrom so as to preserve the station in case of accident to the surface mark.

undershoot—A data omission which occurs during the data capture process, particularly during manual digitizing. In an undershoot situation, linear data falls short of reaching an intersection. An undershoot may be corrected by a snap software function.

undevelopable—A surface, such as a sphere, that cannot be flattened to form a plane without compressing or stretching some part of it.

undo—(digital) The ability to cancel a previous command to the computer.

undulation of the geoid—See **geoid height.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

unidimensional magnification— Transformation of one rectangle into another of different proportions.

Universal Polar Stereographic (UPS) grid—A military grid system based on the polar stereographic projection, applied to maps of the Earth's polar regions north of 84°N and south of 80°S latitudes.

Universal Rectifier (UR)—Processor used to rectify imagery in support of DMA production. The primary function of the UR is to generate rectified point position data bases. See also **Universal Rectifier Segment.**

Universal Rectifier Segment (UR/S)—Segment of DMA's DPS which processes imagery in soft copy format to produce Point Positioning Data Bases. See also **Universal Rectifier; Digital Production System.**

Universal Space Rectangular (USR) Coordinate System—A right-handed orthogonal coordinate system with its origin at the center of the Earth, positive *x*-axis in the equatorial plane and passing through the 0° meridian, positive *y*-axis in the equatorial plane and passing through 90° east meridian, and positive *z*-axis along the rotational axis of the Earth and passing through the North Pole.

Universal Time (UT)—(JCS) 1. The basis for coordinated dissemination of time signals, counted from 0000 at midnight. Also called **Coordinated Universal Time (CUT); universal time coordinated.** 2. In celestial navigation applications, the time which gives the exact rotational orientation of the Earth obtained from CUT by applying increments determined by the US Naval Observatory. 3. The official civil time of the United Kingdom. Formerly called ***Greenwich Mean Time.***

Universal Transverse Mercator (UTM) coordinates—Quantities which designate the location of a point on the Universal Transverse Mercator grid.

Universal Transverse Mercator (UTM) grid—(JCS) A military grid system based on the transverse Mercator projection, applied to maps of the Earth's surface extending to 84°N and 80°S latitudes.

universal analog photographic rectification system—An electronic rectification system permitting the rapid transfer of detail from trimetrogon or any other type of aerial photography, to include panoramic coverage. The system consists of four basic components: input scanner, computer, console, and *x-y* plotter. Also called **electronic sketchmaster.**

universal instrument—See **altazimuth instrument.**

universal level—See **circular level.**

universal plotting sheet—A plotting sheet on which either the latitude or longitude lines are omitted, and are to be drawn in by the user, making it possible to quickly construct a plotting sheet for any part of the Earth's surface. See also ***plotting chart; position plotting sheet.***

universal sketchmaster—A type of sketchmaster in which vertical or oblique photographs may be utilized.

universal time coordinated (UTC)—Also called **Coordinated Universal Time.** See Universal Time.

universal transforming printer—A specially designed printer for making glass diapositives in which a known camera distortion is eliminated or compensated for, or in some cases introduced. The glass diapositives may be reproduced at scale, enlarged, or reduced.

unperturbed orbit—See **normal orbit.**

unstable-type gravimeter—A gravity meter which utilizes a moving system which approaches a point of instability such that small changes in gravity produce relatively large motions of the system. See also **astatized gravimeter.**

updating—The capability to make changes or add new information to existing data.

upper branch—That half of a meridian or celestial meridian from pole to pole which passes through a place or its zenith.

upper culmination—See **culmination.**

upper limb—That half of the outer edge of a celestial body having the greatest altitude, in contrast with the lower limb, that half having the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

least altitude.

upper motion—(surveying) Rotation of the upper plate of a repeating instrument.

upper transit—Transit of a celestial body over the upper branch of the celestial meridian. Also called **superior transit**. See also **culmination.**

user interface—The method by which the human operator communicates with the various data base and applications modules.

user requirement analysis (URA)—The study of the needs of the potential system users.

UT0 time—The mean universal time derived from observations of time of star transits. Since the directions of the meridians change with time because of the motion of the pole, the UT0 time will, thus, be affected and will, therefore, be irregular. See also **UT1 time; UT2 time.**

UT1 time—The true angular rotation of the Earth about its instantaneous spin axis in the mean equatorial system of dates. UT1 is obtained from UT0 by correcting UT0 for the difference between the instantaneous and mean longitude at the observing station. Since UT1 is keyed precisely to the instantaneous rotation of the Earth, which is not strictly uniform, UT1 does not progress uniformly. See also **UT0 time; UT2 time.**

UT2 time—The mean angular motion of the Earth, freed of predictable periodic variation but still affected by irregular variations and secular variations. UT2 is obtained from UT1 by correcting UT1 for seasonal variations in rotation rate. See also **UT0 time; UT1 time; universal time coordinated.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

V

vacuum box—The frame, containing its own vacuum unit, which encloses the mold for the forming of plastic relief maps.

Vaisala Comparator—A precise distance measuring instrument that uses optical interferometry for making measurements. It can measure distances as great as 864 meters with an accuracy of 1 part in 10^7.

value—See **absolute value; adjusted value; most probable value; observed value; true value.**

vanishing line—The straight line on a photograph upon which lie all the vanishing points of all systems of parallel lines parallel to one plane.

vanishing point—The image, in the plane of a photograph, of the point toward which a system of parallel lines in the object space converges.

variable contour interval—A nonuniform contour interval. It may result from the use of cartographic source materials which do not contain a constant contour interval or from adapting the contour interval to specific types of terrain for the optimum portrayal of relief features.

variable length field—A field whose length is determined by the amount of storage needed to store its contents. Useful for character strings and coordinate strings, both of which are highly variable in length.

variable length records—These records may have a variable number of fields (data elements) or the fields may be of varying length, or both. Variable-length records typically have delimited fields and/or byte counts to facilitate processing. See also **fixed-length records.**

variable perspective camera system— A system which, in its simplest form, consists of a standard-type view camera, a large aperture front-surface mirror of spherical configuration, and an easel used in the rectification of highly tilted long focal-length photographs, and the transformation of maps and charts from one projection to another. When the camera component is replaced with a projector, it becomes possible to expedite the rectification of lunar photography taken by terrestrial observatories.

variable ratio pantograph—**See pantograph.**

variance(σ^2)—The most commonly used measure of dispersion or error in statistical analysis. It is also called the mean squared deviation. Variance is calculated by first taking the sum of the squared deviations from the mean for each of n observations. Then the sum is divided by n-1.

variation of coordinate method— A method of adjusting measurements in which the coordinates of geodetic points are varied so as to best fit the observations and retain mathematical homogeneity. See **observation equations; variation of parameters.**

variation of latitude—A small change in the astronomic latitude of points on the Earth, due to variation of the poles.

variation of parameters—**1.** An interactive method to solving complex equations by successively closer approximations of the variables, usually employing the more significant terms of the first derivatives of mathematically precise functions that must equal zero only when the correct values of the variables are used to compute the functions. **2.** The observation equations method of least squares adjustment is sometimes referred to as the **variation of parameters method of least squares adjustment** because the parameters of observation equations are determined by variation of parameters.

variation of the poles—A small variation of the location of the instantaneous axis of rotation of the Earth with respect to the physical surface thereof. Also called **polar motion.** See also **conventional international origin.**

variational inequality—An inequality in the Moon's motion, due mainly to the tangential component of the Sun's attraction.

variation—See **magnetic variation.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

variometer—An instrument for comparing magnetic forces, especially of the Earth's magnetic field.

vectograph—A stereoscopic photograph composed of two superimposed images that polarize light in planes 90° apart. When these images are viewed through Polaroid spectacles with the polarization axes at right angles, an impression of depth is obtained.

Vector Product Format (VPF)—A geo-relational data structure designed for the provision of vector products on CD-ROM media and manipulated in Geographic Information Systems environments.

Vector Product Standard (VPS)—A suite of user system oriented standards encompassing VPF digitizing conventions, tiling, feature attributing and coding. See also **Vector Product Format (VPF).**

vector data—Data which represents each cartographic feature by an entity description (feature code) and a spatial extent (geographic position). Geographic position may be two-dimensional (horizontal position only) or three-dimensional (including elevation). Features are categorized as point, line, or area features. The position of a point feature is described by a single coordinate pair (or triplet for three dimensional data). The spatial extent of a line feature is described by a string of coordinates of points lying along the line, while the extent of an area feature is described by treating its boundary as a line feature. Vector data may be stored in a sequential, a chain node, or a topological data structure.

vector map product (VMP)—A generic term used to describe an electronic map display product, in vector form. The VMP could be line segments (vectors), points or area polygon delineated areas, displayed as such or symbolized. The features may be attributed as to surface material, structural composition, or radar significance. For visual displays color fill software may be required.

vector quantization—A lossy compression technique in which blocks of data are matched to a limited set of defined blocks called codewords.

vector refresh display—A cathode ray tube on which the image is displayed as a vector and which must be refreshed by a new pass of the electron beam about 30 times a second.

vectorial angle—The angle between the fixed line to which the direction is referred and the radius vector. See also **polar coordinates.**

vector—1. A quantity possessing both magnitude and direction. 2. (data structure) A data structure which emphasizes position. Homogeneous units are points, lines, and polygons. 3. A directed line segment, with magnitude commonly represented by the coordinates for the pair of end points. 4. (Computer Programming/Mathematics) Vector data refers to data in the form of an array with one dimension.

velocity correction—A correction applied to the speed of light to obtain the true speed in consideration of humidity, temperature, and altitude for use in shoran operations.

Vening Meinesz formulas—Formulas for computing deflections of the vertical from gravity data.

verbal scale—See **equivalent scale.**

verify—1. To determine whether a transcription of data or other operation has been accomplished accurately. 2. To confirm the accuracy, portrayal, and completeness of digital data with respect to established standards or specifications.

vernal equinox—That point of intersection of the ecliptic and the celestial equator, occupied by the Sun as it changes from south to north declination, on or about 21 March. Also called **first of Aries; first point of Aries; March equinox. See also mean equinox.**

vernier closure—The difference between the initial and final vernier readings during the survey operation of closing the horizon.

vernier—A short, auxiliary scale situated alongside the graduated scale of an instrument, by means of which fractional parts of the smallest division of the primary scale can be measured accurately. See also **contact vernier; direct vernier; folding vernier; optical vernier; retrograde vernier.**

vertex (vertices)—The highest point. The vertices of a great circle are the points nearest

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the poles. Also called **apex.**

vertex of curve—See **point of intersection.**

Vertical Obstruction Data (VOD)—Data file containing discrete position and height information of manmade and natural objects extending above the earth's surface at heights that could constrain airplane and cruise missile flights at levels established by mission planners. Vertical obstructions include manmade obstructions such as radio towers, smokestacks, bridges, powerlines, and other objects above specified minimum elevations. VOD is superseded by the Probabilistic Vertical Obstruction Data (PVOD). See also **Probabilistic Vertical Obstruction Data.**

vertical angle—1. An angle in a vertical plane. 2. (surveying) One of the directions which form a vertical angle is usually either the direction of the vertical (zenith), and the angle is termed the ***zenith distance***; *or the line of* intersection of the vertical plane in which the angle lies with the plane of the horizon, and the angle is termed the **angle of elevation** *or* **angle of depression**, or simply the **altitude** (*plus or minus, as the case may be*). *The vertical* angle between two directions, neither of which lies in the plane of the horizon or coincides with the vertical, is usually obtained from the combination of two vertical angles as defined above.

vertical angulation—The process of obtaining differences of elevation by means of observed vertical angles, combined with lengths of lines. In geodetic work, trigonometric leveling is used with the same meaning.

vertical axis—(theodolite, transit) The line through the center of the instrument about which the alidade rotates. For an instrument in complete adjustment, this axis occupies a vertical position, passes through the center of the horizontal circle, and is perpendicular to its plane.

vertical bridging—See **bridging.**

vertical circle—1. A great circle of the *celestial sphere, through the zenith and nadir.* Vertical circles are perpendicular to the horizon. 2. A graduated disk mounted on an instrument in such a manner that the plane of its graduated surface can be placed in a vertical plane. It is *primarily used for measuring vertical angles in* astronomic and geodetic work.

vertical collimator—A telescope so mounted that its collimation axis can be made to coincide with the vertical (*or direction of the plumb line*). The vertical collimator serves as an optical plumb line; it may be designed for use in placing a mark on the ground directly under an instrument on a high tower or in centering an instrument on a high tower directly over a mark on the ground. Also called **optical plummet.**

vertical comparator—(pendulum) A stand designed for the support of a pendulum, a bar of known length, and two micrometer microscopes, so placed with reference to one another that the length of the pendulum can be measured.

vertical control datum—Any level surface (as, for example, mean sea level) taken as a surface of reference from which to reckon elevations. Also called **vertical datum; *vertical geodetic datum.*** See also **datum level; reference level; reference plane.**

vertical control net—See **survey net, definition 2.**

vertical control point—See **control point; control station.**

vertical control—The measurements taken by surveying methods for the determination of elevation only with respect to an imaginary level surface, usually mean sea level. See also **survey net, definition 2.**

vertical coordinates—The vertical distance of a point above or below a reference datum. Points may be plus or minus according to whether the point is above or below the datum.

vertical curve—A parabolic curve used to connect grades of different slope, and used at the vertex of a grade to avoid the sudden change in direction in passing from one grade to the other. This method of grade change is usually used when there is an algebraic difference of more than 0.2 percent in the two opposing grades.

vertical datum—See **vertical control datum.**

vertical deformation—In relative orientation, the cumulative model warpage affecting the vertical datum from *x-tilt error and y-tilt error.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

vertical exaggeration—1. The change in a model surface created by proportionally raising the apparent height of all points above the base level while retaining the same base scale. 2. The increase of the vertical scale over the horizontal scale of a terrain model or plastic relief map.

vertical extension—See **extension**, definition 2.

vertical geodetic datum—See **vertical control datum.**

vertical intensity—The magnetic intensity of the vertical component of the Earth's magnetic field, reckoned positive if downward, negative if upward.

vertical parallax—See ***y*-parallax.**

vertical pass point—See **supplemental elevation.**

vertical photograph—An aerial photograph taken with the axis of the camera being maintained as closely as possible to a truly vertical position with the resultant photograph laying approximately in a horizontal plane.

vertical plane—1. Any plane passing through a point on the Earth and containing the zenith and nadir of that point; also a plane containing a plumb line. 2. (surveying) A plane at right angles to a horizontal plane and within which angles and distances are observed.

vertical sketchmaster—A type of sketchmaster in which vertical photographs are utilized.

vertical stereotriangulation—That portion of stereotriangulation concerned with the establishment of vertical data. Vertical stereotriangulation is often limited or precluded as an operation due to the more rigid accuracy standards established for vertical positions than for horizontal positions.

vertical-angle bench mark (*VABM*)—A bench mark with elevation established by vertical angle methods. See also **bench mark.**

vertical-angle station—See **supplemental station.**

vertical—The line perpendicular to the geoid at any point. It is the direction in which the force of gravity acts. See also **local vertical; mass attraction vertical; normal,** definition 3.

vibration—A single movement of a pendulum in either direction, to or fro. See also **oscillation.**

Video Point Positioning Data Base (VPPDB)—A DMA analog video disc product which provides a monoscopic precise positioning capability which is more versatile and easier to use than the Analytical Photogrametric Positioning System (APPS). VPPDB uses a personal computer to allow real time coordinate readout; designed for ease of use and minimal training time.

video disc [MCGT]—A 30.5 cm (12") Constant Angular Velocity (CAV) analog video disc capable of storing video images on each side (equivalent to approximately 200 average-sized map sheets at two different fields of view). See also **CD-ROM.**

video map—Any one of a series of cartographic products that are captured, stored, and displayed using analog video signals.

viewfinder—(aerial camera) An auxiliary device which shows the field of view of a camera. It is used in the taking of vertical aerial photography to correct crab angle and maintain forward lap (end lap).

vignetting—1. (photography) A gradual reduction in density of parts of a photographic image due to the stopping of some of the rays entering the lens. Thus, a lens mounting may interfere with the extreme oblique rays. An antivignetting filter is one that gradually decreases in density from the center toward the edges; it is used with many wide-angle lenses to produce a photograph of uniform density by cutting down the overexposure of the center of the photograph. 2. (lithography) A photographic process which portrays a solid color in a screen which shades off gradually into the unprinted paper. Open water is often shown by this method.

virtual gravity—The force of gravity on an atmospheric parcel, reduced by centrifugal force due to the motion of the parcel relative to the Earth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

virtual image—An image that cannot be shown on a surface but is visible, as in a mirror.

virtual PPI reflectoscope (VPR) chart—A type of radar chart.

visibility chart—A special-purpose map or other graphic showing which areas can be seen and those which cannot be seen from a given observation point.

visible horizon—See **apparent horizon.**

visual acuity—A measure of the ability of the human eye to separate details in viewing an object. The reciprocal of the minimum angular separation, in minutes of arc, of two lines of detail which can be seen separately.

visual range—The limiting range of a light determined after taking into account both the geographic range and the luminous range. The geographic range is the maximum distance at which the curvature of the Earth permits a light to be seen from a particular height of eye without regard to the luminous intensity of the light. The luminous range is determined from the known nominal luminous range, called the **nominal range**, and the existing visibility conditions.

voting (TERCOM)—The technique of having a vehicle fly over the terrain covered by three unique but complementary maps within a finite distance, comparing the calculated positional accuracy within each of the maps and determining whether or not to update the vehicle navigation system based on how closely the three positional accuracies compare with each other.

vulgar establishment—See **establishment of the port.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

W

wading rod—A rod, graduated in feet and tenths of feet, used for stream gaging in shallow water.

wall map—A special-purpose map of a large area designed to be displayed on a wall.

wander—See **apparent precession.**

want of correspondence—See ***y*-parallax.**

warped model—Any spatial model which, due to photographic distortions or orientation errors, has a model datum which is deformed or otherwise incapable of being leveled. See also **flat model.**

water leveling—A method of obtaining relative elevations by observing heights with respect to the surface of a body of still water.

water stage recorder—An automatic recording instrument which records the rise and fall of the water surface at a stream gaging station.

wavelength—Quantitative specification of kinds of radiant energy. See also **dominant wavelength.**

waving the rod—In leveling, a technique whereby the rodman slowly pivots the leveling rod toward and away from the instrument position. The least reading obtainable is the proper one to be recorded.

weather map—(JCS) A map showing the weather conditions prevailing, or predicted to prevail, over a considerable area. Usually, the map is based upon weather observations taken at the same time at a number of stations. Also called **surface charts; synoptic chart.** See also **map.**

weber—(geomagnetism) Unit of magnetic flux. In the mks system, 1 weber = 1 joule/amp = 1 $kgm^2/amp\ sec^2$.

wedge—(optics) A refracting prism of very small deviation, such as those used in the eyepieces of some stereoscopes. Also called **optical wedge.**

weight features—The process of systematically increasing the value of a particular data element or elements so as to give that element more significance in the analysis or calculations.

weighted mean—A value obtained by multiplying each of a series of values by its assigned weight and dividing the sum of those products by the sum of the weights.

weight—The relative value of an observation, source, or quantity when compared with other observations, sources, or quantities of the same or related quantities. The value determined by the most reliable method is assigned the greatest weight.

Werner map projection—A particular case of the Bonne map projection, in which the standard parallel is at the pole, and the tangent cone becomes a tangent plane. Any one geographic meridian is chosen as the central meridian and represented by a straight line, divided to exact scale. The geographic parallels are represented by circular arcs, also divided to exact scale, and the other meridians are curved lines.

west point—See **prime vertical plane.**

westing—See **departure**, definition 1.

wide-angle lens—A lens having an angle of coverage between 75° and 100°. A lens whose focal length is equal to approximately one-half the diagonal of the format.

wiggling-in on line—See **double centering.**

wiggling-in—A survey procedure used when it is necessary to establish a point, exactly on line between two control points neither of which can be occupied. It is essentially a trial-and-error technique where repeated fore and back readings are taken and the instrument shifted after each pair of readings until exactly in line with the stations. Also called **ranging-in.**

wind rose—1. A diagram showing the relative frequency of winds blowing from different directions. It may also show average speed or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

frequency of occurrence of various speeds from different directions. 2. A diagram showing the average relationship between winds from different directions and the occurrence of other meteorological phenomena.

window—1. An interface component for computers through which objects and actions are presented to users. 2. A rectangular frame with a specified size and location on the screen of an interactive graphics system, and within which a rectangular portion, or window, of the data is displayed.

wing photograph—A photograph taken by one of the side or wing lenses of a multiple-lens camera.

wing point—Three easily identified points along each side of an aerial photograph, one near each corner and one near the middle. Used in the extension of radial control in making controlled mosaics.

winter solstice—1. That point on the ecliptic occupied by the Sun at maximum southerly declination. Also called **first point of Capricornus; December solstice.** 2. That instant at which the Sun reaches the point of maximum southerly declination, on or about 22 December.

wipe-on printing plates—A printing plate which has a light-sensitive coating applied by the user.

wire drag—A sounding device consisting of weighted wires which are maintained at a given depth by floats, and then dragged over any desired course.

witness corner—1. A marker set on a property line leading to a corner; used where it is impractical to maintain a monument at the corner itself. 2. A monumented survey point usually on the line of survey near a corner established as a reference when the corner is so situated as to render its monumentation or ready use impracticable.

witness mark—A mark placed at a known distance and direction from a property corner or survey station to aid in its recovery and identification. Also called **witness post; witness stake.**

witness point—A monumented station on a line of the survey, employed to perpetuate an important location without special relation to any regular corner, except that the bearing or distance may be known.

witness post—See **witness mark.**

witness stake—See **witness mark.**

Woodward base-line measuring apparatus—See **iced-bar apparatus.**

World Geographic Reference System (GEOREF)—A worldwide position reference system that maybe applied to any map or chart graduated in latitude and longitude (with Greenwich as prime meridian) regardless of projection. It provides a method of expressing positions in a form suitable for reporting and plotting. The primary use is for interservice and interallied reporting of aircraft and air target positions.

working pendulum—A pendulum which is used (swung) in a determination of the intensity of gravity.

working-in on a line—See **double centering.**

World Data Bank II (WDBII)—A CIA-produced digital representation of the world divided into five geographic areas. Each area contains vector data on coastlines (including islands and lakes), rivers, and international boundaries. WDBII was digitized at scales ranging from 1:1 million to 1:4 million.

World Geodetic System (WGS)—A consistent set of parameters describing the size and shape of the Earth, the positions of a network of points with respect to the center of mass of the Earth, transformations from major geodetic datums, and the potential of the Earth (usually in terms of harmonic coefficients).

World Mean Elevation Data (WMED)—A DMA data base of minimum, maximum, and mean terrain elevations. Provides coarse resolution elevation data with continuous worldwide coverage, which may be used to support military planning and command and control systems.

World Vector Shoreline (WVS)—A digital data file developed to support map display systems, GIS, and weapon systems at 1:250,000 scale which contains shorelines, international boundaries, and country names of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the world.

world polyconic grid—A grid system in which a grid network is mathematically derived from elements of a polyconic projection.

write once, read many (WORM)—1. A laser disc system for custom data base creation. The disc can be encoded only once by the user but may be read an unlimited number of times. WORM discs are not compatible with a CD-ROM disc player. 2. A worm is a software program that copies itself from one system to another, usually without the systems users knowledge, it may also cause harm to a system, e.g., by destroying files.

wrong-reading—A descriptive term for an image which is a reverted or mirror image of the original. Other terms, such as reverse reading, etc., are sometimes used to identify image direction, but are not recommended because of possible confusion in film negative-positive relationship.

WWV time—Accurately controlled time signals transmitted from stations (radio) WWV in Colorado and WWVH in Hawaii. These stations broadcast UTC (universal time coordinated) which is offset from A1 by a variable amount determined annually before the start of the year. The epoch is shifted in increments of 100 milliseconds if it departs too far from UT2.

wye (Y) level—A leveling instrument having the telescope, with attached spirit level, supported in Y-shaped bearings. The telescope can be rotated about its longitudinal axis (collimation axis) and it can be lifted and reversed, end for end, for testing and adjustment. Also called **Y level.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

X

x-axis—1. (JCS) A horizontal axis in a system of rectangular coordinates; that line on which distances to the right or left (east or west) of the reference line are marked, especially on a map, chart, or graph. 2. The line joining the opposite fiducial marks in the direction which is most nearly parallel to the line of flight.

x-coordinate—See **abscissa.**

x-correction—The correction to an *x-direction.*

x-direction—An observed direction in a triangulation figure for which an approximate value is obtained and treated like an observed direction in the adjustment of the figure. The work of a least squares adjustment of a *triangulation figure* sometimes requires the use of an *x-direction,* for which an approximate value is obtained by an inverse position computation, by the solution of the three-point problem, or by other means, and then using the *x*-direction in the adjustment and obtaining a correction (*x*-correction) for it, which makes it consistent with the adjusted values of the observed *directions.*

x-displacement—A component of image displacement. When a point image is to be located by coordinates with respect to rectangular axes, *x*-displacement represents the distance moved in the *x-direction.*

x-motion—In a *stereoplotting instrument,* that linear adjustment approximately parallel to a *line connecting* two projector stations; the path of this adjustment is, in effect, coincident with the flight line between the two relevant exposure stations.

x-parallax—See **absolute stereoscopic parallax.**

x-scale—(JCS) On an *oblique photograph,* the scale along a line parallel to the true horizon.

x-tilt—The component of tilt about the *x*-axis, which is most nearly in the direction of flight. Also called list.

x-y scaler—An instrument that provides *x*- and *y*-coordinates in digital form from analog data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Y

y-axis—1. *(JCS) A vertical axis in a system* of rectangular coordinates; that line on which distances above or below (north or south of) a reference line are marked. especially on a map, chart, or graph. 2. The line which is perpendicular to the *x*-axis and passes through the origin.

y-coordinates—See **ordinates.**

y-displacement—A component of image displacement. When a point image is to be located by coordinates with respect to rectangular axes, *y*-displacement represents the distance moved in the *y*-direction.

y-level—See **wye level.**

y-parallax—The difference between the perpendicular distances of the two images of a point on a pair of photographs from the vertical plane containing the air base. The existence of *y*-parallax is an indication of tilt in either or both photographs, or a difference in flying heights, and interferes with stereoscopic examination of the pair. Also called **vertical parallax; want of correspondence.**

y-scale—(JCS) On an oblique photograph, the scale along the line of the principal vertical or any other line, inherent or plotted, which, on the ground, is parallel to the principal vertical.

y-swing method—See **one-swing method.**

y-tilt—See **pitch.**

yaw—1. (air navigation) The rotation of an aircraft about its vertical axis so as to cause the aircraft's longitudinal axis to deviate from the flight line. Also called **crab.** 2. (photogrammetry) The rotation of a camera or a photograph coordinate system about either the photograph *z*-axis or the exterior *z*-axis. 3. In some photogrammetric instruments and in analytical applications, the symbol κ may be used.

yaw angle—See **angle of yaw.**

year—The period of about 365 1/4 solar days required for one revolution of the Earth around the Sun. See also **anomalistic year; calendar year; eclipse year; fictitious year; great year; sidereal year; tropical year.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Z

z-axis—In a three-dimensional rectangular coordinate system, the axis of reference that is perpendicular to both the *x*- and *y*-axes at their point of intersection.

z-motion—Movement of a stereoplotting projector in a vertical direction.

z-scale—(JCS) On an oblique photograph, the scale used in calculating the height of an object. Also, the name given to this method of height determination.

z-time—See **Universal Time.**

z-value data points—Data points providing a value perpendicular or normal to a specified surface (elevation).

zenith—That point of the celestial sphere vertically overhead. See also **geocentric zenith; geodetic zenith.**

zenith angle—See **zenith distance.**

zenith camera—A special camera so designed that its optical axis may be pointed accurately toward the zenith. It is used for the determination of astronomic positions by photographing the position of the stars. See also **photographic zenith tube.**

zenith distance—The vertical angle between the zenith and the object which is observed or defined. Zenith distance is the complement of the altitude. Also called **zenith angle.** See also **coaltitude.**

zenith telescope—A portable instrument adapted for the measurement of small differences of zenith distance, and used in the determination of astronomic latitude.

zenithal chart—See **azimuthal chart.**

zenithal map projection—See **azimuthal map projection.**

zonal harmonics—The set of spherical harmonics that are functions of latitude only and therefore do not affect the rotational symmetry of the surface about its polar axis.

zone time—The local mean time of a reference zone. See also **time zone.**

zoom system—See **pancratic system.**

zulu time—See ***Universal Time.***

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

4. LIST OF ABBREVIATIONS, ACRONYMS, AND INITIALS

A/T	Along Track
AAFIF	Automated Air Facilities Information File
ABC	Airborne Control
ADAS	Automated Digital Annotation System
ADM	Aeronautical Data Maintenance
ADRG	ARC Digitized Raster Graphics
AFS	Aerial Film Speed
AM/FM	Automated Mapping/Facilities Management
AMSD	Adjusted Mapping Support Data
ANMS	Automated Notice to Mariners System
ANSI	American National Standards Institute
APPS	Analytical Photogrammetric Positioning System
APR	Airborne Profile Recorder
ARAPS	Area Requirements and Product Status
ARC	Equal Arc Second Raster Chart/Map
ASCII	American National Standard Code for Information Interchange
ASW	Antisubmarine Warfare
ATC	Air Target Chart
ATM	Air Target Materials
ATMP	Air Target Materials Program
ATRAN	Automatic Terrain Recognition and Navigation
ATTG	Automated Tactical Target Graphic
AVHRR	Advanced Very High Resolution Radiometer
AWAR	Area Weighted Average Resolution
AWSDB	Advanced Weapons and Systems Data Base
BAD	Bathymetric Archive Data
BC	Bottom Contour Chart
BGN	Board on Geographic Names
BM	Benchmark
BMD	Bathymetric Model Data
BNC	Bathymetric Navigation Chart
BNPC	Bathymetric Navigation Planning Chart
BRAC	Bathymetric Recovery Area Chart
C^3I	Command, Control, Communications and Intelligence
CAC	Computer Assisted Cartography; Compressed Aeronautical Chart
CAD	Computer Aided Design/ Drafting; Computer Assisted Design/Drafting
CAD/CAM	Computer Aided Drafting/Design; Computer Aided Manufacturing
CADD	Computer Aided Design/Drafting; Computer Assisted Design/Drafting
CALS	Computer Aided Logistics Support Processing System
CAM	Computer Aided Mapping; Cartographic Automatic Mapping; Computer Assisted
CCM	Cross-Country Movement Map
CCN	Continental Control Network
CCT	Computer Compatible Tape
CD-I	Compact Disc-Interactive
CD-ROM	Compact Disc-Read Only Memory
CDB	Cartographic Data Base
CDR	Critical Design Review
CE	Circular Error
CEDD	Committee on the Exchange of Digital Data
CEP	Circular Error Probable
CERL	Construction Engineering Research Laboratory
CG	City Graphic
CGS	Centimeter-Gram-Second

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CHIPS	Classified Hydrographic Information Processing System
CHUM	Chart Update Manual
CIO	Conventional International Origin
CLD	Center Line Data
CM	Configuration Management
CMAS	Circular Map Accuracy Standard
CO	Coastal Chart
COMIREX	Committee on Imagery Requirements and Exploitation
CONOPS	Concept of Operations
COTAT	Correlation Tracking and Triangulation
COTS	Commercial Off-the-Shelf
CPE	Circular Probable Error
CRG	City Route Graphic
CUT	Coordinated Universal Time
DAFIF	Digital Aeronautical Flight Information File
DAPPS	DMA Automated Publication Production System
DBDB	Digital Bathymetric Data Base
DC/S	Digital Comparator Segment
DCAC	Digital Concepts Analysis Center
DCDB	Digital Cities Data Base
DCR	Design Concept Review
DCW	Digital Chart of the World
DDM	Distributed Data Management
DE/S	Data Extraction Segment
DEM	Digital Elevation Matrix; Digital Elevation Model
DFAD	Digital Feature Analysis Data
DGI	Digital Geographic Information
DGIWG	Digital Geographic Information Working Group
DHI	Defense Hydrographic Initiative
DHQ	Mean Diurnal High Water Inequality
DI/S	Data Integration Segment
DIGEST	Digital Geographic Information Exchange Standard
DLG	Digital Line Graph
DLG-E	Digital Line Graph-Enhanced
DLMB	Digital Landmass Blanking System
DLMS	Digital Landmass System
DLQ	Mean Diurnal Low Water Inequality
DMA	Defense Mapping Agency
DMAAC	Defense Mapping Agency Aerospace Center
DMACSC	Defense Mapping Agency Combat Support Center
DMAHTC	Defense Mapping Agency Hydrographic/Topographic Center
DMARC	Defense Mapping Agency Reston Center
DMASC	Defense Mapping Agency Systems Center
DMATSC	Defense Mapping Agency Technical Services Center
DME	Distance Measuring Equipment
DMS	Defense Mapping School
DNC	Digital Nautical Chart
DoD	Department of Defense
DoD MC&G Programs	Department of Defense Mapping, Charting, & Geodesy Requirements and
DoD WGS	Department of Defense World Geodetic System
DOS	Disk Operating System
DPPDB	Digitized/Digital Point Positioning Data Base
DPS	Digital Production System
DRLMS	Digital Radar Landmass Simulator
DS/S	Data Services Segment
DTAD	Digital Terrain Analysis Data

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

DTC	Data Transfer Cartridge
DTD	Document Type Definition
DTED	Digital Terrain Elevation Data
DTM	Digital Terrain Model; Digital Terrain Matrix; Data Transfer Modules
DVOF	Digital Vertical Obstruction File
ECDIS	Electronic Chart Display and Information System
EDC	EROS Data Center
EDME	Electronic Distance Measuring Equipment
EFL	Effective Focal Length
EGA	Enhanced Graphics Adapter
EMD	Electronic Map Data; Electronic Map Display
EOD	Erasable Optical Disc
EPI	Electronic Position Indicator
ERDAS	Earth Resources Data Analysis System
EROS	Earth Resources Observation System
ESNA	Electrical Survey Net
ETL	See TEC
FAC	Feature Attribute Code
FACC	Feature and Attribute Coding Catalog
FACS	Feature/Attribute Coding Standard
FADT	Feature Analysis Data Table
FAF	Fast Access Format
FAT	Factory Acceptance Test
FE/S	Feature Extraction Segment
FGDC	Federal Geographic Data Committee
FICCDC	Federal Interagency Coordinating Comittee on Digital Cartography
FID	Feature Identification Data
FIPS	Federal Information Processing Standards
FIPS PUBS	Federal Information Processing Standards (FIPS) Publication
FOOB	FIREFINDER Operational Data Base
G&G	Geodetic and Geophysical
GAT	Greenwich Apparent Time
GB	Gigabyte
GBF/DIME	Geographic Base File/Dual-Independent-Map-Encoding
GCT	Greenwich Civil Time
GEOREF	Geographic Reference
GGF	Ground Gain Forward
GGS	Ground Gained Sideways
GHA	Greenwich Hour Angle
GIS	Geographic Information System
GKS	Graphic Kernal System
GLORIA	Geological Long Range Inclined ASDIC
GM	Gravity Model
GMT	Greenwich Mean Time
GNC	Global Navigation Chart
GNPS	Geographic Names Processing System
GOSIP	Government Open Systems Interconnection Protocol
GP	Gridded Photograph
GPS	Global Positioning System
GRASS	Geographic Resources Analysis Support System
GRS 80	Geodetic Reference System 80
GSD	Ground Scale Distance; Ground Sample Distance
GST	Greenwich Sidereal Time
GUI	Graphical User Interface

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

HA	Harbor and Approach Chart
HDDT	High Density Digital Tape
HDT	High Density Tape
HE/S	Hardcopy Exploitation Segment
HHW	Higher High Water
HHWI	Higher High Water Interval
HIHAN	Hydrographic Information Handling
HITS	Hull Integrity Test Site Chart
HLW	Higher Low Water
HLWI	Higher Low Water Interval
HSDC	High Speed Digital Chart
HW	High Water; Hardware
HWR&C	High Water Full and Change
ICS	Interactive Compilation System
IFOV	Instantaneous Field of View
IGES	Initial Graphic Exchange Specification
IGSN 71	International Gravity Standardization Net 1971
IHO	International Hydrographic Organization
ILW	International Low Water
IMC	Image Motion Compensation
IMO	International Maritime Organization
IMU	Inertial Measurement Unit
IOC	Initial Operating Capability
ISO	International Standards Organization
ITD	Interim Terrain Data
JNC	Jet Navigation Chart
JOG	Joint Operations Graphic
JOG/A	Joint Operations Graphic/Air
JOG/G	Joint Operations Graphic/Ground
JOG/R	Joint Operations Graphic/Radar
KB	Kilobyte
KBS	Knowledge Based System
LHW	Lower High Water
LHWI	Lower High Water Interval
LLW	Lower Low Water
LOC	Lines of Communication
LOLA	Long Line Azimuth
LORAC	Long Range Accuracy
LORAN	Long Range Navigation
LOROP	Long Range Oblique Photography
LW	Low Water
MACDIF	Map and Chart Data Interchange Format
MAD	Magnetic Anomaly Detection
MADPC	MAD Planning Chart
MB	Megabyte
MBR	Minimum Bounding Rectangle
MC&G	Mapping, Charting & Geodesy
MCGFDES	Mapping, Charting, & Geodesy Feature Data Exchange Standard
MCS	Modern Catalog System
MGRS	Military Grid Reference System
MHHW	Mean Higher High Water

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

MHW	Mean High Water
MHWS	Mean High Water Springs
MIL-STD	Military Standard
MIM	Military Installation Map
MINI-TOPO	Minimally Redundant Topology
MK 85	MARK 85
MK 90	MARK 90
MLLW	Mean Lower Low Water
MLLWS	Mean Lower Low Water Springs
MLW	Mean Low Water
MLWS	Mean Low Water Springs
MMSD	Metric Mapping Support Data
MOE	MAD Operational Effectiveness
MP	Middle Point
MSI	Multispectral Imagery
MSL	Mean Sea Level
MSS	Multispectral Scanner
MTL	Mean Tide Level
MUM	Multi Use Manuscript
NAD 27	North American Datum of 1927
NAD 83	North American Datum of 1983
NAVINFONET	Navigation Information Network
NCGIA	National Center for Geographic Information and Analysis
NFS	Navigational Filmstrip
NIST	National Institute for Standards and Technology
NNSS	Navy Navigation Satellite System
NOS	National Ocean Survey
NRL DET SCC	Naval Research Laboratory Detachment Stennis Space Center
NSC	Non-Submarine Contact
NSWC	Naval Surface Weapons Center
NVPUB	Navigation Publication
ONC	Operational Navigation Chart
OPAREA	Operating Area
OS	Operating System
OSA	Open Systems Architecture
PAPI	Precise Automatic Photogrammetric Intervalometer
PBM	Permanent Benchmark
PBNZC	Precise Bathymetric Navigation Zone Chart
PC	Pilotage Chart; Point of Curvature
PCC	Point of Compound Curvature
PDMM	Pulse Doppler Map Matching
PDR	Preliminary Design Review
PG/S	Product Generation Segment
PHIGS	Programmer's Hierarchical Interactive Graphics System
PI	Point of Intersection
PIP	Precise Installation Position
PM/S	Production Management System
PMD	Projected Map Display
POSIX	Portable Operating System Interface
PPD	Point Positioning Data
PPDB	Point Positioning Data Base
PPI	Plan Position Indicator
PRSL	Precise Radar Significant Location
PSR	Pre-Ship Readiness Review
PT	Point of Tangency

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PTADB	Planning Terrain Analysis Data Base
PVC	Point of Vertical Curve
PVI	Point of Vertical Tangent
PVOD	Probabilistic Vertical Obstruction Data
PVT	Point of Vertical Tangent
PZT	Photographic Zenith Tube
R&PRG	DoD MC&G Requirements and Programs Review Group
RDT&E	Research, Development, Test and Evaluation
RF	Representative Fraction
RFP	Radar Fix Point
RGB	Red, Green, Blue
RIM	Radar Intelligence Map
RL	Rome Laboratories
RML	Random Microlenticular
RMP	Raster Map Product
ROCS	Range Only Correlation System
RPS	Raster Product Standard
RRC	Radar Return Code
RSAC	Radar Significance Analysis Code
RSPL	Radar Significant Power Line
RTAD	Relocatable Target Assessment Data
SA	Special Area
SA/S	Source Acquisition Segment
SAGA	Short Arc Geodetic Adjustment
SAR	Synthetic Aperture Radar
SCP	Secondary Control Point
SDTS	Spatial Data Transfer Standard; Spatial Data Transfer Specification
SECOR	Sequential Collation of Range
SGD	Symbolized Graphic Data
SGML	Standard Generalized Markup Language
SHA	Sidereal Hour Angle
SHIRAN	S-Band High-Precision Short-Range Navigation
SHORAN	Short Range Navigation
SINS	Ships Inertial Navigation System
SIS	Sensor Image Simulator; Standard Indexing System
SLAR	Side-Looking Airborne Radar
SLF	Standard Linear Format
SMAC	Scene Matching Area Correlator
SONAR	Sound Navigation
SOSUS	Sound Surveillance Systems Chart
SP/S	Source Preparation Segment
SPOT	Système Probatoire d'Observation de la Terre
SQL	Structured Query Language; Standard Query Language
SWATH	Small Wetted Area Twin Hull
TA	Terrain Analysis
TAP	Terrain Analysis Product
TAS	Terrain Analysis System
TB	Terabyte
TBM	Temporary Benchmark
TCOC	Transverse Cylindrical Orthomorphic Chart
TEC	Topographic Engineering Center
TERCOM	Terrain Contour Matching
TES/EMPS	Terrain Edit Station/Elevation Matrix Processing System
TIGRIS	Topologically Integrated Geographic and Resource Information System
TIN	Triangulated Irregular Network

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TLM	Topographic Line Map
TM	Thematic Mapper; Topographic Map
TMP	Target Materials Program
TP	Turning Point
TPC	Tactical Pilotage Chart
TPR	Terrain Profile Recorder
TS	Tangent to Spiral; Tesselated Sheroid
TTADB	Tactical Terrain Analysis Data Base
TTD	Tactical Terrain Data
U.S.G.S.	U.S. Geological Survey
UPS	Universal Polar Stereographic
UR	Universal Rectifier
UR/S	Universal Rectifier Segment
URA	User Requirement Analysis
USATEC	U.S. Army Topographic Engineering Center
USR	Universal Space Rectangular
UT	Universal Time
UTC	Universal Time Coordinated
UTM	Universal Transverse Mercator
VABM	Vertical Angle Benchmark
VD	MC&G Video Disc
VGA	Video Graphics Array
VMP	Vector Map Product
VOD	Vertical Obstruction Data
VPF	Vector Product Format
VPPDB	Video Point Positioning Data Base
VPR	Virtual PPI Reflectoscope
VPS	Vector Product Standard
WDBII	World Data Bank II
WGS	World Geodetic System
WMED	World Mean Elevation Data
WMRM	Write Many, Read Many
WORM	Write Once, Read Many
WSC	DMA Warrior Support Center
WVS	World Vector Shoreline

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5. MC&G ORGANIZATIONS

American National Standards Institute (ANSI)—An association formed by government and industry to produce and disseminate widely used industrial standards.

Board on Geographic Names (BGN)—An agency of the U.S. Government, first established by Executive Order in 1890 and currently functioning under Public Law 242-80, 25 July 1947. Nine departments and agencies enjoy Board membership. DoD is one of these members; DMA provides the BGN Executive Secretary and a staff for work on the Board's Foreign Names Committee. Conjointly with the Secretary of the Interior, the Board provides for "uniformity in geographic nomenclature and orthography throughout the Federal Government." It develops policies and romanization systems under which names are derived and it standardizes geographic names for use on maps and in textual materials.

Committee on Imagery Requirements and Exploitation (COMIREX)—The U.S. intelligence community's management committee on imagery requirements and exploitation.

Committee on the Exchange of Digital Data (CEDD)—An international hydrographic organization working group tasked with developing standards for the exchange of hydrographic and bathymetric data.

Construction Engineering Research Laboratory (CERL)—Laboratory under the U.S. Army Corps of Engineers. CERL developed GRASS, a raster GIS. See also **GRASS; GIS.**

Defense Mapping Agency (DMA)—A Department of Defense agency in charge of enhancing national security and supporting the Office of Secretary of Defense, Joint Chiefs of Staff, Unified and Specified Commands, Military Departments, and other users. DMA produces and distributes mapping, charting and geodetic (MC&G) products, services and training, and advises on such matters. It provides nautical charts and marine navigational data to worldwide merchant marine and private vessel operators. The agency maintains liaison with civil agencies and national and international mapping, charting and geodetic activities.

Defense Mapping School (DMS)—A Defense Mapping Agency school designed to train DoD, military, civilian, and allied personnel in MC&G techniques, procedure, and products.

DMA Aerospace Center (DMAAC)—A DMA production center located in St. Louis, MO.

DMA Combat Support Center (DMACSC)—A DMA component responsible for distribution of DMA products to the user community. There are several depots located around the world.

DMA Hydrographic/Topographic Center (DMAHTC)—A DMA production center located in Brookmont, MD.

DMA Systems Center (DMASC)—The DMA component responsible for research and development issues, including developing and integrating new MC&G products and production systems.

DMA Warrior Support Center (WSC)—The DMA facility which provides an environment for current, future, and potential DMA customers to participate in the definition, development, test and evaluation and application of DMA MC&G products and data. The focus of the WSC is to conduct research and development and prototyping activities that promote interoperability and MC&G standardization. Additionally, the WSC promotes the optimum production and use of DMA products, data, and services for combat forces by responding directly to their needs.

DoD MC&G Requirements and Program Review Group (R&PRG)—A Department of Defense (DoD) review group comprised of representatives from the DoD agencies and the military services that advises Assistant Secretary of Defense (ASD) on MC&G matters.

Federal Geographic Data Committee (FGDC)—The InterAgency steering committee which superseded the FICCDC. The FGDC provides oversight and policy guidance on the Office of Management and Budget's (OMB) Circular A-16 activities. In particular, the committee promotes the coordinated development, use, sharing, and dissemination of surveying mapping, and related spatial data. Circular A-16 assigns responsibility to various agencies to provide leadership in the coordination of certain categories of surveying,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

mapping, and related spatial data. See also **Federal InterAgency Coordinating Committee on Digital Cartography.**

Federal InterAgency Coordinating Committee on Digital Cartography (FICCDC)—An InterAgency steering committee established in 1983 by the Office of Management and Budget. The purpose is to recommend procedures and programs which would facilitate the coordination of Federal agencies' digital cartographic activities, and establish and promulgate standards and specifications for the production of digital cartographic data. Name changed in 1990 to the Federal Geographic Data Committee. See also **Federal Geographic Data Committee.**

International Hydrographic Organization (IHO)—The coordinating body for international nautical charting practices and procedures. Establishes minimum standards and specifications for the production of nautical charts and the collection of related data. Membership is open to all countries of the world.

International Maritime Organization (IMO)—The Specialized Agency for the United Nations responsible for maritime safety and efficiency of navigation. It provides machinery for cooperation among governments in the field of governmental regulations and practices relating to technical matters of all kinds affecting shipping engaged in international trade.

International Standards Organization (ISO)—An international organization responsible for digital standards for media, format, etc.

National Center for Geographic Information and Analysis (NCGIA)—A consortium of three universities (The University of California at Santa Barbara, The State University of New York at Buffalo, and The University of Maine at Orono) receiving funding from the National Science Foundation (NSF) to focus research and model educational curriculums in spatial analysis and spatial statistics; spatial relationships and data base structures; artificial intelligence and expert systems; visualization; and social, economic and institutional issues in the field of geographic information systems (GIS).

National Ocean Service (NOS)—Major component of NOAA, charged with maintaining the National Geodetic Reference System; producing nautical and aeronautical charts and related publications surveying and monitoring environmental resources, including coastal pollution; researching the physical characteristics of the oceans; and managing marine estuaries and sanctuaries. Formerly National Ocean Survey.

Naval Environmental Prediction Research Facility (NEPRF, NOARL)—A research and development organization under the Navy during the 1980s. NEPRF, NORDA, and INO were merged to form the Naval Oceanographic and Atmospheric Research Laboratory (NOARL) in 1989.

Naval Oceanographic and Atmospheric Research Laboratory (NOARL)—The Navy's lead laboratory for research and development in mapping, charting and geodesy (MC&G). NOARL is actively involved in applying digital MC&G data to the support of naval weapons systems and in conducting research to improve these data. NOARL was created in 1989 as a merger of three research units: Naval Ocean Research and Development Activity, the Institute for Naval Oceanography, and the Naval Environmental Prediction Research Facility.

Naval Surface Warfare Center (NSWC)—An industrially funded Navy facility located in White Oak, MD, and Dahlgren, VA, which conducts research and development in a variety of warfare disciplines including MC&G.

Rome Laboratories (RL)—The principal organization charged with conducting Air Force research and development programs related to C^3I (command, control, communications and intelligence). RL mission areas include communications, electromagnetic guidance and control, surveillance of ground and aerospace objects, intelligence data handling, information systems technology, electronic reliability, maintainability, and compatibility.

U.S. Army Topographic Engineering Center (TEC or USATEC)—Army laboratories located in Fort Belvoir, VA, which support the Army's modern weapons and command and control systems through research and development in mapping, charting, terrain analysis, geodesy, remote sensing, point positioning, surveying and land navigation. TEC also conducts research on environmental effects on military equipment, developing topographic support systems, and terrain analysis products.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

See also **Topographic Engineering Center.**

Warrior Support Center (WSC)—See **DMA Warrior Support Center.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

INDEX

A, 3
Abbreviations, 285
Acronyms, 285
B, 26
C, 35
D, 63
E, 79
F, 91
G, 101
H, 116
I, 125
Initials, 285
J, 135
K, 136
L, 137
M, 151
MC & G Organizations, 292
N, 168
O, 174
P, 182
Q, 206
R, 207
S, 223
T, 251
U, 271
V, 274
W, 279
X, 282
Y, 283
Z, 284

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
DMA - MP

Preparing Activity:
DMA - MP

Review Activities:
Air Force - 09
Army - PO
Navy - NO, MC

Project number:
(MCGT-0065)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-HDBK-850	2. DOCUMENT DATE (YYMMDD) 940121

3. DOCUMENT TITLE: Military Handbook Glossary of Mapping, Charting, and Geodetic Terms

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Mapping Agency ATTN: TIJ, ST A-10	b. TELEPHONE (Include Area Code) (1) Commercial (703) 285-9238	(2) AUTOVON 356-9238
c. ADDRESS (Include Zip Code) 8613 Lee Highway Fairfax, VA 22031-2137	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete* 198-29

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.